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Par :
Rémi Barritault

**On the unitary representations of
certain locally Roelcke precompact
permutation groups**

Sous la direction de :
Todor Tsankov, Université Claude Bernard Lyon 1
Christian Rosendal, University of Maryland

Résumé en français

Cette thèse est consacrée à l'étude des représentations unitaires dans un cadre non-localement compact. Dans ce contexte, la première étape consiste à trouver des outils pour pallier l'absence de mesure de Haar. A cet effet, je m'intéresse tout particulièrement aux groupes de permutations. D'une part, contrairement à la classe plus générale des groupes polonais, ces groupes ont une théorie des représentations riche et moins pathologique. D'autre part, cela permet l'entrée en jeu de la théorie des modèles. En effet, le groupe des automorphismes d'une structure du premier ordre est un groupe de permutations, ce qui ouvre la voie à de nombreuses interactions.

Les groupes de permutations dont les représentations sont les mieux comprises sont les groupes oligomorphes. Dans un premier temps, je montre que la théorie des représentations d'un tel groupe est suffisamment riche pour permettre de le reconstruire complètement. Plus précisément, j'adapte à ce cadre la dualité de Tannaka-Krein, originellement développée pour les groupes compacts.

Ensuite, un aspect central de mes travaux est la tâche de la classification des représentations unitaires d'un groupe donné. Dans un travail collaboratif, nous introduisons la notion de dissociation, inspirée du cas oligomorphe. Nous avons pu généraliser des théorèmes de classification et de rigidité des représentations unitaires aux groupes dissociés. Entre autres, nous avons classifié les représentations unitaires des groupes des isométries des espaces d'Urysohn. Nous étudions aussi les conséquences de la dissociation en théorie ergodique, où cette propriété admet une reformulation équivalente.

Enfin, je définis un analogue des groupes universels de Burger-Mozes. Agissant sur l'arbre régulier de valence infinie dénombrable, ces groupes sont obtenus en prescrivant une action locale. Sous l'hypothèse que cette dernière est 2-transitive et oligomorphe, je classifie les représentations irréductibles des groupes universels.

Abstract

This thesis is devoted to the study of unitary representations in a non-locally-compact setting. In this context, the first step is to find alternative tools to cope with the lack of a Haar measure. To this end, I am particularly interested in permutation groups. On the one hand, unlike the more general class of Polish groups, these groups have a rich and less pathological representation theory. On the other hand, this allows model theory to come into play. Indeed, the automorphism group of a first-order structure is a permutation group, which opens the door to numerous interactions.

The permutation groups whose representations are best understood are the oligomorphic groups. First, I show that the representation theory of such a group is sufficiently rich to allow for its complete reconstruction. More specifically, I adapt Tannaka-Krein duality, initially developed for compact groups, to this setting.

Next, a central aspect of my work is the classification of the unitary representations of a given group. In a collaborative project, we introduce the notion of dissociation, inspired by the oligomorphic case. We have been able to generalize classification and rigidity theorems for unitary representations to dissociated groups. Among other things, we have classified the unitary representations of the isometry groups of Urysohn spaces. We are also studying the consequences of dissociation in ergodic theory, where this property admits an equivalent reformulation.

Finally, I introduce a non-locally compact analogue of the so-called universal Burger-Mozes groups. Acting on the regular tree of countably infinite arity, these groups are obtained by prescribing a local action. Under the assumption that this local action is 2-transitive and oligomorphic, I classify the irreducible representations.

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Introduction

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In this thesis, we study unitary representations of some non-locally-compact topological groups. In this setting, many of the crucial tools used in the classical theory, such as the Haar measure, do not carry over to the class of all Polish groups. In order to keep enough control over the representation theory of the groups at play, we will restrict our attention to closed permutation groups. These Polish groups can all be seen as arising from model theory, which will provide us with alternative tools for the study of their unitary representations. We will also work with a relaxed notion of compactness called Roelcke precompactness which retains many of its geometric properties.

The aim of this introduction is to present the groups in question and go over some landmark results in abstract harmonic analysis. Throughout, we will compare the situation in two different settings: the classical setting of locally compact groups and the setting of infinite closed permutation groups. This will also provide sufficient context to discuss the main results of this thesis.

1.1 Locally Roelcke-precompact closed permutation groups

1.1.1 Polish groups and permutation groups

A metric d on a topological group G is *compatible* if it generates the topology of G and *left-invariant* if $d(gh, gh') = d(h, h')$ for all $g, h, h' \in G$. Compatible metrics will always exist for the groups we will consider as per the following characterization. See [Ber74, Th 6.3] for a proof of the following fundamental result.

Theorem 1.1.1 (Birkhoff-Kakutani). *Let G be a Hausdorff topological group which admits a countable basis of neighborhoods of the identity. Then G is metrizable by a left-invariant metric.*

A *Polish group* is a topological group whose topology is separable and generated by some complete metric. All Polish groups are Hausdorff and second countable and it is a classical fact that a subgroup of a Polish group is Polish if and only if it is closed [Kec95, Ex. 9.6]. **Throughout this introduction, we will make the standing assumption that all groups considered are Hausdorff and second countable.** In particular, they will always be metrizable by the above theorem.

Remark 1.1.2. While the Birkhoff-Kakutani theorem provides both left-invariant metrics and right-invariant metrics, bi-invariant compatible metrics do not always exist. Similarly, if G is a Polish group, there exist a left-invariant compatible

metric and a complete compatible metric. However, there usually does not exist a compatible metric that is both left-invariant and complete.

Polish groups are a natural generalization of locally compact second countable groups. Numerous examples of groups naturally appearing in various domains of mathematics fall into the class of non-locally-compact Polish groups. To cite only a few examples, the following groups admit a Polish group topology:

1. locally compact groups,
2. the unitary group $\mathcal{U}(\mathcal{H})$ of the separable infinite dimensional Hilbert space $\mathcal{H}$,
3. the group $\text{Aut}([0, 1], \lambda)$ of bi-measurable bijections of $[0, 1]$ that preserve the Lebesgue measure λ , identified up to null-sets,
4. the group $\mathfrak{S}_\infty$ of all the bijections of a countable set,
5. the group $\text{Aut}(\mathbf{T})$ of all the automorphisms of the regular tree of countable infinite arity,
6. the homeomorphism group of any compact metrizable space,
7. the isometry group of any separable and complete metric space,
8. full groups of countable Borel equivalence relations.

In this thesis, we will be interested in studying the unitary representations of such groups. This type of study, known as abstract harmonic analysis, is typically conducted on locally compact groups. Jumping from the class of locally compact groups to that of Polish groups is a huge conceptual leap. Many of the tools that are crucial in the classical case are lost in the process and large parts of the classical theory fail for the class of all Polish groups. We will thus have to restrict our attention to better disposed intermediate classes of groups.

One of the main issues is the absence of Haar measure, a central tool used both for constructing and analyzing representations. This role will be fulfilled by another structural specificity: a neighborhood basis at the identity consisting of open subgroups. A topological group that has one is called *non-Archimedean*.

Let Ω be a countable set. We will denote by $\text{Sym}(\Omega)$ the group of all the permutations of Ω . Viewing Ω as a discrete topological space and $\text{Sym}(\Omega)$ as a subset of Ω^Ω , we endow $\text{Sym}(\Omega)$ with the product topology. It is called the *permutation group topology* and makes $\text{Sym}(\Omega)$ a Polish group.

Definition 1.1.3. A *closed permutation group* on a countable set Ω is a closed subgroup $G \leq \text{Sym}(\Omega)$. For every subset A of Ω , we will denote by G_A the *pointwise stabilizer* of A and by $G_{\{A\}}$ the *setwise stabilizer* of A , that is:

$$G_A := \{g \in G : \forall a \in A, g(a) = a\} \quad \& \quad G_{\{A\}} := \{g \in G : g(A) = A\}.$$

Since we require our permutation groups to be closed, their topology as a subgroup of $\text{Sym}(\Omega)$ is Polish. Moreover, given $G \leq \text{Sym}(\Omega)$, the subgroups G_A where A ranges over the finite subsets of Ω are open and form a basis for

the neighborhoods of the identity in G hence the topological group G is non-Archimedean. The converse is true: every non-Archimedean Polish group admits a presentation as a closed permutation group. See [BK96, Th. 1.5.1] for a proof of the following.

Proposition 1.1.4. *Let G be a Polish group and let Ω be an infinite countable set. The following properties are equivalent.*

- (i) G is non-Archimedean.
- (ii) There exists a topological group embedding with closed image $G \hookrightarrow \text{Sym}(\Omega)$.

Remark 1.1.5. We insist on the fact that a closed permutation groups is not just a non-Archimedean Polish group, but one together with a specified embedding into $\text{Sym}(\Omega)$ for some countable set Ω . Every non-Archimedean Polish group has many very different presentations as a permutation group.

1.1.2 Roelcke precompactness and oligomorphicity

Whether local or global, compactness not only ensures the existence of a Haar measure but also smooths out the geometry. While our groups will typically not be locally compact, compactness will not be fully discarded in our study. Instead, we will consider the following weaker notion.

Definition 1.1.6. Let G be a topological group. A subset E of G is *Roelcke precompact* if for every open neighborhood U of the identity in G there exists a finite subset $F \subseteq G$ such that $E \subseteq UFU$. The group G is *Roelcke precompact* if it is Roelcke precompact as a subset of itself and it is *locally Roelcke-precompact* if it admits a Roelcke precompact non-empty open subset.

Roelcke precompactness retains many of the geometrical property of compactness. In particular, the representation theory of Roelcke precompact closed permutation groups is very similar to that of compact groups, see Theorems 1.2.11 and Theorem 1.2.23. This notion is not of any interest in a locally compact setting: a locally compact group is Roelcke precompact if and only if it is compact. This is a direct consequence of the fact that, in a topological group, products of compact subsets are compact. While products of Roelcke precompact subsets fail to be Roelcke precompact in general, they are if the group is locally Roelcke precompact [Zie21, Prop. 12].

Example 1.1.7. The following topological groups are Roelcke-precompact Polish groups:

1. compact groups,
2. the unitary group $\mathcal{U}(\mathcal{H})$ of the separable Hilbert space $\mathcal{H}$ [Usp98],
3. the groups $\text{Homeo}^+([0, 1])$ and $\text{Homeo}^+(\mathbb{S}^1)$ of order-preserving homeomorphisms of the unit segment and the unit circle respectively [Ros22, Ex. 3.12],
4. the group $\text{Aut}([0, 1], \lambda)$ of bi-measurable bijections of the unit segment which preserve the Lebesgue measure, identified up to null-sets [Gla12],

5. $\text{Sym}(\Omega)$ [RD81, Ex. 9.14].
6. the group $\text{Aut}(\mathbb{Q}, <)$ of order-preserving bijections of the rationals [Tsa12, Th. 2.4].

The following groups are locally Roelcke-precompact Polish groups:

1. Roelcke precompact Polish groups,
2. locally compact groups,
3. the groups $\text{Homeo}_{\mathbb{Z}}^+(\mathbb{R})$ of order-preserving homeomorphisms of the real line which commute with integral shifts [Ros22, Ex. 3.12],
4. the isometry group $\text{Iso}(\mathbb{H}^\infty)$ of the separable hyperbolic space $\mathbb{H}^\infty$,
5. the automorphism group $\text{Aut}(\mathbf{T})$ of the regular tree $\mathbf{T}$ of countable infinite arity (Theorem 5.2.16 of Chapter 5),
6. the isometry group of integral Urysohn space $\mathbf{U}_{\mathbb{Z}}$, for the permutation group topology (Lemma 3.4.3 of Chapter 3).

Roelcke precompactness has a dynamical characterization which, specified to permutation groups, yields the following notion.

Definition 1.1.8. An action of a group G on a countable set Ω is *oligomorphic* if for every $n \in \mathbb{N}$, the diagonal action $G \curvearrowright \Omega^n$ has only finitely many orbits. A permutation group G on a countable set Ω is *oligomorphic* if the canonical action $G \curvearrowright \Omega$ is oligomorphic.

Proposition 1.1.9. *Every oligomorphic permutation group is Roelcke precompact.*

Proof. Let $G \leq \text{Sym}(\Omega)$ be an oligomorphic permutation group on a countable set Ω . Let U be an open neighborhood of the identity in G . There exists $A \subseteq \Omega$ finite such that $G_A \subseteq U$. Write $A = \{a_1, \dots, a_n\}$ in an injective way and let $\bar{a} := (a_1, \dots, a_n) \in \Omega^n$. Consider the diagonal action $G \curvearrowright \Omega^{2n}$. For all $g, h \in G$, we have $G_A g G_A = G_A h G_A$ if and only if the orbits $G \cdot (\bar{a}, g(\bar{a}))$ and $G \cdot (\bar{a}, h(\bar{a}))$ are the same. Since $G \curvearrowright \Omega$ is oligomorphic, there are only finitely many such orbits. There thus exists $F \subseteq G$ finite such that $G = G_A F G_A$. A fortiori, $G = U F U$. $\square$

The above result almost has a converse: every Roelcke precompact non-Archimedean Polish group is the inverse limit of an inverse sequence of oligomorphic closed permutation groups [Tsa12, Th. 2.4].

Example 1.1.10. The following actions are oligomorphic:

1. $\text{Sym}(\Omega) \curvearrowright \Omega$,
2. $\text{Aut}(\mathbb{Q}, <) \curvearrowright \mathbb{Q}$,
3. $\text{Aut}(\mathbf{T}, x) \curvearrowright B_{\mathbf{T}}(x, n)$ where x is a vertex of $\mathbf{T}$, $n \in \mathbb{N}$, $\text{Aut}(\mathbf{T}, x)$ is the group of all the automorphisms of $\mathbf{T}$ that fix x and $B_{\mathbf{T}}(x, n)$ is the ball of radius n around x in $\mathbf{T}$.

While permutation groups have many open subgroups, an important feature of Roelcke precompact ones is that they do not have too many.

Proposition 1.1.11 ([Tsa12]). *Let G be a Roelcke precompact non-Archimedean group. The number of distinct open subgroups of G is countable.*

Proof. Let $(U_n)_{n \in \mathbb{N}}$ be a countable basis of neighborhoods of the identity consisting of open subgroups and let $n \in \mathbb{N}$. Every open subgroup of G which contains U_n is a finite union of double cosets of U_n by Roelcke precompactness of G . The double cosets of U_n being disjoint open subsets of G , there can only be countably many of them by the standing assumption of second countability. In particular, there are only countably many open subgroups containing U_n . Since every open subgroup of G contains some U_n , the proposition follows. $\square$

1.1.3 Model Theory: $\aleph_0$ -categoricity and Fraïssé limits

Our main source of both examples of closed permutation groups and tools for studying their representations is model theory. Let us recall the basic definition.

A *signature* is a collection $\mathcal{L}$ of symbols. The symbols come in two separate types, *relation symbols* and *function symbols*. Each symbol also has an associated integer, called its *arity*. In other words, $\mathcal{L} = ((R_i, n_i)_{i \in I}, (f_j, n_j)_{j \in J})$ where for all $i \in I$ and $j \in J$, R_i is a relation symbol of arity n_i and f_j is a function symbol of arity n_j . We will mostly consider *relational signatures*, i.e. signature with no function symbol ($J = \emptyset$). An $\mathcal{L}$ -structure $\mathfrak{M}$ consists of a set, called the *domain* of $\mathfrak{M}$, together with interpretations of each of the symbols in $\mathcal{L}$. More precisely, $\mathfrak{M} = (M, (R_i^{\mathfrak{M}})_{i \in I}, (f_j^{\mathfrak{M}})_{j \in J})$ where M is a set and for all $i \in I$ and $j \in J$, $R_i^{\mathfrak{M}}$ is a subset of M^{n_i} and $f_j^{\mathfrak{M}}$ is a function $M^{n_j} \rightarrow M$. In practice, we will identify $\mathfrak{M}$ and its domain, writing e.g. $x \in \mathfrak{M}$ and $f_j^{\mathfrak{M}}: \mathfrak{M}^{n_j} \rightarrow \mathfrak{M}$ instead of $x \in M$ and $f_j^{\mathfrak{M}}: M^{n_j} \rightarrow M$.

A *substructure* of $\mathfrak{M}$ is a subset $N \subseteq M$ such that $f_j^{\mathfrak{M}}(N^{n_j}) \subseteq N$ for every $j \in J$. It is an $\mathcal{L}$ -structure when equipped with the restrictions of the interpretations of $\mathfrak{M}$. It is *finitely generated* if there exist a finite subset $F \subseteq N$ such that the smallest substructure of $\mathfrak{M}$ containing F is N . If $\mathcal{L}$ is relational, every subset of M is a substructure of $\mathfrak{M}$.

An *embedding* between two $\mathcal{L}$ -structures $\mathfrak{M}$ and $\mathfrak{N}$ is an injective map $\phi: \mathfrak{M} \rightarrow \mathfrak{N}$ such that:

$$\left[x \in R_i^{\mathfrak{M}} \iff \phi(x) \in R_i^{\mathfrak{N}} \right] \quad \& \quad \left[\phi(f_j^{\mathfrak{M}}(y)) = f_j^{\mathfrak{N}}(\phi(y)) \right]$$

for all $i \in I, x \in \mathfrak{M}^{n_i}$ and $j \in J, y \in \mathfrak{M}^{n_j}$. An *isomorphism* is a surjective embedding and an *automorphism* of $\mathfrak{M}$ is an isomorphism $\mathfrak{M} \rightarrow \mathfrak{M}$. The strength of this formalism is that it captures very different situations under the same terminology.

Example 1.1.12.

1. The signature of *graphs* is the relational signature $\mathcal{L}_{\text{Graph}} := (\sim, 2)$ where $\sim$ is a relation of arity 2. Every graph $G = (V, E)$ with vertex set V and edge set E is an $\mathcal{L}_{\text{Graph}}$ -structure by interpreting $\sim$ as the edge relation in G , i.e.

$\sim^G = E$. A substructure of G is exactly an *induced subgraph* i.e. a subgraph of the form $(V', E \cap (V' \times V'))$ for some $V' \subseteq V$. The key point is that one cannot discard any edge between vertices of V' . An automorphism of G as an $\mathcal{L}_{\text{Graph}}$ -structure is exactly a graph automorphism.

2. The signature of *ordered set* is the relational signature $\mathcal{L}_< := (<, 2)$ where $<$ is a relation of arity 2. Again, every partially ordered set $(E, <)$ is an $\mathcal{L}_<$ -structure. A substructure of $(E, <)$ is exactly a subset $E' \subseteq E$ together with the induced partial order $< \cap (E' \times E')$. An automorphism of $(E, <)$ is exactly an increasing bijection $E \rightarrow E$.
3. The signature $\mathcal{L}_{\text{Group}}$ of groups is the signature $((1, 1), ((\cdot, 2), (-^1, 1)))$ where:
 - 1 is a relation symbol of arity 1,
 - $\cdot$ is a function symbol of arity 2,
 - $^{-1}$ is a function symbol of arity 1.

Every group G can be naturally an $\mathcal{L}_{\text{Group}}$ -structure of domain G by interpreting 1 as the singleton containing the neutral element of 1_G of G , $\cdot$ as the group multiplication $G \times G \rightarrow G$ and $^{-1}$ as the inversion map $G \rightarrow G$. A substructure of G is exactly a subgroup of G and an automorphism of G as an $\mathcal{L}_{\text{Group}}$ -structure is exactly an automorphism of G as a group.

4. The signature of rational metric spaces is the relational signature $\mathcal{L}_{\mathbb{Q}} := (d_q, 2)_{q \in \mathbb{Q}}$ where d_q is a relation symbol of arity 2 for every $q \in \mathbb{Q}$. A metric space (X, d) such that $d(X \times X) \subseteq \mathbb{Q}$ is naturally an $\mathcal{L}_{\mathbb{Q}}$ -structure of domain X by interpreting d_q as $d_q^X := \{(x, y) \in X : d(x, y) = q\}$ for every $q \in \mathbb{Q}$. A substructure of X is a subset of X together with the induced metric. An automorphism of X as an $\mathcal{L}_{\mathbb{Q}}$ -structure is exactly an isometry of (X, d) .

Every countable structure $\mathfrak{M}$ comes equipped with the permutation group $\text{Aut}(\mathfrak{M}) \leq \text{Sym}(\mathfrak{M})$ of all the automorphisms of $\mathfrak{M}$, called the *automorphism group* of $\mathfrak{M}$. It is easily seen that $\text{Aut}(\mathfrak{M})$ is always closed in $\text{Sym}(\mathfrak{M})$ hence a closed permutation group. Conversely, if G is a closed permutation group on a countable set Ω , there is countable signature $\mathcal{L}_G$ and an $\mathcal{L}_G$ -structure on Ω such that $G = \text{Aut}(\Omega)$. The construction, called the canonical structure associated to $G \leq \text{Sym}(\Omega)$, goes as follows: for every $n \in \mathbb{N}$, $\mathcal{L}_G$ contains a relation symbol of arity n for each orbit in the diagonal action $G \curvearrowright \Omega^n$.

This simple remark makes available to us all the terminology and tools model theory has to offer. In particular, oligomorphicity is captured by the model theoretic notion of $\aleph_0$ -categoricity. Formally, a countable structure is $\aleph_0$ -categorical if it is fully determined by its first-order theory. To avoid more lengthy definitions, we give a few examples instead of giving a precise definition of the first-order theory of a structure.

Example 1.1.13. Up to isomorphism,

1. the countable set Ω without structure is unique. Its automorphism group is $\text{Sym}(\Omega)$.

2. the rationals $(\mathbb{Q}, <)$ with their usual order form the unique countable linear order which is dense and without end, i.e. such that:

$$[\forall x < y \in \mathbb{Q}, \exists z \in \mathbb{Q}, x < z < y] \quad \& \quad [\forall x \in \mathbb{Q}, \exists y, z \in \mathbb{Q}, y < x < z].$$

3. the countable atomless boolean algebra $\mathcal{A}_\infty$ is unique. Its automorphism group is topologically isomorphic to $\text{Homeo}(\{0, 1\}^\mathbb{N})$.
4. the countable vector space over a finite field $\mathbb{F}$ is unique. Its automorphism group is $GL_\infty(\mathbb{F})$.
5. there exists a unique countable graph $(R, \sim)$ such that for all finite subsets $A, B \subseteq R$ with $A \cap B = \emptyset$, there exists $x \in R \setminus (A \cup B)$ which is adjacent to every element of A but no element of B . The graph $(R, \sim)$ is called the *Rado graph* or the *random graph*.

In each case, we stated a property of the structure, expressible in its natural signature, which makes it unique up to isomorphism. This is $\aleph_0$ -categoricity.

The following classical theorem also readily implies that every finite structure is $\aleph_0$ -categorical. See e.g. [Hod93, 7.3.1] for a proof.

Theorem 1.1.14 (Ryll-Nardzewski). *Let $\mathfrak{M}$ be a countable structure in a countable signature. The following properties are equivalent:*

- (i) $\mathfrak{M}$ is $\aleph_0$ -categorical,
- (ii) the action $\text{Aut}(\mathfrak{M}) \curvearrowright \mathfrak{M}$ is oligomorphic.

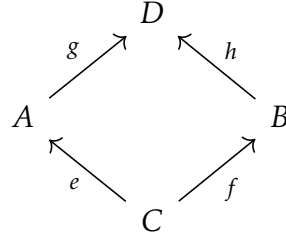
We now describe Fraïssé theory, commonly used for building interesting countable structures, including $\aleph_0$ -categorical ones.

Let $\mathcal{L}$ be a countable signature and let $\mathfrak{M}$ be an $\mathcal{L}$ -structure. The *age* of $\mathfrak{M}$, denoted $\text{Age}(\mathfrak{M})$, is the class of all finitely generated substructures of $\mathfrak{M}$, identified up to isomorphism. In other words, (the isomorphism class of) a finitely generated $\mathcal{L}$ -structure F is in $\text{Age}(\mathfrak{M})$ if and only if F embeds into $\mathfrak{M}$. If $\mathcal{F}$ is a class of isomorphism classes of $\mathcal{L}$ -structures and F is an $\mathcal{L}$ -structure, we will write " $F \in \mathcal{F}$ " instead of "the isomorphism class of F appears in $\mathcal{F}$ ". More generally, we will not make a difference between a class of structures $\mathcal{F}$ and the class $\mathcal{F}'$ of isomorphism classes of elements of $\mathcal{F}$. Whenever cardinality is mentioned, it is about $\mathcal{F}'$. Ages enjoy the following properties.

Definition 1.1.15. Let $\mathcal{F}$ be a class of finitely generated structures in a countable signature $\mathcal{L}$.

- $\mathcal{F}$ has the *Hereditary Property* (HP) if for every $A \in \mathcal{F}$ and every finitely generated substructure $B \subseteq A$, $B \in \mathcal{F}$.
- $\mathcal{F}$ has the *Joint Embedding Property* (JEP) if for all $A, B \in \mathcal{F}$, there exists $C \in \mathcal{F}$ in which both A and B embed.

Fraïssé theory is about going the other way: every countable class $\mathcal{F}$ of finitely generated structures that satisfies HP and JEP is the age of some countable structure [Hod93, Th. 7.1.1]. However, different structures can have the same age. Fraïssé recovered a uniqueness condition by requiring an additional property. We will say that a class $\mathcal{F}$ of finitely generated $\mathcal{L}$ -structures has the *Amalgamation Property* (AP) if for all $A, B, C \in \mathcal{F}$ and all embeddings $e: C \rightarrow A$ and $f: C \rightarrow B$, there exist $D \in \mathcal{F}$ and embeddings $g: A \rightarrow D$ and $h: B \rightarrow D$ such that the following diagram commutes:



If we can moreover ensure that $g(A) \cap h(B) = g \circ e(C) = h \circ f(C)$, we say that F has the *Strong Amalgamation Property* (SAP).

Definition 1.1.16.

1. A *Fraïssé class* is a countable class $\mathcal{F}$ of finitely generated $\mathcal{L}$ -structures that satisfies HP, JEP and AP.
2. A structure $\mathfrak{M}$ is *ultrahomogeneous* if every isomorphism between two finitely generated substructures of $\mathfrak{M}$ extends to an automorphism of $\mathfrak{M}$.

See [Hod93, Th. 7.1.2] for a proof of the following result.

Theorem 1.1.17 (Fraïssé's Theorem). *Let $\mathcal{F}$ be a Fraïssé class. There exists a countable structure $\mathbb{F}$, unique up to isomorphism, which is ultrahomogeneous and such that $\text{Age}(\mathbb{F}) = \mathcal{F}$. The structure $\mathbb{F}$ is called the *Fraïssé limit* of $\mathcal{F}$.*

Example 1.1.18.

1. The Fraïssé limit of finite sets without structure is the countably infinite set Ω , with automorphism group $\text{Sym}(\Omega)$.
2. The Fraïssé limit of finite linear orders is $(\mathbb{Q}, <)$.
3. The Fraïssé limit of finite boolean algebras is the countable atomless algebra $\mathcal{A}_\infty$.
4. The Fraïssé limit of finite graphs is the Rado graph.
5. For every prime number p , the Fraïssé limit of finite fields of characteristic p is the algebraic closure $\overline{\mathbb{F}_p}$ of the field of cardinality p .
6. For every finite field $\mathbb{F}_q$, the Fraïssé limit of finite dimensional vector spaces over $\mathbb{F}_q$ is the countable vector space over $\mathbb{F}_q$.
7. The Fraïssé limit of finite metric spaces with all distances in $\mathbb{Q}$ is called the *rational Urysohn space*.

The main advantage of this construction is that one can choose a model theoretic property beforehand and then fine tune the input to aim for a limit that satisfies it. We illustrate this fact with the notions of transitivity, algebraicity, and $\aleph_0$ -categoricity.

Let $\mathfrak{M}$ be a countable structure and let $G := \text{Aut}(\mathfrak{M})$. The structure $\mathfrak{M}$ is *k-transitive* if the action $G \curvearrowright \mathfrak{M}$ is *k-transitive*, i.e. if for every $a_1, \dots, a_k, b_1, \dots, b_k \in \mathfrak{M}$ such that for all $i \neq j$, $a_i \neq a_j$ and $b_i \neq b_j$, there exists $g \in \text{Aut}(\mathfrak{M})$ such that $g(a_i) = b_i$ for every $i \leq k$. It has *no algebraicity* if for every finite subset $A \subseteq \mathfrak{M}$ and every $a \in \mathfrak{M} \setminus A$, the orbit $G_A \cdot a$ is infinite. Given a class of finitely generated structures $\mathcal{F}$ and $k \in \mathbb{N}$, we will denote by $\mathcal{F}_k$ the class of structures $F \in \mathcal{F}$ that are generated by *k* distinct elements and no less. If $\mathcal{L}$ is relational, $\mathcal{F}_k$ is the class of those $F \in \mathcal{F}$ such that $|F| = k$.

Proposition 1.1.19. *Let $\mathcal{F}$ be a Fraïssé class in a relational signature $\mathcal{L}$, let $\mathbb{F}$ be its Fraïssé limit and let $k \in \mathbb{N}$. The following properties hold.*

1. *If $\mathcal{L}$ is finite, $\mathbb{F}$ is $\aleph_0$ -categorical.*
2. *$\mathbb{F}$ has no algebraicity if and only if $\mathcal{F}$ has SAP.*
3. *If $|\mathcal{F}_k| = 1$ and $\text{Aut}(F) = \text{Sym}(F)$ for the $F \in \mathcal{F}_k$, then $\mathbb{F}$ is *k-transitive*.*

Proof.

1. See [Hod93, Th. 7.4.1].
2. See [Cam90, Prop. 2.15].
3. Let $a_1, \dots, a_k, b_1, \dots, b_k \in \mathbb{F}$ be such that for all $i \neq j$, $a_i \neq a_j$ and $b_i \neq b_j$, let $A := \{a_1, \dots, a_k\}$, and let $B := \{b_1, \dots, b_k\}$. Since $|\mathcal{F}_k| = 1$, the structures A and B are isomorphic and the second assumption says that the map $A \rightarrow B$ which sends a_i to b_i for every $i \leq k$ is an isomorphism. By ultrahomogeneity of $\mathbb{F}$, this map extends to an automorphism of $\mathbb{F}$, which finishes the proof.

□

In Chapter 5, we will be particularly interested permutation groups that are 2-transitive and oligomorphic. Equivalently, we will want countable structures that are both 2-transitive and $\aleph_0$ -categorical. The above theorem provides a recipe for building such groups. Indeed, one can start with signature $\mathcal{L}$ consisting of a single relation symbol R of arity $k \in \mathbb{N}$. Every Fraïssé limit in this language is $\aleph_0$ -categorical per Item 1 above. Then, one can consider the class $\mathcal{F}$ of $\mathcal{L}$ -structures F for which R^F has the following property: for all $f_1, \dots, f_k \in F$ and $\sigma \in \text{Sym}(\{1, \dots, k\})$, if $(f_1, \dots, f_k) \in F$, then the f_i 's are pairwise distinct and also $(f_{\sigma(1)}, \dots, f_{\sigma(k)}) \in F$. Every subclass of $\mathcal{F}$ satisfies Item 3 above hence, if moreover Fraïssé, gives rise to an $\aleph_0$ -categorical and *k-transitive* Fraïssé limit. Natural examples include the *k-uniform* hypergraphs from [EP02].

In Chapters 3 and 4, all of the permutation groups studied will have no algebraicity. Indeed, natural amalgamation processes in both graphs and metric spaces are based on taking disjoint unions hence automatically yield SAP.

In all of this thesis, the vast majority of the groups considered can never be locally compact, in particular because of algebraicity. See the following strong negation of having no algebraicity.

Proposition 1.1.20. *Let G be a locally compact permutation group on a countable set Ω . There exists a finite set $A \subseteq \Omega$ such that every orbit of $G_A \curvearrowright \Omega$ is finite.*

Proof. There exists a finite subset $A \subseteq \Omega$ such that G_A is compact. For every $a \in A$, $G_{A \cup \{a\}}$ is an open subgroup of G_A . Since G_A is compact, it must have finite index. Now, $G_A/G_{A \cup \{a\}}$ is compact for the quotient topology, as the continuous image of the compact group G_A . Since $G_{A \cup \{a\}}$ is open, $G_A/G_{A \cup \{a\}}$ is moreover discrete hence finite. Now, $G_A/G_{A \cup \{a\}}$ naturally identifies with $G_A \cdot a$ which must then also be finite. $\square$

In Chapter 3, we will focus on *Urysohn spaces*. These countable metric spaces, generalizing Item 7 of Example 1.1.18, can all be obtained as Fraïssé limits.

Definition 1.1.21.

1. A *distance set* is a subset Δ of $\mathbb{R}_+$ which contains 0 and is either
 - a subsemigroup of $\mathbb{R}_+$
 - or the intersection of some subsemigroup of $\mathbb{R}_+$ with a segment of the form $[0, M]$ where $M \in \Delta$.
2. If Δ is a distance set, a Δ -*metric space* is a metric space (X, d) such that $d(X \times X) \subseteq \Delta$.

For every countable distance set Δ , the class $\mathcal{F}_\Delta$ of finite Δ -metric spaces is a Fraïssé class [EGLMM21, Lem. 2.6] and its Fraïssé limit is called the Δ -*Urysohn space*, denoted $\mathbf{U}_\Delta$.

A metric space (X, d) is *homogeneous* if every isometry between two finite subsets of (X, d) extends to a surjective isometry of (X, d) . If Δ is a distance set, (X, d) is Δ -*universal* if every finite Δ -metric space isometrically embeds into (X, d) . Fraïssé's Theorem in this specific case has the following consequence:

Theorem 1.1.22. *Let Δ be a countable distance set. $\mathbf{U}_\Delta$ is the unique countable Δ -metric space which is homogeneous and Δ -universal.*

A similar statement holds for $\Delta = \mathbb{R}$, as proved by Urysohn in his posthumous paper [Ury27] well before Fraïssé's work. In this case, one has to replace "countable" with "separable and complete".

Example 1.1.23. Here are some natural examples of distance sets.

1. For $\Delta = \{0\}$, $\mathbf{U}_\Delta$ is a single point.
2. For $\Delta = \{0, r\}$ with $r > 0$, $\mathbf{U}_\Delta$ is discrete and $\text{Iso}(\mathbf{U}_\Delta) = \text{Sym}(\mathbf{U}_\Delta)$.
3. For $\Delta = \{0, r, 2r\}$ with $r > 0$, $\mathbf{U}_\Delta$ identifies with the Rado graph by putting an edge between elements at distance r from each other.
4. The cases most commonly studied are $\Delta = \mathbb{Q}$ and $\Delta = \mathbb{Z}$.

In Chapter 3, we fully classify the unitary representations of Urysohn spaces. See Section 1.2.2, Theorem 1.2.14 in this Introduction for the main statement. More Fraïssé limits are studied in Chapter 4.

1.1.4 Coarse geometry

Geometric group theory plays a key role in dynamics, especially since Gromov's monograph [Gro87], and interacts with representation theory in many ways. Let us briefly describe the classical setting as well as how it generalizes to Polish groups.

Given a finitely generated group G together with a finite generating set $A \subseteq G$, the associated *word metric* δ_A on G is the graph distance computed in the Cayley graph of (G, A) . While δ_A depends on A , the *large scale geometry* it defines on G does not. More precisely, if B is another finite generating set for G , the distances δ_A and δ_B are *quasi-isometric* in the sense that there exists a constant $K > 0$ such that $\frac{\delta_B}{K} - K \leq \delta_A \leq K \cdot \delta_B + K$.

More generally, two pseudo-metric spaces (X, d) and (X', d') are *quasi-isometric* if there exists a map $\phi: X \rightarrow X'$ and a constant $K > 0$ such that:

- (i) for all $x, y \in X$, $\frac{d(x, y)}{K} - K \leq d'(\phi(x), \phi(y)) \leq K \cdot d(x, y) + K$,
- (ii) and for every $z \in X'$, $d(z, \text{Im } \phi) \leq K$.

Being quasi-isometric is an equivalence relation on metric space and the class of a given space is called its *quasi-isometry type*.

It follows from the previous discussion that a finitely generated group G comes equipped with a canonical quasi-isometry type, namely that of (G, δ_A) for any finite generating set A . A quasi-isometry type is a very precise notion of large scale geometry. It gives rise to a notion of bounded subsets but contains much more data.

This construction can be generalized far beyond finitely generated group. Every topological group can be given a canonical notion of bounded subsets and a large class of Polish groups even have a canonical quasi-isometry type. This is the aim of Rosendal's *coarse geometry* (see the monograph [Ros22] for a complete exposition).

An *écart* on a topological group G is a map $\delta: G^2 \rightarrow \mathbb{R}_+$ such that for all $x, y, z \in G$, $\delta(x, x) = 0$ and $\delta(x, z) \leq \delta(x, y) + \delta(y, z)$. The δ -diameter of a subset $A \subseteq G$ is $\delta(A) := \sup\{\delta(a, b) : a, b \in A\}$. The écart δ is *left-invariant* if $\delta(gx, gy) = \delta(x, y)$ for all $g, x, y \in G$.

Definition 1.1.24. Let G be a topological group. A subset $A \subseteq G$ is *coarsely bounded* (with respect to the left-coarse structure on G) if $\delta(A) < +\infty$ for every continuous left-invariant écart δ on G . The group G is coarsely bounded if it is coarsely bounded as a subset of itself.

Example 1.1.25. Let G be a topological group and let $A \subseteq G$. The above notion of large scale geometry generalizes classical situations in the following sense.

1. If G is countable and discrete, A is coarsely bounded if and only if it is finite.
2. More generally, if G is locally compact, A is coarsely bounded if and only if A is relatively compact in G .
3. If G is locally Roelcke-precompact, A is coarsely bounded if and only if A is Roelcke precompact in G [Zie21].

In the best cases, there exists a single continuous left-invariant écart δ on G that generates the left-coarse structure on G in the sense that for every $A \subseteq G$,

$$A \text{ is coarsely bounded} \iff \delta(A) < +\infty.$$

Such an écart is called *coarsely proper*. Recalling the Birkhoff-Kakutani Theorem 1.1.1, one can build a compatible metric from a countable basis of neighborhoods of the identity. In a very similar way, one can build a proper écart from a "countable basis at infinity", in fact even just from a sequence of coarsely bounded sets $(A_n)_{n \in \mathbb{N}}$ that covers G [Ros22, Th. 2.38].

However, this is just short of what is needed to define a quasi-isometry type. For this purpose, one needs a *maximal écart*, i.e. a continuous left-invariant écart δ on G such that every other such écart δ' satisfies $\delta' \leq K \cdot \delta + K$ for some constant $K \geq 0$. If δ is a maximal écart on G , it is easily seen to be coarsely proper. Moreover, the quasi-isometry type of (G, δ) does not depend on the choice of a maximal écart δ hence is called the *quasi-isometry type of G* . When G is Polish, there exists a maximal écart exactly when G is generated by a coarsely bounded subset [Ros22, Th. 2.73]. The situation is then a direct generalization of the case of finitely generated groups, as the maximal metric can even be obtained as an actual word metric [Ros22, Prop. 2.72].

Example 1.1.26.

1. If G is a finitely generated group and A is a finite generating set, the word metric δ_A is maximal [Ros22, Prop. 2.72]. In particular, the quasi-isometry type of G is that of its Cayley graph.
2. If G is coarsely bounded (e.g. compact or Roelcke precompact), every left-invariant continuous écart is bounded and maximal. In particular, G is quasi-isometric to a point.
3. Let $\mathbf{T}$ be the regular tree of countably infinite arity and let d be the graph distance in $\mathbf{T}$. Given any $x \in \mathbf{T}$, the écart on $\text{Aut}(\mathbf{T})$ given by $\delta(g, h) := d(g(x), h(x))$ is maximal. Moreover, the quasi-isometry type of $\text{Aut}(\mathbf{T})$ is that of $(\mathbf{T}, d)$. This is a common situation, see the Milnor-Schwarz Lemma for coarse geometry [Ros22, Th. 2.77].

In practice, we will only use the notion of coarsely bounded sets, and not the finer notion of quasi-isometry types in this thesis.

1.2 Unitary representations

Now that the groups we will be working with have been introduced, we turn to the kind of actions we will be studying; unitary representations. We will go over some key results of the classical theory and see if and how they can be translated to the permutation group setting. Along the way, we will also state the main results of this thesis.

All Hilbert spaces considered in this thesis are complex. Given a Hilbert space $\mathcal{H}$, we will denote by $\mathcal{B}(\mathcal{H})$ the space of bounded operators $\mathcal{H} \rightarrow \mathcal{H}$ and by $\mathcal{U}(\mathcal{H})$

the unitary group of $\mathcal{H}$. The inner product of $\mathcal{H}$ will be denoted $\langle \cdot, \cdot \rangle$, assumed linear in the first variable and conjugate-linear in the second. The associated norm will be denoted $\|\cdot\|$.

Definition 1.2.1. A *unitary representation* of a topological group G is a linear and isometric action of G on $\mathcal{H}$, denoted $\pi: G \curvearrowright \mathcal{H}$. In this notation, π stands for the group morphism $G \longrightarrow \mathcal{U}(\mathcal{H})$. It is *non-zero* if $\mathcal{H} \neq 0$.

There are several ways one could attempt to define continuity of unitary representations, one for each topology $\mathcal{U}(\mathcal{H})$ can be endowed with. Choosing the operator norm on $\mathcal{U}(\mathcal{H})$ is tempting but is too restrictive for our purposes. Indeed, it rules out the most fundamental examples of the theory, such as the translation action $\mathbb{R} \curvearrowright L^2(\mathbb{R}, \lambda)$ where λ is the Lebesgue measure. A more reasonable candidate is the *strong operator topology* (s.o.t.), defined by $T_n \longrightarrow T$ if $\|T_n \xi - T \xi\| \longrightarrow 0$ for every $\xi \in \mathcal{H}$. A third option is the *weak operator topology* (w.o.t.), defined by $T_n \longrightarrow T$ if $|\langle T_n \xi, \eta \rangle - \langle T \xi, \eta \rangle| \longrightarrow 0$ for all $\xi, \eta \in \mathcal{H}$. It turns out that w.o.t. and s.o.t. induce the same topology on $\mathcal{U}(\mathcal{H})$ which is widely considered the right one for defining continuity of unitary representations.

Definition 1.2.2. A unitary representation $\pi: G \curvearrowright \mathcal{H}$ of a topological group G is *continuous* if the following equivalent properties are satisfied:

- (i) For every $\xi \in \mathcal{H}$, the map $G \longrightarrow (\mathcal{H}, \|\cdot\|)$ given by $g \mapsto \pi(g)\xi$ is continuous.
- (ii) For all $\xi, \eta \in \mathcal{H}$, the map $f_{\xi, \eta}^\pi: G \longrightarrow \mathbb{C}$ given by $g \mapsto \langle \pi(g)\xi, \eta \rangle$, called a *matrix coefficient*, is continuous.

In the special case of permutation groups, continuity can be reformulated as follows. In fact, some authors such as Ol'sanskiĭ use Item (ii) below as the very definition of continuous representations in this case. See e.g. [Tsa12, Lem. 3.1] for a proof.

Lemma 1.2.3 (Fundamental Lemma of non-Archimedean Harmonic Analysis). *Let G be a non-Archimedean topological group and let $(U_i)_{i \in I}$ be a basis of neighborhoods of 1_G consisting of open subgroups. Let $\pi: G \curvearrowright \mathcal{H}$ be a unitary representation of G and, for every $i \in I$, let $\mathcal{H}_i := \{\xi \in \mathcal{H}, \forall u \in U_i, \pi(u)\xi = \xi\}$. The following properties are equivalent:*

- (i) *the representation π is continuous,*
- (ii) *$\bigcup_{i \in I} \mathcal{H}_i$ is dense in $\mathcal{H}$.*

Convention. In all of this thesis, "representation" will always mean "continuous unitary representation" in the sense of Definition 1.2.2.

Every topological group G has at least one continuous unitary representation: its trivial action on $\mathbb{C}$. We will denote it by $\mathbf{1}_G$, called the *trivial representation* of G . One of the main ways to produce interesting representations is through the Koopman construction.

Definition 1.2.4. Let G be a topological group and let (X, μ) be a measure space. Suppose that G acts on (X, μ) by measure-preserving transformations and that the action map $G \times X \rightarrow X$ is measurable. The associated *Koopman representation* is the (not necessarily continuous) unitary representation $\kappa: G \curvearrowright L^2(X, \mu)$ given by $(\kappa(g)\xi)(x) = \xi(g^{-1}x)$ for all $g \in G$, $x \in X$, and $\xi \in L^2(X, \mu)$.

It thus appears that topological groups which naturally act on many measure spaces are particularly well suited for our purposes. This partly explains why we restrain our attention to locally compact groups and permutation groups.

Example 1.2.5. Let G be a second countable topological group. All the Koopman representations below are continuous.

1. If G is locally compact, let λ be a left Haar measure on G . The Koopman representation associated to the action $G \curvearrowright (G, \lambda)$ by left translation, denoted $\lambda_G: G \curvearrowright L^2(G, \lambda)$, is called the *regular representation* of G .
2. If G is a permutation group on a countable set Ω , the action $G \curvearrowright \Omega$ preserves the counting measure. The associated Koopman representation is the action

$$G \curvearrowright \ell^2(\Omega).$$

3. If U is an open subgroup of G , then G/U is countable and discrete. Moreover, the action $G \curvearrowright G/U$ by left translation preserves the counting measure. The associated Koopman representation is the action

$$\lambda_{G/U}: G \curvearrowright \ell^2(G/U),$$

called the *quasi-regular representation* associated to U .

Note that Item 3 is particularly relevant for non-Archimedean groups, as they have many open subgroups. Moreover, in the first two Items above, the representation is *faithful* in the sense that the morphism $G \rightarrow \mathcal{U}(\mathcal{H})$ is injective. We will state a much stronger result later in the form of the Gel'fand-Raikov Theorem 1.2.8.

The other main way one produces representations is through *induction*. Suppose that G is a topological group, H is a subgroup of G and π is a representation of H . If there exists a Borel measure on G/H whose measure class is preserved under the action of G by left translation, σ can be induced into a representation of G . Evidently, induction is most useful when G has many subgroups H such that G/H has such a measure. If G is locally compact, every closed subgroup has one [BdV08, Th. B.1.4]. While this fails for Polish groups, the counting measure on G/U where U is an open subgroup is actually G -invariant. Again, permutation groups having plenty open subgroups, the central technique of induction is particularly relevant for them. Let us give a precise definition of induction in the special case of open subgroups, the only one we will use in this thesis.

Let U be an open subgroup of a separable topological group G . Then G/U is countable discrete and can be endowed with the counting measure. If moreover $\sigma: U \curvearrowright \mathcal{H}$ is a unitary representation of U , let $E := \{\xi: G \rightarrow \mathcal{H} \mid \forall u \in U, \forall g \in G, \xi(gu) = \sigma(u^{-1}) \cdot \xi(g)\}$. Then, for all $\xi, \eta \in E$, the maps defined on G by

$$g \mapsto \langle \xi(g), \eta(g) \rangle \quad \& \quad g \mapsto \|\xi(g)\|$$

are constant on every left translate of U . Fix a transversal $(g_i)_{i \in I}$ of G/U and define:

$$\mathcal{E} := \{\xi \in E : \sum_{i \in I} \|\xi(g_i)\|^2 < +\infty\}.$$

Then $\mathcal{E}$ is a Hilbert space with the following inner product:

$$\forall \xi, \eta \in \mathcal{E}, \langle \xi, \eta \rangle := \sum_{i \in I} \langle \xi(g_i), \eta(g_i) \rangle.$$

In this setting, we will write $\xi \in \ell^2(G/U)$ to refer to the fact that if $\xi \in \mathcal{E}$, then $\sum_{i \in I} \|\xi(g_i)\|^2 < +\infty$. The *induced representation* $\text{Ind}_U^G(\sigma)$ is the action $G \curvearrowright \mathcal{E}$ defined by

$$\forall g, x \in G, \forall \xi \in \mathcal{E}, (g \cdot \xi)(x) = \xi(g^{-1}x).$$

Note that this construction does not depend on the choice of a transversal $(g_i)_{i \in I}$.

Example 1.2.6. Let G be a topological group and let U be an open subgroup. Inducing to G the trivial representation 1_U of U yields the quasi-regular representation $\lambda_{G/U} : G \curvearrowright \ell^2(G/U)$.

There are three basic operations one can perform on representations. The most common operation is the *direct sum*. Given two representations $\pi : G \curvearrowright \mathcal{H}$ and $\pi' : G \curvearrowright \mathcal{H}'$ of G , the diagonal action $\pi \oplus \pi' : G \curvearrowright \mathcal{H} \oplus \mathcal{H}'$ is also a representation of G . More generally, if $(\pi_i)_{i \in I}$ is a collection of representations of G , we will denote by $\bigoplus_{i \in I} \pi_i : G \curvearrowright \bigoplus_{i \in I} \mathcal{H}_i$ the action of G on the Hilbert sum of the $\mathcal{H}_i$'s. For $n \in \mathbb{N} \cup \{\infty\}$, we will denote by $n \cdot \pi$ the representation $\bigoplus_{i=1}^n \pi$. A representation of this form is called a *multiple* of π . Another natural operation is *tensor product*. Given two representations $\pi : G \curvearrowright \mathcal{H}$ and $\pi' : G \curvearrowright \mathcal{H}'$, the group G also acts on the Hilbert tensor-product $\mathcal{H} \otimes \mathcal{H}'$, giving rise to the representation $\pi \otimes \pi' : G \curvearrowright \mathcal{H} \otimes \mathcal{H}'$. The last and probably least common is the *dual representation*. If $\pi : G \curvearrowright \mathcal{H}$ is a representation of G , its dual representation is the action of G on the dual $\mathcal{H}^*$ of $\mathcal{H}$ given by $(g \cdot \phi)(\xi) = \phi(\pi(g^{-1})\xi)$ for all $g \in G$, $\phi \in \mathcal{H}^*$ and $\xi \in \mathcal{H}$.

A *morphism of representations* (also called *intertwining operator*) between π and π' is a bounded operator $T : \mathcal{H} \rightarrow \mathcal{H}'$ which is G -equivariant, i.e. such that $T \circ \pi(g) = \pi'(g) \circ T$ for every $g \in G$. We will write $\text{Hom}(\pi, \pi')$ for the space of morphisms between π and π' . An *isomorphism* of representations between π and π' is an element of $\text{Hom}(\pi, \pi')$ that is moreover a surjective isometry. The representations π and π' are *isomorphic* if there exists an isomorphism between them, in which case we write $\pi \simeq \pi'$.

A subspace $\mathcal{K}$ of $\mathcal{H}$ is *G-invariant* if $\pi(g)\xi \in \mathcal{K}$ for all $\xi \in \mathcal{K}$ and $g \in G$. A *subrepresentation* of π is the action of G on a closed G -invariant subspace $\mathcal{K} \subseteq \mathcal{H}$. It is *proper* if $\mathcal{K}$ is a proper subspace of $\mathcal{H}$. In this case, the orthogonal $\mathcal{K}^\perp$ of $\mathcal{K}$ in $\mathcal{H}$ is also G -invariant and $\pi \simeq \pi|_{\mathcal{K}} \oplus \pi|_{\mathcal{K}^\perp}$ where $\pi|_{\mathcal{K}}$ and $\pi|_{\mathcal{K}^\perp}$ are the actions of G on $\mathcal{K}$ and $\mathcal{K}^\perp$ respectively. We say that π is *irreducible* if the only closed G -invariant subspaces of $\mathcal{H}$ are $\mathcal{H}$ and 0 . The *unitary dual* of G is the set of equivalence classes of irreducible representations of G , denoted $\widehat{G}$. They are captured by the following classical result, see [Fol95, Th. 3.5] for a proof.

Theorem 1.2.7 (Schur's Lemma). *Let G be a topological group, let $\pi : G \curvearrowright \mathcal{H}$ be a representation of G , and let $\pi(G)' := \{T \in \mathcal{B}(\mathcal{H}) : \forall g \in G, \pi(g) \circ T = T \circ \pi(g)\}$ be the commutant of π . The following properties are equivalent:*

- (i) π is irreducible,
- (ii) $\pi(G)' = \mathbb{C} \cdot \text{id}_{\mathcal{H}}$.

We have already seen that locally groups and permutation groups all admit a faithful representation. In other words, representations separate the points. In fact, a much stronger statement holds in both cases: *irreducible* representations separate the points. For locally compact groups, this is the classical Gel'fand-Raikov Theorem. It was adapted to permutation groups by Tsankov.

Theorem 1.2.8 ([GR43, Tsa12]). *Let G be either a locally compact group or a permutation group over a countable set. For every $g \in G \setminus \{1_G\}$, there exists an irreducible unitary representation π of G such that $\pi(g) \neq \text{id}$.*

Remark 1.2.9. The Gel'fand-Raikov Theorem fails particularly badly for the Polish group $\text{Homeo}^+([0, 1])$ of order-preserving homeomorphisms of $[0, 1]$. Indeed, Megrelishvili showed that it has no non-trivial representation, not even on a reflexive Banach space [Meg01]. Even more pathological is the Polish group $L^0([0, 1], \mathbb{S}^1)$ of measurable maps from $[0, 1]$ to the unit circle. It has a faithful unitary representation on $L^2([0, 1])$ but no non-trivial irreducible unitary representation [Gla98].

1.2.1 Compact groups and Roelcke precompact closed permutation groups

The irreducible representations play an important role in abstract harmonic analysis. In the best cases, they are the elementary building blocks of all representations. The ideal cases are topological groups G for which every representation *splits as a sum of irreducibles*, i.e. is isomorphic to a representation of the form $\bigoplus_{i \in I} \pi_i$ where π_i is irreducible for every $i \in I$. This holds in particular for compact groups.

Theorem 1.2.10 (Peter-Weyl [PW27]). *Let G be a compact group and let λ be a Haar measure on G . The following properties hold:*

1. *Every representation of G splits as a sum of irreducible subrepresentations.*
2. *Every irreducible representation of G is finite dimensional.*
3. *Every irreducible representation of G is isomorphic to a subrepresentation of $\lambda_G: G \curvearrowright L^2(G, \lambda)$.*

Another class of groups whose representation theory is very tame is that of Roelcke precompact closed permutation groups.

Theorem 1.2.11 ([Tsa12]). *Let G be a Roelcke-precompact non-Archimedean Polish group.*

1. *Every representation of G splits as a sum of irreducible subrepresentations.*
2. *Every irreducible representation of G is of the form $\text{Ind}_{C(V)}^G(\pi)$ where V is an open subgroup of G and σ is an irreducible representation of V which factorizes through a finite quotient of V .*

3. Every irreducible representation of G is isomorphic to a subrepresentation of $\lambda_{G/U}: G \curvearrowright \ell^2(G/U)$ for some open subgroup $U \leq G$.

Comparing both theorems, it appears that the representation theory of Roelcke precompact closed permutation groups is very close to that of compact groups. In particular, both only have countably many irreducible representations up to isomorphism (recall Proposition 1.1.11), all of which appear as subrepresentations of some canonical representation. We insist on the fact that Item 1 in either theorems above is a very strong statement that fails for most groups outside of these two classes (see Section 1.2.5). A major difference, however, is that irreducible representations of Roelcke-precompact closed permutation groups are often infinite dimensional.

Example 1.2.12. Let G be a Roelcke-precompact non-Archimedean Polish group. It appears from Item 2 of the above Theorem that the representation theory of G is governed by that of finite groups living as quotients in G . In many cases¹ where $G = \text{Aut}(\mathfrak{M})$ for some $\aleph_0$ -categorical structure $\mathfrak{M}$, the finite groups in question are the automorphism groups of finite substructures of $\mathfrak{M}$. That is $V = G_A$, $C(V) = G_{\{A\}}$, and $C(V)/V = \text{Aut}(A)$ for some finite substructure $A \subseteq \mathfrak{M}$. Whenever this situation arises, every irreducible representation of G is isomorphic to a subrepresentation of $\ell^2(\mathfrak{M}^n)$ for some $n \in \mathbb{N}$ [Tsa12, Lem. 6.2]. This holds for all of the groups appearing in Example 1.1.13. More explicitly:

1. For $G = \text{Sym}(\Omega)$, the finite groups are the finite symmetric groups $\mathfrak{S}_n$ for every $n \in \mathbb{N}$.
2. For $G = \text{Aut}(\mathbb{Q}, <)$, every finite subset $A \subseteq \mathbb{Q}$ is *rigid*, in the sense that it admits no non-trivial order-preserving bijection. In other words, $G_{\{A\}} = G_A$ for every finite subset $A \subseteq \mathbb{Q}$. Thus, the only finite group involved in Item 2 of Theorem 1.2.11 is the trivial group $\{1\}$.
3. For G the automorphism group of the countable atomless boolean algebra, the finite groups are the automorphism groups of finite boolean algebras,
4. For $G = \text{GL}_\infty(\mathbb{F}_q)$, the finite groups are the finite general linear groups $\text{GL}_n(\mathbb{F}_q)$ for every $n \in \mathbb{N}$.

There are other strong similarities in the representations theories of these two classes of groups. In particular, Ibarlucía showed that Roelcke-precompact Polish groups have Kazhdan's Property (T), see Subsection 1.2.4 of this Introduction. Another similarity is that they are fully determined by their representations through duality theories which is the main result of Chapter 2, see Subsection 1.2.3 of this Introduction. Before that, we describe a large class of groups that also satisfies the conclusion of Theorem 1.2.11.

1.2.2 Dissociated groups

In collaboration with C. Jahel and M. Joseph, we started the systematic study of a class of closed permutation groups whose representation theory behaves as in

¹This happens when $\mathfrak{M}$ is $\aleph_0$ -categorical, ultrahomogeneous and has weak elimination of imaginaries

Example 1.2.12. This permutation group notion, which we called *dissociation*, is motivated by the Roelcke precompact case and is studied in Chapters 3 and 4.

Suppose G is a closed permutation group on some countable set Ω and let $\pi: G \curvearrowright \mathcal{H}$ be a representation of G . For every finite subset $A \subseteq \Omega$, we will denote by

$$\mathcal{H}_A := \{\xi \in \mathcal{H} : \forall g \in G_A, \pi(g)\xi = \xi\}$$

the closed subspace of G_A -invariant vectors and by p_A the orthogonal projection onto $\mathcal{H}_A$. In this setting, the fundamental lemma of non-Archimedean harmonic analysis (Lemma 1.2.3) rephrases as follows:

$$\bigcup_A \mathcal{H}_A \text{ is dense in } \mathcal{H}.$$

Dissociation is about how these invariant subspaces are arranged relatively to each other. It can be seen as a uniform imprimitivity system in the sense of Mackey [Mac49].

Definition 1.2.13. [BJJ24] Let G be a closed permutation group on a countable set Ω .

1. A representation $\pi: G \curvearrowright \mathcal{H}$ of G is *dissociated* if for all finite subsets $A, B \subseteq \Omega$,

$$\mathcal{H}_A \perp_{\mathcal{H}_{A \cap B}} \mathcal{H}_B,$$

$$\text{i.e. } p_A p_B = p_{A \cap B}.$$

2. The permutation group G is *dissociated* if every representation of G is dissociated.

A weak version of this phenomenon was first observed for Roelcke precompact closed permutation groups [JT22, Prop. 3.2] as a consequence of Theorem 1.2.11. We have extracted this property and shown that it, alone, is enough to recover classification theorems. The following is one of the main results of Chapter 3.

Theorem 1.2.14 ([BJJ24]). *Let $G \leq \text{Sym}(\Omega)$ be a dissociated closed permutation group.*

1. *Every representation of G splits as a sum of irreducible subrepresentations.*
2. *Every irreducible unitary representation of G is induced from an irreducible representation of the finite group $G_{\{A\}}/G_A$ for some finite subset $A \subseteq \Omega$.*
3. *Every irreducible representation of G is isomorphic to a subrepresentation of $G \curvearrowright \ell^2(\Omega^n)$ for some $n \in \mathbb{N}$.*

Example 1.2.15. The following closed permutation groups are dissociated.

1. $\text{Aut}(\mathbb{Q}, <) \leq \text{Sym}(\mathbb{Q})$,
2. the automorphism groups of the Rado graph,
3. the isometry group $\text{Iso}(\mathbf{U}_{\mathbb{Q}}) \leq \text{Sym}(\mathbf{U}_{\mathbb{Q}})$ of the rational Urysohn space $\mathbf{U}_{\mathbb{Q}}$,

4. more generally, for every countable distance set Δ , the isometry group of the Δ -Urysohn space.

Alternatively, this notion can also be defined within ergodic theory. Indeed, one can consider actions of a permutation group on a standard probability space (X, μ) . If the action preserves the measure, we say that it is a *p.m.p. action*. Just as in the context of unitary representations, one can define dissociation of p.m.p. actions using invariant σ -algebras and conditional independence. In fact, those two definitions of dissociation are equivalent [BJJ25, Th. 1.5]. Naturally, dissociation also imposes strong rigidity properties on the ergodic theory of the group. For example, Stuck-Zimmer rigidity of p.m.p. actions has been observed: If $G \subseteq \text{Sym}(\Omega)$ is closed, proper, and dissociated, every ergodic p.m.p. action of G is either essentially free or essentially transitive [JJ25, Th. 1.4]. Moreover, adapting methods of Tsankov for Roelcke precompact groups, we obtained a de Finetti Theorem for dissociated groups [BJJ25, Cor. 3.6]: every ergodic p.m.p. generalized Bernoulli shift $G \curvearrowright ([0, 1]^\Omega, \mu)$ of a dissociated transitive closed permutation group $G \leq \text{Sym}(\Omega)$ is i.i.d..

We have also developed general methods for proving dissociation and building dissociated groups. These have a strong model theoretic flavor and are well-suited for application within Fraïssé theory (see Section 1.1.3).

Our most important class of new examples are the Urysohn spaces. See Chapter 3 for the proof of the following result and Chapter 4 for more examples of dissociated groups.

Theorem 1.2.16 ([BJJ24]).

For every countable distance set Δ , the closed permutation group $\text{Iso}(\mathbf{U}_\Delta)$ is dissociated.

Remark 1.2.17. Dissociation as defined above is tailored for permutation groups that have no algebraicity. A more general definition of dissociation can be given to take algebraicity into account, as in [JT22, Prop. 3.2] which motivated the introduction of dissociation. In fact, many of our results in representation theory hold in a higher generality. In Chapter 5, we use such methods to classify the irreducible representations of some subgroups of $\text{Aut}(\mathbf{T})$. See Corollary 5.3.18 therein for an illustration.

1.2.3 Duality Theories

An important justification for dedicating efforts to the study of unitary representations is that they encode a lot of information about the group. Sometimes, the group can be fully reconstructed from the abstract data of its representation theory. If the representation theory of a topological group G is nice enough, one can build from its unitary representations a *dual object* which fully characterizes G . Let us describe the classical duality theories that arise from harmonic analysis and state the main result of Chapter 2: a duality theory for Roelcke precompact non-Archimedean Polish groups.

It all starts from the Gel'fand-Raikov Theorem 1.2.8. Indeed, let G be a locally compact group and let $C_b(G)$ be the space of continuous bounded complex valued functions in G , endowed with the uniform convergence topology. A *matrix*

coefficient of G if a map $G \rightarrow \mathbb{C}$ of the form $f_{\xi,\eta}^\pi: g \mapsto \langle \pi(g)\xi, \eta \rangle$ where $\pi: G \curvearrowright \mathcal{H}$ is a representation of G and $\xi, \eta \in \mathbb{C}$. We will denote by $\text{Hilb}(G)$ the closure in $C_b(G)$ of the space of matrix coefficients of G . Considering the three standard operations on representations, one can see that $\text{Hilb}(G)$ is a $\mathbb{C}$ -algebra that is stable under the involution induced by complex conjugation. Being closed in $C_b(G)$, it is a C^* -algebra of functions on G . Moreover, the Gel'fand-Raikov Theorem 1.2.8 in particular says that $\text{Hilb}(G)$ separates the points of G : for all $g \neq h \in G$, there exists $f \in \text{Hilb}(G)$ such that $f(g) \neq f(h)$. As a consequence, through Gel'fand duality, $\text{Hilb}(G)$ characterizes G as a topological space. However, the algebraic structure is lost in this approach. In various cases, duality theories have been developed to fully recover G by adding finer data from representation theory.

The best known instance of this is the Pontryagin–van Kampen duality theory for locally compact Abelian groups [Pon34, van35]. If G is Abelian, its irreducible representations all have dimension 1 by Schur's Lemma (Theorem 1.2.7). Thus, its unitary dual $\widehat{G}$ naturally identifies with $\text{Hom}(G, \mathbb{S}^1)$, the continuous group morphisms from G to the unit circle. If G is moreover locally compact, then $\widehat{G}$ endowed with the compact-open topology, is also a locally compact Abelian group. Pontryagin and van Kampen showed that every Abelian locally compact group is canonically isomorphic to its bi-dual. The extra data in this approach is the group structure on $\widehat{G}$, which boils down to how the tensor product behaves on irreducible representations.

Another duality theory was later developed for compact groups. Independently and with different approaches, Tannaka and Kreĭn gave methods to canonically recover any compact group from the abstract data of its unitary representations [Tan39, Kre41]. The main difference from the previous case is that irreducible representations of compact groups can have arbitrary finite dimension. Still, the finer data included in the dual objects defined by Tannaka and Kreĭn also boil down to the behavior of the tensor product. More precisely, it contains data of how tensor products of pairs of irreducibles decompose. The main ingredients in these duality theories are the Peter-Weyl Theorem 1.2.10 and the Haar measure. Coping with the lack of Haar measure with model theoretic tools, we extend in Chapter 2 both Tannaka's and Kreĭn's duality theories to Roelcke-precompact non-Archimedean Polish groups. Let us state the main result.

Let G and H be Roelcke precompact non-Archimedean Polish groups. We will denote by $\mathbf{Rep}(G)$ the category whose objects are the representations of G that split as a finite sum of irreducibles and whose morphisms are the intertwining operators. Observe also that every group morphism $\phi: G \rightarrow H$ induces a functor $\mathbf{Rep}(H) \rightarrow \mathbf{Rep}(G)$ that sends a representation π of H to the representation $\pi \circ \phi$ of G . We can thus view $\mathbf{Rep}(G)$ as an object in a category where the arrows are defined by abstracting the properties of these functors (see Definition 2.5.2). The main result of Chapter 2 is the following, which also holds for Abelian or compact groups by the classical duality theories.

Theorem 1.2.18. *The contravariant functor $\mathbf{Rep}$ that sends a Roelcke-precompact non-Archimedean Polish group to the category of its unitary representations is full and faithful onto its image, i.e. a duality.*

1.2.4 Interactions with large scale geometry

The results in Subsection 1.2.1 reveal a notable phenomenon: forbidding non-trivial large-scale geometry, i.e. considering coarsely bounded groups, tends to tame down the representation theory. Indeed, a locally compact group is coarsely bounded if and only if it is compact, in which case the Peter-Weyl Theorem 1.2.10 holds. Similarly, a locally Roelcke-precompact closed permutation group is coarsely bounded if and only if it is Roelcke precompact, in which case Theorem 1.2.11 holds. The situation in both cases is the best one can ever hope for, as every representation splits as a sum of irreducible ones. On the contrary, representations of $SL(2, \mathbb{R})$ or $\text{Aut}(\mathbf{T}_d)$ only decompose as direct integrals of irreducibles, their unitary duals are uncountable, and representations of general Polish groups do not decompose into irreducibles at all.

More direct interactions between large scale geometry and abstract harmonic analysis come in the form of rigidity properties. We will now discuss two of the most common such properties.

The Howe-Moore Property

The Howe-Moore property is about the vanishing at infinity of matrix coefficients. While classically defined for locally compact groups, it generalizes to every topological group. To this end, we will use the geometrical framework discussed in Subsection 1.1.4. We insist on the fact that when G is locally compact, all the definitions below coincide with the classical ones.

Definition 1.2.19. Let G be a topological group.

1. A function $f: G \rightarrow \mathbb{C}$ *vanishes at infinity* if for every $\varepsilon > 0$, the set $\{g \in G: |f(g)| \geq \varepsilon\}$ is coarsely bounded. We will denote by $C_0(G)$ the set of all such continuous functions.
2. The group G has *the Howe-Moore property* if for every unitary representation $\pi: G \curvearrowright \mathcal{H}$ without non-zero invariant vector, all the matrix coefficients arising from π vanish at infinity.

Examples of locally compact groups with the Howe-Moore property include connected simple real Lie groups with finite center [HM79] or the groups $\text{Aut}(\mathbf{T}_d)$ where for every $d \geq 3$, $\mathbf{T}_d$ is the d -regular tree [LM92].

Even in our more general setting, this notion can often be rephrased using locally compact objects. Given a topological group G , its *Roelcke completion* $\mathcal{R}(G)$ is the completion of G with respect to the meet of the left and right uniformities on G . A more elementary description of this object is as follows. Let δ_L be any left-invariant metric on G that generates its topology and set for all $g, h \in G$:

$$\delta_{L \wedge R}(g, h) = \inf\{\delta_L(g, k) + \delta_L(k^{-1}, h^{-1}) : k \in G\}.$$

The metric $\delta_{L \wedge R}$ is no longer left-invariant but it still generates the topology of G . Moreover, $\mathcal{R}(G)$ is the completion of the metric space $(G, \delta_{L \wedge R})$. It is straightforward that a Polish group G is Roelcke precompact if and only if $\mathcal{R}(G)$ is compact. A deeper result due to Zielinski is that G is locally Roelcke-precompact if and only if $\mathcal{R}(G)$ is locally compact [Zie21].

In this case, we define $C_0(\mathcal{R}(G))$ in the classical way, as the set of continuous functions $f: \mathcal{R}(G) \rightarrow \mathbb{C}$ such that for every $\varepsilon > 0$, the set $\{x \in \mathcal{R}(G): |f(x)| \geq \varepsilon\}$ is compact.

Back to unitary representations, every matrix coefficient of G is *Roelcke-uniformly continuous*, i.e. uniformly continuous with respect to $\delta_{L \wedge R}$. In particular, every matrix coefficients $G \rightarrow \mathbb{C}$ uniquely extends to a continuous function $\mathcal{R}(G) \rightarrow \mathbb{C}$.

Proposition 1.2.20. *Let G be a locally Roelcke-precompact Polish group. The following properties are equivalent:*

- (i) G has the Howe-Moore property,
- (ii) for every unitary representation $\pi: G \curvearrowright \mathcal{H}$ with no non-zero invariant vector, all the matrix coefficients of G arising from π vanish at infinity on the locally compact space $\mathcal{R}(G)$.

In this thesis, we will prove the following statements regarding the Howe-Moore property.

Theorem 1.2.21.

1. Let G be a dissociated closed permutation group on a countable set Ω . Then G has the Howe-Moore Property if and only if G_a is coarsely bounded for every $a \in \Omega$. In particular, $\text{Iso}(\mathbf{U}_\Delta)$ has the Howe-Moore property for every countable distance set Δ .
2. Let $\mathbf{T}$ be the regular tree of countably infinite arity. The group

$$\text{Aut}^+(\mathbf{T}) := \{g \in \text{Aut}(\mathbf{T}): d(g(x), x) \text{ is even for every vertex } x \in \mathbf{T}\}$$

has the Howe-Moore Property.

Item 1 is proved in Chapter 4, Theorem 4.3.11. Item 2 is a special case of Theorem 5.1.4 in Chapter 5.

Kazhdan's Property (T)

Another important rigidity property of geometrical flavor is Kazhdan's Property (T). While it could also be defined using coarse geometry, we will stick to the classical definition.

Definition 1.2.22. Let G be a topological group.

1. Let Q be a compact subset of G , let $\varepsilon > 0$, and let $\pi: G \curvearrowright \mathcal{H}$ be a representation of G . A (Q, ε) -invariant vector is a vector $\xi \in \mathcal{H}$ such that

$$\sup_{g \in Q} \|\pi(g)\xi - \xi\| \leq \varepsilon \|\xi\|.$$

2. A representation of G has *almost invariant vectors* if it has non-zero (Q, ε) -invariant vectors for all Q and ε .

3. The group G has *Kazhdan's Property (T)* if every representation of G that has almost invariant vectors has an invariant vector.

Among the countless applications of Property (T), Margulis used it to construct the first expander graphs. Simple real Lie groups of real rank at least two, such as $\mathrm{SL}(n, \mathbb{R})$ for $n \geq 3$, have Property (T) while $\mathbb{Z}$, $\mathbb{R}$, $\mathrm{SL}(2, \mathbb{R})$ or $\mathrm{Aut}(\mathbf{T}_d)$ do not. All compact groups obviously have it. Ibarlucía showed that Roelcke precompactness retain this geometrical property.

Theorem 1.2.23 ([Iba21]). *Every Roelcke precompact Polish group has Kazhdan's Property (T).*

The above result does not rely on a classification of Theorem, as none is available in this generality. The non-Archimedean case was first obtained by Evans and Tsankov [ET16] by using the classification recalled in Theorem 1.2.11. The representation theory of dissociated groups being just as rigid as that of Roelcke closed permutation groups (Theorem 1.2.14), we obtain the following in Chapter 3 by adapting the method used in [ET16]

Theorem 1.2.24. *Every dissociated closed permutation group has Kazhdan's Property (T).*

1.2.5 Type I groups and decomposition into irreducibles

Given a representation $\pi: G \curvearrowright \mathcal{H}$ of a topological group G , the data about how π decomposes into subrepresentations is encoded in the von Neumann algebra $\pi(G)'$ of bounded operators $\mathcal{H} \rightarrow \mathcal{H}$ that commute with every element of $\pi(G)$. Here is an illustration of this idea:

Proposition 1.2.25. *Let $\pi: G \curvearrowright \mathcal{H}$ be a representation of a topological group G . Let $\mathcal{K}$ be a closed subspace of $\mathcal{H}$ and let p be the orthogonal projection of $\mathcal{H}$ onto $\mathcal{K}$.*

1. *The subspace $\mathcal{K}$ is G -invariant if and only if $p \in \pi(G)'$.*
2. *If $p \in \pi(G)'$, the subrepresentation $G \curvearrowright \mathcal{K}$ of π is irreducible if and only if p is minimal in $\pi(G)'$.*
3. (Schur's Lemma) *The representation π is irreducible if and only if $\pi(G)' = \mathbb{C} \cdot \mathrm{id}$.*
4. *The representation π is multiplicity free if and only if $\pi(G)'$ is commutative.*

Every von Neumann algebra decomposes into elementary ones called factors which are classified into types depending on their complexity. The tame factors are those of Type I, which are the von Neumann algebras of the form $\mathcal{B}(\mathcal{H})$ for a Hilbert space $\mathcal{H}$ of finite or countable dimension. An arbitrary von Neumann algebra is said to be of Type I if it decomposes into Type I factors only.

Dixmier, Glimm, and Mackey used this formalism to develop the notion of *Type I groups* for gauging the complexity of the representation theory of a group.

Definition 1.2.26. A topological group G is said to be of Type I if $\pi(G)'$ is a Type I von Neumann algebra for every unitary representation π of G .

In the locally compact case, this notion has numerous reformulations and has strong consequences about the decomposability of arbitrary representations into irreducible ones. Indeed, every representation of a locally compact group G decomposes into a *direct integral* of irreducible representations. The caveat is that this decomposition is highly non-unique. If G is of Type I, however, there exists a canonical decomposition into irreducibles. Moreover, an actual characterization of G being of Type I is that the Mackey structure on $\widehat{G}$ makes it a standard Borel space [Gli61, Th. 2].

It thus appears that the notion of Type I forms a dividing line between the groups whose representation theory is tractable and those for which it is too wild to be fully understood. Indeed, understanding the representation theory of a locally compact group G of Type I reduces to understanding the standard Borel space $\widehat{G}$. While this can still be a formidable task that often remains beyond our current reach, the situation in the non-type I case is beyond any hope. Indeed, if G is not of Type I, then $\widehat{G}$ is a pathological Borel space [Dix96, Sec. 9.5.6] and not even enough to fully understand the representation theory of G . These considerations led Auslander and Moore to assert that “*one of the fundamental questions one can raise about any locally compact group or C^* -algebra is that of determining when it is type I*” [AM66, p. 1].

Let us now give some examples. First, every topological group such that each of its unitary representations splits as a sum of irreducible subrepresentations is of Type I. Indeed, in this ideal situation regarding decomposability, the uniqueness condition mentioned above is automatically verified. In particular, finite groups are of Type I and more generally, so are compact groups by the Peter-Weyl Theorem. Another large class of groups that is of Type I for fundamental reasons is that of Abelian groups. In fact, Thoma fully answered the Type I question in the discrete case: a countable discrete group is of Type I if and only if it is virtually Abelian [Tho68]. Next, if G is neither compact nor Abelian, it often has irreducible representations of infinite dimension hence deciding the Type I question can become much more difficult. A vast amount of efforts has been dedicated to this question. To only cite a few, nilpotent connected locally compact groups, real algebraic groups over a local field of characteristic 0, and semi-simple Lie groups are of Type I. See [Bd20, Th. 6.E.19] for the references and more examples. On the other hand, the Mautner group is a solvable connected Lie group which is not of Type I (due to Mautner, unpublished, see [Bag78]).

Another case of interest is that of groups acting on trees. The group $\text{Aut}(\mathbf{T}_d)$ of automorphisms of the regular tree $\mathbf{T}_d$ of arity $d \in \mathbb{N}$ is a locally compact group of Type I [Ol’77]. More generally, the Type I question for closed subgroups of $\text{Aut}(\mathbf{T}_d)$ has attracted a substantial amount of attention in the last three decades (a monograph [FN91], a conjecture [Neb99], a thesis [Sem23] and other recent approaches [HR19, CKM23]). The striking result below due to Nebbia is a milestone in that study. In this statement, the CCR property is a strengthening of the Type I property and the other assumptions are no more than non-triviality hypotheses.

Theorem 1.2.27 ([Neb99]). *Let $G \leq \text{Aut}(\mathbf{T}_d)$ be closed, unimodular and acting transitively on the vertices of $\mathbf{T}_d$. If G is CCR, then G acts transitively on the boundary of $\mathbf{T}_d$.*

In the same paper, Nebbia conjectured that the converse also holds, i.e. that having a very tame representation theory is the same as having a very transitive action on the tree. While Nebbia's conjecture remains open in its full generality, significant progress was achieved for the *Burger-Mozes universal groups*. Given $\mathbf{T}_d$ the regular tree of finite arity $d \in \mathbb{N}_{\geq 3}$, one can color the unoriented edges of $\mathbf{T}_d$ with labels in $\{1, \dots, d\}$. Then, every pair (g, x) where g is an automorphism of $\mathbf{T}_d$ and x is a vertex of $\mathbf{T}_d$ induces a bijection of $\{1, \dots, d\}$ by keeping track of how g sends the edges around x to the edges around $g(x)$. This bijection is called the *local action* of g at x . Now, given a group $F \leq \text{Sym}(\{1, \dots, d\})$, one can consider the subgroup $\mathbf{U}(F)$ of $\text{Aut}(\mathbf{T}_d)$ consisting of all the automorphisms g of $\mathbf{T}_d$ such that for every vertex x in $\mathbf{T}_d$, the local action of g at x belongs to F .

Theorem 1.2.28 ([Ol'80, HR19]). *Let $\mathbf{T}_d$ be the regular tree of finite arity $d \in \mathbb{N}_{\geq 3}$ and let $F \leq \text{Sym}(\{1, \dots, d\})$. The following properties are equivalent:*

- (i) $\mathbf{U}(F)$ is of Type I,
- (ii) the action of $\mathbf{U}(F)$ on the boundary of $\mathbf{T}_d$ is transitive,
- (iii) the action $\mathbf{U}(F) \curvearrowright \mathbf{T}_d$ is metrically 2-transitive,
- (iv) the action $F \curvearrowright \{1, \dots, d\}$ is 2-transitive.

Ol'sanskiĭ's contribution to the above theorem is the classification of all the irreducible unitary representations of $\mathbf{U}(F)$ for every 2-transitive $F \curvearrowright \{1, \dots, d\}$, from which he deduced that $\mathbf{U}(F)$ is of Type I [Ol'77]. It corresponds to (iv) $\Rightarrow$ (i). Houdayer and Raum proved (i) $\Rightarrow$ (ii). The other implications are more elementary.

As for non-locally-compact groups, the abstract definition of Type I still makes perfect sense. However, the reformulations that hold in the locally compact case do not carry over at all for general Polish groups. Moreover, there is no integral decomposition theory for Polish groups.

Still, for a nice enough intermediate class, surely some of the theory should adapt. While this task has yet to be undertaken, the Type I question for non-locally-compact Polish groups does appear in the literature on a case by case basis. In all of these cases, a group is shown to be of Type I after its unitary representations are classified (at least all the irreducible ones). The earliest instance is the work of Lieberman, who fully classified the representation of $\mathfrak{S}_\infty$ [Lie72]. Ol'sanskiĭ later developed the semi-group method, used it to reprove Lieberman's result, and classified the representations of other classical groups such as $\mathcal{U}(\mathcal{H})$ and its hyperbolic and symplectic variants, general linear groups over finite fields, see the survey [Ol'91]. Most of these groups are Roelcke-precompact non-Archimedean Polish groups, and those classification results were reobtained as special cases of Theorem 1.2.11 by Tsankov. Ol'sanskiĭ also classified the irreducible unitary representations of the non-locally-compact Polish group $\text{Aut}(\mathbf{T})$ where $\mathbf{T}$ is the regular tree of countably infinite arity, concluding again that $\text{Aut}(\mathbf{T})$ is of Type I [Ol'80].

In Chapter 5, we introduce an infinite-arity analogue of universal Burger-Mozes groups, replacing $\mathbf{T}_d$ with $\mathbf{T}$ and $F \curvearrowright \{1, \dots, d\}$ with a closed permutation group $\Gamma \curvearrowright \Omega$ on some countably infinite set Ω . Using model theoretic tools,

we extend Ol'sanskii's classification of the irreducible unitary representations of $\text{Aut}(\mathbf{T})$ to $\mathbf{U}(\Gamma)$ where $\Gamma \curvearrowright \Omega$ is 2-transitive and oligomorphic and conclude that $\mathbf{U}(\Gamma)$ is of type I. Let us give some more details.

The study of the irreducible unitary representations of any subgroup G of $\text{Aut}(\mathbf{T})$ splits into two very distinct parts.

Definition 1.2.29. Let G be a closed subgroup of $\text{Aut}(\mathbf{T})$.

1. An irreducible representation $G \curvearrowright \mathcal{H}$ is *spherical* if there exists a vertex $x \in \mathbf{T}$ such that $\mathcal{H}_x \neq 0$.
2. A representation $G \curvearrowright \mathcal{H}$, irreducible or not, is *aspherical* if $\mathcal{H}_x = 0$ for every vertex $x \in \mathbf{T}$.

Ol'sanskii's classification of spherical irreducible representations of $\text{Aut}(\mathbf{T})$ extends with the same proof to every metrically 2-transitive subgroup of $\text{Aut}(\mathbf{T})$ [Ol'80, Sec. 5.3–5.5]. In particular, it applies to $\mathbf{U}(\Gamma)$ provided that $\Gamma \curvearrowright \Omega$ is 2-transitive. In fact, the spherical part of the unitary dual of $\mathbf{U}(\Gamma)$ is independent of the choice of the 2-transitive closed permutation group Γ . We still give a brief description of these representations for the interested reader that is familiar with the GNS construction. Fix $x \in \mathbf{T}$, let $c \in [-1, 1]$ and denote by d the graph distance in $\mathbf{T}$. The map $\varphi_c: \mathbf{U}(\Gamma) \rightarrow \mathbb{C}$ defined by $\varphi_c(g) = c^{d(g(x), x)}$ is positive definite. Moreover, the representation π_c obtained from φ_c by the GNS construction is irreducible and spherical. In fact, every spherical irreducible representation of $\mathbf{U}(\Gamma)$ is isomorphic to π_c for some unique $c \in [-1, 1]$. It also follows from Ol'sanskii's results that the process of restricting representations of $\text{Aut}(\mathbf{T})$ to $\mathbf{U}(\Gamma)$ induces a one-to-one correspondence between the spherical irreducible representations of $\text{Aut}(\mathbf{T})$ and those of $\mathbf{U}(\Gamma)$.

The following is the main result of Chapter 5 (Theorem 5.1.1 therein).

Theorem 1.2.30. Let $\Gamma \curvearrowright \Omega$ be an oligomorphic and 2-transitive closed permutation group.

1. Every spherical irreducible representation of $\mathbf{U}(\Gamma)$ is the restriction of a uniquely determined spherical irreducible representation of $\text{Aut}(\mathbf{T})$.
2. Every aspherical unitary representation of $\mathbf{U}(\Gamma)$ splits as a sum of aspherical irreducible subrepresentations.
3. Every aspherical irreducible representation of $\mathbf{U}(\Gamma)$ is of the form $\text{Ind}_V^{\mathbf{U}(\Gamma)}(\sigma)$ where $V \leq \mathbf{U}(\Gamma)$ is an open subgroup and σ is an irreducible representation of V which factorizes through a finite quotient of V .
4. The group $\mathbf{U}(\Gamma)$ is of Type I.

This theorem is very similar to Theorems 1.2.10 and 1.2.11, which classify the representations of compact groups and Roelcke precompact non-Archimedean Polish groups respectively. However, the representation theory of $\mathbf{U}(\Gamma)$ significantly differs from that of Roelcke precompact non-Archimedean Polish groups because of the spherical case. Indeed, Roelcke precompact non-Archimedean Polish groups only have countably many irreducible representations and all their representations split as sums of irreducibles. The same holds for $\mathbf{U}(\Gamma)$ if one

forbids sphericity but both parts fail in general. Recall also that Roelcke pre-compact non-Archimedean Polish groups have Property (T) (Theorem 1.2.23) whereas $\mathbf{U}(\Gamma)$ does not.

Chapter 2

Tannaka-Krein Duality

The content of this chapter is the same as that of the article [Bar25].

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2.1 Introduction

A fundamental question in abstract harmonic analysis is: How much information about a topological group can one recover from its representation theory? In this paper, we are interested in cases where the group can be fully reconstructed.

More precisely, a *unitary representation* of a topological group G is a continuous group morphism π from G to the unitary group $\mathcal{U}(\mathcal{H}_\pi)$ of a complex Hilbert space $\mathcal{H}_\pi$. Continuity means here that all the maps of the form $f_{\xi,\eta}^\pi : g \mapsto \langle \pi(g)\eta, \xi \rangle$ for $\xi, \eta \in \mathcal{H}_\pi$, called the *matrix coefficients* of π , are continuous. π is said *irreducible* if $\mathcal{H}_\pi$ admits no non-trivial G -invariant closed subspace. In the well-behaved cases, every unitary representation of G decomposes uniquely into an aggregate of irreducible subrepresentations. Harmonic analysis on such a group G reduces to the study of its *unitary dual* $\widehat{G}$, namely the set of isomorphism classes of irreducible unitary representations of G . In various subcases, duality theories have been established, allowing for the abstract reconstruction of G from $\widehat{G}$.

A fundamental instance of this is the case of locally compact Abelian groups with the celebrated Pontryagin–van Kampen duality theory [Pon34, van35]. Indeed, if G is locally compact and Abelian, its irreducible representations all have dimension 1 and $\widehat{G}$ can be identified with $\text{Hom}(G, \mathbb{S}^1)$, the continuous group morphisms from G to the unit circle. In particular, $\widehat{G}$ is also an Abelian group that is moreover locally compact when endowed with the compact-open topology. Finally, G is canonically isomorphic to its bi-dual.

Another context where representation theory is very tame is the compact case. Indeed, recall the Peter-Weyl Theorem [PW27] which states, in particular, that every representation of a compact group splits in an essentially unique way as a sum of finite dimensional irreducible subrepresentations. This ideal situation allowed Tannaka and Kreĭn to develop, independently and with different approaches, duality theories for compact groups.

Recall that the *Gel'fand spectrum* of an involutive algebra $\mathcal{B}$ is the set of non-zero multiplicative linear functionals $\mathcal{B} \rightarrow \mathbb{C}$ that commute with the involution. It is well known that continuity is automatic when $\mathcal{B}$ is a C^* -algebra. Kreĭn considered the algebra $\mathcal{B}_0(G)$ generated by the matrix coefficients of a general topological group G arising from its finite dimensional representations, which is not complete for the supremum norm in general, and proved the following technical but crucial result: positive linear functionals on $\mathcal{B}_0(G)$ are automatically continuous [Kre41]. As a consequence, the Gel'fand spectrum of $\mathcal{B}_0(G)$ and that of its completion coincide (the involution considered here is the one induced by complex conjugation). With some more data from the representation theory of G ,

Kreĭn gives this compact space a group structure. It becomes a compact topological group, sometimes called the *Bohr compactification* bG of G , and if G is compact then G and bG are canonically isomorphic.

Tannaka obtained similar duality results from a different perspective. He associated to a compact group G a monoid $\mathbf{T}(G)$ of *operations* on the class of representations of G . More explicitly, an element of $\mathbf{T}(G)$ is a family of operators $(u_\pi)_\pi$ where π ranges over all the finite dimensional representations of G and u_π is an operator on the same Hilbert space as π . Moreover, the family must commute with representation morphisms and preserve the common operations on representations: sum, tensor product and conjugation. The monoid law is pointwise composition. This structure can be endowed with a natural topology and is in fact a compact group canonically isomorphic to G [Tan39].

The wider class of *Roelcke-precompact* groups seems to retain a lot of the geometrical properties of compact groups. A topological group G is *Roelcke-precompact* if it is precompact in the Roelcke uniformity, in other words if for every open neighborhood U of the identity, there exists a finite subset F of G such that $G = UFU$. Roelcke-precompact Polish groups are receiving an increasing amount of interest. Cameron started the investigation by carrying out an extensive study of the dynamical properties of Roelcke-precompact permutation groups [Cam90]. Uspenksij [Usp98, Usp01] and Glasner [Gla12] showed that well known groups, such as the unitary group of the separable Hilbert space or $\text{Aut}(\mu)$ for an atomless standard Borel probability measure, are Roelcke-precompact and deduced strong properties such as minimality. [Iba16] and the pair of papers [BYT16, BYIT18] studied compactifications of such groups. More recently, Ibarlucia showed that Roelcke-precompact Polish groups have Kazhdan's Property (T) [Iba21], like compact groups. Moreover, the Roelcke-precompact Polish groups that are *non-Archimedean*, i.e. that admit a basis of identity neighborhoods consisting of open subgroups, have seen their unitary representations fully classified [Tsa12] in a way that reminds a lot of the Peter-Weyl Theorem. Indeed, let $\mathfrak{M}_G$ be the set of left translates of open subgroups of G , called the *open cosets*. Then G acts continuously on $\mathfrak{M}_G$ seen as a countable (see Item 1 of Remark 2.2.6) discrete space and it gives rise to a unitary representation:

$$\Lambda_G: G \curvearrowright \ell^2(\mathfrak{M}_G).$$

This canonical construction captures all the representation theory of G , reminding of the left-regular representation of a compact group. More precisely, the following holds:

Fact 2.1.1. *Let G be a Roelcke-precompact non-Archimedean Polish group.*

1. *Every unitary representation of G splits a sum of irreducible subrepresentations.*
2. *Every irreducible unitary representation of G is isomorphic to a subrepresentation of Λ_G .*
3. *The irreducible unitary representations of G separate points. Equivalently, Λ_G is faithful.*
4. *$\widehat{G}$ is countable.*

The first item is contained in [Tsa12, Th. 1.3]. The second item is a consequence of the same result and is proved as Corollary 2.2.9 below. The third is a straightforward consequence of (1) and (2), and is an analogue of the Gel'fand-Raikov Theorem for locally compact groups. The last one is a consequence of (2) and the fact that, $\mathfrak{M}_G$ being countable, $\ell^2(\mathfrak{M}_G)$ is separable.

The main source of examples for such groups is model theory. Indeed, by the classical Ryll-Nardzewski Theorem (see e.g. [Hod93, Th. 7.3.1]) the automorphism group of any $\aleph_0$ -categorical model theoretical structure is part of this class. This includes the group S_∞ of all permutations of the countable set, the group $\text{Aut}(\mathbb{Q}, <)$ of order preserving bijections of $\mathbb{Q}$, $\text{Homeo}(2^\mathbb{N})$, $\text{Aut}(R)$ the automorphism group of the random graph, or the group $\text{GL}(\infty, q)$ of linear automorphisms of the countably infinite vector space over the finite field $\mathbb{F}_q$.

There are however fundamental properties that are lost in the Roelcke-precompact case: their irreducible representations can have infinite dimension and there is no Haar measure available. Using dynamical and model-theoretical properties of these groups, we are still able to carry out similar constructions as Tanaka's and Kreĭn's. In particular, we obtain an analogue of Kreĭn's technical result, which is the key to both dualities in both contexts.

To that aim, we will consider several algebras of functions on G . They all live in $C_b(G)$, the normed algebra of bounded continuous complex-valued maps on G . $\mathcal{B}(G)$ will denote the set of matrix coefficients of G arising from all the unitary representation of G while $\mathcal{B}_0(G)$ will only contain those arising from finite sums of irreducible representations. Considering sums, tensor products and conjugates of representations, it is easily seen that $\mathcal{B}(G)$ is a subalgebra of $C_b(G)$, closed under complex conjugation. So is $\mathcal{B}_0(G)$ by Lemma 2.2.15. Let $\mathcal{A}$ be the linear span of the indicator maps of open cosets in G . This is in turn a subalgebra of $\mathcal{B}_0(G)$ closed under complex conjugation per Proposition 2.2.16 below. We insist on the following inclusions:

$$\mathcal{A} \subseteq \mathcal{B}_0(G) \subseteq \mathcal{B}(G) \subseteq C_b(G),$$

and on the fact that $\mathcal{A}$ is dense in $\mathcal{B}(G)$ (see Proposition 2.2.16). Note that in the compact case, this definition of $\mathcal{B}_0(G)$ coincides with the previous one. It seems Kreĭn's intent was to exhibit a somewhat *minimal* dual object, whose definition requires as little data as possible (in his case, the algebra $\mathcal{B}_0(G)$ linearly generated by a countable basis, with no need to remember the topology). Here, we will work with the smaller and more intrinsic algebra $\mathcal{A}$, which does not even need any harmonic analysis to be defined. Moreover, at the intersection of the compact case and ours, i.e. for G profinite and separable, $\mathcal{A} = \mathcal{B}_0(G)$. For all these reasons, and for the useful connections between $\mathcal{A}$ and model theory (see e.g. Corollary 2.3.3), we believe this is the right algebra to consider in this context. One of the main results of this paper is the following analogue of Kreĭn's technical result (restated and proved as Theorem 2.3.4):

Theorem 2.1.2. *Let G be Roelcke-precompact non-Archimedean Polish group. Multiplicative linear functionals $\mathcal{A} \rightarrow \mathbb{C}$ are automatically positive and continuous.*

Again, it implies that the Gel'fand spectra of $\mathcal{A}$ and its completion (i.e. the completion of $\mathcal{B}(G)$) coincide. Since this spectrum is compact, it cannot be isomorphic to G in full generality. However, it can be endowed with a structure

of *semi-topological $\ast$ -monoid*. This structure is known as the *Hilbert compactification* $\mathbf{H}(G)$ of G and has received quite some attention [GM14, BYT16]. Along the way, we obtain an identification of the Hilbert compactification of G as the monoid $\mathbf{P}(\mathfrak{M}_G)$ of partial elementary maps with algebraically closed domains on $\mathfrak{M}_G$ seen as a homogeneous model theoretical structure (complete definitions are given in section 1). When G is the automorphism group of some $\aleph_0$ -categorical structure $\mathfrak{M}$, then $\mathfrak{M}_G$ is very similar to $\mathfrak{M}^{\text{eq}}$ hence this identification can be seen as a reformulation of [BYIT18, Th. 02].

Tannaka's approach also bears fruit in this context, providing another realization of $\mathbf{H}(G)$. Indeed, consider the category $\mathbf{Rep}(G)$ of unitary representations of G that are isomorphic to a finite sum of irreducible representations. Let $\mathbf{T}(G)$ be the monoid of families of operators $(u_\pi)_{\pi \in \mathbf{Rep}(G)}$ that commute with sums, tensor products and representation morphisms (complete definitions are given in section 2.4 Definition 2.4.5). We obtained the following (restated and proved as Theorem 2.4.6):

Theorem 2.1.3. *Let G be a Roelcke-precompact non-Archimedean Polish group. Then $\mathbf{T}(G)$ is a compact semi-topological $\ast$ -monoid canonically isomorphic to the Hilbert compactification of G .*

Finally, we establish that the Hilbert compactification fully remembers the original group. Indeed, a Roelcke-precompact non-Archimedean Polish group G is homeomorphic to the set of invertible elements of $\mathbf{H}(G)$ via the canonical map that sends g in G to the *evaluation map at g* . Moreover, we can form the category whose objects are the $\mathbf{Rep}(G)$ for every Roelcke-precompact non-Archimedean Polish group G . With the right arrows (the *admissible* functors, see Definition 2.5.2), we have the following (corresponding to Theorems 2.5.1 and 2.5.3 respectively):

Theorem 2.1.4.

1. *Let G be a Roelcke-precompact non-Archimedean Polish group. The canonical map $G \longrightarrow \mathbf{H}(G)$ is a homeomorphic embedding that respects the algebraic structures. Its image is the set of invertibles of $\mathbf{H}(G)$.*
2. *The contravariant functor $\mathbf{Rep}$ that sends a Roelcke-precompact non-Archimedean Polish group to the category of its unitary representations is full and faithful, i.e. a duality.*

Numerous treatments, variants and generalizations of the Tannaka–Kreĭn duality can be found in the literature. An extensive exposition of the theory, recounting both Kreĭn's and Tannaka's approaches successively, appears in [HR70]. A more modern take on Tannaka's point of view, phrased in the language of category theory, which also lists developments and applications, figures in [JS91].

Tannaka–Kreĭn duality has sprouted roots in different domains of study. One is category theory, with the works of Grothendieck, Deligne and Saavedra-Rivano among others, who introduced *Tannakian formalism* to reverse the process of the classical duality. The idea is that to a *Tannakian category* $\mathcal{C}$ can be associated an algebraic group G such that $\mathcal{C}$ is equivalent to $\mathbf{Rep}(G)$ (see for instance [SR72]).

More recently, Tannaka–Kreĭn duality theory has been reappropriated in a prolific way for the study of *quantum groups* and *knot theory*. It also appears in mathematical physics, related to the *superselection principle* in quantum field theory. See the survey [Vaĭ15].

This article is organized as follows. In the first section, we introduce the notations and some necessary preliminary results. The second section is inspired from Kreĭn’s approach to the duality and contains a proof of Theorem 2.1.2. In section 2.4, where the formalism follows that of [JS91], we switch to Tannaka’s point of view and use the automatic continuity property to prove Theorem 2.1.3. In the last section, we prove Theorem 2.1.4.

2.2 Preliminaries

In this section, we set the global framework by fixing notations and stating useful general facts. We start with dynamical and model-theoretical notions.

Let G be a *Polish group* i.e. a separable and completely metrizable topological group. For subgroups U, V of G , we will write

$$U \backslash G / V = \{UgV, g \in G\}.$$

This is a partition of G . Note that if G is non-Archimedean, then it is Roelcke-precompact if and only if $U \backslash G / U$ is finite for every open subgroup U of G .

We will need the dynamical notion of algebraicity, which coincides with the model-theoretical one in the $\aleph_0$ -categorical case but not in general. It is only assumed that the definition of first-order structure, substructure and $\aleph_0$ -categoricity, in the sense of [Hod93], are known. Anyway, the only structure that is actually involved here, $\aleph_G$, is relational, so a substructure of $\aleph_G$ is nothing more than a subset. All the structures considered will be assumed countable. In particular, by an $\aleph_0$ -categorical structure, we always mean a countable one.

Definition 2.2.1. Let G be the automorphism group of some first-order structure $\aleph$ and $A \subseteq \aleph$.

1. We denote by G_A the pointwise stabilizer of A and by $G_{(A)}$ the setwise stabilizer of A .
2. If A is finite, an element $a \in \aleph$ is said to be *algebraic over A* if the orbit $G_A \cdot a$ is finite. We denote by $\text{acl}(A)$ the *algebraic closure of A* i.e. the set of all elements of $\aleph$ that are algebraic over A . If A is infinite, the algebraic closure of A is

$$\text{acl}(A) = \bigcup \{\text{acl}(B), B \subseteq A \text{ finite}\}.$$

3. A is said to be *algebraically closed* if $\text{acl}(A) = A$.

Recall that acl is a closure operator, in particular it is non-decreasing and satisfies $\text{acl}^2 = \text{acl}$.

Definition 2.2.2. Let $\aleph$ be a first-order structure. A *partial elementary map* φ on $\aleph$ is an elementary map between substructures of $\aleph$, where elementary means a

bijection that preserves all formulae in the language of $\mathfrak{M}$ with parameters in the domain of φ .

We will see these maps as subsets of $\mathfrak{M} \times \mathfrak{M}$. Thus, for partial elementary maps s, s' on $\mathfrak{M}$, $s \subseteq s'$ means s' extends s and $s \cup s'$ denotes the unique common extension of s and s' to the substructure of $\mathfrak{M}$ generated by $\text{dom}(s) \cup \text{dom}(s')$, if it exists.

We will denote by $\mathbf{P}(\mathfrak{M})$ the set of partial elementary maps on $\mathfrak{M}$ with algebraically closed domain and by $\mathbf{F}(\mathfrak{M})$ the set of partial elementary maps on $\mathfrak{M}$ with finite domain. Note that those sets are stable under *composition where defined* and inversion.

Finally, $\mathfrak{M}$ is *homogeneous* if every element of $\mathbf{F}(\mathfrak{M})$ extends to a *total* automorphism, i.e. an element of $\text{Aut}(\mathfrak{M})$.

We will be needing the following combinatorial result, a variant of B.H. Neumann's Lemma appearing as [BBMN76, Th.1]:

Theorem 2.2.3 (B.H. Neumann's Lemma). *Let G be a group acting on a set X such that all orbits are infinite. For all finite subsets $A, B \subseteq X$, there exist $g \in G$ such that $g(A) \cap B = \emptyset$.*

More precisely, the following translation of the above Theorem will be key in establishing the main result of section 2.3.

Lemma 2.2.4. *Let G be the automorphism group of some homogeneous countable structure $\mathfrak{M}$ and assume it is Roelcke-precompact. Let $s, s_1, \dots, s_n$ be partial elementary map on $\mathfrak{M}$ with finite domain such that:*

$$\forall i \leq n, \text{dom}(s_i) \not\subseteq \text{acl}(\text{dom}(s))$$

Then, there exist $g \in G$ that extends s but none of the s_i 's.

Proof. By homogeneity of $\mathfrak{M}$, we can pick $g_0 \in G$ extending s . Fix also $x_i \in \text{dom}(s_i) \setminus \text{acl}(\text{dom}(s))$ for every $i \leq n$. Using Roelcke-precompactness of G , Lemma 2.4 in [ET16] tells us that the action

$$G_{\text{acl}(\text{dom}(s))} \curvearrowright \mathfrak{M} \setminus \text{acl}(\text{dom}(s))$$

only has infinite orbits. Now B.H. Neumann's Lemma above gives u in $G_{\text{acl}(\text{dom}(s))}$ such that:

$$u(\{x_1, \dots, x_n\}) \cap \{g_0^{-1}(s_1(x_1)), \dots, g_0^{-1}(s_n(x_n))\} = \emptyset.$$

Then $g = g_0 u$ extends s but none of the s_i 's □

Remark 2.2.5. Let $\mathfrak{M}$ be a homogeneous structure. $\mathbf{P}(\mathfrak{M})$ is a *semi-topological *-monoid compactification* of $G := \text{Aut}(\mathfrak{M})$, i.e.:

- it is stable under *composition where defined* with a neutral element: $\text{Id}_{\mathfrak{M}}$,
- it has an involutive anti-automorphism:

$$u \mapsto u^* = \{(y, x), (x, y) \in u\},$$

- it can be endowed with a compact topology that makes composition separately continuous and the involution continuous,

- there is a continuous action $G \curvearrowright \mathbf{P}(\mathfrak{M})$, namely by composition on the left,
- there is a homeomorphic embedding $G \hookrightarrow \mathbf{P}(\mathfrak{M})$ with dense image, namely natural inclusion, such that on the image of G the involution restricts to inversion and the monoid law restricts to the product.

An extensive survey regarding such objects can be found in [GM14] and this precise construction is inspired from [BYIT18, Prop. 3.8]. While additional assumptions appear in [BYIT18], the same proofs apply. We reproduce them for completeness.

To define the topology, consider $K := \mathfrak{M} \cup \{\infty\}$ the one-point compactification of the discrete structure $\mathfrak{M}$. Elements $p \in K^K$ can be seen as partial maps $\mathfrak{M} \rightarrow \mathfrak{M}$, undefined wherever $p(x) = \infty$. The domain of p corresponds to the set $p^{-1}(\mathfrak{M})$.

K^K is endowed with the product topology, which is compact and Hausdorff. Subbasic neighborhoods of an element $u \in \mathbf{P}(\mathfrak{M})$ are of two kinds:

$$O_A^u = \{v \in \mathbf{P}(\mathfrak{M}), \forall x \in A, v(x) = u(x)\},$$

where $A \subseteq \text{dom}(u)$ is finite, and

$$U_{B,C}^u = \{v \in \mathbf{P}(\mathfrak{M}), \forall x \in B, [x \in \text{dom}(v) \Rightarrow v(x) \notin C]\},$$

where $B \subseteq \mathfrak{M} \setminus \text{dom}(u)$ and $C \subseteq \mathfrak{M}$, both finite.

Clearly, $G \subseteq \mathbf{P}(\mathfrak{M})$. Moreover, $\text{Aut}(\mathfrak{M})$ does not see the open sets of the second kind in the above description, hence the inclusion is indeed a homeomorphic embedding. In fact, $\mathbf{P}(\mathfrak{M})$ is exactly the closure of G hence compact.

Indeed, fix $u \in \mathbf{P}(\mathfrak{M})$. Let U be a neighborhood of u . By the above description of the topology, we can assume that there exists $A, B, C \subseteq \mathfrak{M}$ finite such that $A \subseteq \text{dom}(u)$, $B \subseteq \mathfrak{M} \setminus \text{dom}(u)$ and:

$$U = O_A^u \cap U_{B,C}^u.$$

Since u is elementary and $\mathfrak{M}$ is homogeneous, there exists $g \in G$ such that $u|_A = g|_A$. Note that $\text{acl}(A) \subseteq \text{acl}(\text{dom}(u)) = \text{dom}(u)$, hence $B \subseteq \mathfrak{M} \setminus \text{acl}(A)$. By definition, the (well defined) action $G_A \curvearrowright \mathfrak{M} \setminus \text{acl}(A)$ only has infinite orbits. Using B.H. Neumann's Lemma 2.2.3, there exists $h \in G_A$ such that $h(B) \cap g^{-1}(C) = \emptyset$. Then $gh \in U \cap G$, proving that $\mathbf{P}(\mathfrak{M}) \subseteq \overline{G}$.

For the reverse inclusion, let $u \in \overline{G}$. Considering open sets of the first kind above, finite restrictions of u all extend to automorphism of $\mathfrak{M}$: u is thus an elementary map. Toward a contradiction, suppose that there exists some element $a \in \text{acl}(\text{dom}(u)) \setminus \text{dom}(u)$. Fix $A \subseteq \text{dom}(u)$ finite such that $a \in \text{acl}(A)$ and let $g_0 \in G$ be any extension of $u|_A$. Set $C := g_0 G_A \cdot a$, which is a finite subset of $\mathfrak{M}$. Then $u \in O_A^u \cap U_{\{a\},C}^u$ while $G \cap O_A^u \cap U_{\{a\},C}^u = g_0 G_A \cap U_{\{a\},C}^u = \emptyset$, a contradiction to $u \in \overline{G}$.

Remark 2.2.6. Let G be a Roelcke-precompact non-Archimedean Polish group. We recall well-known facts about the structure $\mathfrak{M}_G$ and give sketches of proofs.

1. $\mathfrak{M}_G$ is countable. Indeed, let (U_n) be a countable basis of open neighborhoods of 1_G consisting of open subgroups and let $D \subseteq G$ be a countable dense subset. Let U be an arbitrary open subgroup of G . There exists $n \in \mathbb{N}$ such that $U_n \subseteq U$. Since G is Roelcke-precompact, there is $F \subseteq D$ finite such that $U = U_n F U_n$. This only allows for countably many distinct open subgroups. Moreover, G/U is a collection of disjoint non-empty open sets. Since G is separable, G/U must be countable.

2. G naturally acts on the left of $\mathfrak{M}_G$. This action is continuous and the associated morphism $G \rightarrow \text{Sym}(\mathfrak{M}_G)$ is a homeomorphic embedding. Because G and $\text{Sym}(\mathfrak{M}_G)$ are both Polish, G must be closed in $\text{Sym}(\mathfrak{M}_G)$. Thus, after naming the orbits of the diagonal actions $G \curvearrowright \mathfrak{M}_G^n$ for every $n \in \mathbb{N}$, we can view $\mathfrak{M}_G$ as a discrete first-order homogeneous structure such that:

$$G = \text{Aut}(\mathfrak{M}_G).$$

3. Let gU, hV be cosets of G . Then, $gU \cap hV \neq \emptyset$ if and only if there is $f \in G$ such that $fU = gU$ and $fV = hV$. In particular, if $gU \cap hV$ is non-empty, then it is a left-translate of $U \cap V$.
4. Suppose $G = \text{Aut}(\mathfrak{M})$ for some countable and homogeneous structure $\mathfrak{M}$. There is a natural map $F(\mathfrak{M}) \rightarrow \mathfrak{M}_G$. Indeed, for s in $F(\mathfrak{M})$, fix an extension $g_0 \in G$. Then, a given automorphism g of $\mathfrak{M}$ extends s if and only if $g \in g_0 G_{\text{dom}(s)}$ and this coset does not depend on the choice of g_0 . Similarly, if $\text{dom}(s) = A \subseteq B$ where A and B are finite, there is a bijection between G_A/G_B and $\{s' \in F(\mathfrak{M}), s \subseteq s' \text{ and } \text{dom}(s') = B\}$, namely $s' \mapsto g_0^{-1} g G_B$ where g is any extension of s' (this map does depend on the choice of g_0).

Switching to representation theory, we will write $\text{Hilb}(G)$ for the closure of $\mathcal{B}(G)$ in $C_b(G)$ with respect to the norm topology. This is called the Hilbert algebra of G . The *Hilbert compactification* $\mathbf{H}(G)$ of G is the Gel'fand spectrum of $\text{Hilb}(G)$, i.e. the weak*-compact space of non-zero multiplicative linear functionals on $\text{Hilb}(G)$ that commute with complex conjugation. Recall that these are automatically continuous since $\text{Hilb}(G)$ is a C^* -algebra.

Given two representations π, π' of G , a morphism of representations between π and π' , or *intertwining operator*, is a bounded operator

$$h: \mathcal{H}_\pi \rightarrow \mathcal{H}_{\pi'}$$

that commutes with the action of G :

$$\forall g \in G, h \circ \pi(g) = \pi'(g) \circ h.$$

The set of such operators will be denoted by $\text{Hom}(\pi, \pi')$ or $\text{Hom}_G(\mathcal{H}_\pi, \mathcal{H}_{\pi'})$. An *isomorphism* between π and π' is an element $h \in \text{Hom}(\pi, \pi')$ that is also a surjective isometry. $\widehat{G}$ will denote the set of isomorphism classes of irreducible representation of G .

We now state the following classical theorem for later references. See for instance [Fol95, Th. 3.5]

Theorem 2.2.7 (Schur's Lemma). *Let G be a topological group and π a continuous unitary representation of G . Then π is irreducible if and only if $\text{Hom}(\pi, \pi) = \mathbb{C} \cdot 1$.*

The following are consequences of the results in [Tsa12]. For the definition and properties of *induced representations*, we refer the reader to section 6 of [Fol95]. It is usually defined in the locally compact setting using invariant measures but the basic results still hold in this context with the same proofs, and the formalism is even simpler. Indeed, our quotient spaces are all discrete hence can be endowed with the counting measure. The distinction between isomorphic representations is often omitted.

Lemma 2.2.8. *Let G be a Roelcke-precompact non-Archimedean Polish group and let π be an irreducible representation of G . There exists an open subgroup V of G such that π is isomorphic to a subrepresentation of the associated quasi-regular representation $\lambda_{G/V}: G \curvearrowright \ell^2(G/V)$.*

Proof. Using Theorem 4.2 from [Tsa12], π is isomorphic to an induced representation of the form $\text{Ind}_H^G(\sigma)$, where H is an open subgroup of G and σ is an irreducible representation of H that factors through a finite quotient of H . More explicitly, there exists V an open normal subgroup of H of finite index such that σ is the pullback of an irreducible representation of the finite group H/V .

It is a fact that the quasi-regular representations $\lambda_{G/V}: G \curvearrowright \ell^2(G/V)$ and $\lambda_{H/V}: H \curvearrowright \ell^2(H/V)$ are linked in the following way:

$$\lambda_{G/V} = \text{Ind}_V^G(\mathbf{1}_V) = \text{Ind}_H^G(\text{Ind}_V^H(\mathbf{1}_V)) = \text{Ind}_H^G(\lambda_{H/V}), \quad (2.1)$$

where $\mathbf{1}_V$ denotes the trivial one-dimensional representation of V . The first and last equalities are basic properties of induction and the second one is given by the theorem on induction by stages (see [Fol95, Theorem 6.14]). Now, applying the Peter-Weyl Theorem to the finite group H/V , we get that σ is a subrepresentation of $\lambda_{H/V}$. Since induction preserves subrepresentations, we deduce from (2.1) that $\pi = \text{Ind}_H^G(\sigma)$ is a subrepresentation of $\lambda_{G/V} = \text{Ind}_H^G(\lambda_{H/V})$. $\square$

Corollary 2.2.9. *Let G be a Roelcke-precompact non-Archimedean Polish group. Every irreducible representation of G is a subrepresentation of*

$$\Lambda_G: G \curvearrowright \ell^2(\mathfrak{M}_G),$$

which is faithful.

Proof. Clearly, $\lambda_{G/V}$ is a subrepresentation of Λ_G for every $V \leq G$ open. The faithfulness of Λ_G is easily seen directly and is also a consequence of the fact that closed permutation groups satisfy the Gel'fand-Raikov Theorem [Tsa12, Th. 1.1]. $\square$

The following lemmas are probably well known but we sketch a proof for lack of a reference in such generality.

Lemma 2.2.10. *Let G be a topological group, let $\pi_1, \dots, \pi_n$ be irreducible representations of G and let $\pi := \pi_1 \oplus \dots \oplus \pi_n$. If σ is a non-zero subrepresentation of π , then σ contains an irreducible representation isomorphic to π_i for some i .*

Proof. Let $\mathcal{H}, \mathcal{H}_1, \dots, \mathcal{H}_n$ be the underlying spaces of σ and $\pi_1, \dots, \pi_n$ respectively. Let $p, p_1, \dots, p_n$ be the orthogonal projections on $\mathcal{H}, \mathcal{H}_1, \dots, \mathcal{H}_n$ respectively. Then $p_1 + \dots + p_n = 1$ hence $p = pp_1 + \dots + pp_n$. There must exist i such that $pp_i \neq 0$. Then pp_i is a non-zero intertwining operator between π_i and σ . Applying Lemma 1.A.5 in [Bd20], we get the existence of a subrepresentation of σ (the closure of $\text{im } pp_i$) which is isomorphic to a subrepresentation of π_i (the orthogonal of $\ker pp_i$). This suffices to conclude since p_i is irreducible. $\square$

Lemma 2.2.11. *Let G be a topological group and let $\pi: G \curvearrowright \mathcal{H}$ be a representation of G that splits as a finite sum of irreducible subrepresentations. There is a unique decomposition*

$$\mathcal{H} = \bigoplus_{\lambda \in \widehat{G}} \mathcal{H}_\lambda$$

into closed invariant subspaces where for every $\lambda \in \widehat{G}$, all the irreducible subrepresentations of $\mathcal{H}_\lambda$ have isomorphism type λ .

This is called the isotypical decomposition of π .

Proof. For existence, write $\pi = \pi_1 \oplus \dots \oplus \pi_n$ for some irreducible representations $\pi_i: G \curvearrowright \mathcal{H}_i$ and simply let

$$\mathcal{H}_\lambda := \bigoplus_{i \in I, \pi_i \in \lambda} \mathcal{H}_i.$$

It has the desired property by Lemma 2.2.10.

As for uniqueness, suppose another such decomposition is given:

$$\mathcal{H} = \bigoplus_{\lambda \in \widehat{G}} \mathcal{H}_\lambda = \bigoplus_{\lambda \in \widehat{G}} \mathcal{H}'_\lambda.$$

Let $\lambda \in \widehat{G}$. It suffices to show that $\mathcal{H}_\lambda \subseteq \mathcal{H}'_\lambda$ which in turn reduces to showing that if $\pi_i \in \lambda$ then $\mathcal{H}_i \subseteq \mathcal{H}'_\lambda$. So let $i \leq n$ be such that $\pi_i \in \lambda$. Let p be the orthogonal projection on $\mathcal{H}_i$ and, for every $\lambda \in \widehat{G}$, let p_λ be the projection on $\mathcal{H}'_\lambda$. Then $\sum_{\lambda \in \widehat{G}} p_\lambda = 1$ and:

$$p = \sum_{\lambda \in \widehat{G}} p p_\lambda. \quad (2.2)$$

But if $\mu \neq \lambda$, then $p p_\mu = 0$. Otherwise, by Lemma 1.A.5 of [Bd20], it implies the existence of a non-zero subrepresentation of π_i (the closure of $\text{im } p p_\mu$) which is isomorphic to a subrepresentation of $G \curvearrowright \mathcal{H}'_\mu$ (the orthogonal of $\ker p p_\lambda$). By choice of $\mathcal{H}'_\mu$ and irreducibility of π_i , we get $\pi_i \in \mu$, a contradiction. Equation (2.2) becomes $p = p p_\lambda$, i.e. $\mathcal{H}_i \subseteq \mathcal{H}'_\lambda$. □

Lemma 2.2.12. *Let G be a topological group, let π be an irreducible representation of G and let $n \in \mathbb{N}$. Every subrepresentation of $n \cdot \pi := \bigoplus_{1 \leq i \leq n} \pi$ is isomorphic to $k \cdot \pi$ for some $k \leq n$.*

Proof. Let σ be a subrepresentation of $n \cdot \pi$. Using Lemma 2.2.10 and Zorn's Lemma, we see that σ is also isomorphic to a multiple of π , say $\sigma \simeq k \cdot \pi$ for some possibly infinite cardinal k . However, $\text{Hom}(\sigma, \sigma) = k^2 \cdot \text{Hom}(\pi, \pi) \simeq k^2 \cdot \mathbb{C}$ embeds into $\text{Hom}(n \cdot \pi, n \cdot \pi) \simeq n^2 \cdot \text{Hom}(\pi, \pi) \simeq n^2 \cdot \mathbb{C}$. Thus $k \leq n$. The converse is trivial. □

Lemma 2.2.13. *Let G be a topological group and let $\pi: G \curvearrowright \mathcal{H}$ be a representation of G that splits as finite sum of irreducibles. Every subrepresentation of π also splits as a finite sum of irreducibles.*

Proof. Let $\sigma \subseteq \pi$ and consider the isotypical decomposition of π given by Lemma 2.2.11: $\mathcal{H} = \bigoplus_{\lambda \in \widehat{G}} \mathcal{H}_\lambda$. Let p be the orthogonal projection on the space $\mathcal{K} \subseteq \mathcal{H}$ underlying σ and, for every $\lambda \in \widehat{G}$, let p_λ be the orthogonal projection on $\mathcal{H}_\lambda$. Then $1 = \sum_{\lambda \in \widehat{G}} p_\lambda$ hence

$$p = \sum_{\lambda, \mu \in \widehat{G}} p_\mu p p_\lambda. \quad (2.3)$$

If $\mu, \lambda \in \widehat{G}$, and $p_\mu p p_\lambda \neq 0$, then $\text{Hom}_G(\mathcal{H}_\lambda, \mathcal{H}_\mu) \neq 0$. In particular, by Lemma 1.A.5 in [Bd20], there is a non-zero subrepresentation of $\mathcal{H}_\lambda$ isomorphic to a subrepresentation of $\mathcal{H}_\mu$. Using Lemma 2.2.10 and the properties of the isotypical decomposition, we get that $\lambda = \mu$. Thus, Equation (2.3) becomes $p = \sum_{\lambda \in \widehat{G}} p_\lambda p p_\lambda$ and we get $\mathcal{K} = \bigoplus_{\lambda \in \widehat{G}} \mathcal{K} \cap \mathcal{H}_\lambda$. We conclude with Lemma 2.2.12. $\square$

Lemma 2.2.14. *Let G be a Roelcke-precompact non-Archimedean Polish group and let V be an open subgroup of G . The associated quasi-regular representation $G \curvearrowright \ell^2(G/V)$ splits as a finite sum of irreducible subrepresentations.*

Proof. Let H be the commensurator of V , meaning:

$$H = \{g \in G, [V, V \cap gVg^{-1}] < \infty \quad \& \quad [gVg^{-1}, V \cap gVg^{-1}] < \infty\}.$$

The hypotheses on G imply that V has finite index in H [Tsa12, Lem. 2.7]. Hence,

$$V' := \bigcap_{h \in H} hVh^{-1},$$

is in fact a finite intersection of conjugates of V . Thus, it is open and has finite index in V by definition of H .

We have obtained $V' \subseteq V$ open and, as a subgroup of H , normal with finite index. In particular, H/V' is a finite group hence the quasi-regular representation $\lambda_{H/V'}$ of H is finite dimensional. It thus splits as a finite sum of irreducible subrepresentations:

$$\lambda_{H/V'} = \bigoplus_{1 \leq i \leq n} \sigma_i.$$

Then, by the basic properties of induction:

$$\lambda_{G/V'} = \text{Ind}_H^G(\lambda_{H/V'}) = \bigoplus_{1 \leq i \leq n} \text{Ind}_H^G(\sigma_i)$$

Finally, each of the $\text{Ind}_H^G(\sigma_i)$ is irreducible by [Tsa12, Prop. 4.1]. Moreover, since $V' \subseteq V$ with finite index, $\lambda_{G/V}$ is a subrepresentation of $\lambda_{G/V'}$ hence also splits as a finite sum of irreducibles by Lemma 2.2.13. $\square$

Corollary 2.2.15. *Let G be a Roelcke-precompact non-Archimedean Polish group and let π_1, π_2 be two irreducible representations of G . Then $\pi_1 \otimes \pi_2$ decomposes as a finite sum of irreducible subrepresentations.*

Proof. Using Lemma 2.2.8, we can find two open subgroups V_1 and V_2 of G such that π_i is a subrepresentation of λ_{G/V_i} , $i = 1, 2$.

Since there is a natural surjective map

$$(V_1 \cap V_2) \backslash G / (V_1 \cap V_2) \longrightarrow V_1 \backslash G / V_2$$

and G is Roelcke-precompact, $V_1 \backslash G / V_2$ is finite.

Write $V_1 \backslash G / V_2 = \{V_1 f_1 V_2, \dots, V_1 f_n V_2\}$ for some $f_1, \dots, f_n \in G$ (in an injective way). Then:

$$\begin{aligned} \ell^2(G/V_1) \otimes \ell^2(G/V_2) &\simeq \ell^2(G/V_1 \times G/V_2) \\ &\simeq \bigoplus_{1 \leq i \leq n} \ell^2(G \cdot (V_1, f_i V_2)) \\ &\simeq \bigoplus_{1 \leq i \leq n} \ell^2\left(G / \left(V_1 \cap f_i V_2 f_i^{-1}\right)\right). \end{aligned}$$

In particular, we have shown that $\pi_1 \otimes \pi_2$ is a subrepresentation of a finite sum of quasi-regular representations. We conclude using Lemmas 2.2.14 and 2.2.13. $\square$

Recall that $\mathcal{A}$ is the linear span of the indicator maps of open cosets in G . From the above observations follow the claims made about $\mathcal{A}$ in the introduction:

Proposition 2.2.16. *Let G be a Roelcke-precompact non-Archimedean Polish group. Then $\mathcal{A}$ is a subalgebra of $\mathcal{B}_0(G)$ which is closed under the involutions induced by inversion in G and complex conjugation. Hence the following chain of algebras*

$$\mathcal{A} \subseteq \mathcal{B}_0(G) \subseteq \mathcal{B}(G)$$

with $\mathcal{A}$ dense in $\mathcal{B}(G)$.

Proof. Let gV be an open coset in G and let $\mathbb{1}_{gV}: G \rightarrow \{0, 1\}$ be the indicator map of gV . Denote by $\delta_V, \delta_{gV} \in \ell^2(G/V)$ the Dirac functions at V and gV respectively, and by $\lambda_{G/V}: G \curvearrowright \ell^2(G/V)$ the quasi-regular representation of G . Then, for every $h \in G$,

$$\mathbb{1}_{gV}(h) = \langle \lambda_{G/V}(h) \delta_V, \delta_{gV} \rangle.$$

This, in combination with Lemma 2.2.14, shows that $\mathbb{1}_{gV}$ belongs to $\mathcal{B}_0(G)$. Thus $\mathcal{A} \subseteq \mathcal{B}_0(G)$.

Using Item 3 of Remark 2.2.6, $\mathcal{A}$ is indeed a subalgebra of $\mathcal{B}_0(G)$. It is easily seen on the generators that $\mathcal{A}$ is stable under both of the involutions mentioned.

As for density, by Item 1 of Fact 2.1.1, every matrix coefficient of G can be approximated by linear combinations of matrix coefficients arising from irreducible representations. In turn, by Lemma 2.2.8, every matrix coefficient arising from an irreducible representation of G also comes from a representation of the form $\ell^2(G/V)$ for some open subgroup $V \leq G$. Finally, given V an open subgroup of G , the family $(\delta_{gV})_{gV \in G/V}$ is the canonical Hilbert basis of $\ell^2(G/V)$. Thus, linear combinations of the matrix coefficients associated to these vectors, which all appear in the list of generators of $\mathcal{A}$, can approximate all matrix coefficients coming from $G \curvearrowright \ell^2(G/V)$. This shows the density of $\mathcal{A}$ in $\mathcal{B}(G)$. $\square$

In the next sections, we give two different characterizations of $\mathbf{H}(G)$ for a Roelcke-precompact non-Archimedean Polish group G . For a detailed study of the Hilbert compactification of such groups, see [BYIT18].

2.3 Application to Kreĭn's duality

In [Kre41], Kreĭn associated to a compact group G a dual object consisting of the algebra $\mathcal{B}_0(G)$ of finite dimensional matrix coefficients of G together with a specific *structured* linear basis. More precisely, fix a compact group G and choose $(\pi_\lambda)_{\lambda \in \widehat{G}}$ a family of representatives of the isomorphism classes for the irreducible representations of G . Recalling that these are finite dimensional, fix an orthonormal basis $(e_1^\lambda, \dots, e_{d_\lambda}^\lambda)$ of the underlying Hilbert space of each representative. For every $\lambda \in \widehat{G}$ and $g \in G$, the matrix of the operator $\pi_\lambda(g)$ in the basis (e_i^λ) is given by:

$$[\pi_\lambda(g)]_{(e_i^\lambda)} = (f_{i,j}^\lambda(g))_{1 \leq i,j \leq d_\lambda},$$

where $f_{i,j}^\lambda(g) = \langle \pi_\lambda(g)e_j^\lambda, e_i^\lambda \rangle$ for every $\lambda \in \widehat{G}$ and all $i, j \leq d_\lambda$. It is a consequence of the Peter-Weyl Theorem that the following is a linear basis of $\mathcal{B}_0(G)$:

$$B = \left((f_{i,j}^\lambda(g))_{1 \leq i,j \leq d_\lambda} \right)_{\lambda \in \widehat{G}}.$$

Kreĭn's dual object is the pair $(\mathcal{B}_0(G), B)$. The properties of such a pair can be abstracted and such objects are now called *Kreĭn algebras*. Note that the arrangement of the basis in definite squares is part of the data. In particular, an isomorphism of Kreĭn algebras is an isomorphism of algebras that moreover exchanges the bases while preserving the coordinates. Of course, different choices for (π_λ) and the bases yield isomorphic Kreĭn algebras.

To recover G from this object, one of the key steps in Kreĭn's duality is that positive linear functionals on $\mathcal{B}_0(G)$ are automatically continuous. As a consequence, the Gel'fand spectrum $\mathbf{S}_{\mathcal{B}_0(G)}$ of $\mathcal{B}_0(G)$ coincides with that of its completion. Since this completion is precisely $C(G)$ [PW27], one can recover the compact topological space G through Gel'fand duality. Furthermore, the composition law is recovered from the structured basis. Indeed, it can be redefined by setting, for all $\phi, \psi \in \mathbf{S}_{\mathcal{B}_0(G)}$ and every element $f_{i,j}^\lambda$ of the basis:

$$\phi \cdot \psi(f_{i,j}^\lambda) = \sum_{1 \leq k \leq d_\lambda} \phi(f_{i,k}^\lambda) \psi(f_{k,j}^\lambda).$$

In this context, the natural map $G \longrightarrow \mathbf{S}_{\mathcal{B}_0(G)}$ is a topological group isomorphism.

We describe a similar construction in the Roelcke-precompact non-Archimedean and Polish case where we emphasize on the following major differences:

1. Irreducible representations are often *infinite* dimensional,
2. $\mathcal{B}(G)$ is not dense in $C(G)$ in general.

The key result of this section is the automatic continuity of multiplicative linear functionals on $\mathcal{A}$, an analogue of the central technical lemma in the establishment of the original duality. We also obtain a model theoretic description of $\mathbf{H}(G)$ as $\mathbf{P}(\mathfrak{M}_G)$.

Recalling Fact 2.1.1, we consider the "universal" representation Λ_G given by the action $G \curvearrowright \ell^2(\mathfrak{M}_G)$ and form the matrix coefficients obtained from the canonical Hilbert basis $(\delta_x)_{x \in \mathfrak{M}_G}$ of $\ell^2(\mathfrak{M}_G)$. These are all the maps of the form

$$f_{x,y}: G \longrightarrow \mathbb{C}, \quad g \mapsto \langle g \cdot \delta_y, \delta_x \rangle = \begin{cases} 1 & \text{if } g(y) = x, \\ 0 & \text{otherwise.} \end{cases}$$

for $x, y \in \mathfrak{M}_G$. Note that these are exactly the indicator maps of open cosets in G together with the zero function, hence generate $\mathcal{A}$. Indeed, if $x = hU$ and $y = kV$, then $g(y) = x$ exactly when $U = V$ and $g \in hk^{-1}(kUk^{-1})$.

Alternatively, $\mathcal{A}$ can be described in terms of finite partial elementary maps on $\mathfrak{M}_G$. Indeed, define for every $s \in F(\mathfrak{M}_G)$ a map

$$e_s: G \longrightarrow \mathbb{C}, \quad g \mapsto \begin{cases} 1 & \text{if } g \text{ extends } s, \\ 0 & \text{otherwise.} \end{cases}$$

It follows from Item 4 in Remark 2.2.6 that these are also exactly the indicator maps of open cosets in G . Note that for $s, s' \in F(\mathfrak{M}_G)$, we have:

$$e_s \cdot e_{s'} = \begin{cases} e_{s \cup s'} & \text{if } s \cup s' \in F(\mathfrak{M}_G), \\ 0 & \text{otherwise.} \end{cases} \quad (2.4)$$

A special case of interest is when G can be presented as the automorphism group of some $\aleph_0$ -categorical structure $\mathfrak{M}$ that admits *weak elimination of imaginaries*. For the formal definition, we refer the reader to section 4.2 of [Hod93]. Informally, an $\aleph_0$ -categorical structure $\mathfrak{M}$ weakly eliminates imaginaries if we can recover all open subgroups of $G = \text{Aut}(\mathfrak{M})$ from the action $G \curvearrowright \mathfrak{M}$ up to finite index. More precisely, if $\mathfrak{M}$ is an $\aleph_0$ -categorical structure, the following property can be taken as a definition of *weakly eliminating imaginaries* (see [Tsa12] Lemma 5.1): For every open subgroup U of G , there exists a unique algebraically closed finite substructure A of $\mathfrak{M}$ such that

$$G_A \subseteq U \subseteq G_{(A)}.$$

Note that $G_{(A)}/G_A$ is isomorphic to $\text{Aut}(A)$, hence finite. In particular, G_A has finite index in U . It is a classical fact that all the examples from the introduction weakly eliminate imaginaries (see for example section 4.2 of [Hod93]).

In that case, we can extract a canonical linear basis from the previous generating family of $\mathcal{A}$. Namely, defining $e'_s: G \longrightarrow \{0, 1\}$ as above for every $s \in F(\mathfrak{M})$, we have the following:

Proposition 2.3.1. *Let $\mathfrak{M}$ be a countable $\aleph_0$ -categorical structure that admits weak elimination of imaginaries and let $G = \text{Aut}(\mathfrak{M})$. Then*

$$B := (e'_s, s \in F(\mathfrak{M}), \text{dom}(s) = \text{acl}(\text{dom}(s)))$$

is a linear basis of $\mathcal{A}$ (and a subfamily of the previous generating family).

Proof. First, recall that $\aleph_0$ -categorical structures are homogeneous and have finite algebraicity: algebraic closures of finite sets are finite. Hence, using Item 4 in Remark 2.2.6, the elements of B are indeed indicator functions of open cosets, namely of all cosets associated to subgroups of the form $G_{\text{acl}(A)}$ for $A \subseteq \aleph$ finite.

Now if U is an open subgroup of G , we can find by weak elimination of imaginaries a finite algebraically closed substructure $A \subset \aleph$ such that $V := G_{\text{acl}(A)} \subseteq U$ with finite index. Then, writing $U = u_1 V \sqcup \dots \sqcup u_n V$ for some $u_1, \dots, u_n \in U$, we have for every $g \in G$:

$$\mathbb{1}_{gU} = \mathbb{1}_{gu_1 V} + \dots + \mathbb{1}_{gu_n V},$$

where $\mathbb{1}_X$ denotes the indicator function of the set $X \subseteq G$. In particular, B generates $\mathcal{A}$.

It remains to see that B is free. Assume we have an equation of the form

$$\lambda_1 e'_{s_1} + \dots + \lambda_n e'_{s_n} = 0 \quad (2.5)$$

for some $\lambda_1, \dots, \lambda_n \in \mathbb{C}$ and distinct $s_1, \dots, s_n \in F(\aleph)$ with algebraically closed domains. Let i be such that $\text{dom}(s_i)$ is minimal for inclusion. Using Lemma 2.2.4, there exists $g \in G$ that extends s_i without extending s_j for any $j \neq i$ such that $\text{dom}(s_j) \not\subseteq \text{dom}(s_i)$. Note that if $j \neq i$ is such that $\text{dom}(s_j) \subseteq \text{dom}(s_i)$ then the domains are equal by choice of i and there exists $a \in \text{dom}(s_i)$ such that

$$s_j(a) \neq s_i(a) = g(a).$$

Thus g only extends s_i . Evaluating the above linking equation (2.5) in g yields $\lambda_i = 0$. We conclude by induction on n . $\square$

The main object of study in this section is the Gel'fand spectrum of $\mathcal{A}$, denoted $\mathbf{S}_{\mathcal{A}}$, namely the non-zero multiplicative linear functionals on $\mathcal{A}$ that commute with complex conjugation. In the following lemma, we do not need to assume that G is Roelcke-precompact.

Lemma 2.3.2. *Let G be a non-Archimedean Polish group.*

Let $\phi: \mathcal{A} \rightarrow \mathbb{C}$ be a non-zero multiplicative linear functional. There exists a unique partial elementary map u_ϕ on $\aleph_G$ with algebraically closed domain such that:

$$\forall s \in F(\aleph_G), \phi(e_s) = \begin{cases} 1 & \text{if } u_\phi \text{ extends } s, \\ 0 & \text{otherwise.} \end{cases} \quad (2.6)$$

Proof. First note that for every $s \in F(\aleph_G)$, $\phi(e_s) \in \{0, 1\}$ since $e_s^2 = e_s$. Then, let

$$u_\phi = \bigcup_{\substack{s \in F(\aleph_G) \\ \phi(e_s) = 1}} s.$$

We will show that u_ϕ satisfies the claim.

– u_ϕ is a function:

Let $x \in \aleph_G$ be such that there exist $s, s' \in F(\aleph_G)$ with $x \in \text{dom}(s) \cap \text{dom}(s')$ and $\phi(e_s) = 1 = \phi(e_{s'})$. By multiplicativity of ϕ , we have

$$\phi(e_s e_{s'}) = \phi(e_s) \phi(e_{s'}) = 1.$$

In particular, $e_s e_{s'} \neq 0$ which implies the existence of $g \in G$ that extends both s and s' . Thus $s(x) = s'(x)$ and u_ϕ is a partial function $\aleph_G \rightarrow \aleph_G$.

- u_ϕ is a partial elementary map:

Since $\mathfrak{M}_G$ is a relational structure, every set in $\mathfrak{M}_G$ is a substructure. Thus, it is enough to show that each restriction of u_ϕ to a finite set extends to an automorphism of $\mathfrak{M}_G$. To that aim, let $A = \{a_1, \dots, a_n\} \subseteq \text{dom}(u_\phi)$ be finite. By definition of u_ϕ , there exist $s_1, \dots, s_n$ in $F(\mathfrak{M}_G)$ such that for all $i \leq n$, $a_i \in \text{dom}(s_i)$ and $\phi(e_{s_i}) = 1$. Up to replacing s_i with $s_{i|A}$, we can assume $\text{dom}(s_i) \subseteq A$ for every $i \leq n$. Using the multiplicativity of ϕ , we get:

$$\phi(e_{s_1} \cdots e_{s_n}) = \phi(e_{s_1}) \cdots \phi(e_{s_n}) = 1.$$

In particular, $e_{s_1} \cdots e_{s_n} \neq 0$ and there exists $g \in G$ a common extension of $s_1, \dots, s_n$ hence of $u_{\phi|A}$. Note also that, as

$$e_{s_1} \cdots e_{s_n} = e_{s_1 \cup \dots \cup s_n} = e_{u_{\phi|A}},$$

we have that $\phi(e_{u_{\phi|A}}) = 1$.

- $\text{dom}(u_\phi)$ is algebraically closed:

Let $A \subseteq \text{dom}(u_\phi)$ be finite and $b \in \text{acl}(A)$. As noted above, $\phi(e_{u_{\phi|A}}) = 1$ and there exists $g_0 \in G$ that extends $u_{\phi|A}$. An element $g \in G$ that coincides with u_ϕ on A lies in $g_0 G_A$ hence must send b in the finite set $g_0 G_A \cdot b$. In other terms, writing

$$g_0 G_A \cdot b = \{g_0(b), \dots, g_n(b)\}$$

for some $g_1, \dots, g_n \in G$ such that the $g_i(b)$ are pairwise distinct, we have:

$$e_{u_{\phi|A}} = e_{g_0|A \cup \{b\}} + \dots + e_{g_n|A \cup \{b\}}.$$

Applying ϕ to the above equation gives by linearity:

$$1 = \phi(e_{g_0|A \cup \{b\}}) + \dots + \phi(e_{g_n|A \cup \{b\}}).$$

Since ϕ is $\{0, 1\}$ valued at idempotents, there exists $i \leq n$ such that $\phi(e_{g_i|A \cup \{b\}}) = 1$. This shows that b lies in $\text{dom}(u_\phi)$ which is thus algebraically closed.

Now that we have built u_ϕ , recalling that $\phi(e_{u_{\phi|A}}) = 1$ for every finite subset A of $\mathfrak{M}_G$, the lemma is proved. $\square$

In the Roelcke-precompact and Polish case, this construction yields an identification of the Gel'fand spectrum $\mathbf{S}_A$ of A :

Corollary 2.3.3. *Let G be a Roelcke-precompact non-Archimedean Polish group. The map*

$$\begin{aligned} \mathbf{S}_A &\longrightarrow \mathbf{P}(\mathfrak{M}_G) \\ \phi &\longmapsto u_\phi \end{aligned}$$

is a homeomorphism.

Proof. It is straightforward from the previous result that the map is injective. We now show surjectivity, i.e. that if $u \in \mathbf{P}(\mathfrak{M}_G)$ is fixed, the conditions

$$\forall s \in F(\mathfrak{M}_G), \phi(e_s) = \begin{cases} 1 & \text{if } u \text{ extends } s, \\ 0 & \text{otherwise,} \end{cases}$$

always define a multiplicative linear functional $\phi_u : \mathcal{A} \rightarrow \mathbb{C}$. To that aim, assume we have an equation of the form

$$\sum_{i \leq n} \lambda_i e_{s_i} = 0 \quad (2.7)$$

for some $s_1, \dots, s_n \in F(\mathfrak{M}_G)$ and $\lambda_i \in \mathbb{C}$. Recalling the identification of G as $\text{Aut}(\mathfrak{M}_G)$ from Item 2 of Remark 2.2.6, we set up for the application of Lemma 2.2.4.

We define the following sets:

$$A = \bigcup \{\text{dom}(s_i), i \leq n, \text{dom}(s_i) \subseteq \text{dom}(u)\}$$

and

$$I = \{i \leq n, \text{dom}(s_i) \not\subseteq \text{acl}(A)\}.$$

By Lemma 2.2.4, there exists $g_0 \in G$ that extends $u|_A$ but none of the s_i for $i \in I$. We claim that it satisfies the following:

$$\forall i \leq n, [s_i \subseteq g_0 \iff s_i \subseteq u]. \quad (2.8)$$

Indeed, assume that $s_i \not\subseteq u$. Then either $\text{dom}(s_i)$ is included in $\text{dom}(u)$ or not. In the first case, there must exist $a \in \text{dom}(s_i)$ such that

$$s_i(a) \neq u(a) = g_0(a),$$

hence $s_i \not\subseteq g_0$. In the second case, we also have $\text{dom}(s_i) \not\subseteq \text{acl}(A)$ otherwise $\text{dom}(s_i) \subseteq \text{acl}(\text{dom}(u)) = \text{dom}(u)$. Hence $i \in I$ and $s_i \not\subseteq g_0$. The converse is clear by construction.

Now, evaluating (2.7) in g_0 yields:

$$0 = \sum_{i \leq n} \lambda_i e_{s_i}(g_0) = \sum_{s_i \subseteq g_0} \lambda_i = \sum_{s_i \subseteq u} \lambda_i.$$

Thus ϕ_u is well defined and linear. Complex conjugation is then automatically preserved because $\mathcal{A}$ is spanned by real-valued maps. We check the multiplicativity of ϕ_u on the spanning subset $\{e_s, s \in F(\mathfrak{M}_G)\}$. Recalling (2.4), just observe that, for all $s, s' \in F(\mathfrak{M}_G)$, we have:

$$\begin{aligned} \phi_u(e_{s \cup s'}) = 1 & \iff [s \cup s' \in F(\mathfrak{M}_G) \text{ and } s \cup s' \subseteq u] \\ & \iff s, s' \subseteq u \iff \phi_u(e_s) = 1 = \phi_u(e_{s'}). \end{aligned}$$

This shows multiplicativity. Finally, it is non-zero since the empty map $\emptyset$ belongs to $F(\mathfrak{M}_G)$ and we always have $\emptyset \subseteq u$.

It is easily checked that $u \mapsto \phi_u$ and $\phi \mapsto u_\phi$ are inverse to each other. To finish the proof, recall that $\mathbf{P}(\mathfrak{M}_G)$ is compact and Hausdorff. Since $\mathbf{S}_\mathcal{A}$ is Hausdorff

too, we can use Poincaré's Theorem and it suffices to show that $\mathbf{P}(\mathfrak{M}_G) \longrightarrow \mathbf{S}_{\mathcal{A}}$ is continuous.

Recall that $\{e_s, s \in F(\mathfrak{M}_G)\}$ are idempotents and linearly span $\mathcal{A}$. In particular, $\phi(e_s) \in \{0, 1\}$ for all $\phi \in \mathbf{S}_{\mathcal{A}}$ and $s \in F(\mathfrak{M}_G)$ hence the following sets form a subbasis for the topology on $\mathbf{S}_{\mathcal{A}}$:

$$O_{s,\varepsilon} = \{\phi \in \mathbf{S}_{\mathcal{A}}, \phi(e_s) = \varepsilon\}, \quad \text{for } s \in F(\mathfrak{M}_G) \text{ and } \varepsilon \in \{0, 1\}.$$

Now, let $s \in F(\mathfrak{M}_G)$ and $\varepsilon \in \{0, 1\}$. We treat the case $\varepsilon = 0$, the other is similar. We have:

$$\begin{aligned} \phi \in O_{s,0} &\iff \phi(e_s) = 0 \\ &\iff s \not\leq u_\phi \\ &\iff \exists x \in \text{dom}(s), [x \notin \text{dom}(u_\phi) \text{ or } u_\phi(x) \neq s(x)] \\ &\iff u_\phi \in \bigcup_{x \in \text{dom}(s)} \{v \in \mathbf{P}(\mathfrak{M}_G), x \in \text{dom}(v) \Rightarrow v(x) \notin \{s(x)\}\}. \end{aligned}$$

Recalling the definition of the topology on $\mathbf{P}(\mathfrak{M}_G)$ from Remark 2.2.5, we see that the desired continuity holds. $\square$

We can now turn to the key result of this section. Recall that a *positive linear functional* on an involutive algebra $\mathcal{B}$ is a linear map $\phi: \mathcal{B} \longrightarrow \mathbb{C}$ such that for every $a \in \mathcal{B}$, $\phi(a^*a) \geq 0$, where $*$ denotes the involution of $\mathcal{B}$. As mentioned in the introduction, the next Theorem is an adaptation of a technical but crucial lemma from the compact duality: Kreĭn proved [Kre41] that for a compact group K , positive linear functionals on $\mathcal{B}_0(K)$ are automatically continuous. Bochner [Boc42] gave an alternative and more elementary proof, based on generalized Fourier analysis and uniform approximation. It also appears as Theorem 30.2 in [HR70] with Bochner's proof and more context. We prove here that multiplicative linear functionals on $\mathcal{A}$ are automatically continuous. Note that we make a stronger assumption on the functionals but work in a smaller algebra.

The key tools from the compact case, namely the Haar measure and the finite dimension, are not available here. Instead, we rely on dynamical properties of Roelcke-precompact groups, in the form of Lemma 2.2.4.

Theorem 2.3.4. *Let G be a Roelcke-precompact non-Archimedean Polish group and let $\phi: \mathcal{A} \longrightarrow \mathbb{C}$ be multiplicative and linear. Then ϕ is continuous and has operator norm at most 1.*

Proof. As in both [Kre41] and [Boc42], we will show a strong positivity property for ϕ , namely that:

$$\forall f \in \mathcal{A}, [f \geq 0 \Rightarrow \phi(f) \geq 0],$$

where $f \geq 0$ means $f(g) \geq 0$ for every $g \in G$.

To that aim, fix $f \in \mathcal{A}$. There exist $s_1, \dots, s_n \in F(\mathfrak{M}_G)$ and $\lambda_1, \dots, \lambda_n \in \mathbb{C}$ such that $f = \sum_{i=1}^n \lambda_i e_{s_i}$. Recalling the identification $G = \text{Aut}(\mathfrak{M}_G)$ from Item 2 of Remark 2.2.6, we set up for the application of Lemma 2.2.4.

Let $A = \bigcup \{\text{dom}(s_i), i \leq n, \text{dom}(s_i) \subseteq \text{dom}(u_\phi)\}$ which is finite and let $I = \{i \leq n, \text{dom}(s_i) \not\subseteq \text{acl}(A)\}$. By Lemma 2.2.4, there exists $g_0 \in G$ that extends $u_\phi|_A$

but none of the s_i for $i \in I$. Exactly as in the proof of Corollary 2.3.3, g_0 in fact satisfies:

$$\forall i \leq n, [s_i \subseteq g_0 \iff s_i \subseteq u_\phi].$$

From this and the properties of u_ϕ , it follows that:

$$\phi(f) = \sum_{i \leq n} \lambda_i \phi(e_{s_i}) = \sum_{s_i \subseteq u_\phi} \lambda_i = \sum_{s_i \subseteq g_0} \lambda_i = f(g_0).$$

In particular, if $f \geq 0$, then $\phi(f) \geq 0$. Hence ϕ is positive in a strong sense which is well-known to imply continuity. Indeed, for every $f \in \mathcal{A}$, the functions $\|f\|_\infty \cdot 1 \pm f$ are in $\mathcal{A}$ and only take positives values. Hence $\phi(\|f\|_\infty \cdot 1 \pm f) \geq 0$ i.e.:

$$|\phi(f)| \leq \phi(\|f\|_\infty \cdot 1) = \|f\|_\infty.$$

□

A direct consequence of this result is that the map from Corollary 2.3.3 lifts to a homeomorphism $\mathbf{H}(G) \longrightarrow \mathbf{P}(\mathfrak{M}_G)$. This homeomorphism gives $\mathbf{H}(G)$ a semi-topological $*$ -monoid structure, transported from $\mathbf{P}(\mathfrak{M}_G)$.

Question 2.3.5. *Let G be a Roelcke-precompact non-Archimedean Polish group. Are multiplicative and linear functionals on $\mathcal{B}_0(G)$ (or $\mathcal{B}(G)$) automatically continuous ?*

The Hilbert compactification comes with a semi-topological monoid structure. The composition law is a form of convolution. Sometimes seen as a variation of the *Arens product*, it also appears in [Boc42]. One way to build it is as follows:

For $\phi \in \mathbf{H}(G)$, one can define a map $\underline{\phi}: \mathcal{A} \longrightarrow \mathcal{A}$ by setting

$$\forall f \in \mathcal{A}, \forall g \in G, \underline{\phi}(f)(g) = \phi(g^{-1} \cdot f)$$

To see that it is well defined, note that for $g \in G$ and $s \in F(\mathfrak{M}_G)$, we have $\phi(g^{-1} \cdot e_s) = \phi(e_{g^{-1} \circ s}) = 1$ if u_ϕ extends $g^{-1} \circ s$ and 0 otherwise. In other terms,

$$\underline{\phi}(e_s) = \begin{cases} e_{s \circ (u_\phi^{-1})} & \text{if } \text{dom}(s) \subseteq \text{dom}(u_\phi), \\ 0 & \text{otherwise.} \end{cases} \quad (2.9)$$

Thus, in both cases, $\underline{\phi}(e_s)$ is an element of $\mathcal{A}$. Next, given ϕ and ψ in $\mathbf{H}(G)$, we can define a map $\phi \cdot \psi: \mathcal{A} \longrightarrow \mathbb{C}$ by setting $\phi \cdot \psi = \phi \circ \underline{\psi}$.

Then, given $s \in F(\mathfrak{M}_G)$, using (2.9) we obtain:

$$\phi \cdot \psi(e_s) = \phi \circ \underline{\psi}(e_s) = \begin{cases} 1 & \text{if } s \subseteq u_\phi \circ u_\psi, \\ 0 & \text{otherwise.} \end{cases}$$

By Corollary 2.3.3, $\phi \cdot \psi$ is an element of $\mathbf{S}_\mathcal{A}$ which, by Theorem 2.3.4, uniquely extends to an element of $\mathbf{H}(G)$. Moreover, we obtained along the way that the map $\phi \mapsto u_\phi$ sends convolution in $\mathbf{H}(G)$ to composition where defined in $\mathbf{P}(\mathfrak{M}_G)$.

The Hilbert compactification also has an involution $\phi \mapsto \phi^*$, which is given by:

$$\forall f \in \text{Hilb}(G), \phi^*(f) = \overline{\phi(f^*)},$$

where $f^*(g) = \overline{f(g^{-1})}$ for all $f \in \text{Hilb}(G)$ and $g \in G$. It is preserved by the map $\mathbf{P}(\mathfrak{M}_G) \longrightarrow \mathbf{H}(G)$. Indeed, if $s \in F(\mathfrak{M}_G)$, we have $e_s^* = e_{s^*}$ and:

$$\phi^*(e_s) = \overline{\phi(e_s^*)} = \phi(e_{s^*}) = \begin{cases} 1 & \text{if } s \subseteq u_\phi^*, \\ 0 & \text{otherwise,} \end{cases}$$

hence $u_{\phi^*} = u_\phi^*$.

Consequently, we can identify the Hilbert compactification of G with $\mathbf{P}(\mathfrak{M}_G)$. We have in fact just proved the following. Because of the similarity between $\mathfrak{M}_G$ and $\mathfrak{M}^{\text{eq}}$ when $\mathfrak{M}$ is $\aleph_0$ -categorical, it can be seen as a reformulation of [BYIT18, Th. 0.2], obtained with a different approach.

Theorem 2.3.6. *The map $\mathbf{H}(G) \longrightarrow \mathbf{P}(\mathfrak{M}_G)$ is an isomorphism of semi-topological monoidal compactifications of G .*

Since G can be fully reconstructed from its Hilbert compactification (see e.g. Theorem 2.5.1 below), the pair $(\mathcal{A}, B)$, where $B = (\mathbb{1}_{gU})_{gU}$ with gU ranging over the open cosets of G , is a dual object to G much alike a Kreĭn algebra in the compact case. Again, the topology on $\mathcal{A}$ does not need to be included in the data.

It is interesting to note that it is possible to define this structure on $\mathcal{B}_0(G)$, even more similarly to what Kreĭn did.

To that aim, fix $(\pi_\lambda)_{\lambda \in \widehat{G}}$ a family of representatives of the isomorphism classes of irreducible representations of G . Using [Tsa12, Th. 1.3], every π_λ is of the form $\text{Ind}_{H_\lambda}^G(\sigma_\lambda)$ where H_λ is an open subgroup of G and σ_λ is a finite dimensional irreducible representation of H_λ . Then let $(e_1, \dots, e_{d_\lambda})$ be an orthonormal basis of $\mathcal{H}_{\sigma_\lambda}$. We can lift this basis to a Hilbert basis of $\mathcal{H}_{\pi_\lambda}$. Indeed, choose $(g_c)_{c \in G/H_\lambda}$ a family of representatives of the cosets of H_λ and define:

$$\xi_{i,c}^\lambda : G \longrightarrow \mathcal{H}_{\sigma_\lambda}, \quad g \mapsto \begin{cases} \sigma_\lambda(h^{-1})e_i & \text{if } g = g_ch \text{ with } h \in H_\lambda, \\ 0 & \text{otherwise,} \end{cases}$$

where $1 \leq i \leq d_\lambda$ and $c \in G/H_\lambda$.

These maps form a Hilbert basis of $\mathcal{H}_{\pi_\lambda}$ and can be used to define the product law on $\mathbf{S}_{\mathcal{B}_0(G)}$. Indeed, the family of matrix coefficients that arises using these vectors, the $f_{i,c,j,c'}^\lambda : g \mapsto \langle \pi_\lambda(g)\xi_{j,c'}, \xi_{i,c} \rangle$, are naturally arranged in block matrices:

$$\left(\left(f_{i,c,j,c'}^\lambda \right)_{\substack{1 \leq i, j \leq d_\lambda \\ c, c' \in G/H_\lambda}} \right)_{\lambda \in \widehat{G}}.$$

In analogy with the compact case, these span a dense subalgebra of $\mathcal{B}_0(G)$.

Given **continuous** $\phi, \psi \in \mathbf{S}_{\mathcal{B}_0(G)}$, we claim that $\phi \cdot \psi$ behaves on the generating family as follows:

$$\phi \cdot \psi(f_{i,c,j,c'}^\lambda) = \sum_{\substack{1 \leq k \leq d_\lambda \\ c'' \in G/H_\lambda}} \phi(f_{i,c,k,c''}^\lambda) \psi(f_{k,c'',j,c'}^\lambda).$$

To see that the sum converges, it suffices, by the Cauchy-Schwarz inequality, to show that for fixed $i \leq d_\lambda$ and $c \in G/H_\lambda$,

$$\sum_{\substack{1 \leq k \leq d_\lambda \\ c'' \in G/H_\lambda}} \left| \phi(f_{i,c,k,c''}^\lambda) \right|^2 < +\infty$$

The key is that the matrix coefficients at play here are all bounded by 1 and the collection of their support have locally finite intersection: $f_{i,c,k,c''}^\lambda(g) = 0$ as soon as $g \notin g_{c''} H_\lambda g_c^{-1}$. Thus, for any finite subcollection of indices, since ϕ is multiplicative and preserves complex conjugation:

$$\sum_{k,c''} \left| \phi(f_{i,c,k,c''}^\lambda) \right|^2 = \phi \left(\sum_{k,c''} |f_{i,c,k,c''}^\lambda|^2 \right) \leq \|\phi\| \cdot \left\| \sum_{k,c''} |f_{i,c,k,c''}^\lambda|^2 \right\|_\infty \leq d_\lambda$$

This construction yields the same operation as the above variants. Indeed, for all λ, i, j, c, c' and all $g, h \in G$, we have:

$$\pi(h) \xi_{j,c'}^\lambda = \sum_{k,c''} \langle \pi(h) \xi_{j,c'}^\lambda, \xi_{k,c''}^\lambda \rangle \xi_{k,c''}^\lambda = \sum_{k,c''} f_{k,c'',j,c'}^\lambda(h) \xi_{k,c''}^\lambda,$$

hence:

$$\begin{aligned} f_{i,c,j,c'}^\lambda(gh) &= \langle \pi_\lambda(g) \pi_\lambda(h) \xi_{j,c'}^\lambda, \xi_{i,c}^\lambda \rangle \\ &= \sum_{k,c''} f_{k,c'',j,c'}^\lambda(h) \langle \pi_\lambda(g) \xi_{k,c''}^\lambda, \xi_{i,c}^\lambda \rangle \\ &= \sum_{k,c''} f_{i,c,k,c''}^\lambda(g) f_{k,c'',j,c'}^\lambda(h). \end{aligned}$$

Thus, for $\phi, \psi \in \mathbf{S}_{\mathcal{B}_0(G)}$ and λ, i, j, c, c' ,

$$\begin{aligned} \psi \cdot \psi(f_{i,c,j,c'}^\lambda) &= \phi \circ \psi(f_{i,c,j,c'}^\lambda) \\ &= \phi(g \mapsto \psi(g^{-1} \cdot f_{i,c,j,c'}^\lambda)) \\ &= \phi \left(g \mapsto \sum_{k,c''} f_{i,c,k,c''}^\lambda(g) \psi(f_{k,c'',j,c'}^\lambda) \right) \\ &= \sum_{k,c''} \phi(f_{i,c,k,c''}^\lambda) \psi(f_{k,c'',j,c'}^\lambda). \end{aligned}$$

The involution can also be defined in terms of the $(f_{i,c,j,c'}^\lambda)$. Indeed, for all λ, i, j, c, c' , we have:

$$f_{i,c,j,c'}^{\lambda*} = f_{j,c',i,c}^\lambda.$$

Thus, a positive answer to Question 2.3.5 would make the pair $(\mathcal{B}_0(G), B)$, where B is the countable structured family of matrix coefficients $(f_{i,c,j,c'}^\lambda)$, a dual object that contains enough data for the full reconstruction of G , even without remembering the topology on $\mathcal{B}_0(G)$.

2.4 The Tannaka monoid

In [Tan39], Tannaka associated to a compact group G a monoid $\mathbf{T}(G)$ of *operations* on the class of representations of G . More explicitly, an element of $\mathbf{T}(G)$ is

a family of operators $(u_\pi)_\pi$ where π ranges over all the finite dimensional representations of G and u_π is an operator on same the Hilbert space as π . Moreover, the family must commute with intertwining operators and preserve the common operations on representations: sum, tensor product and conjugation. The details can be found in [Che99] or [HR70]. A more recent treatment of this approach using efficient terminology from category theory appears in [JS91]. Inspired by this take on Tannaka's duality, we carry a similar construction for Roelcke-precompact non-Archimedean Polish groups. While Tannaka showed in the compact case that $\mathbf{T}(G)$ is a compact group canonically isomorphic to G , we will obtain a compact semi-topological $*$ -monoid canonically isomorphic to the Hilbert compactification.

Given a topological group G , we will denote by $\mathbf{Rep}(G)$ the category whose objects are the representations of G that split as a finite sum of irreducibles and whose morphisms are the intertwining operators. Then $\mathcal{B}_0(G)$ is exactly the algebra of matrix coefficients of G arising from representations in $\mathbf{Rep}(G)$. Recall that $\widehat{G}$ denotes the unitary dual of G , i.e. the set of equivalence classes of irreducible unitary representations of G .

Proposition 2.4.1. *Let G be a Roelcke-precompact non-Archimedean Polish group. The following properties hold:*

1. $\mathbf{Rep}(G)$ is stable under tensor product of representations.
2. Only countably many equivalence classes of representations appear in $\mathbf{Rep}(G)$.

Proof.

1. This is Lemma 2.2.15.
2. This follows from the fact that $\widehat{G}$ is countable, stated as Item 4 of Fact 2.1.1.

□

Let $\mathbf{U}_G$ be the forgetful functor from $\mathbf{Rep}(G)$ to the category of Hilbert spaces and let $\mathcal{Nat}(\mathbf{U}_G)$ denote the class of natural transformations of $\mathbf{U}_G$. Explicitly, an element of $\mathcal{Nat}(\mathbf{U}_G)$ is a family $u = (u_\pi)_{\pi \in \mathbf{Rep}(G)}$ where for every representation $\pi \in \mathbf{Rep}(G)$, u_π is a bounded operator on $\mathcal{H}_\pi$ that commutes with intertwining operators. In other words, such that for every pair of representations $\pi_1, \pi_2 \in \mathbf{Rep}(G)$ and every intertwining operator $h \in \text{Hom}(\pi_1, \pi_2)$, the following diagram is commutative:

$$\begin{array}{ccc} \mathcal{H}_{\pi_1} & \xrightarrow{u_{\pi_1}} & \mathcal{H}_{\pi_1} \\ h \downarrow & & \downarrow h \\ \mathcal{H}_{\pi_2} & \xrightarrow{u_{\pi_2}} & \mathcal{H}_{\pi_2} \end{array}$$

Note that every element g of G naturally induces a natural transformation of $\mathbf{U}_G$, namely $(\pi(g))_{\pi \in \mathbf{Rep}(G)}$.

It is easily seen that $\mathcal{Nat}(\mathbf{U}_G)$ is a complex algebra under coordinatewise sum and composition. Moreover, elements of $\mathcal{Nat}(\mathbf{U}_G)$ behave well with subrepresentations:

Lemma 2.4.2. *Let G be a Roelcke-precompact non-Archimedean Polish group and let $u \in \text{Nat}(\mathbf{U}_G)$.*

1. *Let $\pi \in \mathbf{Rep}(G)$ and let F be a closed subspace of $\mathcal{H}_\pi$ that is stable under the action of G . Then F is also stable under u_π .*
2. *Let $\pi_1, \pi_2 \in \mathbf{Rep}(G)$. Then:*

$$u_{\pi_1 \oplus \pi_2} = u_{\pi_1} \oplus u_{\pi_2}.$$

Proof.

1. Let $p_F: \mathcal{H}_\pi \rightarrow \mathcal{H}_\pi$ be the orthogonal projection on F . Then F is stable under the action of G if and only if p_F commutes with the action of G . In particular, if F is G -stable then u_π must also commute with p_F , i.e. $u_\pi(F) \subseteq F$.
2. Let $\pi = \pi_1 \oplus \pi_2$. By definition, $\mathcal{H}_\pi = \mathcal{H}_{\pi_1} \oplus \mathcal{H}_{\pi_2}$. Moreover, the inclusion maps $\mathcal{H}_{\pi_1} \rightarrow \mathcal{H}_\pi$ are intertwining operators. Thus, the following diagram must be commutative, which proves the claim:

$$\begin{array}{ccc} \mathcal{H}_{\pi_1} & \xrightarrow{u_{\pi_1}} & \mathcal{H}_{\pi_1} \\ \downarrow & & \downarrow \\ \mathcal{H}_\pi & \xrightarrow{u_\pi} & \mathcal{H}_\pi \\ \uparrow & & \uparrow \\ \mathcal{H}_{\pi_2} & \xrightarrow{u_{\pi_2}} & \mathcal{H}_{\pi_2} \end{array}$$

□

The following is essentially Proposition 4 in [JS91]. The proof is the same but we reproduce it for completeness. If $\mathcal{H}$ is Hilbert space, $\mathcal{L}(\mathcal{H})$ will denote the space of bounded operators $\mathcal{H} \rightarrow \mathcal{H}$ and $\mathcal{L}_1(\mathcal{H})$ the subset of those operators with norm at most 1.

Proposition 2.4.3. *Let G be a Roelcke-precompact non-Archimedean Polish group. For every $\lambda \in \widehat{G}$, fix $\pi_\lambda: G \rightarrow \mathcal{U}(\mathcal{K}_\lambda)$ a representative of λ . Then, the restriction map*

$$q: \text{Nat}(\mathbf{U}_G) \rightarrow \prod_{\lambda \in \widehat{G}} \mathcal{L}(\mathcal{K}_\lambda), \quad u \mapsto (u_{\pi_\lambda})_{\lambda \in \widehat{G}}$$

is a bijective correspondence that respects the algebra structures. In particular, $\text{Nat}(\mathbf{U}_G)$ is a set.

Proof. Lemma 2.4.2 together with the fact that our representations are finite sums of irreducibles show that an element u is determined by its image under q . In particular, q is injective. For surjectivity, fix $(t_\lambda) \in \prod_{\lambda \in \widehat{G}} \mathcal{L}(\mathcal{K}_\lambda)$. We will define a pre-image u for (t_λ) .

Let $\pi: G \rightarrow \mathcal{U}(\mathcal{H})$ be an element of $\mathbf{Rep}(G)$. Recalling Lemma 2.2.11, there is a unique *isotypical* decomposition $\mathcal{H} = \bigoplus_{\lambda} \mathcal{H}_{\lambda}$ into closed invariant subspaces where for every $\lambda \in \widehat{G}$, all the irreducible subrepresentations of $\mathcal{H}_{\lambda}$ are isomorphic to π_{λ} .

Claim. *There is a canonical G -equivariant isomorphism:*

$$\Psi_{\lambda}: \mathcal{K}_{\lambda} \otimes \mathrm{Hom}_G(\mathcal{K}_{\lambda}, \mathcal{H}_{\lambda}) \rightarrow \mathcal{H}_{\lambda},$$

namely, the one induced by the bilinear map

$$\mathcal{K}_{\lambda} \times \mathrm{Hom}_G(\mathcal{K}_{\lambda}, \mathcal{H}_{\lambda}) \rightarrow \mathcal{H}_{\lambda}, \quad (\xi, h) \mapsto h(\xi).$$

Proof of the claim. Because $\mathrm{Hom}(\mathcal{K}_{\lambda}, \mathcal{H}_{\lambda})$ is finite dimensional (this an application of Schur's Lemma, detailed below), the subtleties of working with *Hilbert tensor products* rather than *algebraic* ones vanish and the verifications are routine. Still, we give all the details for completeness.

Since $\pi \in \mathbf{Rep}(G)$, $\mathcal{H}_{\lambda}$ splits as a finite sum of subrepresentations $\mathcal{H}_1 \oplus \cdots \oplus \mathcal{H}_n$ where for every $i \leq n$, $G \curvearrowright \mathcal{H}_i$ is isomorphic to π_{λ} , witnessed by an isomorphism of representations $h_i: \mathcal{K}_{\lambda} \rightarrow \mathcal{H}_i$.

– Ψ_{λ} is well defined:

Factually, Ψ_{λ} is well defined on the *algebraic* tensor product of $\mathcal{K}_{\lambda}$ and $\mathrm{Hom}_G(\mathcal{K}_{\lambda}, \mathcal{H}_{\lambda})$. Let $(\xi_i)_{i \in I}$ be a Hilbert basis of $\mathcal{K}_{\lambda}$. Also,

$$\mathrm{Hom}_G(\mathcal{K}_{\lambda}, \mathcal{H}_{\lambda}) = \bigoplus_{1 \leq i \leq n} \mathrm{Hom}_G(\mathcal{K}_{\lambda}, \mathcal{H}_i)$$

where, by Schur's Lemma, $\mathrm{Hom}_G(\mathcal{K}_{\lambda}, \mathcal{H}_i) = \mathbb{C} \cdot h_i$ for every $i \leq n$. Thus, $(h_i)_{1 \leq i \leq n}$ is an orthonormal basis of the finite dimensional Hilbert space $\mathrm{Hom}_G(\mathcal{K}_{\lambda}, \mathcal{H}_{\lambda})$ and:

$$(\xi_i \otimes h_j)_{\substack{i \in I \\ 1 \leq j \leq n}}$$

is a Hilbert basis for $\mathcal{K}_{\lambda} \otimes \mathrm{Hom}_G(\mathcal{K}_{\lambda}, \mathcal{H}_{\lambda})$ that is moreover contained in the algebraic tensor product. Finally, for every $i \in I$ and $j \leq n$, we have:

$$\|\Psi_{\lambda}(\xi_i \otimes h_j)\| = \|h_j(\xi_i)\| = \|\xi_j\| = 1.$$

Thus, Ψ_{λ} is bounded (and even isometric) hence extends to the Hilbert tensor product.

– Ψ_{λ} is surjective:

For every $j \leq n$, $\mathcal{H}_j = h_j(\mathcal{K}_{\lambda}) = \Psi_{\lambda}(\mathcal{K}_{\lambda} \otimes h_j) \subseteq \mathrm{im} \Psi_{\lambda}$ hence $\mathcal{H}_{\lambda} = \mathcal{H}_1 \oplus \cdots \oplus \mathcal{H}_n \subseteq \mathrm{im} \Psi_{\lambda}$.

– Ψ_{λ} is injective:

First, note that:

$$\begin{aligned} \mathcal{K}_{\lambda} \otimes \mathrm{Hom}_G(\mathcal{K}_{\lambda}, \mathcal{H}_{\lambda}) &= \mathcal{K}_{\lambda} \otimes \left(\bigoplus_{1 \leq i \leq n} \mathrm{Hom}_G(\mathcal{K}_{\lambda}, \mathcal{H}_i) \right) \\ &= \bigoplus_{1 \leq i \leq n} \mathcal{K}_{\lambda} \otimes \mathrm{Hom}_G(\mathcal{K}_{\lambda}, \mathcal{H}_i). \end{aligned}$$

Now, suppose $x \in \ker \Psi_\lambda$. Using the above decomposition, write $x = x_1 + \dots + x_n$ where $x_i \in \mathcal{K}_\lambda \otimes \text{Hom}_G(\mathcal{K}_\lambda, \mathcal{H}_i)$ for every $i \leq n$. Then:

$$0 = \Psi_\lambda(x) = \Psi_\lambda(x_1) + \dots + \Psi_\lambda(x_n).$$

Since $\Psi_\lambda(x_i) \in \mathcal{H}_i$ for every $i \leq n$, and because the $\mathcal{H}_i$'s are pairwise orthogonal, we have $\Psi_\lambda(x_i) = 0$ for every $i \leq n$. We thus reduce to the case $x = x_1$. By Schur's Lemma, $\text{Hom}_G(\mathcal{K}_\lambda, \mathcal{H}_1) = \mathbb{C} \cdot h_1$ hence x is of the form $\xi \otimes h_1$ for some $\xi \in \mathcal{K}_\lambda$. Then:

$$0 = \Psi_\lambda(x) = h_1(\xi).$$

Since h_1 is isometric, $\xi = 0$ and $x = 0$.

- Ψ_λ is G -equivariant: By continuity, it suffices to check it on simple tensors. For all $g \in G$, $\xi \in \mathcal{K}_\lambda$ and $h \in \text{Hom}_G(\mathcal{K}_\lambda, \mathcal{H}_\lambda)$, we have:

$$g \cdot (\xi \otimes h) = g \cdot \xi \otimes g \cdot h = \pi_\lambda(g)\xi \otimes h,$$

hence:

$$\Psi_\lambda(g \cdot (\xi \otimes h)) = h(\pi_\lambda(g)\xi) = \pi(g)h(\xi) = \pi(g)\Psi_\lambda(\xi \otimes h).$$

□_{claim}

Using Ψ_λ , we can define u_π as follows:

$$u_\pi = \bigoplus_\lambda \left[\Psi_\lambda \circ (t_\lambda \otimes 1) \circ \Psi_\lambda^{-1} \right]$$

To see that it defines a natural transformation of $\mathbf{U}_G$, fix $\pi, \pi' \in \mathbf{Rep}(G)$ and write $\mathcal{H} = \mathcal{H}_\pi$ and $\mathcal{H}' = \mathcal{H}_{\pi'}$. Let $h \in \text{Hom}(\pi, \pi')$. Recalling the classical general fact that $\text{Hom}(\pi_\lambda, \pi_\mu) = 0$ if $\lambda \neq \mu \in \widehat{G}$, we see that h preserves the isotypical decompositions of π and π' :

$$\forall \lambda \in \widehat{G}, h(\mathcal{H}_\lambda) \subseteq \mathcal{H}'_\lambda$$

with the above notations. Moreover, these decompositions are also preserved by u_π and $u_{\pi'}$ by construction. Hence, we can reduce to the case $\mathcal{H} = \mathcal{H}_\lambda$ and $\mathcal{H}' = \mathcal{H}'_\lambda$ for a single $\lambda \in \widehat{G}$, which leaves us with the following diagram:

$$\begin{array}{ccccccc} \mathcal{H}_\lambda & \xrightarrow{\Psi_\lambda^{-1}} & \mathcal{K}_\lambda \otimes \text{Hom}_G(\mathcal{K}_\lambda, \mathcal{H}_\lambda) & \xrightarrow{t_\lambda \otimes 1} & \mathcal{K}_\lambda \otimes \text{Hom}_G(\mathcal{K}_\lambda, \mathcal{H}_\lambda) & \xrightarrow{\Psi_\lambda} & \mathcal{H}_\lambda \\ \downarrow h & & \downarrow 1 \otimes (h \circ \cdot) & & \downarrow 1 \otimes (h \circ \cdot) & & \downarrow h \\ \mathcal{H}'_\lambda & \xrightarrow{\Psi_\lambda'^{-1}} & \mathcal{K}_\lambda \otimes \text{Hom}_G(\mathcal{K}_\lambda, \mathcal{H}'_\lambda) & \xrightarrow{t_\lambda \otimes 1} & \mathcal{K}_\lambda \otimes \text{Hom}_G(\mathcal{K}_\lambda, \mathcal{H}'_\lambda) & \xrightarrow{\Psi_\lambda'} & \mathcal{H}'_\lambda \end{array}$$

To conclude, we need to prove that the greatest square is commutative. But the three smallest squares are easily seen to be commutative, which is enough. □

Consider the weak operator topology on $\mathcal{L}(\mathcal{H})$ for every representation of G . Then we can endow $\prod_{\lambda \in \widehat{G}} \mathcal{L}(\mathcal{H}_\lambda)$ with the product topology. $\mathcal{Nat}(\mathbf{U}_G)$ can also be endowed with the coarsest topology for which the projection maps $u \mapsto u_\pi$ are continuous for every representation π of G . Then q is also a homeomorphism for those topologies.

Lemma 2.4.4. *Let G be a Roelcke-precompact non-Archimedean Polish group, let $u \in \mathcal{Nat}(\mathbf{U}_G)$ and let $\pi, \sigma \in \mathbf{Rep}(G)$.*

1. *If σ and π are isomorphic, then $\|u_\sigma\| = \|u_\pi\|$.*
2. *If σ is a subrepresentation of π , then $\|u_\sigma\| \leq \|u_\pi\|$.*

Proof.

1. In this case, there exists a surjective isometry $h \in \text{Hom}(\sigma, \pi)$. Then:

$$\|u_\pi\| = \|h \circ u_\sigma \circ h^{-1}\| = \|u_\sigma\|.$$

2. Follows directly from the previous point and Item 2 of Lemma 2.4.2.

□

Tensor product is the last ingredient we need to add in order to define our version of the Tannaka monoid:

Definition 2.4.5. The *Tannaka monoid* $\mathbf{T}(G)$ of a Roelcke-precompact non-Archimedean Polish group G is the set of elements $u \in \mathcal{Nat}(\mathbf{U}_G)$ that are non-zero and commute with tensor product, i.e. such that:

$$u_1 = \text{Id}_{\mathbb{C}} \quad \text{and} \quad \forall \pi, \pi' \in \mathbf{Rep}(G), \quad u_{\pi \otimes \pi'} = u_\pi \otimes u_{\pi'}$$

where 1 denotes the trivial representation of G on $\mathbb{C}$. The operation is coordinatewise composition: for all $u, v \in \mathbf{T}(G)$, $(u \circ v)_\pi = u_\pi \circ v_\pi$. It also admits an involution $u \mapsto u^*$ given by coordinatewise adjunction. It is endowed with the induced topology as a subspace of $\mathcal{Nat}(\mathbf{U}_G)$.

Theorem 2.4.6. *Let G be a Roelcke-precompact non-Archimedean Polish group. Then $\mathbf{T}(G)$ is a compact semi-topological $*$ -monoid isomorphic to the Hilbert compactification of G .*

Proof. First, it is easily seen that composition is separately continuous in $\mathbf{T}(G)$ as it is in $\mathcal{L}(\mathcal{H})$ with the weak operator topology for every Hilbert space $\mathcal{H}$. Similarly, the involution $*$ of $\mathbf{T}(G)$ is also continuous.

Next, we will define a map $\Phi: \mathbf{T}(G) \rightarrow \mathbf{H}(G)$. Fix $u \in \mathbf{T}(G)$ in order to define a continuous linear multiplicative functional $\phi: \text{Hilb}(G) \rightarrow \mathbb{C}$. For every $\pi \in \mathbf{Rep}(G)$ and all $\xi, \eta \in \mathcal{H}_\pi$, denote by $f_{\xi, \eta}^\pi: g \mapsto \langle \pi(g)\eta, \xi \rangle$ the associated matrix coefficient. We first define ϕ on $\mathcal{B}_0(G)$ by setting:

$$\phi(f_{\xi, \eta}^\pi) = \langle u_\pi(\eta), \xi \rangle. \quad (2.10)$$

To see that it correctly defines a linear map $\mathcal{B}_0(G) \rightarrow \mathbb{C}$, fix representations $\pi_1, \dots, \pi_n$ in $\mathbf{Rep}(G)$ with respective underlying spaces $\mathcal{H}_1, \dots, \mathcal{H}_n$ and let $\xi_1, \eta_1 \in \mathcal{H}_1, \dots, \xi_n, \eta_n \in \mathcal{H}_n$ and $\lambda_1, \dots, \lambda_n \in \mathbb{C}$ such that:

$$\lambda_1 f_{\xi_1, \eta_1}^{\pi_1} + \dots + \lambda_n f_{\xi_n, \eta_n}^{\pi_n} = 0. \quad (2.11)$$

Then, let $\pi = \oplus_{i \leq n} \pi_i$, $\eta = \eta_1 + \dots + \eta_n$ and $\xi = \lambda_1 \xi_1 + \dots + \lambda_n \xi_n$. Equation (2.11) becomes:

$$\forall g \in G, \quad 0 = \langle \pi(g)\eta, \xi \rangle = \langle \eta, \pi(g^{-1})\xi \rangle,$$

i.e. η lies in $(\pi(G)\xi)^\perp$, a closed subspace of $\mathcal{H}$ stable under the action of G . Using Lemma 2.4.2, we also have $u_\pi(\eta) \in (\pi(G)\xi)^\perp$ and even:

$$0 = \langle u_\pi(\eta), \xi \rangle_{\mathcal{H}} = \lambda_1 \langle u_{\pi_1}(\eta_1), \xi_1 \rangle_{\mathcal{H}_1} + \dots + \lambda_n \langle u_{\pi_n}(\eta_n), \xi_n \rangle_{\mathcal{H}_n}.$$

This shows that ϕ is well defined on $\mathcal{B}_0(G)$. It clearly is non-zero since u is non-zero.

To see that ϕ is multiplicative, fix $f_{\xi, \eta}^\pi, f_{\xi', \eta'}^{\pi'}$ in $\mathcal{B}_0(G)$. We have:

$$\begin{aligned} \phi(f_{\xi, \eta}^\pi) \phi(f_{\xi', \eta'}^{\pi'}) &= \langle u_\pi(\eta), \xi \rangle \langle u_{\pi'}(\eta'), \xi' \rangle \\ &= \langle u_\pi(\eta) \otimes u_{\pi'}(\eta'), \xi \otimes \xi' \rangle \\ &= \langle u_\pi \otimes u_{\pi'}(\eta \otimes \eta'), \xi \otimes \xi' \rangle \\ &= \langle u_{\pi \otimes \pi'}(\eta \otimes \eta'), \xi \otimes \xi' \rangle \\ &= \phi(f_{\xi \otimes \xi', \eta \otimes \eta'}^{\pi \otimes \pi'}) \\ &= \phi(f_{\xi, \eta}^\pi f_{\xi', \eta'}^{\pi'}), \end{aligned}$$

where we used the fact that $u \in \mathbf{T}(G)$ to obtain the fourth equality.

Next, we claim that ϕ is bounded. By Theorem 2.3.4, ϕ is bounded by 1 on $\mathcal{A}$. Thus, for every open subgroup $V \leq G$ and every $g, h \in G$,

$$\left| \langle u_{\lambda_{G/V}} \delta_{gV}, \delta_{hV} \rangle \right| = |\phi(\mathbb{1}_{hVg^{-1}})| \leq \|\mathbb{1}_{hVg^{-1}}\| = 1.$$

Since $u_{\lambda_{G/V}}$ is continuous by definition and $(\delta_{gV})_{gV \in G/V}$ is a Hilbert basis for the space $\ell^2(G/V)$, the above calculation shows that $\|u_{\lambda_{G/V}}\| \leq 1$. Now, if $\pi \in \mathbf{Rep}(G)$, one can use Lemma 2.2.8 to find open subgroups $V_1, \dots, V_n \leq G$ such that π is isomorphic to a subrepresentation of $\lambda_{G/V_1} \oplus \dots \oplus \lambda_{G/V_n}$. Then, by Lemma 2.4.4, we have:

$$\|u_\pi\| \leq \|u_{\lambda_{G/V_1} \oplus \dots \oplus \lambda_{G/V_n}}\| = \|u_{\lambda_{G/V_1}} \oplus \dots \oplus u_{\lambda_{G/V_n}}\| \leq 1.$$

Thus, for every $f_{\xi, \eta}^\pi$ in $\mathcal{B}_0(G)$, we have:

$$\left| \phi(f_{\xi, \eta}^\pi) \right| = |\langle u_\pi \eta, \xi \rangle| \leq \|u_\pi\| \cdot \|\xi\| \cdot \|\eta\| \leq 1 \cdot \|f_{\xi, \eta}^\pi\|_\infty,$$

and ϕ is bounded by 1.

The map ϕ being multiplicative, linear and bounded on $\mathcal{B}_0(G)$, it uniquely extends to an element of $\mathbf{H}(G)$ and hence $\Phi: \mathbf{T}(G) \rightarrow \mathbf{H}(G)$ is well defined. The identity (2.10) above also shows that Φ is continuous as $\mathbf{H}(G)$ is endowed with the pointwise convergence (=weak*) topology.

To define the reciprocal $\Psi: \mathbf{H}(G) \rightarrow \mathbf{T}(G)$, fix $\phi \in \mathbf{H}(G)$ and $\pi \in \mathbf{Rep}(G)$. Recall that ϕ is automatically bounded since $\text{Hilb}(G)$ is a C^* -algebra. Then, the map

$$\mathcal{H}_\pi \times \mathcal{H}_\pi \longrightarrow \mathbb{C}, \quad (\xi, \eta) \longmapsto \phi(f_{\xi, \eta}^\pi)$$

is a bounded sesquilinear form. Indeed, for all $\xi, \eta \in \mathcal{H}$:

$$|\phi(f_{\xi, \eta}^\pi)| \leq \|\phi\| \cdot \|f_{\xi, \eta}^\pi\|_\infty = \|\phi\| \cdot \|\eta\| \cdot \|\xi\|.$$

Using the Riesz Representation Theorem, there exists a unique bounded operator $u_\pi \in \mathcal{L}(\mathcal{H}_\pi)$ such that:

$$\forall \xi, \eta \in \mathcal{H}_\pi, \phi(f_{\xi, \eta}^\pi) = \langle u_\pi(\eta), \xi \rangle. \quad (2.12)$$

It is straightforward to show that $(u_\pi)_{\pi \in \mathbf{Rep}(G)}$ lies in $\mathbf{T}(G)$ and that (2.12) implies the continuity of the map $\Psi: \mathbf{H}(G) \rightarrow \mathbf{T}(G)$. Equations (2.10) and (2.12) together imply that those maps are inverse to each other, hence homeomorphisms.

To finish the proof, it only remains to show that the $*$ -monoid structures are preserved. To see this, fix $u, v \in \mathbf{T}(G)$, and let $\pi \in \mathbf{Rep}(G)$. For all $\xi, \eta \in \mathcal{H}_\pi$ the following holds:

$$\begin{aligned} \Phi(u \circ v)(f_{\xi, \eta}^\pi) &= \langle u_\pi \circ v_\pi(\eta), \xi \rangle \\ &= \Phi(u)(f_{\xi, v_\pi(\eta)}^\pi) \\ &= \Phi(u) \left[g \mapsto \langle v_\pi(\eta), \pi(g^{-1})\xi \rangle \right] \\ &= \Phi(u) \left[g \mapsto \Phi(v)(f_{\xi, \pi(g^{-1})\eta}^\pi) \right] \\ &= \Phi(u) \left[g \mapsto \Phi(v)(g^{-1} \cdot f_{\xi, \eta}^\pi) \right] \\ &= \Phi(u) \cdot \Phi(v)(f_{\xi, \eta}^\pi). \end{aligned}$$

Finally, since multiplicative linear functionals on $\mathcal{A}$ automatically preserve conjugation, Φ preserves the involution. $\square$

2.5 Additional properties of $\mathbf{H}(G)$

We end this article by recovering previously known basic properties of $\mathbf{H}(G)$ and by turning the previous result into a proper duality, i.e a contra-variant functor that is full and faithful.

First, recall that there is a natural map $\iota: G \rightarrow \mathbf{H}(G)$ that sends $g \in G$ to the *evaluation map at g*. Viewing $\mathbf{H}(G)$ as $\mathbf{T}(G)$, $\iota(g)$ is given by:

$$\forall \pi \in \mathbf{Rep}(G), \iota(g)_\pi = \pi(g)$$

Finally, viewing $\mathbf{H}(G)$ as $\mathbf{P}(\mathfrak{M}_G)$ and identifying G with $\text{Aut}(\mathfrak{M}_G)$ by Item 2 of Remark 2.2.6, ι becomes the canonical inclusion $G \hookrightarrow \mathbf{P}(\mathfrak{M}_G)$. It allows for an abstract reconstruction of G from $\mathbf{H}(G)$:

Theorem 2.5.1. *Let G be a non-Archimedean Roelcke-precompact Polish group. The natural map*

$$\iota: G \rightarrow \mathbf{H}(G)$$

is a homeomorphic embedding such that:

$$\forall g, h \in G, \iota(gh) = \iota(g) \cdot \iota(h) \quad \& \quad \iota(g^{-1}) = \iota(g)^*.$$

Moreover, $\iota(G)$ is exactly the set of invertible elements of $\mathbf{H}(G)$. In other words, G is canonically isomorphic, as a topological group, to the set of invertible elements of $\mathbf{H}(G)$.

Proof. This is contained in Remark 2.2.5 after identifying G with $\text{Aut}(\mathfrak{M}_G)$ by Item 2 of Remark 2.2.6. $\square$

Finally, we can reformulate the previous results as an equivalence of categories, or more precisely a duality. Let $\mathbf{RPnA}$ be the category whose objects are the Roelcke-precompact non-Archimedean Polish groups and arrows the continuous group morphisms. Note that if $p: G \rightarrow H$ is a continuous group morphism between topological groups, it induces a functor $\tilde{p}: \mathbf{Rep}(H) \rightarrow \mathbf{Rep}(G)$ by setting $\tilde{p}(\pi) = \pi \circ p$ for every $\pi \in \mathbf{Rep}(H)$ and $\tilde{p}(h) = h$ for every morphism h between representations of H . The functor $\tilde{p}$ has additional properties, it is *admissible* in the following sense:

Definition 2.5.2. Let G, H be topological groups. A functor

$$F: \mathbf{Rep}(H) \rightarrow \mathbf{Rep}(G)$$

is said to be *admissible* if it satisfies the following properties:

- (i) F sends the trivial one dimensional representation of H to the trivial one dimensional representation of G .
- (ii) For every $\pi \in \mathbf{Rep}(H)$, $F(\pi)$ has the same underlying Hilbert space as π .
- (iii) For all $\pi, \pi' \in \mathbf{Rep}(H)$ and $h \in \text{Hom}(\pi, \pi')$, $F(h) = h$.
- (iv) F commutes with tensor product of representations, i.e. for all $\pi, \pi' \in \mathbf{Rep}(G)$,

$$F(\pi \otimes \pi') = F(\pi) \otimes F(\pi').$$

The collection of such functors is stable under composition and contains the identity maps. We can thus form the category $\mathbf{Rep}(\mathbf{RPnA})$ whose class of objects is $(\mathbf{Rep}(G))_{G \in \mathbf{RPnA}}$ and whose arrows are the admissible functors. We obtain the following, which is also true for compact (resp. locally compact Abelian) groups by the usual duality theories of Pontryagin–van Kampen and Tannaka-Krein.

Theorem 2.5.3. *The canonical contravariant functor*

$$\mathbf{Rep}: \mathbf{RPnA} \rightarrow \mathbf{Rep}(\mathbf{RPnA})$$

$$G \mapsto \mathbf{Rep}(G)$$

$$p \mapsto \tilde{p}$$

is a duality (i.e. surjective and fully faithful).

Proof. It is clear by construction that $\mathbf{Rep}$ is surjective with regard to objects. To prove that it is faithful, let $G, H \in \mathbf{RPnA}$ and assume that $p, q: G \rightarrow H$ are such that $\tilde{p} = \tilde{q}$. Then, for every $\pi \in \mathbf{Rep}(G)$, we have $\pi \circ p = \tilde{p}(\pi) = \tilde{q}(\pi) = \pi \circ q$. Since non-Archimedean Polish groups satisfy the Gel'fand-Raikov Theorem (Fact 2.1.1 Item (3)), it implies that $p = q$.

Finally, we prove that $\mathbf{Rep}$ is full. Let $G, H \in \mathbf{RPnA}$ and fix an admissible functor $F: \mathbf{Rep}(H) \rightarrow \mathbf{Rep}(G)$. Let us prove that there exists $p: G \rightarrow H$ such that $F = \tilde{p}$. First, we show that F induces a map $p_F: \mathbf{T}(G) \rightarrow \mathbf{T}(H)$. Indeed, let $u \in \mathbf{T}(G)$ and $\pi \in \mathbf{Rep}(H)$. Since F is admissible, we can define $p_F(u)$ at π by setting $p_F(u)_\pi = u_{F(\pi)}$. Now, let $\pi, \pi' \in \mathbf{Rep}(H)$ and let $h \in \text{Hom}(\pi, \pi')$. Since F is a functor, $F(h)$ lies in $\text{Hom}(F(\pi), F(\pi'))$. Hence, by definition of $\mathbf{T}(G)$, the following diagram is commutative:

$$\begin{array}{ccc}
 \mathcal{H}_{F(\pi)} & \xrightarrow{u_{F(\pi)}} & \mathcal{H}_{F(\pi)} \\
 F(h) \downarrow & & \downarrow F(h) \\
 \mathcal{H}_{F(\pi')} & \xrightarrow{u_{F(\pi')}} & \mathcal{H}_{F(\pi')}
 \end{array}$$

Since F is admissible, the above diagram is *equal* to the following:

$$\begin{array}{ccc}
 \mathcal{H}_\pi & \xrightarrow{p_F(u)_\pi} & \mathcal{H}_\pi \\
 h \downarrow & & \downarrow h \\
 \mathcal{H}_{\pi'} & \xrightarrow{p_F(u)_{\pi'}} & \mathcal{H}_{\pi'}
 \end{array}$$

This shows that $p_F(u) \in \mathcal{N}at(\mathbf{U}_H)$.

By admissibility of F and since $u \in \mathbf{T}(G)$:

$$p_F(u)_1 = u_{F(1)} = u_1 = \text{Id}_{\mathbb{C}},$$

Moreover, for all $\pi, \pi' \in \mathbf{Rep}(H)$, using the fact that u is in $\mathbf{T}(G)$ again,

$$p_F(u)_{\pi \otimes \pi'} = u_{F(\pi \otimes \pi')} = u_{F(\pi) \otimes F(\pi')} = u_{F(\pi)} \otimes u_{F(\pi')} = p_F(u)_\pi \otimes p_F(u)_{\pi'}.$$

Hence $p_F(u)$ is an element of $\mathbf{T}(H)$. Clearly, p_F is continuous and respects the $*$ -monoid structures. Recalling the last statement of Theorem 2.5.1, we see that p_F restricts to a continuous group morphism $p: G \longrightarrow H$ that induces F . $\square$

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Chapter 3

The Urysohn spaces

The content of this chapter is the same as that of the article [BJJ24], a joint work with Colin Jahel and Matthieu Joseph.

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3.1 Introduction

Let $\mathbb{Q}\mathbf{U}$ be the rational Urysohn space, which is the unique countable metric space satisfying the following two conditions:

1. (*universality*) every countable metric space with rational distances embeds isometrically into $\mathbb{Q}\mathbf{U}$,
2. (*ultrahomogeneity*) every isometry between finite subspaces of $\mathbb{Q}\mathbf{U}$ extends to an isometry of $\mathbb{Q}\mathbf{U}$.

The group $\text{Iso}(\mathbb{Q}\mathbf{U})$ of all isometries of $\mathbb{Q}\mathbf{U}$ onto itself is a Polish group when equipped with the topology of pointwise convergence (with $\mathbb{Q}\mathbf{U}$ viewed as a discrete topological space). It has been the object of intensive study over the last three decades [CV06, HN08, EGLMM21, FLMMM22, Mel08, TZ13b, TZ13a].

A family of similar metric spaces can be defined as follows. A distance set is either a countable additive subsemigroup of the positive reals that contains 0, or the intersection $\mathcal{S} \cap [0, r]$ of such a semigroup $\mathcal{S}$ and a bounded interval with $r \in \mathcal{S}$. To each distance set corresponds a Urysohn Δ -metric space $\mathbf{U}_\Delta$: it is the unique countable metric space which is ultrahomogeneous and universal among countable metric spaces with distances in Δ . Our main result is a classification of the unitary representations of the Polish groups $\text{Iso}(\mathbf{U}_\Delta)$.

Theorem 3.1.1 (see Theorem 3.4.13). *Let $\Delta \subseteq \mathbb{R}_+$ be a countable distance set and $G = \text{Iso}(\mathbf{U}_\Delta)$. Every continuous irreducible unitary representation of G is induced from an irreducible representation of the setwise stabilizer $G_{\{A\}}$ for some finite $A \subseteq \mathbf{U}_\Delta$, which is trivial on the pointwise stabilizer G_A . Moreover, every continuous unitary representation of G is a direct sum of irreducible ones.*

In particular, every irreducible representation of $G = \text{Iso}(\mathbf{U}_\Delta)$ is a subrepresentation of a quasi-regular representation $\ell^2(G/V)$ for some open subgroup $V \leq G$ which in turn is a subrepresentation of $\ell^2(\mathbf{U}_\Delta^n)$ for some $n \in \mathbb{N}$.

As a direct corollary, we obtain the following result. We refer to [Bd20, Sec. 6] for an introduction to topological groups of type I.

Corollary 3.1.2. *The group $\text{Iso}(\mathbf{U}_\Delta)$ is of type I for every distance set $\Delta \subseteq \mathbb{R}_+$.*

Let us mention that a classification of the unitary representations for the isometry group of *ultrametric* variants of $\mathbb{Q}\mathbf{U}$ has recently been achieved by Neretin [Ner22]. Even though the results obtained there have the same flavor, the techniques of proofs are different and Neretin's results do not overlap with ours. Using our description of the unitary representations of $\text{Iso}(\mathbf{U}_\Delta)$ together with some

techniques developed by Bekka [Bek03] and Evans-Tsankov [ET16], we prove in Section 3.5 that $\text{Iso}(\mathbf{U}_\Delta)$ has property (T).

Theorem 3.1.3. *For every distance set $\Delta \subseteq \mathbb{R}_+$, the Polish group $\text{Iso}(\mathbf{U}_\Delta)$ has property (T). More precisely, for every integer $m \geq 1$, one can find a Kazhdan set Q_m of cardinality m such that*

$$\sqrt{2 - \frac{2\sqrt{2m-1}}{m}}$$

is the optimal Kazhdan constant for Q_m .

The group $\text{Iso}(\mathbf{U}_\Delta)$ being topologically simple (Theorem 7.3 of [EGLMM21]), our work provides new examples of topologically simple Polish groups which have property (T) and are

- locally bounded but not locally Roelcke precompact, e.g. $\text{Iso}(\mathbb{Q}\mathbf{U})$,
- locally Roelcke precompact but not Roelcke precompact, e.g. $\text{Iso}(\mathbb{Z}\mathbf{U})$,
- coarsely bounded but not Roelcke precompact, e.g. $\text{Iso}(\mathbb{Q}\mathbf{U}_1)$.

We refer to Lemma 3.4.3 for large scale geometric properties of the groups $\text{Iso}(\mathbf{U}_\Delta)$. Let us mention that if Δ is a finite distance set, the group $\text{Iso}(\mathbf{U}_\Delta)$ is oligomorphic. In that case, Tsankov proved in [Tsa12] that $\text{Iso}(\mathbf{U}_\Delta)$ has property (T). He derived this result as a consequence of the classification of the continuous unitary representations for oligomorphic groups. Generalizing Tsankov's results, Property (T) for the very general class of Roelcke precompact groups was later established by Ibarlucía [Iba21].

In Section 3.6, we settle some results around representations of the isometry groups of the Urysohn space $\mathbf{U}$ that are probably well-known but written nowhere in the literature. Those results answer several questions asked by Pestov in [Pes07]. More precisely, we use our classification of the unitary representations of $\text{Iso}(\mathbb{Q}\mathbf{U})$ to prove that $\text{Iso}(\mathbf{U})$ admits no non-trivial unitary representation. In fact, much more is true: $\text{Iso}(\mathbf{U})$ has no non-trivial representation by isometry on reflexive Banach spaces (Corollary 3.6.3). In particular, $\text{Iso}(\mathbf{U})$ has property (T).

Let us explain the argument that we develop to prove Theorem 3.1.1. Let Ω be a countably infinite set and let $\text{Sym}(\Omega)$ be the group of *all* permutations of Ω . It is endowed with the topology induced from the product one on Ω^Ω (Ω is equipped with the discrete topology). A closed subgroup G of $\text{Sym}(\Omega)$ is called a *closed permutation group*. A neighborhood basis of the identity consisting of clopen subgroups is given by *pointwise stabilizers* of finite subsets, which are defined for $A \subseteq \Omega$ finite by $G_A := \{g \in G : \forall a \in A, g(a) = a\}$. The *setwise stabilizer* of a finite set $A \subseteq \Omega$ is the subgroup $G_{\{A\}} := \{g \in G : \forall g \in G, g(A) = A\}$. A closed subgroup $G \leq \text{Sym}(\Omega)$ has *no algebraicity* if for every finite subset $A \subseteq \Omega$, the action $G_A \curvearrowright \Omega \setminus A$ has no finite orbit. Fix a closed permutation group $G \leq \text{Sym}(\Omega)$ and a continuous unitary representation $\pi : G \rightarrow \mathcal{U}(\mathcal{H})$. Given a finite subset $A \subseteq \Omega$, we let $\mathcal{H}_A \subseteq \mathcal{H}$ be the subspace of G_A -invariant vectors. The representation π is *dissociated* if for all finite subsets $A, B \subseteq \Omega$, the subspaces $\mathcal{H}_A$ and $\mathcal{H}_B$ are orthogonal conditionally on $\mathcal{H}_{A \cap B}$. Given three subspaces $\mathcal{H}_1, \mathcal{H}_2, \mathcal{H}_3$ of a Hilbert space $\mathcal{H}$ satisfying $\mathcal{H}_2 \subseteq \mathcal{H}_1 \cap \mathcal{H}_3$, we say that $\mathcal{H}_1$ and $\mathcal{H}_3$ are orthogonal conditionally on $\mathcal{H}_2$ if the subspaces $\mathcal{H}_1 \cap (\mathcal{H}_2)^\perp$ and $\mathcal{H}_3 \cap (\mathcal{H}_2)^\perp$ are orthogonal. For

closed permutation groups with no algebraicity, we classify dissociated unitary representations.

Theorem 3.1.4 (see Theorem 3.3.9). *Let $G \leq \text{Sym}(\Omega)$ be a closed subgroup with no algebraicity. Then every dissociated unitary representation $\pi: G \rightarrow \mathcal{U}(\mathcal{H})$ is isomorphic to a direct sum $\bigoplus_{i \in I} \pi_i$ of irreducible representations, where for every $i \in I$, there exists a finite subset $A_i \subseteq \Omega$ such that π_i is induced from an irreducible representation σ_i of the setwise stabilizer $G_{\{A_i\}}$ which factors through the finite group $G_{\{A_i\}}/G_{A_i}$.*

In order to obtain Theorem 3.1.1, it remains to show that every continuous unitary representation of $\text{Iso}(\mathbf{U}_\Delta)$ is dissociated. Using the Katětov construction, we show in Section 3.4 that $\text{Iso}(\mathbf{U}_\Delta)$ can be approximated in a certain sense by an increasing sequence of oligomorphic groups, which we obtain as isometry groups of Urysohn spaces with *finite* distance sets. Combining this observation with the fact that dissociation holds for every continuous unitary representation of oligomorphic groups (Proposition 3.2 of [JT22]), we prove in Theorem 3.4.12 that dissociation also holds for every unitary representation of $\text{Iso}(\mathbf{U}_\Delta)$. The notion of approximation and the fact that dissociation passes to the limit is made rigorous in Section 3.3.2.

We believe that the methods we develop to obtain dissociation - and therefore classification of unitary representations - through an approximation argument have a considerable potential for further applications. One possible class of structures for which these methods could apply is that of countable relational structures with a *stationary independence relation*. Such structures can be approximated by substructures (see [Mül16]) in a way that generalizes the approximation argument that we use for $\text{Iso}(\mathbf{U}_\Delta)$ in the present article.

Ergodic theoretic consequences. Besides being relevant for classifying unitary representations as explained in Theorem 3.1.4, the notion of dissociation turns out to be appropriate for various other questions in ergodic theory. Let (X, μ) be a standard probability space. A p.m.p. action of a topological group G on (X, μ) is an action of G on X by Borel automorphisms, such that the probability measure μ is G -invariant. A p.m.p. action $G \curvearrowright (X, \mu)$ leads to a unitary representation via the Koopman representation $\kappa: G \rightarrow \mathcal{U}(L_0^2(X, \mu))$ defined by $\kappa(g): f \mapsto f(g^{-1}x)$. Here $L_0^2(X, \mu) = L^2(X, \mu) \ominus \mathbb{C}$. In the context of closed permutation groups, dissociation of the Koopman representation is equivalent to conditional independence of some sub- σ -algebras of the Borel σ -algebra on X . We explain this correspondence here. Let $G \leq \text{Sym}(\Omega)$ be a closed permutation group and $G \curvearrowright (X, \mu)$ a p.m.p. action. For every finite subset $A \subseteq \Omega$, we denote by $\mathcal{F}_A$ the σ -algebra of measurable subsets $Y \subseteq X$ that are G_A -invariant, in the sense that for every $g \in G$, $\mu(gY \Delta Y) = 0$. Say that the p.m.p. action $G \curvearrowright (X, \mu)$ is dissociated¹ if for all finite subsets $A, B \subseteq \Omega$, the σ -algebras $\mathcal{F}_A$ and $\mathcal{F}_B$ are independent conditionally on $\mathcal{F}_{A \cap B}$. Recall that given three sub- σ -algebras $\mathcal{F}_1, \mathcal{F}_2, \mathcal{F}_3$, we say that $\mathcal{F}_1$ and $\mathcal{F}_3$ are independent conditionally on $\mathcal{F}_2$ and write $\mathcal{F}_1 \perp_{\mathcal{F}_2} \mathcal{F}_3$ if for every $Y \in \mathcal{F}_3$, we have $\mathbb{P}(Y \mid \sigma(\mathcal{F}_1, \mathcal{F}_2)) = \mathbb{P}(Y \mid \mathcal{F}_2)$. Conditional independence is related to conditional orthogonality (see [Kal21, Thm. 8.13]) in a way that leads to the following.

¹Dissociation was defined for ergodic p.m.p. actions in [JJ25] as follows: for all $A, B \subseteq \Omega$ finite disjoint, $\mathcal{F}_A$ and $\mathcal{F}_B$ are independent. We do believe that the present definition, which is stronger, is the correct one and we will amend our previous work to reflect this fact.

Fact 3.1.5. *The p.m.p. action $G \curvearrowright (X, \mu)$ is dissociated if and only if its Koopman representation $\kappa: G \rightarrow \mathcal{U}(L_0^2(X, \mu))$ is dissociated.*

In the work [JJ25] of the last two authors, we show how dissociation for p.m.p. actions leads to stabilizer rigidity results *à la* Stuck-Zimmer [SZ20]. Using Theorem 1.4 in [JJ25], the model theoretic properties satisfied by $\text{Iso}(\mathbf{U}_\Delta)$ (see Lemma 3.4.8) and our result that every unitary representation of $\text{Iso}(\mathbf{U}_\Delta)$ is dissociated (see Theorem 3.4.12), we obtain the following result.

Theorem 3.1.6. *Let $\Delta \subseteq \mathbb{R}_+$ be a distance set. Then every ergodic p.m.p. action of $\text{Iso}(\mathbf{U}_\Delta)$ is either essentially free (a conull set of points have a trivial stabilizer) or essentially transitive (one orbit has full measure).*

Let us go through one straightforward consequence of Theorem 3.1.6 concerning invariant random subgroups of $\text{Iso}(\mathbf{U}_\Delta)$. Let G be a Polish group. Let $\text{Sub}(G)$ be the space of all closed subgroups of G . There is a natural σ -algebra on $\text{Sub}(G)$ called the Effros σ -algebra which turns $\text{Sub}(G)$ into a standard Borel space. An *invariant random subgroup* (IRS for short) of G is a Borel probability measure on $\text{Sub}(G)$ which is invariant by conjugation. An IRS ν is concentrated on a conjugacy class if there exists an orbit O of the G -action by conjugation on $\text{Sub}(G)$ such that $\nu(O) = 1$. For more about IRSs of Polish groups, we refer to [JJ25]. A direct consequence of Theorem 3.1.6 is the following.

Corollary 3.1.7. *Let $\Delta \subseteq \mathbb{R}_+$ be a distance set. Then every invariant random subgroup of $\text{Iso}(\mathbf{U}_\Delta)$ is concentrated on an orbit.*

We finish by another ergodic theoretic application of the notion of dissociation, that of de Finetti's theorem. Let $G \leq \text{Sym}(\Omega)$ be a closed permutation group. For every standard Borel space Z , the group G acts on Z^Ω by shifting coordinates: for all $g \in G$ and $(z_\omega)_{\omega \in \Omega} \in Z^\Omega$,

$$g \cdot (z_\omega)_{\omega \in \Omega} = (z_{g^{-1}(\omega)})_{\omega \in \Omega}.$$

Classifying ergodic probability measures on Z^Ω that are invariant under this action is a main problem in exchangeability theory which dates back to de Finetti. When G acts transitively on Ω and μ is an ergodic probability measure on Z^Ω such that the p.m.p. action $G \curvearrowright (Z^\Omega, \mu)$ is dissociated, then μ is clearly a product measure of the form λ^Ω for some Borel probability measure λ on Z . Since we prove that every p.m.p. action (and in fact every unitary representation) of $G = \text{Iso}(\mathbf{U}_\Delta)$ is dissociated, we obtain the following theorem.

Theorem 3.1.8. *Let $\Delta \subseteq \mathbb{R}_+$ be a distance set. Let Z be a standard Borel space. Then the only ergodic probability measures on $Z^{\mathbf{U}_\Delta}$ that are invariant under the shift action $\text{Iso}(\mathbf{U}_\Delta) \curvearrowright Z^{\mathbf{U}_\Delta}$ are product measures of the form $\lambda^{\mathbf{U}_\Delta}$, where λ is a Borel probability measure on Z .*

Remark 3.1.9. A different proof of this result can be obtained via the methods from [BJM25]. For this, one has to check that $\mathbf{U}_\Delta$ is 1-overlap closed. This is done in Section 6 of [AKL14].

3.2 Background on representation theory

In this article, Hilbert spaces are always complex. For such a space $\mathcal{H}$, we denote by $\mathcal{B}(\mathcal{H})$ the space of linear operators $T: \mathcal{H} \rightarrow \mathcal{H}$ which are bounded in the sense that there exists a constant $C > 0$ such that for every $\xi \in \mathcal{H}$, $\|T\xi\| \leq C\|\xi\|$. The strong operator topology on $\mathcal{B}(\mathcal{H})$ is the topology induced by the seminorms $T \mapsto \|T\xi\|$ for $\xi \in \mathcal{H}$. Examples of bounded operators are orthogonal projections. For a closed subspace $\mathcal{K} \subseteq \mathcal{H}$, we denote by $p_{\mathcal{K}} \in \mathcal{B}(\mathcal{H})$ the orthogonal projection onto $\mathcal{K}$. Recall that $p_{\mathcal{K}}\xi$ is the unique element of $\mathcal{K}$ minimizing $\|\xi - \cdot\|$ and that it satisfies $p_{\mathcal{K}} + p_{\mathcal{K}^\perp} = \text{id}_{\mathcal{H}}$. The following result is the Hilbertian version of the classical reverse martingale convergence theorem.

Lemma 3.2.1 (Hilbertian reverse martingale convergence theorem). *Let $(\mathcal{H}_n)_{n \geq 0}$ be a decreasing sequence of closed subspaces of a Hilbert space $\mathcal{H}$ and let $\mathcal{H}_\infty := \bigcap_{n \geq 0} \mathcal{H}_n$. Then $p_{\mathcal{H}_n} \rightarrow p_{\mathcal{H}_\infty}$ in the strong operator topology.*

If $\mathcal{H}_1, \mathcal{H}_2, \mathcal{H}_3$ are three closed subspaces of $\mathcal{H}$ satisfying $\mathcal{H}_2 \subseteq \mathcal{H}_1 \cap \mathcal{H}_3$, we say that $\mathcal{H}_1$ and $\mathcal{H}_3$ are *orthogonal conditionally on $\mathcal{H}_2$* and write $\mathcal{H}_1 \perp_{\mathcal{H}_2} \mathcal{H}_3$ if the subspaces $\mathcal{H}_1 \cap (\mathcal{H}_2)^\perp$ and $\mathcal{H}_3 \cap (\mathcal{H}_2)^\perp$ are orthogonal. The following lemma is a useful (and straightforward) characterization of conditional orthogonality using orthogonal projections.

Lemma 3.2.2. *$\mathcal{H}_1 \perp_{\mathcal{H}_2} \mathcal{H}_3$ if and only if $p_{\mathcal{H}_1} p_{\mathcal{H}_3} = p_{\mathcal{H}_2}$. If this holds, $\mathcal{H}_2 = \mathcal{H}_1 \cap \mathcal{H}_3$.*

Let $\mathcal{U}(\mathcal{H})$ be the unitary group of $\mathcal{H}$. Equipped with the strong operator topology, this is a Polish group. Let G be a topological group. A *unitary representation* of G is a homomorphism from G to the unitary group $\mathcal{U}(\mathcal{H})$. In this article, unitary representations will always be continuous homomorphisms. Since we will be dealing with Polish groups, we may always assume that Hilbert spaces are separable.

Let $\pi: G \rightarrow \mathcal{U}(\mathcal{H})$ be a unitary representation of a topological group G . Given a subgroup $K \leq G$, we will denote by $\mathcal{H}^K$ the closed subspace of $\pi(K)$ -invariant vectors and by $p_K \in \mathcal{B}(\mathcal{H})$ the orthogonal projection onto $\mathcal{H}^K$. For $\xi \in \mathcal{H}$, let $\overline{\text{Conv}}(\pi(G)\xi)$ denote the closed convex hull of $\pi(G)\xi$. The *G-cyclic hull* of ξ is the closure of the linear span of the orbit of ξ under G . Similarly, if $\mathcal{K}$ is a subspace of $\mathcal{H}$, the *G-cyclic hull* of $\mathcal{K}$ is the closure of the linear span of $G \cdot \mathcal{K} = \{\pi(g)\xi, \xi \in \mathcal{K}\}$. The vector ξ (resp. the subspace $\mathcal{K}$) is said to be *G-cyclic in $\mathcal{H}$* if the G-cyclic hull of ξ (resp. of $\mathcal{K}$) is $\mathcal{H}$.

The following useful theorem is due to Alaoglu-Birkhoff [AB39]. To make the treatment comprehensive, we include a proof.

Theorem 3.2.3 (Alaoglu-Birkhoff). *Let $\mathcal{H}$ be a Hilbert space and G be any subgroup of $\mathcal{U}(\mathcal{H})$. For every $\xi \in \mathcal{H}$, $p_G \xi$ is the unique vector of minimal norm in $\overline{\text{Conv}}(\pi(G)\xi)$. In particular, $p_G \xi$ lies in the G-cyclic hull of ξ .*

Proof. Fix $\xi \in \mathcal{H}$. Let $\xi_0 \in \overline{\text{Conv}}(\pi(G)\xi)$ be the unique vector of minimal norm. Since $\overline{\text{Conv}}(\pi(G)\xi)$ is G -invariant, then ξ_0 belongs to $\mathcal{H}^G$. We first show that $p_G \xi = \xi_0$. Notice that $0 \in \overline{\text{Conv}}(\pi(G)(\xi - \xi_0)) = \overline{\text{Conv}}(\pi(G)\xi) - \xi_0$. So we may assume that $\xi_0 = 0$. For every $\varepsilon > 0$, there exists $T_1, \dots, T_n \in G$ and $\lambda_1, \dots, \lambda_n \in \mathbb{R}_+$ such that $\lambda_1 + \dots + \lambda_n = 1$ and

$$\left\| \sum_{i=1}^n \lambda_i T_i \xi \right\| \leq \varepsilon.$$

Therefore, for every $\eta \in \mathcal{H}^G$, we have

$$\begin{aligned} |\langle \xi \mid \eta \rangle| &= \left| \sum_{i=1}^n \lambda_i \langle \xi \mid T_i^{-1} \eta \rangle \right| \\ &= \left| \left\langle \sum_{i=1}^n \lambda_i T_i \xi \mid \eta \right\rangle \right| \\ &\leq \varepsilon \|\eta\|. \end{aligned}$$

Thus, ξ is orthogonal to $\mathcal{H}^G$ and this proves that $p_G \xi = 0$.

Finally, let us prove that ξ_0 is the unique G -invariant vector of $\overline{\text{Conv}}(\pi(G)\xi)$. Fix $\xi_1 \in \overline{\text{Conv}}(\pi(G)\xi) \cap \mathcal{H}^G$. For $i \in \{1, 2\}$, define $\eta_i = \xi - \xi_i$. Define $\eta = \eta_0 - \eta_1 = \xi_1 - \xi_0$ and observe that $\eta \in \mathcal{H}^G$. Fix $t \in \mathbb{R}$. For $T_1, \dots, T_n \in G$ and $\lambda_1, \dots, \lambda_n \in \mathbb{R}_+$ satisfying $\lambda_1 + \dots + \lambda_n = 1$, we have

$$\left\| \sum_{i=1}^n \lambda_i T_i (\eta + t\eta_0) \right\| \leq \|\eta + t\eta_0\|.$$

Moreover, the left-hand side of the inequality can be made arbitrarily close to $\|\eta\|$ since $0 \in \overline{\text{Conv}}(\pi(G)\eta_0) = \overline{\text{Conv}}(\pi(G)\xi) - \xi_0$. Thus, for all $t \in \mathbb{R}$, we have $\|\eta\| \leq \|\eta + t\eta_0\|$. Similarly, for all $t \in \mathbb{R}$, we have $\|\eta\| \leq \|\eta + t\eta_1\|$. We therefore get that $\langle \eta \mid \eta_0 \rangle = \langle \eta \mid \eta_1 \rangle = 0$ and thus $\|\eta\|^2 = \langle \eta \mid \eta_0 - \eta_1 \rangle = 0$. This shows that ξ_0 is the unique element of $\overline{\text{Conv}}(\pi(G)\xi) \cap \mathcal{H}^G$. $\square$

A subrepresentation of a unitary representation $G \longrightarrow \mathcal{U}(\mathcal{H})$ is a closed, G -invariant vector subspace of $\mathcal{H}$. A unitary representation is *irreducible* if its only subrepresentations are $\{0\}$ and $\mathcal{H}$. Note that a closed subspace $\mathcal{K} \subseteq \mathcal{H}$ is G -invariant if and only if $p_{\mathcal{K}}$ and $\pi(g)$ commute for every $g \in G$.

Lemma 3.2.4. *Let G be a topological group and $\pi: G \longrightarrow \mathcal{U}(\mathcal{H})$ a unitary representation. Let $K \leq G$ be a subgroup and $\mathcal{K} \subseteq \mathcal{H}$ be a subrepresentation. Then the following holds*

1. $p_{\mathcal{K}}$ and p_K commutes.
2. If $q_{\mathcal{K}} \in \mathcal{B}(\mathcal{K})$ denotes the orthogonal projection onto $\mathcal{K}^K$, then $(p_K)_{|\mathcal{K}} = q_{\mathcal{K}}$.

Proof. 1. Let $\xi \in \mathcal{H}$. Since $p_{\mathcal{K}}$ is continuous and commutes with $\pi(g)$ for every $g \in G$, it must satisfy $p_{\mathcal{K}}(\overline{\text{Conv}}(\pi(K)\xi)) \subseteq \overline{\text{Conv}}(\pi(K)p_{\mathcal{K}}\xi)$. As $p_{\mathcal{K}}p_K\xi$ is clearly K -invariant, we conclude by Theorem 3.2.3.

2. Fix $\xi \in \mathcal{K}$. By 1, $p_K\xi = p_Kp_{\mathcal{K}}\xi = p_{\mathcal{K}}p_K\xi \in \mathcal{K}$. Therefore, $p_K\xi \in \mathcal{K}^K$. We deduce that $p_K\xi$ is the unique element of $\mathcal{K}^K$ minimizing $\|\xi - \cdot\|$, that is $p_K\xi = q_{\mathcal{K}}\xi$. $\square$

One of the main techniques in representation theory is that of induction. Let G be a topological group and $H \leq G$ an open subgroup. Let $\sigma: H \longrightarrow \mathcal{U}(\mathcal{K})$ be a unitary representation. Let $\mathcal{E}$ be the space of maps $f: G \longrightarrow \mathcal{K}$ such that for every $g \in G$, $h \in H$, $f(gh) = \sigma(h^{-1})f(g)$. Notice that the map $g \mapsto \|f(g)\|$ is constant on

each left H -coset. Denote by $\|f(q)\|$ its value on the coset $q \in G/H$. Let $\mathcal{H}$ be the Hilbert space of all $f \in \mathcal{E}$ such that

$$\sum_{q \in G/H} \|f(q)\|^2 < +\infty.$$

The induced representation $\pi := \text{Ind}_H^G(\sigma)$ is the representation of G on $\mathcal{H}$ defined by

$$\pi(g)f : x \mapsto f(g^{-1}x) \quad \text{for all } g \in G, f \in \mathcal{H}.$$

Since H is open, $\text{Ind}_H^G(\sigma) : G \rightarrow \mathcal{U}(\mathcal{H})$ is indeed continuous. For a map $f \in \mathcal{H}$, we will denote by $\text{supp}(f)$ the set of elements $g \in G$ such that $f(g) \neq 0$. Here are some classical general facts about induced representations. We refer for instance to [Bd20, Sec. 1.F] for more details.

Lemma 3.2.5. *Let G be a topological group and $H \leq G$ an open subgroup. Let $\sigma : H \rightarrow \mathcal{U}(\mathcal{K})$ be a unitary representation and $\pi = \text{Ind}_H^G(\sigma)$. Write $\mathcal{H}$ for the underlying Hilbert space of the representation π . Let $\mathcal{H}_0 := \{f \in \mathcal{H} : \text{supp}(f) \subseteq H\}$.*

1. $\mathcal{H}_0$ is a closed subspace of $\mathcal{H}$ that is stable under action of H .
2. The restriction of π to H and $\mathcal{H}_0$ is canonically isomorphic to σ .
3. $\mathcal{H}_0$ is G -cyclic in $\mathcal{H}$.

A topological group is *non-Archimedean* if it admits a basis of neighborhoods consisting of open subgroups. The class of topological groups under consideration in this article is the class of non-Archimedean Polish groups, which coincides with that of closed permutation groups [BK96, Thm. 1.5.1]. For such groups, the following classical lemma (a proof of which can be found for instance in [Tsa12, Lem. 3.1]) turns out to be very useful.

Lemma 3.2.6. *Let G be a topological group and let $(V_i)_{i \geq 0}$ be a countable basis of neighborhood of the identity consisting of open subgroups. Then for every unitary representation $G \rightarrow \mathcal{U}(\mathcal{H})$, the space $\bigcup_{i \geq 0} \mathcal{H}^{V_i}$ is dense in $\mathcal{H}$.*

Given a closed permutation group $G \leq \text{Sym}(\Omega)$, a continuous unitary representation $\pi : G \rightarrow \mathcal{U}(\mathcal{H})$ and a finite subset $A \subseteq \Omega$, we denote (for conciseness) by $\mathcal{H}_A$ the subspace $\mathcal{H}^{G_A}$, that is, the subspace of G_A -invariant vectors in $\mathcal{H}$. Similarly, we denote by $p_A \in \mathcal{B}(\mathcal{H})$ the orthogonal projection onto $\mathcal{H}_A$. The subspaces $\mathcal{H}_A$ play an important role in the definition of dissociation that is the main topic of the next section.

3.3 Dissociated unitary representations

3.3.1 Dissociation and induced representations

We introduce here a structural property for unitary representations of closed permutation groups that we call dissociation. As the present paper focuses on $\text{Iso}(\mathbb{Q}\mathbf{U})$ which has no algebraicity and weakly eliminates imaginaries (see Lemma 3.4.8), the following definition is tailored for groups satisfying these two assumptions (see Remark 3.3.2). Without these assumptions, dissociation can be defined by mimicking Proposition 3.2 of [JT22]; this will be the topic of a future work.

Definition 3.3.1. Let $G \leq \text{Sym}(\Omega)$ be a closed permutation group. A unitary representation $\pi: G \rightarrow \mathcal{U}(\mathcal{H})$ is *dissociated* if for all finite subsets $A, B \subseteq \Omega$, the subspaces $\mathcal{H}_A$ and $\mathcal{H}_B$ are orthogonal conditionally on $\mathcal{H}_{A \cap B}$.

Remark 3.3.2. Let $G \leq \text{Sym}(\Omega)$ be a closed permutation group acting without fixed points on Ω . Assume that for every unitary representation $\pi: G \rightarrow \mathcal{U}(\mathcal{H})$ and all finite subsets $A, B \subseteq \Omega$, we have $\mathcal{H}_A \perp_{\mathcal{H}_{A \cap B}} \mathcal{H}_B$. Fix any two finite subsets $A, B \subseteq \Omega$. By dissociation of the quasi-regular representation $G \curvearrowright \ell^2(G/\langle G_A, G_B \rangle)$, we get that $\mathcal{H}^{\langle G_A, G_B \rangle} = \mathcal{H}_A \cap \mathcal{H}_B = \mathcal{H}_{A \cap B}$. Thus, $\langle G_A, G_B \rangle = G_{A \cap B}$. Therefore, G has no algebraicity and weakly eliminates imaginaries by Lemma 3.4.5.

Remark 3.3.3. A closed permutation group $G \subseteq \text{Sym}(\Omega)$ is *oligomorphic* if for every $n \in \mathbb{N}$, the diagonal action $G \curvearrowright \Omega^n$ has finitely many orbits. When G is the automorphism group of a first order structure with domain Ω in a countable signature, this is equivalent to the structure being $\aleph_0$ -categorical by the celebrated Ryll-Nardzewski Theorem [Hod93, Thm. 7.3.1]. Examples of such structures include the countable set, the rationals with the usual order, the Rado graph, the countable vector space over a finite field, the countable atomless boolean algebra and Δ -Urysohn spaces where Δ is a finite distance set.

Oligomorphic groups with no algebraicity that admit weak elimination of the imaginaries is an important class of examples of groups for which *all* unitary representations are dissociated (see Proposition 3.2 in [JT22]). Notice that dissociation is obtained in [JT22] as a corollary of the classification due to Tsankov [Tsa12] of unitary representations of oligomorphic groups. We go here in the other direction, by proving that the abstract notion of dissociation implies a classification of the unitary representations. This method applies to new examples such as $\text{Iso}(\mathbf{U}_\Delta)$ for *every* countable distance set Δ .

We prove below that dissociation passes to subrepresentations.

Lemma 3.3.4. Let $G \leq \text{Sym}(\Omega)$ be a closed permutation group. Let $\pi: G \rightarrow \mathcal{U}(\mathcal{H})$ be a unitary representation. If π is dissociated, then every subrepresentation of π is dissociated.

Proof. Let $\mathcal{K} \subseteq \mathcal{H}$ be a subrepresentation of π . Let $A, B \subseteq \Omega$ be two finite subsets. Write $p_A, p_B, p_{A \cap B} \in \mathcal{B}(\mathcal{H})$ for the orthogonal projections onto $\mathcal{H}_A$, $\mathcal{H}_B$ and $\mathcal{H}_{A \cap B}$ and write $q_A, q_B, q_{A \cap B} \in \mathcal{B}(\mathcal{K})$ for the orthogonal projections onto $\mathcal{K}_A$, $\mathcal{K}_B$ and $\mathcal{K}_{A \cap B}$. For $\xi \in \mathcal{K}$, we have that

$$q_A q_B \xi = p_A p_B \xi = p_{A \cap B} \xi = q_{A \cap B} \xi$$

where the first and last equality uses Lemma 3.2.4 and the middle one uses the assumption that π is dissociated. Thus, the subrepresentation $\mathcal{K}$ is dissociated. $\square$

The following theorem is a essential step towards classifying dissociated unitary representations. A similar method has been used by Ol'shanskii in [Ol'80, Lem 2.2] for $G = \text{Sym}(\Omega)$.

Theorem 3.3.5. Let $G \leq \text{Sym}(\Omega)$ be a closed subgroup. Let $\pi: G \rightarrow \mathcal{U}(\mathcal{H})$ be a non-zero unitary representation. If π is dissociated, then π contains a non-zero subrepresentation which is of the form $\text{Ind}_{G_{\{A\}}}^G(\sigma)$ for some finite non-empty subset $A \subseteq \Omega$ and some unitary representation σ of $G_{\{A\}}$ that is trivial on G_A .

Proof. By Lemma 3.2.6, fix $A \subseteq \Omega$ finite with minimal cardinality for the property $\mathcal{H}_A \neq \{0\}$. Then $\mathcal{H}_A$ is a closed subspace of $\mathcal{H}$ stable under the action of $G_{\{A\}}$. Thus restricting π gives rise to a representation σ of $G_{\{A\}}$ on $\mathcal{H}_A$, which is trivial on G_A . Let us show that $\text{Ind}_{G_{\{A\}}}^G(\sigma)$ is a subrepresentation of π .

Claim. For all $g, h \in G$ such that $h^{-1}g \notin G_{\{A\}}$, we have $\pi(g)\mathcal{H}_A \perp \pi(h)\mathcal{H}_A$.

Proof of the claim. Notice that $\pi(g)\mathcal{H}_A = \mathcal{H}_{gA}$ and similarly for h . Since π is dissociated,

$$\mathcal{H}_{gA} \perp_{\mathcal{H}_{gA \cap hA}} \mathcal{H}_{hA}.$$

Assuming $h^{-1}g \notin G_{\{A\}}$, then $gA \neq hA$ and $|gA \cap hA| < |A|$. In particular, we get $\mathcal{H}_{gA \cap hA} = \{0\}$ by minimality of $|A|$ and the claim is proved. $\square_{\text{claim}}$

Next, denote by $\mathcal{K}$ the underlying Hilbert space of $\text{Ind}_{G_{\{A\}}}^G(\sigma)$ and let $(g_i)_{i \in I}$ be a system of representatives of left $G_{\{A\}}$ -cosets in G . The previous claim ensures that the following map is well defined and isometric:

$$\mathcal{K} \longrightarrow \mathcal{H}, \quad f \longmapsto \sum_{i \in I} \pi(g_i)f(g_i).$$

It is easily seen that this map does not depend on the choice of $(g_i)_{i \in I}$ and, hence, that it is G -equivariant. Therefore, $\text{Ind}_{G_{\{A\}}}^G(\sigma)$ is a subrepresentation of π . $\square$

The second step towards classifying dissociated unitary representations is a fine analysis of induced representations of the form $\text{Ind}_{G_{\{A\}}}^G(\sigma)$.

Remark 3.3.6. Note that if $H \trianglelefteq K$ are topological groups, then representations of K that are trivial on H and representations of K/H are really the same thing. We will make implicit use of this observation when stating results and manipulating representations.

In our context, if $G \leq \text{Sym}(\Omega)$ is a closed permutation group and $A \subseteq \Omega$ is finite, then $G_{\{A\}}/G_A$ is a finite group, which naturally identifies as a subgroup of $\text{Sym}(A)$. In particular, every irreducible representation of $G_{\{A\}}$ which is trivial on G_A is finite dimensional.

Lemma 3.3.7. Let $G \leq \text{Sym}(\Omega)$ be a closed permutation group with no algebraicity. Let $A \subseteq \Omega$ be a finite subset and σ a unitary representation of $G_{\{A\}}/G_A$. Consider $\pi = \text{Ind}_{G_{\{A\}}}^G(\sigma)$ and denote by $\mathcal{H}$ its underlying Hilbert space. Then the following holds.

1. For every $B \subseteq \Omega$ finite, we have $\mathcal{H}_B = \{f \in \mathcal{H} : \text{supp}(f) \subseteq \{g \in G : gA \subseteq B\}\}$. In particular, $\mathcal{H}_A = \{f \in \mathcal{H} : \text{supp}(f) \subseteq G_{\{A\}}\}$ and the restriction of π to $\mathcal{H}_A$ and $G_{\{A\}}$ is isomorphic to σ .
2. For every $B \subseteq \Omega$ finite, $p_B \in \mathcal{B}(\mathcal{H})$ is the multiplication by $\mathbb{1}_{\{g \in G : gA \subseteq B\}}$.
3. π is irreducible if and only if σ is.

Proof. 1. Let $f \in \mathcal{H}_B$. Notice that the map $g \mapsto \|f(g)\|$ is constant on each double coset of the form $G_B g G_{\{A\}}$. Since $\|f\|^2 = \sum_{q \in G/G_{\{A\}}} \|f(q)\|^2$ is finite, this shows that for every $g \in \text{supp}(f)$, the double coset $G_B g G_{\{A\}}$ is a disjoint union of finitely many left cosets of $G_{\{A\}}$. Since A is finite, this is equivalent to saying that for every $a \in A$, $G_B g(a)$ is finite. Using that G has no algebraicity, we get that $gA \subseteq B$ and thus $\text{supp}(f) \subseteq \{g \in G : gA \subseteq B\}$.

Conversely, let $f \in \mathcal{H}$ be such that $\text{supp}(f) \subseteq \{g \in G : gA \subseteq B\}$. Fix $h \in G_B$. We need to prove that $\pi(h)f = f$, that is, for every $g \in G$, $f(h^{-1}g) = f(g)$. There are two cases to check.

- If $gA \not\subseteq B$, then $h^{-1}gA \not\subseteq B$ and thus $f(h^{-1}g) = 0 = f(g)$.
- If $gA \subseteq B$, then $h^{-1}g \in G_B g$. But $G_B g = gG_{g^{-1}B} \subseteq gG_A$. So there exists $k \in G_A$ such that $h^{-1}g = gk$. Thus

$$f(h^{-1}g) = \sigma(k^{-1})(f(g)) = f(g)$$

since σ is trivial on G_A .

Applying the above to $B = A$, we indeed obtain $\mathcal{H}_A = \{f \in \mathcal{H} : \text{supp}(f) \subseteq G_{\{A\}}\}$. The last claim then follows from Lemma 3.2.5.

2. Fix $B \subseteq \Omega$ finite and $f \in \mathcal{H}$. Let M be the sum of $\|f(q)\|^2$ for every $q \in G/G_{\{A\}}$ satisfying $q \cap \{g \in G : gA \subseteq B\} = \emptyset$. Using the description of $\mathcal{H}_B$ obtained above, it is clear that for every $f' \in \mathcal{H}_B$, we have $\|f - f'\|^2 \geq M$ with equality if and only if $f' = f \cdot \mathbb{1}_{\{g \in G : gA \subseteq B\}}$, which proves the result.
3. Assume that σ is irreducible. Note that a representation is irreducible if and only if every non-zero vector is cyclic. We first prove the following:

Claim. Every non-zero vector in $\mathcal{H}_A$ is G -cyclic in $\mathcal{H}$.

Proof of the claim. Using Item 1 and Lemma 3.2.5, we get that $G_{\{A\}} \curvearrowright \mathcal{H}_A$ is irreducible. Thus every non-zero vector in $\mathcal{H}_A$ is $G_{\{A\}}$ -cyclic in $\mathcal{H}_A$. But $\mathcal{H}_A$ is G -cyclic in $\mathcal{H}$ by Item 1 and Lemma 3.2.5. $\square_{\text{claim}}$

Fix $f \in \mathcal{H} \setminus \{0\}$. Up to translating f using π , we can assume that $f(e_G) \neq 0$. It follows from Item 2 that p_A is the multiplication by $\mathbb{1}_{G_{\{A\}}}$. In particular, $p_A f \neq 0$. By the claim, $p_A f$ is a G -cyclic vector in $\mathcal{H}$ that lies in the cyclic hull of f by the Alaoglu-Birkhoff theorem. Necessarily, f is also G -cyclic in $\mathcal{H}$ and π is irreducible.

The converse is straightforward since induction preserves subrepresentations, see for instance [Bd20, Cor. E.2.3]. $\square$

Corollary 3.3.8. Let $G \leq \text{Sym}(\Omega)$ be a closed permutation group with no algebraicity. Let $A \subseteq \Omega$ be a finite subset and σ a unitary representation of $G_{\{A\}}/G_A$. Then $\text{Ind}_{G_{\{A\}}}^G(\sigma)$ is dissociated.

Proof. Using Item 2 of Lemma 3.3.7, we readily get that for all finite subsets $B, C \subseteq \Omega$, $p_B p_C = p_{B \cap C}$. Therefore, $\mathcal{H}_B \perp_{\mathcal{H}_{B \cap C}} \mathcal{H}_C$ and thus $\text{Ind}_{G_{\{A\}}}^G(\sigma)$ is dissociated. $\square$

We therefore obtain a complete classification of the dissociated representations of a permutation group with no algebraicity, which is very similar to the classification obtained by Tsankov for oligomorphic groups [Tsa12]. Given a subgroup K of G and an element $g \in G$, we will write $K^g = gKg^{-1}$ and denote by σ^g the unitary representation of K^g given by $\sigma^g(u) = \sigma(g^{-1}ug)$ for every $u \in H^g$.

Theorem 3.3.9. *Let $G \leq \text{Sym}(\Omega)$ be a closed permutation group without algebraicity.*

1. *The dissociated irreducible unitary representations of G are exactly the unitary representations isomorphic to one of the form $\text{Ind}_{G_{\{A\}}}^G(\sigma)$ where A ranges over the finite subsets of Ω and σ over the irreducible representations of the finite group $G_{\{A\}}/G_A$.*
2. *Two such irreducible representations $\text{Ind}_{G_{\{A\}}}^G(\sigma)$ and $\text{Ind}_{G_{\{B\}}}^G(\tau)$ are isomorphic if and only if there exists $g \in G$ such that $gA = B$ and $\sigma^g \simeq \tau$.*
3. *Every dissociated unitary representation of G splits as direct a sum of irreducible subrepresentations.*

Proof.

1. Every irreducible unitary representation of G has this form by Theorem 3.3.5. Moreover, all of these representations are irreducible by Item 3 in Lemma 3.3.7.
2. Let A, B be finite subsets of Ω and σ, τ be irreducible representations of $G_{\{A\}}/G_A$ and $G_{\{B\}}/G_B$ respectively. Assume that $\pi = \text{Ind}_{G_{\{A\}}}^G(\sigma)$ and $\pi' = \text{Ind}_{G_{\{B\}}}^G(\tau)$ are isomorphic, with respective underlying Hilbert spaces $\mathcal{H}$ and $\mathcal{K}$. Then $\mathcal{H}_B \simeq \mathcal{K}_B$ which is non-zero by Item 1 in Lemma 3.3.7, hence there exists $g \in G$ such that $gA \subset B$. By symmetry, $|A| = |B|$ and $gA = B$. Moreover, using Item 1 of Lemma 3.3.7 again and the fact that $G_{\{A\}}^g = G_{\{gA\}} = G_{\{B\}}$, we get

$$\tau \simeq (\pi' : G_{\{B\}} \curvearrowright \mathcal{K}_B) \simeq (\pi : G_{\{B\}} \curvearrowright \mathcal{H}_B) = (\pi^g : G_{\{A\}} \curvearrowright \mathcal{H}_A) \simeq \sigma^g.$$

3. Let $\pi : G \rightarrow \mathcal{U}(\mathcal{H})$ be a non-zero dissociated unitary representation of G . By Theorem 3.3.5, π contains a non-zero subrepresentation of the form $\text{Ind}_{G_{\{A\}}}^G(\sigma)$ for some finite $A \subseteq \Omega$ and some unitary representation σ of $G_{\{A\}}$ that factors through $G_{\{A\}}/G_A$. Since induction preserves subrepresentations and every unitary representation of a finite group contains an irreducible subrepresentation, we may assume that σ is irreducible. Therefore $\text{Ind}_{G_{\{A\}}}^G(\sigma)$ is irreducible by Item 3 of Lemma 3.3.7. Finally, one concludes using Zorn's Lemma and the fact dissociation passes to subrepresentations (Lemma 3.3.4). $\square$

3.3.2 Obtaining dissociation via approximating sequences

Let Ω be a countably infinite set and $G \leq \text{Sym}(\Omega)$ be a closed permutation group. Let $\Omega' \subseteq \Omega$ be an infinite subset and $H \leq \text{Sym}(\Omega')$ a closed permutation group.

An *extension embedding* is an embedding of topological groups $\theta: H \hookrightarrow G$ such that for all $x \in X$, $h \in H$, $\theta(h)(x) = h(x)$. Fix an increasing sequence $\Omega_0 \subseteq \Omega_1 \subseteq \dots \subseteq \Omega$ of infinite subsets with $\Omega = \bigcup_{n \geq 0} \Omega_n$, a sequence of closed permutation groups $G_n \leq \text{Sym}(\Omega_n)$ and a sequence of extension embeddings $\theta_n: G_n \hookrightarrow G_{n+1}$. Notice that for each $n \geq 0$, we can naturally define an extension embedding $\iota_n: G_n \hookrightarrow \text{Sym}(\Omega)$ as follows: for all $g \in G_n$ and $x \in \Omega$, set

$$\iota_n(g)(x) := \begin{cases} g(x) & \text{if } x \in \Omega_n, \\ (\theta_{m-1} \circ \dots \circ \theta_n(g))(x) & \text{if } x \in \Omega_m \text{ for some } m \geq n. \end{cases} \quad (3.1)$$

It is clear that ι_n is well-defined and that this is an extension embedding. Notice moreover that $\iota_n(G_n)$ is a subgroup of $\iota_{n+1}(G_{n+1})$ for every $n \geq 0$.

Definition 3.3.10. Let $G \leq \text{Sym}(\Omega)$ be a closed permutation group. An *approximating sequence* for G is the data of an increasing sequence $\Omega_0 \subseteq \Omega_1 \subseteq \dots \subseteq \Omega$ of infinite subsets with $\Omega = \bigcup_{n \geq 0} \Omega_n$, a sequence of closed permutation groups $G_n \leq \text{Sym}(\Omega_n)$ and a sequence of extension embeddings $\theta_n: G_n \hookrightarrow G_{n+1}$ such that $\bigcup_{n \geq 0} \iota_n(G_n)$ is a dense subgroup of G , where $\iota_n: G_n \rightarrow \text{Sym}(\Omega)$ is the extension embedding defined in (3.1).

In the sequel, approximating sequences will be denoted by $G_0 \hookrightarrow G_1 \hookrightarrow \dots \hookrightarrow G$, the extension embeddings $G_n \hookrightarrow G_{n+1}$ by θ_n and the extension embeddings $G_n \hookrightarrow \text{Sym}(\Omega)$ by ι_n .

Lemma 3.3.11. Let $G \leq \text{Sym}(\Omega)$ be a closed permutation group with an approximating sequence $G_0 \hookrightarrow G_1 \hookrightarrow \dots \hookrightarrow G$. Let $A \subseteq \Omega$ be a finite subset and let $N \geq 0$ be such that $A \subseteq \Omega_N$. Then the sequence $(\iota_n(G_n)_A)_{n \geq N}$ of subgroups is increasing and $\bigcup_{n \geq N} \iota_n(G_n)_A$ is a dense subgroup of G_A .

Proof. Notice that since $A \subseteq \Omega_N$, then for every $n \geq N$ we have $\iota_n(G_n)_A = \iota_n((G_n)_A)$. Thus, the sequence $(\iota_n(G_n)_A)_{n \geq N}$ is increasing. It is clear that $\bigcup_{n \geq N} \iota_n(G_n)_A$ is a subgroup of G_A . Let us prove that it is dense. Fix $g \in G_A$. By density of $\bigcup_{n \geq 0} \iota_n(G_n)$ in G , there is a sequence $(g_k)_{k \geq 0}$ of elements in $\bigcup_{n \geq 0} \iota_n(G_n)$ such that $g_k \rightarrow g$. But the sequence $(\iota_n(G_n))_{n \geq 0}$ is increasing, so the g_k 's belong to the union $\bigcup_{n \geq N} \iota_n(G_n)$. Since $g \in G_A$ and $(g_k)_{k \geq 0}$ converges pointwise to g , then g_k fixes A pointwise eventually, which finishes the proof. $\square$

The main result of this subsection is the following, which establishes the fact that dissociation is closed under taking limits of approximating sequences.

Theorem 3.3.12. Let $G \leq \text{Sym}(\Omega)$ be a closed permutation group with an approximating sequence $G_0 \hookrightarrow G_1 \hookrightarrow \dots \hookrightarrow G$. Assume that for every $n \geq 0$, every unitary representation of the permutation group $G_n \leq \text{Sym}(\Omega_n)$ is dissociated. Then every unitary representation of $G \leq \text{Sym}(\Omega)$ is dissociated.

Proof. Let $\pi: G \rightarrow \mathcal{U}(\mathcal{H})$ be a unitary representation. Fix $A \subseteq \Omega$ finite and let $N \geq 0$ be such that $A \subseteq \Omega_N$. As usual, $\mathcal{H}_A$ denotes the subspace of all vectors $\xi \in \mathcal{H}$ such that $\pi(g)\xi = \xi$ for every $g \in G_A$. We denote by p_A the orthogonal projection onto $\mathcal{H}_A$. For $n \geq N$, we let $\mathcal{H}_A^n$ be the set of all $\xi \in \mathcal{H}$ such that $\pi(g)\xi = \xi$ for every $g \in \iota_n(G_n)_A$. We denote by p_A^n the orthogonal projection onto $\mathcal{H}_A^n$.

Claim. $p_A^n \rightarrow p_A$ in the strong operator topology.

Proof of the claim. First, $(\iota_n(G_n)_A)_{n \geq N}$ is increasing, so $(\mathcal{H}_A^n)_{n \geq N}$ is decreasing. We claim that $\bigcap_{n \geq N} \mathcal{H}_A^n = \mathcal{H}_A$. We indeed have $\mathcal{H}_A \subseteq \bigcap_{n \geq N} \mathcal{H}_A^n$ since $\iota_n(G_n)_A$ is a subgroup of G_A for every $n \geq N$. For the converse inclusion, fix $\xi \in \bigcap_{n \geq N} \mathcal{H}_A^n$. Then for every $g \in \bigcup_{n \geq N} \iota_n(G_n)_A$, we have $\pi(g)\xi = \xi$. By Lemma 3.3.11 and continuity of π , we get that $\pi(g)\xi = \xi$ for every $g \in G_A$ and thus $\xi \in \mathcal{H}_A$. By the Hilbertian reverse martingale theorem (see Lemma 3.2.1), we conclude that $p_A^n \rightarrow p_A$ in the strong operator topology. $\square_{\text{claim}}$

Fix $A, B \subseteq \Omega$ finite. We want to prove that $\mathcal{H}_A \perp_{\mathcal{H}_{A \cap B}} \mathcal{H}_B$. For this, let us show that $p_A p_B = p_{A \cap B}$. Fix $N \geq 0$ such that A and B belongs to Ω_N . For every $n \geq N$, the unitary representation $\pi_n: G_n \xrightarrow{\iota_n} \iota_n(G_n) \xrightarrow{\pi} \mathcal{U}(\mathcal{H})$ is continuous. Moreover, since for every $n \geq N$ we have $\iota_n(G_n)_A = \iota_n((G_n)_A)$ (and similarly for B and $A \cap B$), we get that $\mathcal{H}_A^n$ is the subspace of vectors that are invariant by $(G_n)_A$ (and similarly for $\mathcal{H}_B^n$ and $\mathcal{H}_{A \cap B}^n$). So by assumption, $\mathcal{H}_A^n \perp_{\mathcal{H}_{A \cap B}^n} \mathcal{H}_B^n$. In other words, $p_A^n p_B^n = p_{A \cap B}^n$. Since $\|p_A^n\|, \|p_B^n\| \leq 1$ for every $n \in \mathbb{N}$, their product also converges. Thus, $p_A p_B = p_{A \cap B}$, which concludes the proof. $\square$

3.4 The rational Urysohn space and its fellows

3.4.1 Some properties of $\text{Iso}(\mathbf{U}_\Delta)$

In this section we recall some basic definitions related to variants of the rational Urysohn space. These objects are studied for instance in [EGLMM21].

Definition 3.4.1. A *distance set* is

- either a countable additive subsemigroup of the positive reals that contains 0,
- or the intersection $\mathcal{S} \cap [0, r]$ of such a subsemigroup $\mathcal{S}$ and a bounded interval $[0, r]$ with $r \in \mathcal{S}$.

A Δ -metric space is a metric space whose metric takes its value in Δ . Given a distance set $\Delta \subseteq \mathbb{R}_+$, the class of finite Δ -metric spaces forms a Fraïssé class (for a proof, see for instance [EGLMM21, Lem. 2.6]) and we denote by $\mathbf{U}_\Delta$ its Fraïssé limit. This is a Δ -metric space which is called the Urysohn Δ -metric space. It is the unique (up to isometry) countable Δ -metric space satisfying the following two properties:

- (*ultrahomogeneity*) Given any two finite subsets $A, B \subseteq \mathbf{U}_\Delta$ and any isometry $\varphi: A \rightarrow B$, there exists an isometry $\tilde{\varphi}: \mathbf{U}_\Delta \rightarrow \mathbf{U}_\Delta$ which extends φ ,
- (*universality*) Every countable Δ -metric space embeds isometrically into $\mathbf{U}_\Delta$.

Let $\text{Iso}(\mathbf{U}_\Delta)$ be the isometry group of $\mathbf{U}_\Delta$, which is equipped with the Polish topology of pointwise convergence ($\mathbf{U}_\Delta$ is here equipped with the discrete topology).

Example 3.4.2. Here are some natural choices for Δ .

- If $\Delta = \{0, r\}$, then $\mathbf{U}_\Delta$ is the countable discrete set. In this case $\text{Iso}(\mathbf{U}_\Delta) \simeq S_\infty$, the group of all permutations of a countably infinite set.

- If $\Delta = \{0, r, 2r\}$, then $\mathbf{U}_\Delta$ is the Rado graph. In fact, if we put an edge between $x, y \in \mathbf{U}_\Delta$ iff $d(x, y) = r$, then the structure obtained is isomorphic to the Rado graph.
- If $\Delta = \mathbb{Q}_+$, then $\mathbf{U}_\Delta$ is the rational Urysohn space $\mathbb{Q}\mathbf{U}$.
- If $\Delta = \mathbb{Z}_+$, then $\mathbf{U}_\Delta$ is the integral Urysohn space $\mathbb{Z}\mathbf{U}$.
- If $\Delta = \mathbb{Q}_+ \cap [0, 1]$, then $\mathbf{U}_\Delta$ is the rational Urysohn sphere $\mathbb{Q}\mathbf{U}_1$.

We now discuss the geometrical properties of the groups $\text{Iso}(\mathbf{U}_\Delta)$. These properties will not be used in the sequel, but they help situate these groups in the global picture of Polish groups. Recall that a subset E of a topological group G is *Roelcke precompact* if for every neighborhood U of the identity, there exists a finite subset $F \subseteq G$ such that $E \subseteq UFU$. The group G is *Roelcke precompact* if it is Roelcke precompact as a subset of itself. It is *locally Roelcke precompact* if it admits a Roelcke precompact non-empty open subset. Moreover, a subset $E \subseteq G$ is *coarsely bounded* if every left-invariant and continuous écart on G assigns a finite diameter to E .

Lemma 3.4.3. *Let $\Delta \subseteq \mathbb{R}_+$ be a distance set. Then the following hold:*

1. $\text{Iso}(\mathbf{U}_\Delta)$ is locally bounded.
2. $\text{Iso}(\mathbf{U}_\Delta)$ is coarsely bounded if and only if Δ is bounded.
3. $\text{Iso}(\mathbf{U}_\Delta)$ is locally Roelcke precompact if and only if Δ is closed and discrete.
4. $\text{Iso}(\mathbf{U}_\Delta)$ is Roelcke precompact if and only if Δ is finite.

Proof. The proof of Items 1 and 2 is contained in Theorem 6.31 and Examples 6.32 of [Ros22]. A topological group being Roelcke precompact if and only if it is both locally Roelcke precompact and coarsely bounded, Item 4 follows from Items 2 and 3.

Thus, it only remains to prove Item 3. Let $G = \text{Iso}(\mathbf{U}_\Delta)$. First, note that Δ is closed and discrete if and only if $\Delta \cap [0, M]$ is finite for every $M \geq 0$.

Assume that there exists $M > 0$ such that $\Delta \cap [0, M]$ is infinite. Let U be an open neighborhood of the identity in $\text{Iso}(\mathbf{U}_\Delta)$. Up to replacing U with a smaller neighborhood, we can assume that U is of the form G_A for some finite set $A \subseteq \mathbf{U}_\Delta$. Let $b \in \mathbf{U}_\Delta$ be such that $d(b, A) \geq M/2$. By the Katětov construction and ultrahomogeneity, the distances between b and elements of the G_A orbit of b take every value in the infinite set $\Delta \cap [0, M]$. Thus $G_{A \cup \{b\}}$ has infinitely many orbit on $G_A \cdot b$ and G_A is not Roelcke precompact by [Tsa12, Thm. 2.4].

Conversely, assume Δ is closed and discrete. Fix $a \in \mathbf{U}_\Delta$ and let us show that G_a is Roelcke precompact. For every $\delta \in \Delta$, define

$$B(a, \delta) := \{x \in \mathbf{U}_\Delta : d(a, x) \leq \delta\}.$$

Let us prove that each action $G_a \curvearrowright B(a, \delta)$ is oligomorphic. Fix $\delta \in \Delta$ and recall that $\Delta \cap [0, \delta]$ is finite by assumption. For every $n \geq 1$, there are only finitely many isometric types of metric spaces of the form $(a, x_1, \dots, x_n)$ with $x_1, \dots, x_n \in B(a, \delta)$. Moreover, by ultrahomogeneity of $\mathbf{U}_\Delta$, any two such finite metric spaces

which are isometric, are in the same G_a -orbit. Thus $G_a \curvearrowright B(a, \delta)$ is oligomorphic. Finally, G_a is an inverse limit of oligomorphic groups and therefore is Roelcke precompact by [Tsa12, Thm. 2.4]. $\square$

Definition 3.4.4. Let $G \leq \text{Sym}(\Omega)$ be closed permutation group. We say that G has *weak elimination of imaginaries* if for every open subgroup $V \leq G$, there exists a finite subset $A \subseteq \Omega$ such that $G_A \leq V$ and $[V : G_A] < +\infty$.

Recall the following characterization obtained in [JJ25, Lem. 3.5].

Lemma 3.4.5. *Let $G \leq \text{Sym}(\Omega)$ be a closed permutation group. Then the following are equivalent.*

1. G has no algebraicity and admits weak elimination of imaginaries.
2. G acts on Ω without fixed point and for all finite subsets $A, B \subseteq \Omega$, we have $\langle G_A, G_B \rangle = G_{A \cap B}$.

Slutsky essentially proved that these conditions are satisfied for the isometry groups of Urysohn spaces. See [Slu12, Thm. 4.12, Thm. 4.16 and Cor. 4.17], from which we extract the following.

Theorem 3.4.6 (Slutsky, [Slu12]). *Let $\Delta \subseteq \mathbb{R}_+$ be a distance set and let $G = \text{Iso}(\mathbf{U}_\Delta)$. Then for all $A, B \subseteq \mathbf{U}_\Delta$ finite, we have $\langle G_A, G_B \rangle = G_{A \cap B}$.*

Remark 3.4.7. Slutsky's original result [Slu12, Cor. 4.17] (see also [EGLMM21, Thm. 7.7]) is stated by saying that $\langle G_A, G_B \rangle$ is dense in $G_{A \cap B}$. This is indeed what he proves. However, the subgroup $\langle G_A, G_B \rangle$ being open, it is also closed and the equality holds.

Combining Lemma 3.4.5 and Theorem 3.4.6, we obtain the following result.

Lemma 3.4.8. *Let $\Delta \subseteq \mathbb{R}_+$ be a distance set. Then $\text{Iso}(\mathbf{U}_\Delta)$ has no algebraicity and admits weak elimination of imaginaries.*

3.4.2 The Katětov construction and approximation

We explain now how to build $\mathbf{U}_\Delta$ using the standard construction due to Katětov [Kat88]. Let Δ be a distance set and X be a countable Δ -metric space. A Δ -valued Katětov function on X is a map $f : X \rightarrow \Delta$ satisfying

$$\forall x, y \in X, |f(x) - f(y)| \leq d(x, y) \leq f(x) + f(y). \quad (3.2)$$

The Urysohn Δ -metric space $\mathbf{U}_\Delta$ is characterized by the so-called *Urysohn property*:

Lemma 3.4.9. *The Urysohn Δ -metric space $\mathbf{U}_\Delta$ is, up to isometry, the unique countable Δ -metric space X with the Δ -Urysohn property, i.e. such that for every finite subset $Y \subseteq X$ and every Δ -valued Katětov function $f : Y \rightarrow \Delta$, there exists $x \in X$ such that $f = d(x, \cdot)$.*

Let $E_\Delta(X, \omega)$ be the set of Δ -valued Katětov functions $f: X \rightarrow \Delta$ such that there exists a finite subset $F \subseteq X$ satisfying

$$\forall x \in X, f(x) = \min_{y \in F} (f(y) + d(x, y)).$$

Since X is countable, so is $E_\Delta(X, \omega)$. We equip it with the metric d defined for $f, g \in E_\Delta(X, \omega)$ by $d(f, g) = \inf\{f(x) + g(x) : x \in X\}$. Then $E_\Delta(X, \omega)$ is a Δ -metric space which satisfies the following property: every isometry of X extends uniquely to an isometry of $E(X, \omega)$ in such a way that the extension morphism $\text{Iso}(X) \rightarrow \text{Iso}(E(X, \omega))$ is an embedding of topological groups (for a proof, see [Mel08, Prop. 2.5]). Here $\text{Iso}(X)$ and $\text{Iso}(E(X, \omega))$ are equipped with their respective permutation topology.

To construct $\mathbf{U}_\Delta$, start with the empty space X_0 and define inductively $X_{n+1} = E_\Delta(X_n, \omega)$. Identify X_n as an isometric subspace of X_{n+1} via the isometry $X \rightarrow E(X, \omega)$ given by $x \mapsto d(x, \cdot)$ and let $X = \bigcup_{n \geq 0} X_n$. Then X is a countable Δ -metric space which satisfies the Urysohn property by construction. So X is the Urysohn Δ -metric space $\mathbf{U}_\Delta$.

The following lemma is a direct consequence of the Katětov construction we just explained. It is also a straightforward application of a very general theorem due to Müller on Fraïssé structures with a stationary independence relation, see [Mül16].

Lemma 3.4.10. *Let $\Delta, \Lambda \subseteq \mathbb{R}_+$ be two distance sets such that $\Delta \subseteq \Lambda$. Then there exists an isometric embedding $f: \mathbf{U}_\Delta \rightarrow \mathbf{U}_\Lambda$ such that every isometry of $f(\mathbf{U}_\Delta)$ extends to an isometry of $\mathbf{U}_\Lambda$ in such a way that this extension yields an embedding $\text{Iso}(\mathbf{U}_\Delta) \rightarrow \text{Iso}(\mathbf{U}_\Lambda)$ of topological groups.*

We obtain the following approximation property for $\mathbf{U}_\Delta$.

Corollary 3.4.11. *Let Δ be a distance set and $(\Delta_n)_{n \geq 0}$ an increasing sequence of distance sets such that $\bigcup_{n \geq 0} \Delta_n = \Delta$. Then there exists an approximating sequence $G_0 \hookrightarrow G_1 \hookrightarrow \dots \hookrightarrow \text{Iso}(\mathbf{U}_\Delta)$ such that for every $n \geq 0$, the metric space $\Omega_n \subseteq \mathbf{U}_\Delta$ is isometric to $\mathbf{U}_{\Delta_n}$ and $G_n = \text{Iso}(\Omega_n)$.*

Proof. For every $n \in \mathbb{N}$, let $\Omega_n = \mathbf{U}_{\Delta_n}$ and $G_n = \text{Iso}(\mathbf{U}_{\Delta_n})$. The Katětov construction allows us to see Ω_n as an increasing sequence of metric spaces with $\Omega := \bigcup_{n \in \mathbb{N}} \Omega_n$ being (isometric to) $\mathbf{U}_\Delta$. Lemma 3.4.10 gives the embeddings $G_0 \hookrightarrow G_1 \hookrightarrow \dots \hookrightarrow \text{Iso}(\mathbf{U}_\Delta)$.

It only remains to see that $\bigcup_{n \geq 0} G_n$ is a dense subgroup of $\text{Iso}(\mathbf{U}_\Delta)$. To that aim, let $A \subseteq \mathbf{U}_\Delta$ be a finite subset and $g \in \text{Iso}(\mathbf{U}_\Delta)$. It suffices to find $g' \in \bigcup_{n \geq 0} G_n$ such that $g'|_A = g|_A$. Since A is finite, there exists $n \in \mathbb{N}$ such that $A \subseteq \Omega_n = \mathbf{U}_{\Delta_n}$. Then $g|_A$ is a partial isometry of $\mathbf{U}_{\Delta_n}$ which extends to an element $g' \in \text{Iso}(\mathbf{U}_{\Delta_n})$ by ultrahomogeneity of Urysohn spaces. $\square$

We can now use the approximation property obtained in Corollary 3.4.11 to prove the following result.

Theorem 3.4.12. *Let Δ be a distance set. Then every unitary representation of $\text{Iso}(\mathbf{U}_\Delta)$ is dissociated.*

Proof. Let Δ be a distance set. If Δ is finite, then $\text{Iso}(\mathbf{U}_\Delta)$ is Roelcke precompact by Lemma 3.4.3. Since its action on $\mathbf{U}_\Delta$ is transitive, $\text{Iso}(\mathbf{U}_\Delta)$ is oligomorphic by [Tsa12, Prop. 2.4]. It moreover has no algebraicity and weakly eliminates imaginaries by Lemma 3.4.8. Thus, Proposition 3.2 of [JT22] shows that every unitary representation of $\text{Iso}(\mathbf{U}_\Delta)$ is dissociated.

Therefore, we may assume that Δ is countably infinite and fix an enumeration $\Delta = \{\delta_n : n \geq 0\}$ with $\delta_0 = 0$. For every $n \geq 0$, let $M_n := \max\{\delta_0, \dots, \delta_n\}$. Let Δ_n be the intersection of the closed interval $[0, M_n]$ and the subsemigroup generated by $\{\delta_0, \dots, \delta_n\}$. Then $(\Delta_n)_{n \geq 0}$ forms an increasing sequence of finite distance sets. By Corollary 3.4.11, we have an approximating sequence

$$\text{Iso}(\mathbf{U}_{\Delta_0}) \hookrightarrow \text{Iso}(\mathbf{U}_{\Delta_1}) \hookrightarrow \dots \hookrightarrow \text{Iso}(\mathbf{U}_\Delta).$$

Since the Δ_n 's are finite, every unitary representation of $\text{Iso}(\mathbf{U}_{\Delta_n})$ is dissociated by the first case. By Theorem 3.3.12, we obtain that every unitary representation of $\text{Iso}(\mathbf{U}_\Delta)$ is dissociated. $\square$

Combining with the results obtained in Section 3.3, we get the following. Elaborating on Remark 3.3.6, note that for $G = \text{Iso}(\mathbf{U}_\Delta)$ where $\Delta \subseteq \mathbb{R}_+$ is a distance set, $G_{\{A\}}/G_A$ naturally identifies with $\text{Iso}(A)$ for every finite subset $A \subseteq \mathbf{U}_\Delta$.

Theorem 3.4.13. *Let Δ be a distance set and let $G = \text{Iso}(\mathbf{U}_\Delta)$. Then every unitary representation of G is a direct sum $\bigoplus_{i \in I} \text{Ind}_{G_{\{A_i\}}}^G(\sigma_i)$, where A_i are finite subsets of $\mathbf{U}_\Delta$ and σ_i are irreducible unitary representations of the finite groups $\text{Iso}(A_i)$.*

Remark 3.4.14. For a *finite* distance set Δ , we explained that $\text{Iso}(\mathbf{U}_\Delta)$ is oligomorphic, has no algebraicity and eliminates weakly imaginaries. Therefore, Theorem 3.4.12 and Theorem 3.4.13 for Δ finite are special cases of results respectively due to the second author and Tsankov [JT22, Prop. 3.2] and to Tsankov [Tsa12, Cor. 5.2]. For infinite distance set, the results are new.

3.5 Property (T) for $\text{Iso}(\mathbf{U}_\Delta)$

The aim of this section is to use techniques developed by Tsankov in [Tsa12] to prove property (T) for isometry groups of Urysohn Δ -metric spaces. We start by recalling the definition of property (T) for topological groups.

Definition 3.5.1. A topological group G has *property (T)* if there exists a compact subset $Q \subseteq G$ and $\varepsilon > 0$ such that every unitary representation $\pi: G \rightarrow \mathcal{U}(\mathcal{H})$ with a non-zero vector $\xi \in \mathcal{H}$ which is (Q, ε) -invariant in the sense that

$$\sup_{g \in Q} \|\pi(g)\xi - \xi\| \leq \varepsilon \|\xi\|,$$

admits a non-zero invariant vector. The set Q is called a Kazhdan set for G and ε a Kazhdan constant for Q . The couple (Q, ε) is called a Kazhdan pair.

To prove property (T) for $\text{Iso}(\mathbf{U}_\Delta)$ we follow the strategies from Bekka [Bek03] and Evans, Tsankov [ET16]. For this, let us extract the following result from their works (see the proof of Theorem 2 in [Bek03] and the proof of Theorem 1.1 in [ET16]).

Theorem 3.5.2. *Let $G \leq \text{Sym}(\Omega)$ be a closed permutation group. Assume that the following holds.*

- *Every irreducible unitary representation of G is a subrepresentation of $\ell^2(G/V)$ for some open subgroup $V \leq G$.*
- *Every unitary representation of G is a direct sum of irreducible ones.*
- *There exists a finite subset $Q \subseteq G$ generating a free group F , such that for all proper, open subgroup $V \leq G$, the action of F on G/V is free.*

Then G has property (T). More precisely, (Q, ε) is a Kazhdan pair for G , with

$$\varepsilon = \sqrt{2 - \frac{2\sqrt{2|S|} - 1}{|S|}}$$

and ε is the optimal Kazhdan constant for Q .

Notice that the first two items hold when $G = \text{Iso}(\mathbf{U}_\Delta)$. Indeed, by Theorem 3.4.13, every unitary representation of $\text{Iso}(\mathbf{U}_\Delta)$ is a direct sum of irreducible ones, which are of the form $\text{Ind}_{G_{\{A\}}}^G(\sigma)$ for some $A \subseteq \mathbf{U}_\Delta$ finite and some unitary representation σ of $G_{\{A\}}$ which factors through the finite group $F = G_{\{A\}}/G_A$. But such a σ is a subrepresentation of the left-regular representation λ_F of F and therefore we have

$$\begin{aligned} \text{Ind}_{G_{\{A\}}}^G(\sigma) &\leq \text{Ind}_{G_{\{A\}}}^G(\lambda_F) \simeq \text{Ind}_{G_{\{A\}}}^G(\text{Ind}_{G_A}^{G_{\{A\}}}(1_{G_A})) \\ &\simeq \text{Ind}_{G_A}^G(1_{G_A}) \\ &\simeq \ell^2(G/G_A). \end{aligned}$$

Therefore, to prove Theorem 3.1.3, it remains to show the existence of a non-Abelian free subgroup acting freely on every infinite transitive permutation representation of $\text{Iso}(\mathbf{U}_\Delta)$. To do this, we first refer to the methods developed in [FLMMM22] to obtain the following:

Lemma 3.5.3. *Let $\Delta \subseteq \mathbb{R}_+$ be a distance set. Then every countably infinite group admits a free isometric action on $\mathbf{U}_\Delta$.*

The authors of [FLMMM22] proved much stronger versions of the above statement for Δ bounded and $\Delta = \mathbb{Q}_+$ (Lemma 3.11 and Proposition 7.5 in [FLMMM22] respectively). However, their construction in the bounded case can easily be adapted in order to obtain the above statement. Indeed, say that an isometric action $\Gamma \curvearrowright (X, d)$ of a countable group Γ on a metric space (X, d) is *strongly free* if for every $\gamma \neq e_\Gamma, \forall x \in X, d(\gamma \cdot x, x) \geq 1$. Given a distance set $\Delta \subseteq \mathbb{R}_+$ and a countable group Γ , start with the strongly free left-action $\Gamma \curvearrowright (\Gamma, d)$ where d is the discrete metric ($d(\gamma, \gamma') = 1$ if $\gamma \neq \gamma'$ and 0 otherwise). Up to rescaling d , we can always assume that (Γ, d) is a Δ -metric space. The authors of [FLMMM22] defined a variation of the Katětov construction such that at each step, the action of Γ on the adapted Katětov space remains strongly free. This is the content of Item 4 in [FLMMM22, Prop. 3.8] and still holds when we remove the bound in the definition of the adapted Katětov space. Applying this construction iteratively, we recover a strongly free action of Γ on $\mathbf{U}_\Delta$, hence the above lemma.

Now, if one assumes that Γ is torsion free, the action such obtained has the desired property:

Corollary 3.5.4. *Let $\Delta \subseteq \mathbb{R}_+$ be a distance set. Then for every torsion-free countable group Γ , there exists a free isometric action of Γ on $\mathbf{U}_\Delta$ such that for all proper, open subgroup $V \leq \text{Iso}(\mathbf{U}_\Delta)$, the action of Γ on $\text{Iso}(\mathbf{U}_\Delta)/V$ is free.*

Proof. Let $G = \text{Iso}(\mathbf{U}_\Delta)$ and let Γ be a torsion-free countable group. By Lemma 3.5.3, we fix $\Gamma \leq G$ whose action on $\mathbf{U}_\Delta$ is free. Let $V \leq G$ be a proper, open subgroup. Since G has no algebraicity and weakly eliminates imaginaries by Lemma 3.4.5, there exists a unique finite subset $A \subseteq \mathbf{U}_\Delta$ such that $G_A \leq V \leq G_{\{A\}}$. As V is proper, then A is non-empty. Assume that there exists a non-trivial element $\gamma \in \Gamma$ and a coset $gV \in G/V$ such that $\gamma \cdot gV = gV$. Then $\gamma \in gVg^{-1}$ and in particular, γ fixes setwise $g(A)$. Since Γ is torsion-free, some non-trivial power of γ has a fixed point in $g(A)$. This contradicts the freeness of the action $\Gamma \curvearrowright \mathbf{U}_\Delta$. Thus, the action $\Gamma \curvearrowright G/V$ is free. $\square$

This applies in particular to non-Abelian free groups. Therefore, $\text{Iso}(\mathbf{U}_\Delta)$ satisfies every assumptions of Theorem 3.5.2. Thus, we have proved Theorem 3.1.3.

3.6 A discussion on representations of $\text{Iso}(\mathbf{U})$

Let $\mathbf{U}$ be the Urysohn space, which is the unique complete, separable metric space satisfying the following two conditions:

1. (*universality*) every separable metric space embeds isometrically into $\mathbf{U}$,
2. (*ultrahomogeneity*) every isometry between finite subspaces of $\mathbf{U}$ extends to an isometry of $\mathbf{U}$.

One way of constructing $\mathbf{U}$ is by taking the completion of $\mathbb{Q}\mathbf{U}$. Let $\text{Iso}(\mathbf{U})$ be the group of all isometries of $\mathbf{U}$ onto itself. It is a Polish group when equipped with the topology of pointwise convergence (here $\mathbf{U}$ is equipped with the topology induced by its metric). The following result was obtained around a decade ago by Tent and Ziegler but never published. It can be deduced from their techniques developed in [TZ13b] and [TZ13a].

Theorem 3.6.1 (Tent, Ziegler, unpublished). *The isometry group of the Urysohn space $\mathbf{U}$ is topologically simple.*

The proof of the following lemma is straightforward.

Lemma 3.6.2. *Let G, H, K be topological groups. Assume that G is a closed subgroup of H , that H is topologically simple and that every continuous homomorphism $G \rightarrow K$ is trivial. Then every continuous homomorphism $H \rightarrow K$ is trivial.*

We therefore obtain the following consequence on representations of $\text{Iso}(\mathbf{U})$. Recall that a Banach space E is reflexive if the canonical evaluation map from E to its bidual E^{**} is a homeomorphism. If E is a Banach space, let $\text{Iso}(E)$ be the group of linear isometries of E , which we equip with the topology of pointwise convergence.

Corollary 3.6.3. *The isometry group of the Urysohn space $\mathbf{U}$ admits no non-trivial continuous representation by isometry on a reflexive Banach space.*

Proof. Let $\text{Homeo}_+(\mathbb{R})$ be the Polish group of orientation-preserving homeomorphisms of $\mathbb{R}$. By Uspenskij's theorem [Usp90], $\text{Homeo}_+(\mathbb{R})$ embeds into $\text{Iso}(\mathbf{U})$ as a topological group. By a result of Megrelishvili [Meg01], for every reflexive Banach space E , every continuous homomorphism $\text{Homeo}_+(\mathbb{R}) \rightarrow \text{Iso}(E)$ is trivial. Then the same holds for $\text{Iso}(\mathbf{U})$ by Lemma 3.6.2. $\square$

This answer a question of Pestov [Pes07], which follows exactly the same lines of proof to show that the isometry group of the Urysohn sphere has no non-trivial representation by isometry on a reflexive Banach space.

Let (X, μ) be a standard probability space and let $\text{Aut}(X, [\mu])$ be the group of all non-singular bijections of X , i.e. the group of Borel bijections of X which preserves μ -null sets. The weak topology on $\text{Aut}(X, [\mu])$ is the initial topology with respect to the family of functions $T \in \text{Aut}(X, [\mu]) \mapsto \mu(T(A) \Delta T(B)) \in \mathbb{R}$ and $T \in \text{Aut}(X, [\mu]) \mapsto d(\mu \circ T)/d\mu \in L^1(X, \mu)$. This turns $\text{Aut}(X, [\mu])$ into a Polish group. A non-singular near-action of a topological group G is a continuous group homomorphism $G \mapsto \text{Aut}(X, [\mu])$. Using the standard Koopman representation $\text{Aut}(X, [\mu]) \rightarrow \mathcal{U}(L^2(X, \mu))$ defined by $T \mapsto (d(\mu \circ T^{-1})/d\mu)^{1/2} f \circ T^{-1}$, we obtain the following result.

Corollary 3.6.4. *The isometry group of the Urysohn space $\mathbf{U}$ admits no non-trivial non-singular near-action.*

We provide below an independent proof that $\text{Iso}(\mathbf{U})$ has no non-trivial unitary representation, where we use neither topological simplicity of $\text{Iso}(\mathbf{U})$, nor the result that $\text{Homeo}_+(\mathbb{R})$ has no non-trivial unitary representation.

Proof that $\text{Iso}(\mathbf{U})$ has no non-trivial unitary representation. Let $G = \text{Iso}(\mathbf{U})$ and let $\pi: G \rightarrow \mathcal{U}(\mathcal{H})$ be a unitary representation. Let $H = \{g \in G: g(\mathbb{Q}\mathbf{U}) = \mathbb{Q}\mathbf{U}\}$. Notice that H is a continuous homomorphic image of $\text{Iso}(\mathbb{Q}\mathbf{U})$, which is dense by Theorem 1 of [CV06]. By Theorem 3.1.1, $\pi|_H$ is a direct sum $\bigoplus_i \text{Ind}_{H_{\{A_i\}}}^H(\sigma_i)$, where $A_i \subseteq \mathbb{Q}\mathbf{U}$ are finite subsets and σ_i are irreducible representations of $H_{\{A_i\}}/H_{A_i}$.

Fact 3.6.5. *There exists a sequence $(g_n)_{n \geq 0}$ of isometries of $\mathbb{Q}\mathbf{U}$ which converges to $\text{id}_{\mathbb{Q}\mathbf{U}}$ for the topology of pointwise convergence (where $\mathbb{Q}\mathbf{U}$ is equipped with the topology arising from its metric), such that for every $n \geq 0$, the isometry g_n fixes setwise no finite subset of $\mathbb{Q}\mathbf{U}$.*

Proof of Fact 3.6.5. By a result of Cameron and Vershik [CV06, Thm. 6], there exists an isometry g of $\mathbb{Q}\mathbf{U}$ such that $\langle g \rangle$ is transitive on $\mathbb{Q}\mathbf{U}$ and for every $h \in \langle g \rangle$, the displacement $d(x, h(x))$ is constant for $x \in \mathbb{Q}\mathbf{U}$. Fix $x \in \mathbb{Q}\mathbf{U}$ and a sequence x_n of elements in $\mathbb{Q}\mathbf{U} \setminus \{x\}$ which converges to x . By transitivity, we fix for every $n \geq 0$ an element $k_n \in \mathbb{Z}$ such that $g^{k_n}(x_n) = x$ and set $g_n := g^{k_n}$. Then $(g_n)_{n \geq 0}$ is as wanted. $\square$

Assume that A_i is non-empty for some i . Denote by $\mathcal{K}_i$ the Hilbert space of the representation σ_i and by $\mathcal{H}_i$ the Hilbert space of the representation $\text{Ind}_{H_{\{A_i\}}}^H(\sigma_i)$. Fix a unit vector $\xi \in \mathcal{K}_i$ and define $f \in \mathcal{H}_i$ by

$$f(g) = \begin{cases} \sigma(g)\xi & \text{if } g \in H_{\{A_i\}}, \\ 0 & \text{else.} \end{cases}$$

Since for every $n \geq 0$, the isometry g_n doesn't fix A_i setwise, we obtain that

$$\|\pi(g_n)f - f\| = \sqrt{2}.$$

This is in contradiction with the continuity of $\pi|_H$ (where H is here equipped with the topology of pointwise convergence). Hence A_i is empty for every i and π is trivial on H . By density of H in G , the representation π is trivial. $\square$

Remark 3.6.6. In [Tsa12, Cor. 5.5], Tsankov used a similar strategy to prove that $\text{Homeo}_+(\mathbb{R})$ admits no continuous unitary representation. For this, he used the classification of unitary representations of $\text{Aut}(\mathbb{Q}, <)$ he obtained. This is a special case of a more general result about representations by isometries of $\text{Homeo}_+(\mathbb{R})$ on Banach spaces due to Megrelishvili [Meg01].

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Chapter 4

Dissociated Permutation groups

The content of this chapter is the same as that of the article [BJJ25], a joint work with Colin Jahel and Matthieu Joseph.

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4.1 Introduction

In this paper, we gather together under a common name two techniques – *independence and orthogonality* – that were used many times in the past in the study of infinite permutation groups. *Dissociation*, which is the main notion of the present paper, finds its roots in exchangeability theory. In that context, dissociation of a random process corresponds to *independence*. It is a central notion designed to generalize the famous de Finetti’s theorem and has been extensively studied especially in the setting of random arrays [Ald85, Kal05]. In the last decades, this notion of dissociation has percolated in a domain at the intersection of probability and model theory, which studies random structures that are invariant under the automorphism group of some structure (see e.g. [CT18, DJ08, JT22]).

Another technique that has been widely used is that of *orthogonality*, that leads to classification results of unitary representations for many automorphism groups of structures [BJJ24, Lie72, Tsa12]. One of the aims of this paper is to explain how these techniques of independence and orthogonality are one and the same, which we call dissociation.

An important motivation for introducing this notion is the fact that this phenomenon has already been observed for the class of $\aleph_0$ -categorical structures in a weaker form [JT22, Prop. 3.2]. This observation has consequences, some ergodic theoretic [JJ25, JT22, Tsa25], others on unitary representations [Tsa12]. Dissociation beyond the $\aleph_0$ -categorical case was already applied to the Urysohn spaces by the authors [BJJ24]. We now intend to start a more systematic study of this property.

Notation. In this paper, Ω stands for a countably infinite set and $\text{Sym}(\Omega)$ for the group of all permutations of Ω , equipped with the topology of pointwise convergence. A closed permutation group (on Ω) is a closed subgroup G of $\text{Sym}(\Omega)$. For every $A \subseteq \Omega$, we denote by G_A the *pointwise stabilizer* of A and by $G_{\{A\}}$ the *setwise stabilizer* of A , that is:

$$G_A := \{g \in G, \forall a \in A, g(a) = a\} \quad \text{and} \quad G_{\{A\}} := \{g \in G, g(A) = A\}.$$

We emphasize the fact that the data of a permutation group does not reduce to that of a topological group. It is a specific presentation as a subgroup of $\text{Sym}(\Omega)$ and, in particular, Ω itself is part of the data.

Dissociation for unitary representations. Let G be a closed permutation group on Ω . A *unitary representation* of G on a Hilbert space $\mathcal{H}$ is a group homomorphism $\pi: G \rightarrow \mathcal{U}(\mathcal{H})$ such that for all $\xi \in \mathcal{H}$, the map $g \mapsto \pi(g)\xi$ is continuous.

In other words, we require π to be continuous for the strong operator topology on the unitary group $\mathcal{U}(\mathcal{H})$.

For every finite subset $A \subseteq \Omega$, we will denote by

$$\mathcal{H}_A := \{\xi \in \mathcal{H}, \forall g \in G_A, \pi(g)\xi = \xi\}$$

the closed subspace of G_A -invariant vectors and write p_A for the orthogonal projection onto $\mathcal{H}_A$. One key fact about unitary representations of permutation groups is that invariant vectors determine the whole representation (see for instance [Tsa12, Lem. 3.1]), that is to say:

$$\mathcal{H} = \overline{\bigcup \{\mathcal{H}_A, A \subseteq \Omega \text{ finite}\}}.$$

Dissociation is about how the invariant spaces are arranged with respect to each other.

Definition 4.1.1. The unitary representation π is *dissociated* if for all $A, B \subseteq \Omega$ finite, we have $p_A p_B = p_{A \cap B}$. Geometrically, this means that the subspaces $\mathcal{H}_A \cap (\mathcal{H}_{A \cap B})^\perp$ and $\mathcal{H}_B \cap (\mathcal{H}_{A \cap B})^\perp$ are orthogonal.

One notable feature that we will use in the paper is that there is no real need to reduce our attention to finite subsets of Ω . Indeed, for any subset $A \subseteq \Omega$, one can define:

$$\mathcal{H}_A := \overline{\bigcup \{\mathcal{H}_B, B \subseteq A \text{ finite}\}}$$

and we still denote by p_A the orthogonal projection onto $\mathcal{H}_A$. Dissociation is then equivalent to $p_A p_B = p_{A \cap B}$ for all $A, B \subseteq \Omega$, not necessarily finite.

Dissociation for Boolean p.m.p. actions. Let G be a closed permutation group on Ω . A Boolean p.m.p. action of G on a standard probability space (X, μ) is a homomorphism $\alpha: G \rightarrow \text{Aut}(X, \mu)$ which is continuous for the weak topology on $\text{Aut}(X, \mu)$. For every finite subset $A \subseteq \Omega$, let $\mathcal{F}_A$ be the σ -algebra of G_A -invariant measurable subsets, i.e.:

$$\mathcal{F}_A := \{Y \subseteq X, Y \text{ is measurable and } \forall g \in G_A, \mu(gY \Delta Y) = 0\}.$$

The Boolean p.m.p. action α is *dissociated* if for every finite subset $A, B \subseteq \Omega$, the σ -algebras $\mathcal{F}_A$ and $\mathcal{F}_B$ are independent conditionally on $\mathcal{F}_{A \cap B}$.

Dissociated groups. In Section 4.2, we discuss the relation between dissociation for unitary representations and for Boolean p.m.p. actions. We prove that such an action is dissociated if and only if its Koopman representation is dissociated. We then prove that for a closed permutation group G , every unitary representation is dissociated if and only if every Boolean p.m.p. action is dissociated. If this holds, we say that G is *dissociated*.

Compendium on dissociated groups. One of the goals of this paper is to collect in one place some results related to dissociation that were either proved in other papers, e.g. [BJJ24, JJ25, Tsa25, Tsa12], or are proved in the present paper.

Theorem 4.1.2. *Let G be a closed permutation group on Ω . If G is dissociated, then the following statements hold.*

I. Unitary representations:

- 1) **Rigidity of unitary representations** (Corollary 4.3.3). *Every irreducible unitary representation of G is induced from an irreducible representation of the setwise stabilizer $G_{\{A\}} := \{g \in G : g(A) = A\}$ for some finite $A \subseteq \Omega$, which is trivial on the pointwise stabilizer G_A . Moreover, every unitary representation of G is a direct sum of irreducible ones.*
- 2) **Property (T)** (Theorem 4.3.6). *G has Kazhdan's Property (T).*
- 3) **The Howe-Moore property** (Theorem 4.3.11). *If every proper open subgroup of G is coarsely bounded, then G has the Howe-Moore property.*

II. Ergodic theory:

- 1) **Rigidity of non-singular actions** (Theorem 4.4.2). *Every non-singular ergodic action of G is induced by a probability measure-preserving action of the setwise stabilizer $G_{\{A\}}$ for some finite subset $A \subseteq \Omega$. Moreover, every non-singular action of G is a disjoint union of ergodic ones.*
- 2) **Stabilizer rigidity** (Theorem 4.4.3). *If G is a proper subgroup of $\text{Sym}(\Omega)$ which acts transitively on Ω , then any Borel p.m.p. ergodic action of G is either essentially free or essentially transitive.*
- 3) **Rigidity of invariant random processes** (Corollary 4.4.6). *If G acts transitively on Ω , then every random process $(X_\omega)_{\omega \in \Omega}$ whose law is G -invariant and ergodic is i.i.d.*

How to obtain dissociation? To tackle this question, we change our point of view in Section 4.5 and consider closed subgroups of $\text{Sym}(\Omega)$ as groups of automorphisms of countable relational structures and present two general methods for obtaining dissociation. First, through *approximating sequences*, when the automorphism group can be seen as a coherent limit of dissociated groups. Second, by introducing the notion of *strong cofinite amalgamation over countable sets* (abbrev. σ -SAP, see Definition 4.5.9). Our main new result in this section is Theorem 4.5.14, which proves that if a ultrahomogeneous structure satisfies σ -SAP and weakly eliminates imaginaries then its automorphism group is dissociated. The method we use to prove this result was developed by Tsankov to obtain de Finetti-like results [Tsa25, Sec. 4].

New examples of dissociated groups. Finally, we apply in Section 4.6 the methods from Section 4.5 to provide new examples of dissociated groups, such as isometry groups of some metrically homogeneous spaces, automorphism groups of some diversities and more. In particular, we obtain examples going beyond the more widely studied Roelcke-precompact case, some being locally Roelcke-precompact, others being coarsely bounded but not Roelcke-precompact.

Acknowledgments. We thank Todor Tsankov for sharing a first version of the paper [Tsa25] which inspired many results in the present paper. A few results in the present paper (e.g. Theorem 4.4.2) are even direct adaptations of his results. We also thank Yves Benoist and Christian Rosendal for pointing out to us the similarities between the notion of dissociation and Mackey's work on systems of imprimitivity.

4.2 About the definition of dissociation

4.2.1 Dissociation: unitary representations and Boolean p.m.p. actions

Let (X, μ) be a standard probability space and $\mathcal{F}_1, \mathcal{F}_2, \mathcal{F}_3$ be three σ -algebras of measurable sets on X satisfying $\mathcal{F}_2 \subseteq \mathcal{F}_1 \cap \mathcal{F}_3$. We say that $\mathcal{F}_1$ and $\mathcal{F}_3$ are *independent conditionally on $\mathcal{F}_2$* if the following holds: for every $\mathcal{F}_3$ -measurable function $f \in L^2(X, \mu)$, we have $\mathbb{E}[f \mid \mathcal{F}_1] = \mathbb{E}[f \mid \mathcal{F}_2]$, where $\mathbb{E}[\cdot \mid \mathcal{F}_i]: L^2(X, \mu) \rightarrow L^2(X, \mathcal{F}_i, \mu)$ is the conditional expectation.

Let $\mathcal{H}$ be a Hilbert space and $\mathcal{H}_1, \mathcal{H}_2, \mathcal{H}_3$ be three closed subspaces satisfying $\mathcal{H}_2 \subseteq \mathcal{H}_1 \cap \mathcal{H}_3$ and let p_1, p_2, p_3 be the associated orthogonal projections. We say that $\mathcal{H}_1$ and $\mathcal{H}_3$ are *orthogonal conditionally on $\mathcal{H}_2$* , and write:

$$\mathcal{H}_1 \perp_{\mathcal{H}_2} \mathcal{H}_3$$

if $p_1 p_3 = p_2$, i.e. if the subspaces $\mathcal{H}_1 \cap \mathcal{H}_2^\perp$ and $\mathcal{H}_3 \cap \mathcal{H}_2^\perp$ are orthogonal. Note that if $\mathcal{H}_1 \perp_{\mathcal{H}_2} \mathcal{H}_3$ then $\mathcal{H}_2 = \mathcal{H}_1 \cap \mathcal{H}_3$.

These two notions of conditional independence/orthogonality are related by the following classical result.

Lemma 4.2.1. *Let (X, μ) be a standard probability space and $\mathcal{F}_1, \mathcal{F}_2, \mathcal{F}_3$ be three sub- σ -algebras on X satisfying $\mathcal{F}_2 \subseteq \mathcal{F}_1 \cap \mathcal{F}_3$. If $L^2(X, \mathcal{F}_i, \mu)$ denotes the space of square-integrable, $\mathcal{F}_i$ -measurable complex valued functions on (X, μ) (for $i = 1, 2, 3$), then the following are equivalent.*

- (i) $\mathcal{F}_1$ and $\mathcal{F}_3$ are independent conditionally on $\mathcal{F}_2$.
- (ii) $L^2(X, \mathcal{F}_1, \mu)$ and $L^2(X, \mathcal{F}_3, \mu)$ are orthogonal conditionally on $L^2(X, \mathcal{F}_2, \mu)$.

Proof. Write $\mathcal{H}_i := L^2(X, \mathcal{F}_i, \mu)$ and $p_i := \mathbb{E}[\cdot \mid \mathcal{F}_i]: L^2(X, \mu) \rightarrow \mathcal{H}_i$ for $i = 1, 2, 3$. Notice that $\mathcal{F}_2 \subseteq \mathcal{F}_1 \cap \mathcal{F}_3$ implies $\mathcal{H}_2 \subseteq \mathcal{H}_1 \cap \mathcal{H}_3$. Since p_i is the orthogonal projection onto $\mathcal{H}_i$, the following equivalences hold:

$$\begin{aligned} \mathcal{F}_1 \perp_{\mathcal{F}_2} \mathcal{F}_3 &\Leftrightarrow \forall f \in \mathcal{H}_3, p_1 f = p_2 f \\ &\Leftrightarrow p_1 p_3 = p_2 \\ &\Leftrightarrow \mathcal{H}_1 \perp_{\mathcal{H}_2} \mathcal{H}_3. \end{aligned} \quad \square$$

Let (X, μ) be a probability space and $\text{Aut}(X, \mu)$ be the group of measure-preserving bijections of (X, μ) , identified up to measure zero. It is equipped with the weak topology, defined as follows: $T_n \rightarrow T$ if for all measurable $A \subseteq X$, $\mu(T_n(A) \Delta T(A)) \rightarrow 0$.

A Boolean p.m.p. action $\alpha: G \longrightarrow \text{Aut}(X, \mu)$ of a topological group G on (X, μ) is ergodic if $\mathcal{F}_0$ is trivial (i.e. if $\forall Y \in \mathcal{F}_0, \mu(Y) \in \{0, 1\}$). The following lemma connects G -invariant functions with $\mathcal{F}_0$ -measurable functions. While often stated for Borel p.m.p. actions of locally compact groups, its proof readily adapts to Boolean p.m.p. actions of topological groups.

Lemma 4.2.2. *Let G be a topological group, let $\alpha: G \longrightarrow \text{Aut}(X, \mu)$ be a Boolean p.m.p. action. A function $f: X \longrightarrow \mathbb{C}$ is $\mathcal{F}_0$ -measurable if and only if for every $g \in G$, we have $f \circ g = f$ μ -almost everywhere.*

As a consequence, we connect the two notions of dissociation – for unitary representations and Boolean p.m.p. actions – defined in the introduction. First, recall that a Boolean p.m.p. action $\alpha: G \longrightarrow \text{Aut}(X, \mu)$ of a topological group G on a standard probability space (X, μ) naturally defines a unitary representation of G on $L^2(X, \mu)$, called the *Koopman representation* of α , as follows:

$$\forall f \in L^2(X, \mu), \forall g \in G, \forall x \in X, (g \cdot f)(x) = f(\alpha(g^{-1})x).$$

Lemma 4.2.3. *Let $G \leq \text{Sym}(\Omega)$ be a closed permutation group and $\alpha: G \longrightarrow \text{Aut}(X, \mu)$ be a Boolean p.m.p. action. Then α is dissociated if and only if its Koopman representation is dissociated.*

Proof. Let $\mathcal{H} := L^2(X, \mu)$. For every finite subset $C \subseteq \Omega$, Lemma 4.2.2 implies that $\mathcal{H}_C = L^2(X, \mathcal{F}_C, \mu)$. Therefore, Lemma 4.2.1 shows that α is dissociated if and only if κ is. $\square$

The following result is standard using Gaussian actions. For a proof in the context of Boolean p.m.p. actions of Polish groups, we refer to Section 8.2 of [CGM24] and the references therein.

Theorem 4.2.4. *Let G be a Polish group and $\pi: G \longrightarrow \mathcal{U}(\mathcal{H})$ be a unitary representation. There exists a standard probability space (X, μ) and a Boolean p.m.p. action $\alpha: G \longrightarrow \text{Aut}(X, \mu)$ whose (complex) Koopman representation contains π as a subrepresentation.*

We are now ready to prove the following result.

Theorem 4.2.5. *Let G be a closed permutation group. The following are equivalent.*

- (i) *Every unitary representation of G is dissociated.*
- (ii) *Every Boolean p.m.p. action of G is dissociated.*

If this holds, we say that G is dissociated.

Proof. Assume that every unitary representation of G is dissociated. Let $\alpha: G \longrightarrow \text{Aut}(X, \mu)$ be a Boolean p.m.p. action. By assumption, its Koopman representation is dissociated. Therefore, α is dissociated by Lemma 4.2.3

Assume that every Boolean p.m.p. action of G is dissociated. Let $\pi: G \longrightarrow \mathcal{U}(\mathcal{H})$ be a unitary representation. By Theorem 4.2.4, there exists a Boolean p.m.p. action $\alpha: G \longrightarrow \text{Aut}(X, \mu)$ whose Koopman representation $\kappa: G \longrightarrow \mathcal{U}(L^2(X, \mu))$ contains π as a subrepresentation. Since α is dissociated, so is κ by Lemma 4.2.3. But dissociation passes to subrepresentations [BJJ24, Lem. 3.4] so π is also dissociated. $\square$

4.2.2 Lattice of open subgroups and topological simplicity

Let $G \leq \text{Sym}(\Omega)$ be a closed permutation group. We say that G has *no algebraicity* if for every finite subset $A \subseteq \Omega$, the orbits of the pointwise stabilizer G_A on $\Omega \setminus A$ are all infinite. We say that G *weakly eliminates imaginaries* if for every open subgroup $V \leq G$, there exists a finite subset $A \subseteq \Omega$ such that $G_A \leq V$ and $[V : G_A] < +\infty$. The combination of these two properties leads to a precise understanding of the lattice of open subgroups, as explained in the following lemma.

Lemma 4.2.6. *Let G be a closed subgroup of $\text{Sym}(\Omega)$. Assume that G acts without fixed point on Ω . Then the following are equivalent.*

- (i) G has no algebraicity and weakly eliminates imaginaries.
- (ii) For all finite subsets $A, B \subseteq \Omega$, the subgroup $\langle G_A, G_B \rangle$ generated by G_A and G_B is equal to $G_{A \cap B}$.
- (iii) For every open subgroup $V \leq G$, there exists a unique finite subset $A \subseteq \Omega$ such that $G_A \leq V \leq G_{\{A\}}$.

Proof. The equivalence between (i) and (ii) is Lemma 3.5 of [JJ25] while (iii) is simply an efficient reformulation of the above definitions. Indeed, assume G satisfies (i) and let $V \leq G$ be an open subgroup. By weak elimination of imaginaries, there exists $A \subseteq \Omega$ finite such that $G_A \leq V$ with finite index. Necessarily, every element of $V \cdot A$ has a finite orbit under the action of G_A . Since G has no algebraicity, $V \cdot A \subseteq A$ i.e. $V \leq G_{\{A\}}$.

Assume now B is another finite subset of Ω such that $G_B \leq V \leq G_{\{B\}}$. Then $G_A \leq V \leq G_{\{B\}}$ and G_A stabilizes B . In particular, every element of B has a finite orbit under the action of G_A . By the no algebraicity hypothesis, $B \subseteq A$. The reverse inclusion also holds by symmetry.

Conversely, assume (iii). Clearly, G has weak elimination of imaginaries. Now let $A \subseteq \Omega$ be finite and assume $G_A \cdot a$ is finite for some $a \in \Omega$. Then $B := A \cup G_A \cdot a$ is finite and stable under the action of G_A . Thus:

$$G_B \leq G_A \leq G_{\{B\}}.$$

But we obviously also have $G_A \leq G_A \leq G_{\{A\}}$. By uniqueness in (iii), $A = B$ i.e. $a \in A$ and G has no algebraicity. $\square$

In fact, our definition of dissociation is tailored for permutation groups that have no algebraicity and weakly eliminate imaginaries. In particular, we have the following lemma, which appears as Remark 3.2 of [BJJ24].

Lemma 4.2.7. *Let $G \leq \text{Sym}(\Omega)$ be a closed permutation group. Assume that G acts without fixed points on Ω . If G is dissociated, then G has no algebraicity and eliminates weakly imaginaries.*

The proof of the following result was explained to us by David Evans (personal communication).

Proposition 4.2.8. *Let G be a transitive, closed subgroup of $\text{Sym}(\Omega)$. If G has no algebraicity and weakly eliminates imaginaries, then G is topologically simple.*

Proof. Let H be a closed, normal, proper subgroup of G . Expressing that H is proper and closed, and therefore not dense in G , we get an integer $n \geq 1$ and a G -orbit $\Delta \subseteq \Omega^n$ such that H is not transitive on Δ . As $H \trianglelefteq G$, the H -orbits form a G -equivariant equivalence relation $\sim$ on Δ . Fix $\bar{x} \in \Delta$ and write $[\bar{x}]_\sim$ for the equivalence class $H \cdot \bar{x}$ of $\bar{x}$. It is clear that H is contained in the subgroup $V := \{g \in G: g([\bar{x}]_\sim) = [\bar{x}]_\sim\}$ which is open since it contains the pointwise stabilizer $G_{\bar{x}}$. By Lemma 4.2.6, there exists a unique finite subset $A \subseteq \Omega$ such that $G_A \leq V \leq G_{\{A\}}$. Necessarily, $A \neq \emptyset$ otherwise $[\bar{x}]_\sim = \Delta$, a contradiction. Now, $H \leq G_{\{A\}}$ and, by normality, we get:

$$H \leq K := \bigcap_{g \in G} G_{\{g(A)\}}.$$

Let us prove that K is trivial. Since K is normal in G and G acts transitively on Ω , it suffices to prove that K has a fixed point on Ω . Fix any $a \in A$. Since the G_a -orbits on $\Omega \setminus \{a\}$ are infinite, there exists by Neumann's lemma [BBMN76, Thm. 1] an element $g_0 \in G_a$ such that $A \cap g_0 A = \{a\}$. Thus $K \leq G_{\{A\}} \cap G_{\{g_0(A)\}} \leq G_a$, which concludes the proof. $\square$

Corollary 4.2.9. *Let $G \leq \text{Sym}(\Omega)$ be a transitive, closed permutation group. If G is dissociated, then G is topologically simple.*

4.3 Unitary representations of dissociated groups

4.3.1 Classification of unitary representations

In order to state the classification result of dissociated unitary representations that we obtained in [BJJ24], let us first recall the notion of induced representations. Let G be a separable topological group and let $H \leq G$ be an open subgroup. Then G/H is at most countable by separability of G and we endow it with the counting measure. Let $\sigma: H \rightarrow \mathcal{U}(\mathcal{K})$ be a unitary representation of H . Let $\mathcal{EE}$ be the space of maps $f: G \rightarrow \mathcal{K}$ such that for every $g \in G$ and $h \in H$ we have $f(gh) = \sigma(h^{-1})f(g)$. Notice that for all $f, f_1, f_2 \in \mathcal{EE}$, the maps $g \mapsto \langle f_1(g), f_2(g) \rangle$ and $g \mapsto \|f(g)\|$ are constant on each left H -coset. Denote by $\langle f_1, f_2 \rangle(q)$ and $\|f(q)\|$ their respective value on the coset $q \in G/H$. Let $\mathcal{H}$ be the Hilbert space of all $f \in \mathcal{EE}$ such that $\sum_{q \in G/H} \|f(q)\|^2 < +\infty$ with inner product given by

$$\langle f_1, f_2 \rangle = \sum_{q \in G/H} \langle f_1, f_2 \rangle(q).$$

The *induced representation* $\pi := \text{Ind}_H^G(\sigma)$ is the representation of G on $\mathcal{H}$ defined by

$$\pi(g)f: x \mapsto f(g^{-1}x) \quad \text{for all } g \in G, f \in \mathcal{H}.$$

Since H is open, $\text{Ind}_H^G(\sigma): G \rightarrow \mathcal{U}(\mathcal{H})$ is indeed continuous.

Induction is a powerful technique whose importance was promoted by Mackey in his seminal work on representation theory of locally compact groups. It turns out that dissociation can be phrased in terms of systems of imprimitivity, a key notion in Mackey's theory. In what follows, if $\mathcal{H}$ is a Hilbert space, $\mathcal{B}(\mathcal{H})$ will denote the space of bounded operators on $\mathcal{H}$. Let G be a topological group, $(X, \mathcal{A})$

be a standard Borel space and $G \times X \rightarrow X$ be a Borel action. A *system of imprimitivity* is a couple (π, p) , where $\pi: G \rightarrow \mathcal{U}(\mathcal{H})$ is a unitary representation and $p: \mathcal{A} \rightarrow \mathcal{B}(\mathcal{H})$ is a projection-valued measure (see Section 2.5 of [Mac76]) such that for every $A \in \mathcal{A}$ and $g \in G$,

$$\pi(g)p(A)\pi(g^{-1}) = p(gA). \quad (4.1)$$

For a closed permutation group $G \leq \text{Sym}(\Omega)$, there is a natural Borel action, that of G on Ω . Moreover, given a unitary representation $\pi: G \rightarrow \mathcal{U}(\mathcal{H})$, there is a natural projection-valued measure $p_\pi: \mathcal{P}(\Omega) \rightarrow \mathcal{B}(\mathcal{H})$, by setting $p_\pi(A) = p_A$. Notice that equation (4.1) always holds in that case. Moreover, dissociation easily rephrases as follows.

Fact 4.3.1. *A unitary representation $\pi: G \rightarrow \mathcal{U}(\mathcal{H})$ with no non-zero invariant vector of a closed permutation group $G \leq \text{Sym}(\Omega)$ is dissociated if and only if (π, p_π) is a system of imprimitivity.*

In [BJJ24, Thm. 3.9], we revisit Mackey's famous Imprimitivity Theorem in the context of closed permutation groups. Below, if G is a group and H_1, H_2 are two conjugate subgroups of G , say $H_1 = gH_2g^{-1}$ for some $g \in G$, and if σ is a representation of H_1 , σ^g will denote the representation $h \mapsto \sigma(ghg^{-1})$ of H_2 on the same Hilbert space as σ .

Theorem 4.3.2. *Let $G \leq \text{Sym}(\Omega)$ be a closed permutation group without algebraicity.*

1. *The dissociated irreducible unitary representations of G are exactly the unitary representations isomorphic to one of the form $\text{Ind}_{G_{\{A\}}}^G(\sigma)$ where A ranges over the finite subsets of Ω and σ over the irreducible representations of the finite group $G_{\{A\}}/G_A$.*
2. *Two such irreducible representations $\text{Ind}_{G_{\{A\}}}^G(\sigma)$ and $\text{Ind}_{G_{\{B\}}}^G(\tau)$ are isomorphic if and only if there exists $g \in G$ such that $g(B) = A$ and $\sigma^g \simeq \tau$.*
3. *Every dissociated unitary representation of G splits as direct a sum of irreducible subrepresentations.*

In particular, we get the following classification of unitary representations for dissociated groups.

Corollary 4.3.3. *Let $G \leq \text{Sym}(\Omega)$ be a closed permutation group. If G is dissociated, then every irreducible unitary representation of G is induced from an irreducible representation of the setwise stabilizer $G_{\{A\}} := \{g \in G: g(A) = A\}$ for some finite $A \subseteq \Omega$, which is trivial on the pointwise stabilizer G_A . Moreover, every unitary representation of G is a direct sum of irreducible ones.*

If $E \subseteq \mathcal{B}(\mathcal{H})$ for some Hilbert space $\mathcal{H}$, we will denote by $E' := \{h \in \mathcal{B}(\mathcal{H}), \forall k \in E, h \circ k = k \circ h\}$ the *commutant* of E . Recall that a unitary representation $\pi: G \rightarrow \mathcal{U}(\mathcal{H})$ is of type I if the von Neumann algebra $\pi(G)''$ it generates is of type I. A group is of type I if all of its unitary representations are of type I. We refer to [Bd20, Sec. 6] for a modern reference on type I topological groups. Here is a direct corollary of the above classification of unitary representations of dissociated groups.

Corollary 4.3.4. *Every dissociated closed permutation group is of type I.*

Proof. We use the terminology of [Bd20]. Let $G \leq \text{Sym}(\Omega)$ be a dissociated closed permutation group and let π be a unitary representation of G . By Corollary 4.3.3, there exists a set I , cardinals $(n_i)_{i \in I}$ and a family of mutually non-equivalent irreducible representations $(\pi_i)_{i \in I}$ of G such that $\pi = \bigoplus_{i \in I} n_i \pi_i$. Then π is type I by Remark 6.A.13 (2) in [Bd20]. $\square$

4.3.2 Property (T)

A topological group G has *Property (T)* if there exists a compact subset $Q \subseteq G$ and $\varepsilon > 0$ such that every unitary representation $G \rightarrow \mathcal{U}(\mathcal{H})$ with a non-zero vector $\xi \in \mathcal{H}$ satisfying

$$\sup_{g \in Q} \|\pi(g)\xi - \xi\| \leq \varepsilon \|\xi\|$$

admits a non-zero invariant vector. The pair (Q, ε) is called a Kazhdan pair. We describe here a useful criterion for proving Property (T) for closed permutation groups. It is due to Tsankov [Tsa12] but is not explicitly formulated. We will not provide a proof, but a careful look at Section 6 (in particular Lemma 6.3, Proposition 6.4 and Lemma 6.5) of Tsankov's paper allows one to extract the following result.

Theorem 4.3.5. *Let $G \leq \text{Sym}(\Omega)$ be a closed permutation group without algebraicity. Assume that every unitary representation of G is a direct sum of irreducible ones and that every irreducible representation of G is a subrepresentation of $G \curvearrowright \ell^2(\Omega^n)$ for some $n \leq 0$. Then G has property (T).*

Notice that a dissociated group satisfies the assumption of the above theorem. Indeed, if $G \leq \text{Sym}(\Omega)$ is dissociated, then we know by Corollary 4.3.3 that every unitary representation of G is a direct sum of irreducible ones, and that an irreducible one has the form $\text{Ind}_{G_{\{A\}}}^G(\sigma)$ for some $A \subseteq \Omega$ and some irreducible unitary representation $\sigma: G_{\{A\}} \rightarrow \mathcal{U}(\mathcal{K})$ which factors through the finite group $F := G_{\{A\}}/G_A$. If λ_F denotes the left-regular representation of F , it follows from the basic properties of induction (see e.g. [Fol95, Sec. 6]) that:

$$\begin{aligned} \text{Ind}_{G_{\{A\}}}^G(\sigma) &\leq \text{Ind}_{G_{\{A\}}}^G(\lambda_F) \simeq \text{Ind}_{G_{\{A\}}}^G(\text{Ind}_{G_A}^{G_{\{A\}}}(1_{G_A})) \\ &\simeq \text{Ind}_{G_A}^G(1_{G_A}) \\ &\simeq \ell^2(G/G_A) \\ &\leq \ell^2(\Omega^n), \end{aligned}$$

where $n = |A|$. Therefore, Theorem 4.3.5 applies and yields the following result.

Theorem 4.3.6. *Every dissociated closed permutation group has Property (T).*

A topological group has the strong Property (T) if there exists a Kazhdan pair (Q, ε) with Q finite. Evans and Tsankov proved in [ET16] that every dissociated oligomorphic group has strong Property (T).

Question 4.3.7. *Does every dissociated group have strong Property (T)?*

To give a positive answer to this question, it would be sufficient to prove that every dissociated group $G \leq \text{Sym}(\Omega)$ contains a free group of finite rank acting freely on Ω , see e.g. [BJJ24, Thm. 5.2] or [ET16].

4.3.3 The Howe-Moore property

Another important rigidity phenomenon in harmonic analysis is the Howe-Moore property: the vanishing at infinity of the matrix coefficients of its unitary representations. While classically stated for locally compact groups, we will give a general definition for topological groups in Definition 4.3.8.

Historically, this property was proved for several families of locally compact groups. It was first observed by Howe, Moore and Zimmer in some Lie groups¹ as well as in their algebraic counterparts². A third class of groups with this property was identified by Lubotzky, Mozes and independently Pemantle in the context of groups acting on regular trees, a result later refined by Burger and Mozes³. All these groups admit a so called Cartan decomposition, allowing for a unified proof that covers all known examples [Cio15]. While the groups we consider in this paper are never locally compact, the Howe-Moore property can still be defined within the framework of Rosendal's large scale geometry [Ros22]. In this section, we show that some dissociated groups have this property. These new examples are not covered by the unified proof from the classical case.

Recall that an *écart* on a topological group G is a map $\delta: G^2 \rightarrow \mathbb{R}_+$ such that for every $x, y, z \in G$, we have $\delta(x, x) = 0$ and $\delta(x, z) \leq \delta(x, y) + \delta(y, z)$. It is *left-invariant* if $\delta(xy, xz) = \delta(y, z)$ for every $x, y, z \in G$. A subset $B \subseteq G$ is *coarsely bounded* (in the left coarse-structure) if its δ -diameter is finite for every continuous left-invariant écart δ on G . We will write $\mathcal{CB}(G)$ for the set of coarsely bounded subsets of G . The group G is coarsely bounded if $G \in \mathcal{CB}(G)$ and *locally bounded* if $\mathcal{CB}(G)$ contains a non-empty open set. Recall also that if G is locally compact, $\mathcal{CB}(G)$ coincides with the collection of relatively compact subsets of G .

A generalization of compactness that naturally arises in model theory is Roelcke-precompactness. A subset B of a topological group G is Roelcke-precompact if for every neighborhood U of the identity in G , there exists a finite subset $F \subseteq G$ such that $B \subseteq UFU$. A group is *Roelcke-precompact* if it is Roelcke-precompact as a subset of itself and *locally Roelcke-precompact* if it admits a non-empty Roelcke-precompact open subset. The class of Roelcke-precompact groups contains all the automorphism groups of $\aleph_0$ -categorical structures.

Since Roelcke-precompact subsets are always coarsely bounded, Roelcke-precompact groups have a trivial large scale geometry. Locally Roelcke-precompact Polish groups, however, form a particularly nice class of groups for the study of their large scale geometry [Ros22, Chap. 3]. In particular, they are characterized among Polish groups by the fact that their Roelcke completion is locally compact [Zie21]. Moreover, a subset of a locally Roelcke-precompact Polish group is coarsely bounded if and only if it is Roelcke-precompact. Examples include the group $\text{Aut}(\mathbf{T}_\infty)$ of automorphisms of the regular tree of countably-infinite valency or the isometry group of the integral Urysohn space $\mathbb{Z}\mathbf{U}$. See Section 4.6 for more examples.

Now that this framework has been established, the Howe-Moore property naturally generalizes to general topological groups. To that aim, we will say that a function $f: G \rightarrow \mathbb{C}$ *vanishes at infinity* if for every $\varepsilon > 0$, the set $\{x \in G, |f(x)| \geq \varepsilon\}$

¹Connected simple real Lie groups with finite center [HM79].

²Isotropic simple algebraic groups over non-Archimedean local fields [HM79].

³Topologically-simple closed subgroups of $\text{Aut}(\mathbf{T})$ that act 2-transitively on the boundary of a bi-regular tree $\mathbf{T}$ where both arities are at least 3. [BM00b, Prop. 4.2].

is coarsely bounded, and write $C_0(G)$ for the set of all such continuous functions. In what follows, given a unitary representation $G \rightarrow \mathcal{U}(\mathcal{H})$, an *invariant vector* is a fixed point of the action $G \curvearrowright \mathcal{H}$.

Definition 4.3.8. A topological group G has the *Howe-Moore* property if for every unitary representation $G \rightarrow \mathcal{U}(\mathcal{H})$ with no non-zero invariant vector, every matrix coefficient of π vanishes at infinity, i.e. if for every $\xi, \eta \in \mathcal{H}$, the map $g \mapsto \langle \pi(g)\xi \mid \eta \rangle$ belongs to $C_0(G)$.

We recover some properties that are well known when G is a locally compact group.

Lemma 4.3.9. *Let G be a topological group with the Howe-Moore property. Then every proper open subgroup of G is coarsely bounded.*

Proof. Let $U \leq G$ be a proper open subgroup. Consider the associated quasi-regular representation $\lambda_{G/U}: G \rightarrow \mathcal{U}(\ell^2(G/U))$. Note that this representation has non-zero invariant vectors (the constant functions) if and only if U has finite index in G . We deal with the two cases separately.

Assume first that U has infinite index in G . Write δ_U for the Dirac function at U on G/U . Then $\delta_U \in \ell^2(G/U)$ and, by the Howe-Moore property, the set

$$B := \{g \in G \mid |\langle \lambda_{G/U}(g)\delta_U \mid \delta_U \rangle| \geq 1\}$$

is coarsely bounded. Since for every $g \in U$, $\langle \lambda_{G/U}(g)\delta_U \mid \delta_U \rangle = 1$, we have $U \subseteq B$ hence U is coarsely bounded.

Assume now that U has finite index in G and restrict $\lambda_{G/U}$ to the orthogonal complement $\ell_0^2(G/U)$ of the constant functions. By construction, this representation has no non-zero invariant vector. Denoting again by δ_U the Dirac function at U on G/U , define $\xi := \delta_U - 1/[G : U]$. Then $\xi \in \ell_0^2(G/U)$ and by the Howe-Moore property, the set

$$B := \{g \in G, |\langle \lambda_{G/U}(g)\xi \mid \xi \rangle| \geq 1\}$$

is coarsely bounded. Since U is proper, we have for every $g \in U$, $\langle \lambda_{G/U}(g)\xi \mid \xi \rangle = 2 - 2/[G : U] \geq 1$. Thus $U \subseteq B$ hence U is coarsely bounded. $\square$

Since translates and finite unions of coarsely bounded sets are also coarsely bounded, we get the following consequence.

Corollary 4.3.10. *Let G be a non-bounded topological group which has the Howe-Moore property. Then every proper open subgroup of G has infinite index.*

The main result of this section is the following.

Theorem 4.3.11. *Let $G \leq \text{Sym}(\Omega)$ be a closed permutation group which acts without fixed point on Ω . If G is dissociated, then the following properties are equivalent.*

- (i) G has the Howe-Moore property.
- (ii) Every proper open subgroup of G is coarsely bounded.
- (iii) For every non-empty finite subset $A \subseteq \Omega$, $G_{\{A\}}$ is coarsely bounded.
- (iv) For every non-empty finite subset $A \subseteq \Omega$, G_A is coarsely bounded.

(v) For every $a \in \Omega$, G_a is coarsely bounded.

Proof. The implication (i) $\Rightarrow$ (ii) is proved in Lemma 4.3.9. The equivalences (ii) $\Leftrightarrow$ (iii) $\Leftrightarrow$ (iv) $\Leftrightarrow$ (v) are straightforward by recalling that for a dissociated group G acting without fixed point on Ω , every open subgroup of G lies between the pointwise stabilizer and the setwise stabilizer of a unique finite subset of Ω (Lemma 4.2.6 and 4.2.7).

Let us finally prove that (iii) $\Rightarrow$ (i). Notice that a direct sum of representations whose matrix coefficients are all in $C_0(G)$ have the same property. By the classification of unitary representations for dissociated groups (Corollary 4.3.3), it suffices to prove that the matrix coefficients of every irreducible unitary representation of G without non-zero invariant vector belong to $C_0(G)$.

The only irreducible representation of a topological group that has non-zero invariant vectors is the trivial representation on $\mathbb{C}$. So let π be a nontrivial irreducible representation of G on a Hilbert space $\mathcal{H}$. By Corollary 4.3.3, there exists $A \subseteq \Omega$ a finite subset and $\sigma: G_{\{A\}} \rightarrow \mathcal{U}(\mathcal{K})$ an irreducible unitary representation which is trivial on G_A and such that $\pi \simeq \text{Ind}_{G_{\{A\}}}^G(\sigma)$. Note that A is non-empty since π is nontrivial. Fix a transversal $(h_i)_{i \in I}$ for $G/G_{\{A\}}$, so that for all $\xi, \eta \in \mathcal{H}$, the scalar product $\langle \xi | \eta \rangle$ is given by:

$$\langle \xi | \eta \rangle = \sum_{i \in I} \langle \xi(h_i) | \eta(h_i) \rangle.$$

Define

$$D := \{\xi \in \mathcal{H} : \text{supp}(\xi) \text{ is contained in a finite union of left cosets of } G_{\{A\}}\}.$$

Then D is a dense subset of $\mathcal{H}$. Fix a non-zero vector $\xi \in D$ and let $B := \text{supp}(\xi)$. Notice that B is coarsely bounded by assumption (recall that A is non-empty). Then for all $g \in G$, we have

$$\begin{aligned} \langle \pi(g)\xi | \xi \rangle &= \sum_{i \in I} \langle \xi(g^{-1}h_i) | \xi(h_i) \rangle \\ &= \sum_{\substack{i \in I \\ h_i \in B \cap gB}} \langle \xi(g^{-1}h_i) | \xi(h_i) \rangle \end{aligned}$$

Therefore, for every $g \notin BB^{-1}$, $B \cap gB = \emptyset$ and $\langle \pi(g)\xi | \xi \rangle = 0$. Since BB^{-1} is coarsely bounded, we obtain that for every $\xi \in D$, the map $g \mapsto \langle \pi(g)\xi | \xi \rangle$ belongs to $C_0(G)$. Since D is dense in $\mathcal{H}$, this shows that for all $\xi, \eta \in \mathcal{H}$, the matrix coefficient $g \mapsto \langle \pi(g)\xi | \eta \rangle$ belongs to $C_0(G)$. That is, G has the Howe-Moore property. $\square$

Given a countable, additive subsemigroup of $\mathbb{R}_+$, one can consider the Fraïssé limit of finite metric spaces (X, d) such that $d(X \times X) \subseteq \Delta \cup \{0\}$. This is called the Urysohn Δ -metric space and is denoted by $\mathbf{U}_\Delta$. Isometry groups of Urysohn Δ -metric spaces provide new examples of Polish non-locally compact groups satisfying the Howe-Moore property.

Corollary 4.3.12. *Let Δ be a countable, additive subsemigroup of $\mathbb{R}_+$. Then $\text{Iso}(\mathbf{U}_\Delta)$ has the Howe-Moore property.*

Proof. Notice that countable, additive subsemigroup of $\mathbb{R}_+$ are unbounded, so $\text{Iso}(\mathbf{U}_\Delta)$ is locally bounded but not coarsely bounded by [BJJ24, Lem. 4.3]. We proved in [BJJ24, Thm. 4.12] that $\text{Iso}(\mathbf{U}_\Delta)$ is dissociated. Moreover, pointwise stabilizers $\text{Iso}(\mathbf{U}_\Delta)_A$ of non-empty finite subsets $A \subseteq \mathbf{U}_\Delta$ are coarsely bounded by Theorem 6.31 and Example 6.32 of [Ros22]. The conclusion thus follows from Theorem 4.3.11. $\square$

Just as in the locally compact case, the Howe-Moore property has a direct translation in ergodic theory. A Boolean p.m.p. action $\alpha: G \longrightarrow \text{Aut}(X, \mu)$ of a topological group G is *mixing* if for every measurable sets $Y, Z \subseteq X$, the map

$$g \mapsto \mu(Y \cap \alpha(g)Z) - \mu(Y)\mu(Z)$$

belongs to $C_0(G)$. Moreover, a Boolean p.m.p. action $\alpha: G \longrightarrow \text{Aut}(X, \mu)$ is:

- ergodic if and only if its Koopman representation $\kappa_0: G \longrightarrow \mathcal{U}(L^2(X, \mu) \ominus \mathbb{C})$ has no nontrivial invariant vector,
- mixing if and only if for all $\xi, \eta \in L^2(X, \mu) \ominus \mathbb{C}$, the matrix coefficient $g \mapsto \langle \kappa_0(g)\xi \mid \eta \rangle$ of κ_0 belongs to $C_0(G)$.

Thus, a topological group G has the Howe-Moore property if and only if every ergodic Boolean p.m.p. action of G is mixing.

4.4 Ergodic theory of dissociated groups

4.4.1 Non-singular actions

Let (X, ν) be a σ -finite nontrivial measure space. We denote by $\text{Aut}(X, [\nu])$ the group of bimeasurable bijections which preserve the class of ν , where two such bijections are identified if they coincide on a conull set. Since ν is equivalent to a probability measure, we may always assume that ν is a probability measure. Recall that there is a natural embedding $\kappa: \text{Aut}(X, [\nu]) \longrightarrow \mathcal{U}(L^2(X, \nu))$ defined as follows: for all $g \in \text{Aut}(X, [\nu])$ and $f \in L^2(X, \nu)$,

$$\kappa(g)f: x \mapsto \sqrt{\frac{dg_*\nu}{d\nu}}(x)f(g^{-1}x) \quad (4.2)$$

where for every $g \in G$, $g_*\nu$ is the pushforward of ν by g and $dg_*\nu/d\nu$ is the Radon-Nikodym derivative of $g_*\nu$ with respect to ν . The image of κ is closed and we equip $\text{Aut}(X, [\nu])$ with the topology induced from the strong operator topology on $\mathcal{U}(L^2(X, \mu))$. This is a Polish group. For more details on $\text{Aut}(X, [\nu])$ as a topological group, we refer to Section 4 of [DS22].

A *non-singular* action of a topological group G on a probability space (X, ν) is a continuous homomorphism $\alpha: G \longrightarrow \text{Aut}(X, [\nu])$. It is *ergodic* if every measurable set $Y \subseteq X$ such that for all $g \in G$, $\nu(\alpha(g)Y \Delta Y) = 0$ is either null or conull. A non-singular ergodic action $G \longrightarrow \text{Aut}(X, [\nu])$ is of *type I* if ν is purely atomic, of *type II* if there exists a diffuse, σ -finite, G -invariant measure μ which is equivalent to ν and of *type III* otherwise.

Theorem 4.4.1. *Let $G \leq \text{Sym}(\Omega)$ be a closed permutation group. If G is dissociated, then G admits no ergodic non-singular action of type III.*

In fact, a stronger result holds. In order to state it properly, we need the definition of induction in the setting of p.m.p. actions (there is actually a more general definition of induction in the setting of non-singular actions, however it is not relevant for our purpose). Let G be a Polish group and $H \leq G$ be an open subgroup. We endow the countable space G/H with the counting measure. Let $\alpha: H \rightarrow \text{Aut}(X, \mu)$ be a Boolean p.m.p. action of H on a standard probability space (X, μ) . Let $\text{MAlg}(X, \mu)$ be the measure algebra of (X, μ) . Define

$$\mathfrak{A} := \{f: G \rightarrow \text{MAlg}(X, \mu): \forall g \in G, \forall h \in H, f(gh) = \alpha(h^{-1})f(g)\}.$$

Define two binary operations Δ and $\cap$ on $\mathfrak{A}$ as follows: for all $f_1, f_2 \in \mathfrak{A}$

$$f_1 \Delta f_2: g \mapsto f_1(g) \Delta f_2(g)$$

$$f_1 \cap f_2: g \mapsto f_1(g) \cap f_2(g).$$

Let $0_{\mathfrak{A}}$ and $1_{\mathfrak{A}}$ be the functions which are identically equal to $\emptyset$ and X respectively. Notice that for every $f \in \mathfrak{A}$, the map $g \mapsto \mu(f(g))$ is constant on each left H -coset. Denote by $\mu(f(q))$ its value on the coset $q \in G/H$. Define the following map $\tilde{\mu}: \mathfrak{A} \rightarrow [0, +\infty]$ by

$$\tilde{\mu}(f) := \sum_{q \in G/H} \mu(f(q)).$$

It is left to the cautious reader to check that $(\mathfrak{A}, \Delta, \cap, 0_{\mathfrak{A}}, 1_{\mathfrak{A}}, \tilde{\mu})$ is a measure algebra, which is separable in its measure-algebra topology. Therefore, there exists a standard probability space (Y, ν) such that $\mathfrak{A} = \text{MAlg}(Y, \nu)$. The continuous homomorphism $G \rightarrow \text{Aut}(\mathfrak{A})$ given by the G -action by precomposition on $\mathfrak{A}$ thus provides a Boolean p.m.p. action $G \rightarrow \text{Aut}(Y, \nu)$ which is denoted by $\text{Ind}_H^G(\alpha)$ and is called the *Boolean p.m.p. G -action induced by α* .

The proof of Tsankov's classification of non-singular actions of Roelcke-precompact non-Archimedean Polish groups [Tsa25, Thm. 3.4] adapts without effort to the context of dissociated groups. Due to the substantial similarities, we omit the proof of the following result.

Theorem 4.4.2. *Let $G \leq \text{Sym}(\Omega)$ be a closed permutation group. Assume that G is dissociated. Let (X, ν) be a standard probability space and $\alpha: G \rightarrow \text{Aut}(X, [\nu])$ be a non-singular action. Then α is isomorphic to a countable union $\bigsqcup_{i \in I} \text{Ind}_{G_{A_i}}^G(\alpha_i)$, where for every $i \in I$, A_i is a finite subset of Ω and α_i is a Boolean p.m.p. action of G_{A_i} .*

4.4.2 Stabilizer rigidity

This section goes over the main result of [JJ25], where are studied Borel p.m.p. actions of closed permutation groups through the perspective of the stabilizers associated to these actions. Here, by a *Borel p.m.p. action* of a Polish group G we mean a Borel action $G \times X \rightarrow X$ on a standard Borel space X together with a Borel G -invariant probability measure μ . These are sometimes called spatial p.m.p. actions to emphasize the difference between Boolean p.m.p. actions. To

differentiate them, Borel p.m.p. actions will be denoted by $G \curvearrowright (X, \mu)$ whereas Boolean p.m.p. actions by $G \longrightarrow \text{Aut}(X, \mu)$. A Borel p.m.p. action $G \curvearrowright (X, \mu)$ always yields a Boolean p.m.p. action $G \longrightarrow \text{Aut}(X, \mu)$ (continuity is automatic) and we will say that the Borel p.m.p. action $G \curvearrowright (X, \mu)$ is dissociated if the associated Boolean p.m.p. action is. A Borel p.m.p. action $G \curvearrowright (X, \mu)$ is:

- *essentially transitive* if there exists a G -orbit $O \subseteq X$ such that $\mu(O) = 1$ (recall that orbits are Borel [Kec95, Thm. 15.14]).
- *essentially free* if the set $\{x \in X : \forall g \in G \setminus \{1_G\}, g \cdot x \neq x\}$ is conull.

A subgroup $G \leq \text{Sym}(\Omega)$ is primitive if it acts transitively on Ω and there are no G -invariant equivalence relation on Ω apart from equality and $\Omega \times \Omega$. One can easily check that if G has no algebraicity and weakly eliminates imaginaries, then G is primitive (Corollary 3.6 in [JJ25]). We can now state the main result of [JJ25], which is a permutation group variant on Stuck-Zimmer’s Theorem for locally compact groups. In what follows, given a Borel p.m.p. action $G \curvearrowright (X, \mu)$ and $x \in X$, we will denote by $\text{Stab}(x) := \{g \in G, g(x) = x\}$ the stabilizer of x in G .

Theorem 4.4.3 (Theorem 1.4 of [JJ25]). *Let $G < \text{Sym}(\Omega)$ be a proper, closed permutation group. If G has no algebraicity and is primitive, then for every dissociated Borel p.m.p. ergodic action $G \curvearrowright (X, \mu)$, the following hold:*

- *either $\text{Stab}(x) \curvearrowright \Omega$ has a fixed point for μ -almost every $x \in X$ and in this case $G \curvearrowright (X, \mu)$ is essentially free,*
- *or $\text{Stab}(x) \curvearrowright \Omega$ has no fixed point for μ -almost every $x \in X$ and in this case $G \curvearrowright (X, \mu)$ is essentially transitive.*

For dissociated groups, we therefore have the following striking dichotomy, reminiscent of Stuck-Zimmer’s Theorem for higher rank simple Lie group [SZ20].

Theorem 4.4.4. *Let G be a proper, transitive, closed permutation group. If G is dissociated, then every Borel p.m.p. ergodic action of G is either essentially transitive or essentially free.*

Notice that these results concern Borel p.m.p. actions. Indeed, the notions of essential freeness and essential transitivity make no sense for Boolean p.m.p. action. In fact, there even exists two Borel p.m.p. actions of $\text{Sym}(\Omega)$, one being essentially free, the other one being essentially transitive, whose associated Boolean p.m.p. actions are Booleanly isomorphic [HM25, Sec. 4.3].

4.4.3 Exchangeable processes

Let G be a closed permutation group on Ω and let (Z, ζ) be a standard probability space. The *Bernoulli shift* over G with base space (Z, ζ) is the Borel p.m.p. action of G on $(Z, \zeta)^\Omega$ defined for every $g \in G$ and $(z_\omega)_{\omega \in \Omega} \in Z^\Omega$ by

$$g \cdot (z_\omega)_{\omega \in \Omega} = (z_{g^{-1}(\omega)})_{\omega \in \Omega}.$$

For every subset $A \subseteq \Omega$, let $\text{proj}_A : Z^\Omega \longrightarrow Z^A$ be the restriction map and let $\sigma(\text{proj}_A)$ be the σ -algebra generated by proj_A .

Lemma 4.4.5. *Let $G \leq \text{Sym}(\Omega)$ be a closed permutation group without algebraicity and let $G \curvearrowright (Z, \zeta)^\Omega$ be the Bernoulli shift. For every finite subset $A \subseteq \Omega$, the σ -algebra $\mathcal{F}_A$ of G_A -invariant subsets coincides (up to null sets) with $\sigma(\text{proj}_A)$. Consequently, $G \curvearrowright (Z, \zeta)^\Omega$ is dissociated.*

Proof. Fix $A \subseteq \Omega$ finite. For every Borel subset Y of Z^A , it is clear that $\text{proj}_A^{-1}(Y)$ is G_A -invariant. So the σ -algebra generated by proj_A is contained in $\mathcal{F}_A$. For the converse, fix $Y \in \mathcal{F}_A$. Fix H a countable dense subgroup of G_A and let $Y_0 \subseteq Y$ be a subset of full measure such that for every $h \in H$, the set $h \cdot Y_0$ is equal to Y_0 (here this is a true equality, not only up to null sets). For every $x \in Z^A$, define

$$Y_0^x := \{\text{proj}_{A^c}((z_n)_{n \geq 0}) : (z_n)_{n \geq 0} \in Y_0 \text{ and } (z_a)_{a \in A} = x\}.$$

Then Y_0^x is invariant under H for every $x \in Z^A$. Since H is dense in G_A , they have the same orbits on $\Omega \setminus A$. Since G has no algebraicity, all of these orbits are infinite. This implies that H acts ergodically on $(Z, \zeta)^{\Omega \setminus A}$ (see e.g. [KT08, Prop. 2.1] for a proof). Thus, $\zeta^{\Omega \setminus A}(Y_0^x) \in \{0, 1\}$ for every $x \in Z^A$. If E denotes the set of $x \in Z^A$ such that $\zeta^{\Omega \setminus A}(Y_0^x) = 1$, then $Y_0 \Delta \text{proj}_A^{-1}(E)$ has measure 0. This shows the second inclusion.

To prove dissociation of the Bernoulli shift, take $A, B \subseteq \Omega$ finite. By the previous argument, $\mathcal{F}_A$, $\mathcal{F}_B$ and $\mathcal{F}_{A \cap B}$ coincide (up to null sets) respectively with the σ -algebras generated by proj_A , proj_B and $\text{proj}_{A \cap B}$. But it is clear that $\sigma(\text{proj}_A)$ and $\sigma(\text{proj}_B)$ are independent conditionally on $\sigma(\text{proj}_{A \cap B})$, which shows that the Bernoulli shift $G \curvearrowright (Z, \zeta)^\Omega$ is dissociated. $\square$

Corollary 4.4.6. *Let $G \leq \text{Sym}(\Omega)$ be a transitive, closed permutation group without algebraicity and let Z be a standard Borel space. Let μ be a G -invariant, Borel, ergodic probability measure on Z^Ω . Then μ is dissociated if and only if $\mu = \lambda^{\otimes \Omega}$ for some Borel probability measure λ on Z .*

Proof. Suppose μ is a G -invariant ergodic Borel probability measure on Z^Ω , which is dissociated. By G -invariance of μ and transitivity of $G \curvearrowright \Omega$, the one-dimensional marginals of μ are all equal. Let ζ denote one of them. Then for every measurable $A_1, \dots, A_n \subseteq Z$, dissociation yields that $\mu(A_1 \times \dots \times A_n \times Z \times \dots) = \zeta(A_1) \dots \zeta(A_n)$. Therefore μ is a product measure. The converse is given by Lemma 4.4.5. $\square$

Consequently, de Finetti's theorem [dF31] can be considered as the first historical instance of dissociation, for product actions of the group $\text{Sym}(\Omega)$. Indeed, by Corollary 4.4.6, de Finetti's theorem is equivalent to saying that for every $\text{Sym}(\Omega)$ -invariant ergodic probability measure on Z^Ω , the Borel p.m.p. action $\text{Sym}(\Omega) \curvearrowright (Z^\Omega, \mu)$ is dissociated. A similar result has been obtained by Ryll-Nardzewski for the group $\text{Aut}(\mathbb{Q}, <)$ [Ryl57]. It was later generalized to a large class of closed permutation groups, see [JT22] and [Tsa25].

4.5 How to obtain dissociation

4.5.1 Via approximating sequences

In this section, we recall a method to obtain dissociation that we introduced in [BJJ24] for studying the automorphism group of the rational Urysohn space.

Let $\Omega_0 \subseteq \Omega_1$ be two countably infinite sets and let G_0, G_1 be closed permutation groups over Ω_0 and Ω_1 respectively. An *extension embedding* of G_0 into G_1 is an embedding of topological groups $\theta: G_0 \hookrightarrow G_1$ such that for every $g \in G_0$ and $x \in \Omega_0$, we have $\theta(g)(x) = g(x)$.

Given $G \leq \text{Sym}(\Omega)$ a closed permutation group, an *approximating sequence* for G is the data of an increasing sequence $\Omega_0 \subseteq \Omega_1 \subseteq \dots \subseteq \Omega$ of infinite subsets with $\Omega = \bigcup_{n \geq 0} \Omega_n$, a sequence of closed permutation groups $G_n \leq \text{Sym}(\Omega_n)$ and a sequence of extension embeddings $\theta_n: G_n \hookrightarrow G_{n+1}$ such that $\bigcup_{n \geq 0} \iota_n(G_n)$ is a dense subgroup of G . Here, $\iota_n: G_n \hookrightarrow \text{Sym}(\Omega)$ denotes the natural extension embedding obtained by composing the extension embeddings θ_n .

Theorem 4.5.1 ([BJJ24, Thm. 3.12]). *Let $G \leq \text{Sym}(\Omega)$ be a closed permutation group with an approximating sequence $G_0 \hookrightarrow G_1 \hookrightarrow \dots \hookrightarrow G$. Assume that for every $n \geq 0$, the closed permutation group $G_n \leq \text{Sym}(\Omega_n)$ is dissociated. Then the closed permutation group $G \leq \text{Sym}(\Omega)$ is dissociated.*

Remark 4.5.2. Extension embeddings are a stronger version of what are called extensive embedding in [KSW25]. Indeed, we do ask for more than simply extensivity, as we require our embeddings to be topological. Yet another related notion appears in [BaK26], where they discuss extensibility of structures, however the notion of extension embedding they use for their result is even less restrictive than the one in [KSW25]. The topological version we use is necessary for the dynamical results we obtain.

4.5.2 Via an amalgamation property

The methods presented in this section are heavily inspired from Section 4 in [Tsa25].

Tail subspaces and dissociation

In this section, we introduce for every finite subset $A \subseteq \Omega$ a *tail subspace* $\mathcal{T}_A$ associated with a unitary representation of a closed permutation group $G \leq \text{Sym}(\Omega)$. Our definition is inspired by the classical definition of the tail σ -algebra for an exchangeable random process, see Remark 4.5.8. The key idea in this section is to obtain dissociation by proving that the tail space $\mathcal{T}_A$ coincides with the space of G_A -invariant vectors.

Definition 4.5.3. Let $G \leq \text{Sym}(\Omega)$ be a closed permutation group and $A \subseteq \Omega$ a finite subset. The *tail subspace* $\mathcal{T}_A$ associated with a unitary representation $\pi: G \rightarrow \mathcal{U}(\mathcal{H})$ is defined by

$$\mathcal{T}_A := \bigcap_{\substack{C \subseteq \Omega \text{ cofinite} \\ A \subseteq C}} \mathcal{H}_C.$$

Note that by definition, we always have $\mathcal{H}_A \subseteq \mathcal{T}_A$. In the sequel, we will need a finer notion of conditional orthogonality than the one we present in the introduction. Let $\mathcal{H}$ be a Hilbert space. For any closed subspace $\mathcal{G} \subseteq \mathcal{H}$, we denote by $p_{\mathcal{G}}$ the orthogonal projection onto $\mathcal{G}$.

Definition 4.5.4. Let $\mathcal{G}, \mathcal{K}, \mathcal{L}$ be three closed subspaces of $\mathcal{H}$. We say that $\mathcal{G}$ and $\mathcal{L}$ are orthogonal conditionally on $\mathcal{K}$ if one of the following equivalent assertions is satisfied:

- (i) $p_{\mathcal{K}^\perp} \mathcal{G} \perp p_{\mathcal{K}^\perp} \mathcal{L}$.
- (ii) $p_{\mathcal{G}} p_{\mathcal{K}} p_{\mathcal{L}} = p_{\mathcal{G}} p_{\mathcal{L}}$.
- (iii) $p_{\mathcal{L}} p_{\mathcal{K}} p_{\mathcal{G}} = p_{\mathcal{L}} p_{\mathcal{G}}$.

If this holds, we write $\mathcal{G} \perp_{\mathcal{K}} \mathcal{L}$.

Notice that if $\mathcal{K} \subseteq \mathcal{G} \cap \mathcal{L}$, then we recover the definition given in the introduction: $\mathcal{G} \perp_{\mathcal{K}} \mathcal{L}$ is equivalent to $p_{\mathcal{G}} p_{\mathcal{L}} = p_{\mathcal{L}} p_{\mathcal{G}} = p_{\mathcal{K}}$. In this case, we readily have that $\mathcal{K} = \mathcal{G} \cap \mathcal{L}$.

The following theorem provides a weak form of dissociation.

Theorem 4.5.5. Let $G \leq \text{Sym}(\Omega)$ be a closed subgroup without algebraicity. Let $\pi: G \rightarrow \mathcal{U}(\mathcal{H})$ be a unitary representation. Then $\mathcal{H}_A \perp_{\mathcal{T}_{A \cap B}} \mathcal{H}_B$ for all finite subsets $A, B \subseteq \Omega$.

Proof. Fix $A, B \subseteq \Omega$ finite and let $\mathcal{G} := \mathcal{H}_A$, $\mathcal{K} = \mathcal{T}_{A \cap B}$ and $\mathcal{L} = \mathcal{H}_B$. Let us prove that $p_{\mathcal{G}} p_{\mathcal{K}} p_{\mathcal{L}} = p_{\mathcal{G}} p_{\mathcal{L}}$. Define $\mathcal{K}' = \overline{\text{Vect}(\mathcal{G}, \mathcal{K})}$. Notice that if $p_{\mathcal{K}'} p_{\mathcal{L}} = p_{\mathcal{K}} p_{\mathcal{L}}$, then $\mathcal{G} \perp_{\mathcal{K}} \mathcal{L}$. Indeed

$$\begin{aligned} (p_{\mathcal{K}'} p_{\mathcal{L}} = p_{\mathcal{K}} p_{\mathcal{L}}) &\Rightarrow (p_{\mathcal{G}} p_{\mathcal{K}} p_{\mathcal{L}} = p_{\mathcal{G}} p_{\mathcal{K}'} p_{\mathcal{L}} = p_{\mathcal{G}} p_{\mathcal{L}}) \\ &\Rightarrow \mathcal{G} \perp_{\mathcal{K}} \mathcal{L}. \end{aligned}$$

where the rightmost equality holds because $\mathcal{G} \subseteq \mathcal{K}'$. So let us prove that for all $\xi \in \mathcal{L}$,

$$p_{\mathcal{K}'} \xi = p_{\mathcal{K}} \xi. \quad (4.3)$$

Since $\mathcal{K}$ is a subspace of $\mathcal{K}'$, we have $\|p_{\mathcal{K}} \xi\| \leq \|p_{\mathcal{K}'} \xi\|$ and (4.3) is equivalent to $\|p_{\mathcal{K}'} \xi\| = \|p_{\mathcal{K}} \xi\|$. We thus prove the reverse inequality.

Fix a strictly decreasing sequence $C_0 \supseteq C_1 \supseteq \dots$ of cofinite subsets of Ω whose intersection $\bigcap_{n \geq 0} C_n$ is $A \cap B$. For every $n \geq 0$, let $\mathcal{K}_n := \mathcal{H}_{C_n}$. Then $(\mathcal{K}_n)_{n \geq 0}$ forms a decreasing sequence of closed subspaces. Moreover, the intersection $\bigcap_{n \geq 0} \mathcal{K}_n$ is equal to $\mathcal{K}$ and $p_{\mathcal{K}_n} \rightarrow p_{\mathcal{K}}$ in the strong operator topology. Since G has no algebraicity, Neumann's Lemma [BBMN76, Thm. 1] gives for every $n \geq 0$ some element $g_n \in G_B$ such that $g_n(A)$ is contained in C_n . In particular, $\pi(g_n) \mathcal{G} = \pi(g_n) \mathcal{H}_A = \mathcal{H}_{g_n(A)}$ is contained in $\mathcal{K}_n$.

For all $n \geq 0$, define $\mathcal{K}'_n := \pi(g_n) \mathcal{K}'$. Notice that

$$\begin{aligned} \mathcal{K}'_n &= \overline{\text{Vect}(\pi(g_n) \mathcal{G}, \pi(g_n) \mathcal{K})} \\ &= \overline{\text{Vect}(\mathcal{H}_{g_n(A)}, \mathcal{K})}, \end{aligned}$$

where the last equality holds since $\mathcal{K}$ is a $G_{A \cap B}$ -invariant subspace and $g_n \in G_B \leq G_{A \cap B}$. Therefore, $\mathcal{K}'_n$ is a subspace of $\mathcal{K}_n$. For all $\xi \in \mathcal{L}$ and $n \geq 0$, we have

$$\|p_{\mathcal{K}'} \xi\| = \|\pi(g_n) p_{\mathcal{K}'} \xi\| = \|p_{\mathcal{K}'_n} \pi(g_n) \xi\| = \|p_{\mathcal{K}'_n} \xi\| \leq \|p_{\mathcal{K}_n} \xi\| \rightarrow \|p_{\mathcal{K}} \xi\|,$$

which concludes the proof. $\square$

Therefore, if $\mathcal{T}_A = \mathcal{H}_A$ for every finite $A \subseteq \Omega$, then the unitary representation is dissociated. We can actually prove the converse of this statement, which will allow us to get a nice equivalence in Corollary 4.5.7.

Lemma 4.5.6. *Let $G \leq \text{Sym}(\Omega)$ be a closed permutation group. Let $\pi: G \rightarrow \mathcal{U}(\mathcal{H})$ be a dissociated unitary representation. Then for every $A \subseteq \Omega$ finite, we have $\mathcal{T}_A = \mathcal{H}_A$.*

Proof. Let $A \subseteq \Omega$ be a finite subset. Let (C_n) be a decreasing sequence of cofinite subsets of Ω such that $\bigcap_{n \in \mathbb{N}} C_n = A$. Then $\bigcap_{n \in \mathbb{N}} \mathcal{H}_{C_n} = \mathcal{T}_A$ and $p_{C_n} \rightarrow p_{\mathcal{T}_A}$ in the strong operator topology.

Define $B_n := A \cup (\Omega \setminus C_n)$ for every $n \in \mathbb{N}$. Then (B_n) is an increasing sequence of finite sets whose union is Ω . By continuity of π , we get that $p_{B_n} \rightarrow \text{id}$ in the strong operator topology. Moreover, $B_n \cap C_n = A$ for every $n \in \mathbb{N}$. Thus, using dissociation, we have $p_{C_n} p_{B_n} = p_A$ for every $n \in \mathbb{N}$. Note that $p_{C_n} p_{B_n} \rightarrow p_{\mathcal{T}_A}$ in the strong operator topology (these operators being projections, they are uniformly bounded). At the limit, we get $p_{\mathcal{T}_A} = p_A$ and $\mathcal{T}_A = \mathcal{H}_A$. $\square$

Corollary 4.5.7. *Let $G \leq \text{Sym}(\Omega)$ be a closed permutation group without algebraicity. Let $\pi: G \rightarrow \mathcal{U}(\mathcal{H})$ be a unitary representation. Then π is dissociated if and only if for every finite subset $A \subseteq \Omega$, we have $\mathcal{H}_A = \mathcal{T}_A$.*

Remark 4.5.8. Every result obtained in this section has an analogue in the context of Boolean p.m.p. actions. Let G be a closed subgroup of $\text{Sym}(\Omega)$ and $G \rightarrow \text{Aut}(X, \mu)$ be a Boolean p.m.p. action. For every finite subset $A \subseteq \Omega$, define the following σ -algebra

$$\mathcal{T}_A := \bigcap_{C \subseteq \Omega \text{ cofinite}} \sigma(\mathcal{F}_{A \cup B} : B \subseteq C \text{ finite}).$$

When A is the empty set, we call $\mathcal{T}_\emptyset$ the tail σ -algebra of the action. One checks that the proofs of this section can be adapted in a straightforward way to get analogous results in that context, leading to the following result: *if G has no algebraicity, then the Boolean p.m.p. action α is dissociated if and only if for every finite subset $A \subseteq \Omega$, we have $\mathcal{F}_A = \mathcal{T}_A$.*

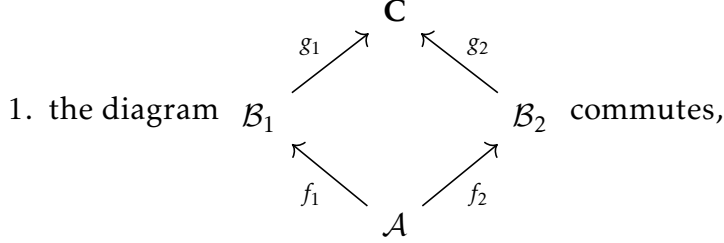
In particular, for a Boolean p.m.p. ergodic, dissociated action, the tail σ -algebra is trivial. Notice also that for $\text{Sym}(\Omega)$ acting on a product probability space $(Z, \zeta)^\Omega$, triviality of the tail σ -algebra and of the σ -algebra of invariant subsets (a.k.a. exchangeable subsets in that context) correspond respectively to Kolmogorov and Hewitt-Savage 0 – 1 laws.

Strong cofinite amalgamation over countable subsets

Recall that, in the context of Fraïssé limits, the non-algebraicity condition we have been working with throughout this paper is equivalent to the Fraïssé class having the *strong amalgamation property* (see Section 2.7 of [Cam90]). We now introduce a reinforcement of this property by allowing infinite amalgamation bases and show it implies dissociation of the limit.

Let $\mathfrak{M}$ be a countable relation structure and let $\sigma\text{Age}(\mathfrak{M})$ be the set of isomorphism classes of countable (that is, finite or infinite) substructures of $\mathfrak{M}$.

Definition 4.5.9. We say that $\mathfrak{M}$ has the *strong cofinite amalgamation property over countable subsets* (abbrev. σ -SAP) if, whenever $\mathcal{A}, \mathcal{B}_1, \mathcal{B}_2 \in \sigma\text{Age}(\mathfrak{M})$ and $f_i: \mathcal{A} \rightarrow \mathcal{B}_i$ are embeddings with $|\text{dom}(\mathcal{B}_i) \setminus \text{dom}(f_i(\mathcal{A}))| < +\infty$, there exist $\mathcal{C} \in \sigma\text{Age}(\mathfrak{M})$ and embeddings $g_i: \mathcal{B}_i \rightarrow \mathcal{C}$ so that the following hold:



2. for every $b_1 \in \text{dom}(\mathcal{B}_1)$ and $b_2 \in \text{dom}(\mathcal{B}_2)$ satisfying $g_1(b_1) = g_2(b_2)$, there exists $a \in \text{dom}(\mathcal{A})$ satisfying $f_1(a) = b_1$ and $f_2(a) = b_2$.

Remark 4.5.10. For *integral* metric spaces, there is a canonical amalgamation over a common non-empty subspace (of any cardinality!). Therefore, several natural Fraïssé limits built out of classes of finite integral metric spaces satisfy σ -SAP. This is for instance the case for the integral Urysohn space $\mathbb{Z}\mathbf{U}$, but other examples will be discussed in Sections 4.6.2 and 4.6.2.

However, it turns out that the rational Urysohn space $\mathbb{Q}\mathbf{U}$ does *not* satisfy σ -SAP. Indeed, there exists an infinite subset $A \subseteq \text{dom}(\mathbb{Q}\mathbf{U})$ and a point $x \in \text{dom}(\mathbb{Q}\mathbf{U}) \setminus A$ such that $d(A, x) = 0$. Let $\mathcal{A}$ be the structure generated by A and let $\mathcal{B}_1 = \mathcal{B}_2$ be the structure generated by $A \sqcup \{x\}$. Then any amalgamation $\mathcal{C}$ would identify $\mathcal{B}_1 \setminus \mathcal{A}$ and $\mathcal{B}_2 \setminus \mathcal{A}$.

Remark 4.5.11. σ -SAP is closely related to *uniform non-algebraicity* introduced by Tsankov in [Tsa25] to obtain de Finetti's style results.

Recall that a structure $\mathfrak{M}$ is *ultrahomogeneous* if every isomorphism between finite substructures of $\mathfrak{M}$ extends to an automorphism of $\mathfrak{M}$.

Proposition 4.5.12. *Let $\mathfrak{M}$ be a countable relational ultrahomogeneous structure. Assume that $\mathfrak{M}$ satisfies σ -SAP and that $\text{Aut}(\mathfrak{M})$ weakly eliminates imaginaries. Then for every unitary representation $\pi: \text{Aut}(\mathfrak{M}) \rightarrow \mathcal{U}(\mathcal{H})$ and every finite subset $A \subseteq \text{dom}(\mathfrak{M})$, the subspaces $\mathcal{H}_A$ and $\mathcal{T}_A$ coincide.*

Proof. Let $G := \text{Aut}(\mathfrak{M})$ and $\Omega := \text{dom}(\mathfrak{M})$. Fix a unitary representation $\pi: G \rightarrow \mathcal{U}(\mathcal{H})$ and a finite subset $A \subseteq \Omega$. The inclusion $\mathcal{H}_A \subseteq \mathcal{T}_A$ always holds, so let us prove the reverse inclusion. We start with the following claim.

Claim 1. $\mathcal{T}_A = \overline{\bigcup_{\substack{B \subseteq \Omega \\ \text{finite}}} \mathcal{T}_A \cap \mathcal{H}_B}.$

Proof of the claim. Recall that the union of the $\mathcal{H}_B$'s for $B \subseteq \Omega$ finite is dense in $\mathcal{H}$ [BJJ24, Lem. 2.6]. Let $\xi \in \mathcal{T}_A$. Then there exists a sequence of finite subsets $(B_n)_{n \geq 0}$ and a sequence of vectors $(\xi_n)_{n \geq 0}$ such that $\xi_n \in \mathcal{H}_{B_n}$ and $\xi_n \rightarrow \xi$. Since $\mathcal{H}_{B_n} \subseteq \mathcal{H}_{A \cup B_n}$, we may assume that $A \subseteq B_n$ for every $n \geq 0$. Define $\eta_n := p_{\mathcal{T}_A} \xi_n$. Clearly, $\eta_n \in \mathcal{T}_A$ and $\eta_n \rightarrow \xi$. So it suffices to prove that $\eta_n \in \mathcal{H}_{B_n}$. But for $g \in G_{B_n}$, one has $\pi(g)\mathcal{T}_A = \mathcal{T}_A$ (because $G_{B_n} \subseteq G_A$) and therefore

$$\pi(g)\eta_n = p_{\pi(g)\mathcal{T}_A} \pi(g)\xi_n = p_{\mathcal{T}_A} \xi_n = \eta_n.$$

□_{claim}

Claim 2. *For every finite subset $B \subseteq \Omega$, we have $\mathcal{T}_A \cap \mathcal{H}_B \subseteq \mathcal{H}_A$.*

Proof of the claim. Without loss of generality, one may assume that $A \subseteq B$. Define $B_1 := B \setminus A$. Let $\xi \in \mathcal{T}_A \cap \mathcal{H}_B$. Let C be a cofinite subset disjoint from B . Since ξ belongs to $\mathcal{T}_A \subseteq \mathcal{H}_C$, there exists an increasing sequence $(C_n)_{n \geq 0}$ of finite subsets of C disjoint from A and vectors $\xi_n \in \mathcal{H}_{A \sqcup C_n}$ such that $\xi_n \rightarrow \xi$. Up to enlarging the C_n 's, one may assume that $\bigcup_{n \geq 0} C_n = C$. Since $\mathfrak{M}$ satisfies σ -SAP, there exists an amalgamation of $C \cup B$ with itself over $C \sqcup A$. By ultrahomogeneity, one readily gets the following.

Fact 4.5.13. *There exists B_2 finite in Ω and an increasing family $(C'_n)_{n \in \mathbb{N}}$ of finite subsets of Ω such that:*

- *there is $g_n \in G_{A \sqcup B_1}$ sending C_n to C'_n ,*
- *there is $h_n \in G_{A \sqcup C'_n}$ sending B_1 to B_2 .*

Let us prove that $\xi \in \mathcal{H}_{A \sqcup B_2}$. We write

$$\begin{aligned} \|\xi - h_n \xi\| &\leq \|\xi - h_n g_n \xi_n\| + \|h_n g_n \xi_n - h_n \xi\| \\ &\leq 2\|\xi - g_n \xi_n\| \\ &= 2\|\xi - \xi_n\| \rightarrow 0 \end{aligned}$$

Thus, since for every n , $h_n \xi$ belongs to $\mathcal{H}_{A \sqcup B_2}$, which is closed, we get that ξ itself belongs to $\mathcal{H}_{A \sqcup B_2}$. Finally, ξ belongs to $\mathcal{H}_{A \sqcup B_1} \cap \mathcal{H}_{A \sqcup B_2} = \mathcal{H}^{\langle G_{A \sqcup B_1}, G_{A \sqcup B_2} \rangle} = \mathcal{H}_A$ and this finishes the proof of the claim. $\square_{\text{claim}}$

Combining Claim 1 and 2, we get that $\mathcal{T}_A \subseteq \mathcal{H}_A$, which finishes the proof. $\square$

As a consequence of the results in this section, we obtain the following theorem.

Theorem 4.5.14. *Let $\mathfrak{M}$ be a countable ultrahomogeneous relational structure. If $\mathfrak{M}$ weakly eliminates imaginaries and satisfies the strong cofinite amalgamation property over countable subsets, then $\text{Aut}(\mathfrak{M})$ is dissociated.*

$\aleph_0$ -categoricity and σ -SAP

We now turn to a first application of Theorem 4.5.14 to $\aleph_0$ -categorical ultrahomogeneous structures in order to recover dissociation for oligomorphic groups with weak elimination of imaginaries and no algebraicity (Theorem 3.4 of [JT22]).

Let $\mathfrak{M}$ be a countable relational structure. We say that two tuples $\bar{x} = (x_1, \dots, x_n)$ and $\bar{y} = (y_1, \dots, y_n)$ on $\mathfrak{M}$ have the same *isomorphism type* if the map $x_i \mapsto y_i$ extends to an isomorphism between the substructures induced by $\bar{x}$ and $\bar{y}$. This defines an equivalence relation on tuples and we denote by $\text{tp}(\bar{x})$ the equivalence class of $\bar{x}$. A *k-type* over $\mathfrak{M}$ is simply the type of a k -tuple in $\mathfrak{M}$. The *tree of isomorphism types* of $\mathfrak{M}$ is a rooted tree defined as follows:

- its vertex set is the collection of all k -types over $\mathfrak{M}$, for every $k \geq 0$,
- for every $k \geq 0$, there is an edge between a $(k+1)$ -type t and a k -type s if and only if there exists $(x_1, \dots, x_{k+1}) \in \mathfrak{M}^{k+1}$ such that $t = \text{tp}(x_1, \dots, x_{k+1})$ and $s = \text{tp}(x_1, \dots, x_k)$.

The following result is a direct reformulation of Ryll-Nardzewski's theorem.

Lemma 4.5.15. *The boundary of the tree of isomorphism types of $\mathfrak{M}$ is compact if and only if $\mathfrak{M}$ is $\aleph_0$ -categorical.*

We now show that $\aleph_0$ -categorical Fraïssé limits satisfy σ -SAP and even more: the cofiniteness assumption is not even necessary in this case.

Lemma 4.5.16. *Let $\mathfrak{M}$ be a countable relational ultrahomogeneous structure which has the strong amalgamation property. If $\mathfrak{M}$ is $\aleph_0$ -categorical, then $\mathfrak{M}$ has the strong amalgamation property over arbitrary subsets.*

Proof. Let $\mathcal{A}, \mathcal{B}_1, \mathcal{B}_2 \in \sigma\text{Age}(\mathfrak{M})$ and embeddings $f_i: \mathcal{A} \hookrightarrow \mathcal{B}_i$ satisfying $|\text{dom}(\mathcal{B}_i) \setminus \text{dom}(f_i(\mathcal{A}))| < +\infty$. If $\mathcal{A}$ is finite, there is nothing to prove since $\mathfrak{M}$ has the strong amalgamation property. So let us assume that $\mathcal{A}$ is infinite. Fix an enumeration $a_1, \dots, a_n, \dots$ of $\text{dom}(\mathcal{A})$ and let $\mathcal{A}_n$ be the structure generated by $\{a_1, \dots, a_n\}$. Let $\bar{x} = (x_1, \dots, x_k)$ be an enumeration of $\text{dom}(\mathcal{B}_1) \setminus \text{dom}(f_1(\mathcal{A}))$ and $\bar{y} = (y_1, \dots, y_l)$ be an enumeration of $\text{dom}(\mathcal{B}_2) \setminus \text{dom}(f_2(\mathcal{A}))$. Let $\mathbf{C}_n$ be an amalgamation of $\mathcal{B}_1 \setminus f_1(\mathcal{A} \setminus \mathcal{A}_n)$ and $\mathcal{B}_2 \setminus f_2(\mathcal{A} \setminus \mathcal{A}_n)$ over $\mathcal{A}_n$ as in Definition 4.5.9. We identify the domain of $\mathbf{C}_n$ with the union of the domains of $\mathcal{B}_1 \setminus f_1(\mathcal{A} \setminus \mathcal{A}_n)$ and $\mathcal{B}_2 \setminus f_2(\mathcal{A} \setminus \mathcal{A}_n)$, and call t_n the type induced by $\mathbf{C}_n$ on $(\bar{x}, \bar{y}, \mathcal{A}_n)$. Let ξ_n be any element of the boundary of the tree of types extending t_n . By compactness (Lemma 4.5.15), (ξ_n) admits a subsequence converging to some point ξ . Any infinite tuple $C = (c_1, c_2, \dots) \in \mathfrak{M}^{\mathbb{N}}$ satisfying $(\text{tp}(c_1, \dots, c_n))_{n \geq 0} = \xi$ is as wanted. Existence of such a tuple is ensured by ultrahomogeneity of $\mathfrak{M}$. $\square$

As a consequence, we recover the result [JT22, Thm. 3.4] of the second author and Tsankov, whose original proof uses the classification of unitary representations for oligomorphic groups due to Tsankov [Tsa12]. Recall that a closed permutation group $G \leq \text{Sym}(\Omega)$ is *oligomorphic* if for every $n \in \mathbb{N}$, the diagonal action $G \curvearrowright \Omega^n$ has only finitely many orbits.

Theorem 4.5.17. *Let $G \leq \text{Sym}(\Omega)$ be a closed subgroup. Assume that G is oligomorphic, has no algebraicity, and weakly eliminates imaginaries. Then G is dissociated.*

Proof. Let $\mathfrak{M}_G$ be the canonical structure associated with G (for a definition, see Section 2.3 of [Cam90]). It is a countable relational ultrahomogeneous structure, which satisfies the strong amalgamation property. Moreover, by the Ryll-Nardzewski Theorem, $\mathfrak{M}_G$ is $\aleph_0$ -categorical (see [Cam90, Th. 2.10]). Thus by Lemma 4.5.16, $\mathfrak{M}_G$ satisfies σ -SAP. Finally, Theorem 4.5.14 allows us to conclude that $\text{Aut}(\mathfrak{M}_G) = G$ is dissociated. $\square$

4.6 New examples of dissociated permutation groups

4.6.1 Weak elimination of imaginaries from Ramsey theorems

This short subsection is here to provide the tools we need in the rest of the section to prove that the described structures admit weak elimination of imaginaries. Simply put, we prove that in the right context, weak elimination of imaginaries is implied by a Ramsey property.

Let $\mathcal{F}$ be a class of structures. For $A, B \in \mathcal{F}$, denote by $\binom{B}{A}$ the set of embeddings of A in B . We say that $\mathcal{F}$ has the *Ramsey Property* if for all $A, B \in \mathcal{F}$ and $k \in \mathbb{N}$, there is a $C \in \mathcal{F}$ such that for every $\chi: \binom{C}{A} \rightarrow \{1, \dots, k\}$, there exists $\phi \in \binom{C}{B}$ such that all embeddings of A with image in $\phi(B)$ have the same image under χ .

Adopting flexible notations, we will say two structures of the form $(C, x_1, \dots, x_n)$ and $(C, y_1, \dots, y_n)$ are isomorphic, and write $(C, x_1, \dots, x_n) \simeq (C, y_1, \dots, y_n)$, if the map $C \cup \{x_1, \dots, x_n\} \rightarrow C \cup \{y_1, \dots, y_n\}$ that fixes C and sends x_i to y_i is an isomorphism. We say that $\mathcal{F}$ is 2-symmetric if the following holds:

$$\forall (C, x, y) \in \mathcal{F}, [(C, x) \simeq (C, y) \Rightarrow (C, x, y) \simeq (C, y, x)].$$

For a class $\mathcal{F}$, we denote by $\mathcal{F}_<$ the class of all ordered elements of $\mathcal{F}$, that is of all expansions of elements of $\mathcal{F}$ by a linear order. If $\mathcal{F}$ is a Fraïssé class, we will denote its Fraïssé limit by $\mathbb{F}$.

Proposition 4.6.1. *Let $\mathcal{F}$ be a 2-symmetric Fraïssé class with the strong amalgamation property. If $\mathcal{F}_<$ has the Ramsey property, then $\mathbb{F}$ weakly eliminates imaginaries.*

Proof. Let us first give a characterization of weak elimination of imaginaries, adapted from [Con17, Cor. 7.6].

Fact 4.6.2. *Let $\mathcal{F}$ be a Fraïssé class and $\mathbb{F}$ its Fraïssé limit such that:*

- (i) $\mathcal{F}$ has the strong amalgamation property,
- (ii) for every finite $D \subseteq \mathbb{F}$ and every $x \in \mathbb{F}$, there is no non-trivial $\text{Aut}(\mathbb{F})_D$ -invariant equivalence relation on $\{x' \in \mathbb{F} : (D, x) \simeq (D, x')\}$.

Then $\text{Aut}(\mathbb{F})$ weakly eliminates imaginaries.

Proof of the fact. Write $G := \text{Aut}(\mathbb{F})$ and let $V \leq G$ be an open subgroup. There exists a finite subset $A \subseteq \Omega$ such that $G_A \leq V$. Write $A = \{a_1, \dots, a_n\}$ in an injective way and let $\bar{a} = (a_1, \dots, a_n)$. Let $\sim$ be the G -invariant equivalence relation on $G \cdot \bar{a}$ defined by $g \cdot \bar{a} \sim h \cdot \bar{a}$ if $g^{-1}h \in V$. Clearly, the stabilizer in G of the $\sim$ -class of $\bar{a}$ is precisely V . Proposition 5.5.3 in [Kru16] yields $I \subset \{1, \dots, n\}$ and a subgroup P of $\text{Sym}(I)$ such that for any $\bar{b} \in G \cdot \bar{a}$, we have

$$\bar{a} \sim \bar{b} \Leftrightarrow \bigvee_{f \in P} \bigwedge_{i \in I} a_i = b_{f(i)}. \quad (4.4)$$

We set $A' = \{a_i : i \in I\}$ and prove that $G_{A'}$ is a finite-index subgroup of V . Since $G_{A'}$ clearly preserves the $\sim$ -class of $\bar{a}$, it is indeed a subgroup of V . Moreover, V acts on I by (4.4) hence the existence of a morphism from V to the finite group $\text{Sym}(I)$. The kernel of this action being precisely $G_{A'}$, the quotient $V/G_{A'}$ is indeed finite. $\square$

Write $G := \text{Aut}(\mathbb{F})$. Towards a contradiction, let us assume that $\mathcal{F}$ does not satisfy (ii). In this case, there exist $(D, x) \subseteq \mathbb{F}$ finite and a non-trivial G_D -invariant equivalence relation $\sim$ on $X := \{x'' \in \mathbb{F} : (D, x) \simeq (D, x'')\}$. By non-triviality, there exist $x', y \in X \setminus \{x\}$ such that $x \sim x' \not\sim y$. By the no-algebraicity hypothesis, $G_{D, x, y} \cdot x'$ is infinite, and we can pick $z \neq z' \in G_{D, x, y} \cdot x' \setminus \{x'\}$. In particular, $(D, y, z) \simeq (D, y, z')$ and $z \sim z' \sim x$.

We now produce a counterexample to the Ramsey property for $\mathcal{F}_<$. Fixing a relation symbol $<$ for the added linear ordering, we will denote by $<^C$ its interpretation in a given structure $C \in \mathcal{F}_<$. Let A be any expansion of (D, y, z) by a linear order where $y <^A z$ and $D <^A y$. Similarly, let B any expansion of (D, z', y, z) by a linear order where $z' <^B y <^B z$ and such that, as ordered expansions, $(D, z', y) \simeq A$. Note that by 2-symmetry B contains two copies of A : (D, y, z) and (D, z', y) . For every $C \in \mathcal{F}_<$, we will produce a bad coloring of $\binom{C}{A}$.

Let us fix $C \in \mathcal{F}_<$. For every copy D_0 of D in C , we denote by $\sim_{D_0}$ the G_{D_0} -invariant equivalence relation on $X_{D_0} := \{a \in C : (D_0, a) \simeq (D, x)\}$ corresponding to $\sim$. We order X_{D_0} in such a way that the classes form intervals, and denote by $<_{D_0}$ this ordering. To color a copy $A_1 = (D_1, x_1, y_1)$ of A in C , compare the ordering of x_1 and y_1 in $<^C$ and $<_{D_1}$. Associate 0 to A_1 if they agree and 1 otherwise.

Let us consider $B_1 = (D_1, z'_1, y_1, z_1)$ a copy of B inside C . Note that since $\mathcal{F}$ is 2-symmetric, B_1 contains two copies of A : $A_1 := (D_1, y_1, z_1)$ and $A_2 := (D_1, z'_1, y_1)$. In the described coloring, those two copies of A can never have the same color. Indeed, $z_1 <_{D_1} y_1$ iff $z'_1 <_{D_1} y_1$ while $z'_1 <^C y_1 <^C z_1$. $\square$

4.6.2 Metrically homogeneous spaces

In this section, we consider some classes of countable metrically homogeneous spaces of infinite diameter for which the isometry group satisfies the assumptions of Theorem 4.5.14. A metric space (X, d) is *metrically homogeneous* if every surjective isometry between two finite subsets of X extends to a surjective isometry of X .

An *integral metric space* is a metric space (X, d) such that $d(X \times X) \subseteq \mathbb{N}$. Every countable integral metric space (X, d) may be considered as a countable relational structure in the language $\mathcal{L} = (R_n)_{n \geq 0}$ where each R_n is a relation of arity 2 whose interpretation R_n^X in X is given by

$$R_n^X := \{(x, y) \in X^2 : d(x, y) = n\}.$$

Notice that X is ultrahomogeneous (as a countable structure) if and only if X is metrically homogeneous. Notice that $\text{Isom}(X, d) = \text{Aut}(X)$. For integral metric spaces, there is a canonical amalgamation over non-empty subspaces that will be useful later.

Definition 4.6.3. Let (X_1, d_1) and (X_2, d_2) be two integral metric spaces such that the intersection $Y := X_1 \cap X_2$ is non-empty. Assume that d_1 and d_2 coincide on $Y \times Y$. Let $X := X_1 \cup X_2$ and define a map $d : X \times X \rightarrow \mathbb{N}$, which restricts to d_1 on $X_1 \times X_1$, to d_2 on $X_2 \times X_2$ and such that for all $x_1 \in X_1 \setminus Y$ and $x_2 \in X_2 \setminus Y$,

$$d(x_1, x_2) = \min\{d(x_1, y) + d(y, x_2) : y \in Y\}.$$

Then (X, d) is an integral metric space. It is called the *metric amalgam* of (X_1, d_1) and (X_2, d_2) over Y and is denoted by $(X_1, d_1) *_Y (X_2, d_2)$.

Integral metric spaces with no small triangle of odd perimeter

Let $p \geq 1$ be an integer. A metric space (X, d) has *no triangle of odd perimeter less than p* if the following holds: for every $(x, y, z) \in X^3$, if $d(x, y) + d(y, z) + d(z, x)$ is odd, then $d(x, y) + d(y, z) + d(z, x) > p$.

Lemma 4.6.4. *Let $p \geq 1$ be an integer. Let (X_1, d_1) and (X_2, d_2) be two integral metric spaces which contain no triangle of odd perimeter less than p . If $X_1 \cap X_2$ is non-empty, then the metric amalgam of (X_1, d_1) and (X_2, d_2) over $X_1 \cap X_2$ contains no triangle of odd perimeter less than p .*

Proof. Let $X := X_1 \cup X_2$ and $Y := X_1 \cap X_2$. Fix $x_1 \in X_1 \setminus Y$, $x_2 \in X_2 \setminus Y$ and $x \in X$ such that $d(x_1, x_2) + d(x_2, x) + d(x, x_1)$ is odd. There are two cases to check.

Assume that $x \in Y$. Let $y \in Y$ be such that $d(x_1, x_2) = d(x_1, y) + d(y, x_2)$. Then $d(x_1, y) + d(y, x_2) + d(x_2, x) + d(x, x_1)$ is odd. Therefore, there is $i \in \{1, 2\}$ such that $d(x_i, y) + d(y, x) + d(x, x_i)$ is odd. But this last quantity is $> p$ since $(x_i, y, x) \in X_i \times X_i \times X_i$. Therefore,

$$\begin{aligned} d(x_1, x_2) + d(x_2, x) + d(x, x_1) &= d(x_1, y) + d(y, x_2) + d(x_2, x) + d(x, x_1) \\ &\geq d(x_i, y) + d(y, x) + d(x, x_i) \\ &> p, \end{aligned}$$

which is what we wanted.

Assume now that $x \in X \setminus Y$. Without loss of generality, we may assume that $x \in X_1 \setminus Y$. Let $y, z \in Y$ be such that $d(x_1, x_2) = d(x_1, y) + d(y, x_2)$ and $d(x_2, x) = d(x_2, z) + d(z, x)$. Then $d(x_1, y) + d(y, x_2) + d(x_2, z) + d(z, x) + d(x, x_1)$ is odd. Therefore, among the following three quantities

$$\begin{aligned} c_1 &:= d(x, x_1) + d(x_1, y) + d(y, x), \\ c_2 &:= d(x, y) + d(y, z) + d(z, x), \\ c_3 &:= d(z, y) + d(y, x_2) + d(x_2, z), \end{aligned}$$

at least one of them is odd. But if c_i is odd, then $c_i > p$. By the triangle inequality, we obtain that

$$d(x_1, x_2) + d(x_2, x) + d(x, x_1) \geq \max(c_1, c_2, c_3) > p,$$

which concludes the proof. $\square$

Lemma 4.6.5. *Let $p \geq 1$ be an integer. Then the class of finite integral metric spaces with no triangle of odd perimeter less than p is a Fraïssé class satisfying the strong amalgamation property.*

Proof. It is clear that this class satisfies the hereditary property. The joint embedding property will follow from the amalgamation property since the language is relational and there is no constant relation. Let us show the strong amalgamation property. For this, fix three finite integral metric spaces (A, d_A) , (B_1, d_1) and (B_2, d_2) with no triangle of odd perimeter less than p . Assume that (B_1, d_1) and (B_2, d_2) contain an isometric copy of A . Let C be the set obtained by identifying both copies of A in the disjoint union $B_1 \sqcup B_2$. We define a new metric d on C as follows. There are two cases to check.

If A is non-empty, then let d be the metric amalgam of (B_1, d_1) and (B_2, d_2) over (A, d_A) . Then (C, d) is a finite integral metric space with no triangle of odd perimeter less than p by Lemma 4.6.4.

If A is empty, then let d be the metric on C which restricts to d_i on $X_i \times X_i$ and such that for all $x_1 \in B_1$, $x_2 \in B_2$,

$$d(x_1, x_2) = \max(p, \text{diam}(B_1, d_1), \text{diam}(B_2, d_2)).$$

Then (C, d) is an integral metric space, with no triangle of odd perimeter less than p .

In both cases, we have found a strong amalgam of (B_1, d_1) and (B_2, d_2) over (A, d_A) , which concludes the proof. $\square$

In the sequel, given an integer $p \geq 1$, we denote by $\mathfrak{M}_p$ the Fraïssé limit of the class of finite integral metric spaces with no triangle of odd perimeter less than p . Notice that when $p = 1$, $\mathfrak{M}_p$ is the integral Urysohn space $\mathbb{Z}\mathbf{U}$.

Lemma 4.6.6. $\mathfrak{M}_p$ satisfies σ -SAP.

Proof. Since the Fraïssé class of finite integral metric spaces with no triangle of odd perimeter less than p satisfies the strong amalgamation property, it remains to prove the existence of a strong cofinite amalgam over a countable metric space. The existence of such an amalgam is provided by Lemma 4.6.4. $\square$

We postpone the proof of the last ingredient to Section 4.6.1.

Lemma 4.6.7. $\text{Iso}(\mathfrak{M}_p)$ weakly eliminates imaginaries.

Proof. The class $\mathcal{F}$ of finite integral metric spaces with no triangle of odd perimeter less than p is clearly 2-symmetric and has the strong amalgamation property by Lemma 4.6.12. Moreover, $\mathcal{F}_<$ has the Ramsey property by [ABH⁺26, Thm. 8.2]. So Proposition 4.6.1 yields the desired conclusion. $\square$

As a consequence of the results obtained in this section, Theorem 4.5.14 applies.

Theorem 4.6.8. $\text{Iso}(\mathfrak{M}_p)$ is dissociated.

We close this section by discussing the topological and geometric nature of $\text{Iso}(\mathfrak{M}_p)$.

Proposition 4.6.9. $\text{Iso}(\mathfrak{M}_p)$ is locally Roelcke-precompact but not coarsely bounded.

Proof. Let X be the distance-1 graph associated with $\mathfrak{M}_p$. It is a straightforward consequence of homogeneity and universality that $\text{Aut}(X) = \text{Iso}(\mathfrak{M}_p)$. Thus, [Ros22, Th. 3.5] applies and $\text{Iso}(M, d)$ is locally Roelcke-precompact. However, it is not bounded because it acts transitively by isometries on the unbounded space $\mathfrak{M}_p$. $\square$

Finally, we can show that this group satisfies the Howe-Moore property.

Proposition 4.6.10. $\text{Iso}(\mathfrak{M}_p)$ has the Howe-Moore property with respect to coarsely bounded sets.

Proof. The proof of [Ros22, Thm. 3.5] shows that for every $a \in \text{dom}(\mathfrak{M}_p)$, the point stabilizer $\text{Iso}(\mathfrak{M}_p)_a$ is Roelcke-precompact and therefore coarsely bounded. Since $\text{Iso}(\mathfrak{M}_p)$ is dissociated, Theorem 4.3.11 implies that $\text{Iso}(\mathfrak{M}_p)$ has the Howe-Moore property with respect to coarsely bounded sets. $\square$

Integral metric spaces with no unit simplex

Let $r \geq 3$ be an integer. A unit $(r - 1)$ simplex in an integral metric space (X, d) is a subset of X consisting of r points that lie mutually at distance 1.

Lemma 4.6.11. *Let $r \geq 3$ be an integer. Let (X_1, d_1) and (X_2, d_2) be two integral metric spaces that contain no unit $(r - 1)$ -simplex. If $X_1 \cap X_2$ is non-empty, then the metric amalgam of (X_1, d_1) and (X_2, d_2) over $X_1 \cap X_2$ contains no unit $(r - 1)$ -simplex.*

Proof. Let $Y := X_1 \cap X_2$ and let $X := (X_1, d_1) *_Y (X_2, d_2)$ be the metric amalgam. Let $F \subseteq X$ be a subset consisting of r points. If $F \subseteq X_1$ or $F \subseteq X_2$, then F is not a unit $(r - 1)$ simplex by assumption. Else, there exist two points $x_1, x_2 \in F$ with $x_1 \in X_1 \setminus Y$ and $x_2 \in X_2 \setminus Y$. Let $y \in Y$ be such that $d(x_1, x_2) = d(x_1, y) + d(y, x_2)$. Then $d(x_1, x_2) > 1$ and thus F is not a unit $(r - 1)$ simplex. $\square$

Lemma 4.6.12. *Let $r \geq 3$ be an integer. Then the class of finite integral metric spaces with no unit $(r - 1)$ -simplex is a Fraïssé class satisfying the strong amalgamation property.*

Proof. The proof is identical to that of Lemma 4.6.5. $\square$

In the sequel, given an integer $r \geq 3$, we denote by $\mathbf{N}_r$ the Fraïssé limit of the class of finite integral metric spaces with no unit $(r - 1)$ -simplex.

Lemma 4.6.13. $\mathbf{N}_r$ satisfies σ -SAP.

Proof. As in Lemma 4.6.6, the proof follows from the fact that the metric amalgam over a non-empty subset preserves the property of having no unit $(r - 1)$ -simplex, see Lemma 4.6.11. $\square$

Again, the last ingredient will be provided by the results in Section 4.6.1.

Lemma 4.6.14. $\text{Iso}(\mathbf{N}_r)$ weakly eliminates imaginaries.

Proof. Again, the class $\mathcal{F}$ of finite integral metric spaces with no unit $(r - 1)$ -simplex is 2-symmetric has the strong amalgamation property by Lemma 4.6.12. Moreover, $\mathcal{F}_<$ has the Ramsey property by [ABH⁺26, Thm. 8.2]. So Proposition 4.6.1 yields the desired conclusion. $\square$

Therefore, Theorem 4.5.14 applies.

Theorem 4.6.15. $\text{Iso}(\mathbf{N}_r)$ is dissociated.

As in the previous section, we obtain the following two results. Since the proofs are identical, we omit them.

Proposition 4.6.16. $\text{Iso}(\mathbf{N}_r)$ is locally Roelcke-precompact but not coarsely bounded.

Proposition 4.6.17. $\text{Iso}(\mathbf{N}_r)$ has the Howe-Moore property with respect to coarsely bounded sets.

4.6.3 Diversities

We now consider a generalization of the concept of metric space. Given a set X , we denote by $\mathcal{P}_{\text{fin}}(X)$ the set of finite subsets of X . A *diversity* is a couple (X, δ) where X is a set and $\delta: \mathcal{P}_{\text{fin}}(X) \rightarrow \mathbb{R}_+$ is a map satisfying:

- $\delta(A) = 0$ if and only if $|A| \leq 1$,
- if $B \neq \emptyset$, then $\delta(A \cup C) \leq \delta(A \cup B) + \delta(B \cup C)$.

We call the map δ a *diversity map*. Note that a metric space always defines a diversity, by seeing the diameter as a diversity map. Conversely, a diversity also defines a metric space but contains more information.

The above axioms imply that a diversity map δ is monotonous: if $A \subseteq B$, then $\delta(A) \leq \delta(B)$. Moreover, it is sublinear on sets with non-empty intersection: if $A \cap B \neq \emptyset$, then $\delta(A \cup B) \leq \delta(A) + \delta(B)$. We say that a diversity is integral (resp. rational) if the diversity map takes only integral (resp. rational) values. In this section, we will discuss diversities constructed by means of a Fraïssé limit. We first focus on diversities with integral values.

Lemma 4.6.18. *The class of finite diversities whose diversity map takes only integral values is a Fraïssé class, the limit of which we denote by $\mathbb{Z}d$.*

We refer to [Hal20, Prop. 3.10] for a proof of this lemma. The only non-obvious part is the amalgamation property. As for integral metric spaces, it turns out that there is a natural way of amalgamating integral diversities (of any cardinality). Let X be a set. A collection $E_1, \dots, E_n$ of finite subsets of X is *connected* if the intersection graph associated with the E_i 's is connected.

Definition 4.6.19. Let (X_1, δ_1) and (X_2, δ_2) be two integral diversities such that the intersection $Y := X_1 \cap X_2$ is non-empty. Assume that d_1 and d_2 coincide on $\mathcal{P}_{\text{fin}}(Y)$. Let $X = X_1 \cup X_2$ and define a map $\delta: \mathcal{P}_{\text{fin}}(X) \rightarrow \mathbb{N}$ as follows:

$$\delta(A) := \min \left\{ \sum_{i=1}^n \delta_{k_i}(E_i) : E_1, \dots, E_n \text{ is connected, } A \subseteq \bigcup_{i=1}^n E_i \text{ and } E_i \subseteq X_{k_i} \right\}.$$

Then (X, δ) is an integral diversity such that the restriction of δ to $\mathcal{P}_{\text{fin}}(X_i) \subseteq \mathcal{P}_{\text{fin}}(X)$ coincides with δ_i for every $i \in \{1, 2\}$. It is called the *diversity amalgam* of (X_1, δ_1) and (X_2, δ_2) over Y .

As before, we use this amalgamation to obtain σ -SAP.

Lemma 4.6.20. *$\mathbb{Z}d$ satisfies σ -SAP.*

Proof. This is a consequence of the existence of the amalgam in Definition 4.6.19 □

Remark 4.6.21. If $\mathbb{Q}d$ denotes the Fraïssé limit of finite diversities whose diversity map takes only rational values, then $\mathbb{Q}d$ does *not* satisfy σ -SAP. The reason is the same as the one discussed in Remark 4.5.10 for the rational Urysohn space $\mathbb{Q}U$.

Again, weak elimination of imaginaries is technical, but the results of Section 4.6.1 allow us to prove that it holds for $\mathbb{Z}d$ and $\mathbb{Q}d$.

Lemma 4.6.22. *$\text{Aut}(\mathbb{Z}d)$ and $\text{Aut}(\mathbb{Q}d)$ weakly eliminate imaginaries.*

Proof. Apply Proposition 4.6.1, the hypotheses of which are satisfied by Example 7.9 and Theorem 6.15 of [HK26]. $\square$

Therefore, Theorem 4.5.14 applies to $\mathbb{Z}d$.

Theorem 4.6.23. *$\text{Aut}(\mathbb{Z}d)$ is dissociated.*

Recall that we denote by $\mathbb{Q}d$ the Fraïssé limit of finite diversities whose diversity map only takes rational values. Even though $\mathbb{Q}d$ does not satisfy σ -SAP, we can still obtain dissociation via approximating sequences. Indeed, by using the analogue of Katětov functions for diversities developed in [BNT17], one can build an approximating sequence for $\mathbb{Q}d$ consisting of dissociated groups and therefore obtain the following result by Theorem 4.5.1.

Theorem 4.6.24. *$\text{Aut}(\mathbb{Q}d)$ is dissociated.*

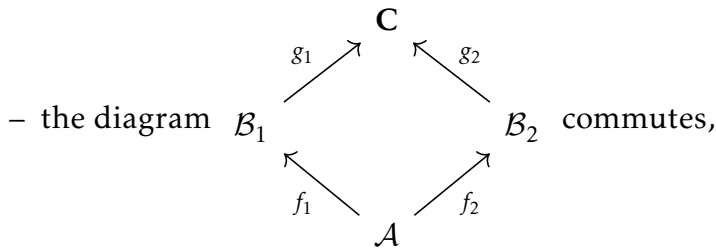
Proposition 4.6.25. *$\text{Aut}(\mathbb{Z}d)$ and $\text{Aut}(\mathbb{Q}d)$ have the Howe-Moore property with respect to coarsely bounded sets.*

Proof. Let G be either $\text{Aut}(\mathbb{Z}d)$ or $\text{Aut}(\mathbb{Q}d)$. Then G is dissociated by Theorems 4.6.23 and 4.6.24. We leave to the cautious reader the proof of the fact that the amalgamation of finite diversities (the above definition applies to *finite* diversities, even if the diversity map takes non-integer values) is a functorial amalgamation in the sense of Rosendal [Ros22, Def. 6.30]. Therefore, by Theorem 6.31 of [Ros22], G_A is coarsely bounded for every non-empty finite subset A of either $\mathbb{Z}d$ or $\mathbb{Q}d$. Finally, Theorem 4.3.11 allows us to conclude that G has the Howe-Moore property with respect to coarsely bounded sets. $\square$

4.6.4 Free amalgamation property

The aim of this section is to prove that the automorphism group of a Fraïssé limit with the free amalgamation property is dissociated, by showing that such structures have σ -SAP.

Definition 4.6.26. A Fraïssé limit $\mathbf{M}$ in a relational language $\mathcal{L}$ satisfies the free amalgamation property (FAP) if for all $\mathcal{A}, \mathcal{B}_1, \mathcal{B}_2 \in \text{Age}(\mathfrak{M})$ and all embeddings $f_i: \mathcal{A} \hookrightarrow \mathcal{B}_i$, there exists $\mathcal{C} \in \text{Age}(\mathfrak{M})$ and embeddings $g_i: \mathcal{B}_i \hookrightarrow \mathcal{C}$ such that



- if there exist $b_i \in \mathcal{B}_i$ satisfying $g_1(b_1) = g_2(b_2)$, then there exists $a \in \mathcal{A}$ satisfying $f_i(a) = b_i$,

- for every relation $R \in \mathcal{L}$ and every tuple $\bar{c}$ of elements in $\mathbf{C}$ such that $R^{\mathbf{C}}(\bar{c})$ holds, then $\bar{c}$ belongs either to $g_1(\mathcal{B}_1)$ or to $g_2(\mathcal{B}_2)$.

We refer to [MT11, Example 2.3] for some examples of Fraïssé limits with the free amalgamation property (FAP). Since all of them are $\aleph_0$ -categorical (and therefore Roelcke-precompact), we give below an example of a Fraïssé limit satisfying (FAP) whose automorphism group is not Roelcke-precompact.

Example 4.6.27. Given a graph Γ , denote by $E(\Gamma)$ the set of its edges. Suppose Γ is endowed with a map $c: E(\Gamma) \rightarrow \mathbb{N}$, thought of as a coloring of its edges. It can then be viewed as a structure in the language $\mathcal{L} = (R_n)_{n \geq 0}$, where each R_n is a relation of arity 2 whose interpretation R_n^Γ is given by

$$R_n^\Gamma := \{(x, y) \in V(\Gamma)^2 : \{x, y\} \in E(\Gamma) \text{ and } c(\{x, y\}) = n\}.$$

Let $\mathcal{C}$ be the class of all structures (Γ, c) such that Γ is a finite graph and $c: E(\Gamma) \rightarrow \mathbb{N}$. Then $\mathcal{C}$ is a Fraïssé class with the free amalgamation property. Let $\mathfrak{M}$ be its Fraïssé limit. Let us show that $\text{Aut}(\mathfrak{M})$ is not Roelcke-precompact. Since $\mathfrak{M}$ is ultrahomogeneous and one-point substructures of $\mathfrak{M}$ are all isomorphic, then $\text{Aut}(\mathfrak{M})$ acts transitively on its domain. However, there are countably many isomorphism types of two-point substructures in $\mathfrak{M}$ which are given by the values of c . Therefore, there are infinitely many orbits for the action of $\text{Aut}(\mathfrak{M})$ on pairs of points. Thus, the action of $\text{Aut}(\mathfrak{M})$ on $\text{dom}(\mathfrak{M})$ is not oligomorphic. This shows that $\text{Aut}(\mathfrak{M})$ is not Roelcke-precompact by [Tsa12, Thm. 2.4].

However, the above example is coarsely bounded as a consequence of the following.

Lemma 4.6.28. *Let $\mathfrak{M}$ be a Fraïssé limit with the free amalgamation property. Then $\text{Aut}(\mathfrak{M})$ is coarsely bounded.*

Proof. Let $\mathcal{C}$ be the Fraïssé class whose limit is $\mathfrak{M}$. It is straightforward to check that the free amalgamation provides a functorial amalgamation of $\mathcal{C}$ over the empty structure $\emptyset$ in the sense of Rosendal [Ros22, Def. 6.30]. Therefore, $\text{Aut}(\mathfrak{M})$ is coarsely bounded by Theorem 6.31 of [Ros22]. $\square$

Lemma 4.6.29. *Let $\mathfrak{M}$ be a Fraïssé limit with the free amalgamation property. Then $\text{Aut}(\mathfrak{M})$ has no algebraicity and weakly eliminates imaginaries.*

Proof. This is the combination of Lemma 2.5 and 2.7 in [MT11] and Lemma 4.2.6. $\square$

Theorem 4.6.30. *Let $\mathfrak{M}$ be a Fraïssé limit satisfying the free amalgamation property. Then $\mathfrak{M}$ satisfies σ -SAP.*

Proof. Fix $\mathcal{A}, \mathcal{B}_1, \mathcal{B}_2 \in \sigma\text{Age}(\mathfrak{M})$ and embeddings $f_i: \mathcal{A} \hookrightarrow \mathcal{B}_i$. Consider $\mathbf{C}$ the free amalgamation of $\mathcal{B}_1$ and $\mathcal{B}_2$ over $\mathcal{A}$, i.e. the union of $\mathcal{B}_1$ and $\mathcal{B}_2$ identifying $\mathcal{A}$ and where all the relations are the ones from $\mathcal{B}_1$ and $\mathcal{B}_2$. $\mathbf{C}$ is well-defined and is in $\sigma\text{Age}(\mathfrak{M})$ as all of its finite substructures are in $\text{Age}(\mathfrak{M})$. $\square$

Theorem 4.6.31. *If $\mathfrak{M}$ is a Fraïssé limit satisfying the free amalgamation property, then $\text{Aut}(\mathfrak{M})$ is dissociated.*

Proof. Let $\mathfrak{M}$ be a Fraïssé limit satisfying the free amalgamation property. Then $\mathfrak{M}$ weakly eliminates imaginaries by Lemma 4.6.29 and satisfies σ -SAP by Theorem 4.6.30. Finally, Theorem 4.5.14 yields that $\text{Aut}(\mathfrak{M})$ is dissociated. $\square$

Chapter 5

Universal Burger-Mozes Groups

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5.1 Introduction

In 1999, Nebbia obtained a striking result by showing that, for permutation groups acting on trees, having a very transitive action on their tree is a necessary condition for having a very tame representation theory [Neb99]. In the same paper, he conjectured that the converse also holds. Nebbia's conjecture remains open in its full generality but it was given a positive answer for a particularly interesting class of groups: the *Burger-Mozes universal groups*. Given $\mathbf{T}_d$ the regular tree of finite arity $d \in \mathbb{N}_{\geq 3}$, one can color the unoriented edges of $\mathbf{T}_d$ with labels in $\{1, \dots, d\}$. Then, every pair (g, x) where g is an automorphism of $\mathbf{T}_d$ and x is a vertex of $\mathbf{T}_d$ induces a bijection of $\{1, \dots, d\}$ by keeping track of how g sends the edges around x to the edges around $g(x)$. This bijection is called the *local action* of g at x . Now, given a group $F \leq \text{Sym}(\{1, \dots, d\})$, one can consider the subgroup $\mathbf{U}(F)$ of $\text{Aut}(\mathbf{T}_d)$ consisting of all the automorphisms g of $\mathbf{T}_d$ such that for every vertex x in $\mathbf{T}_d$, the local action of g at x belongs to F . Ol'sanskiĭ classified the irreducible representations of $\mathbf{U}(F)$ when $F \curvearrowright \{1, \dots, d\}$ is 2-transitive [Ol'80].

In this paper, we will replace $\mathbf{T}_d$ with the regular tree $\mathbf{T}$ of countably infinite degree. This takes us out of the locally compact world but still leaves us in the Polish realm, that is of separable and completely metrizable groups.

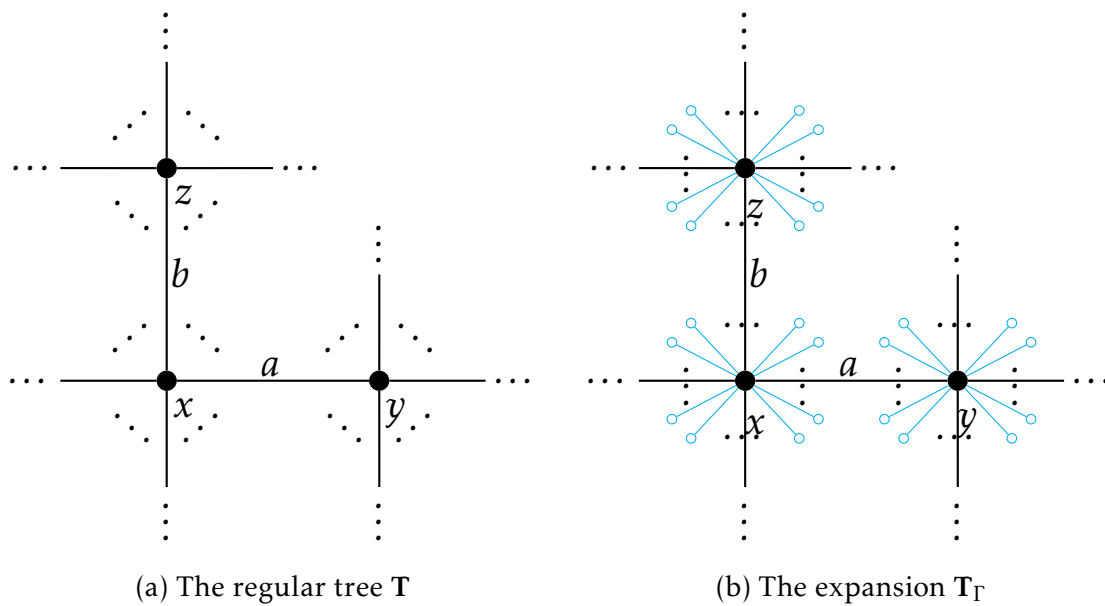


Figure 5.1: The trees $\mathbf{T}$ and $\mathbf{T}_\Gamma$

Representations of Polish groups have been studied at least since the 1970's. One of the first examples of non-locally compact groups whose representations were fully classified is the infinite symmetric group $\mathfrak{S}_\infty$ of all permutations of a countably infinite set [Lie72]. In the years that followed, Ol'sanskiĭ developed the *semigroup* method and used it to classify all the unitary representations of a variety of groups such as the isometry group of the separable (real or complex) Hilbert space, the symplectic variations of it, as well as some infinite dimensional general linear groups over finite fields (see the survey [Ol'91]). More closely related to our work, Ol'sanskiĭ also classified the irreducible representations of $\text{Aut}(\mathbf{T})$, as well as those of $\mathbf{U}(F)$ when $F \curvearrowright \{1, \dots, d\}$ is 2-transitive [Ol'80].

In Section 5.2.1, we introduce¹ the infinite arity analogue of universal Burger-Mozes groups. Indeed, one can carry out a similar construction as the one described above by replacing $\mathbf{T}_d$ with $\mathbf{T}$ and $F \curvearrowright \{1, \dots, d\}$ with $\Gamma \curvearrowright \Omega$ for some infinite closed permutation group Γ on a countable set Ω . We now fix some closed permutation group Γ on a countable set Ω and denote by $\mathbf{U}(\Gamma)$ the subgroup of $\text{Aut}(\mathbf{T})$ obtained in this manner. See Definition 5.2.1.

The goal of this paper is to generalize Ol'sanskiĭ's classification of the irreducible representations of $\text{Aut}(\mathbf{T})$ to $\mathbf{U}(\Gamma)$ under transitivity hypotheses on Γ . The core idea is that control over the representation theory of the *local action* $\Gamma \curvearrowright \Omega$ gives us control over the representation theory of the *global action* $\mathbf{U}(\Gamma) \curvearrowright \mathbf{T}$. While all finite groups are well-behaved in that regard, infinite arity allows for non-locally compact inputs to which the vast majority of the classical theory does not apply. It is thus essential to restrict one's attention to instances of the construction where the representation theory of the local action is well understood.

The class of non-locally-compact permutation groups whose representation theory is best understood is that of oligomorphic closed permutation groups. Recall that the permutation group Γ on Ω is *oligomorphic* if every $n \in \mathbb{N}$, the diagonal action $\Gamma \curvearrowright \Omega^n$ has only finitely many orbits. This can be seen as a relaxation of high transitivity and has topological consequences on Γ akin to compactness. In particular, Tsankov fully classified the unitary representations of these groups [Tsa12] in a way that reminds of the Peter-Weyl Theorem (see Theorem 5.2.28 below).

A convenient way of thinking of closed permutation groups is through model theory. Indeed, countable first-order structures come equipped with their automorphism group and every closed permutation group appears as such. Moreover, if $\mathfrak{M}$ is a countable first-order structure, the classical Ryll-Nardzewski Theorem [Hod93, Th. 7.3.1] asserts that $\text{Aut}(\mathfrak{M}) \curvearrowright \mathfrak{M}$ is oligomorphic if and only if $\mathfrak{M}$ is $\aleph_0$ -categorical. Examples of oligomorphic groups thus include $\text{Sym}(\Omega)$, the group $\text{Aut}(\mathbb{Q}, <)$ of order-preserving bijections of the rationals, or $\text{Homeo}(\{0, 1\}^{\mathbb{N}})$ seen as the automorphism group of countable atomless boolean algebra. A general method for building such structures is via Fraïssé theory. In many cases, a class of finite structures can be amalgamated into a uniquely determined universal countable object, called its *Fraïssé limit*, see e.g. [Hod93, Sec. 7.4]. For instance, the Fraïssé limit of finite orders is $(\mathbb{Q}, <)$ and the Fraïssé limit of finite graphs, called the Rado graph, is also $\aleph_0$ -categorical. See the monograph

¹Another generalization of Burger-Mozes groups in infinite arity called *box products* has already been introduced [Smi17]. Using a different formalism, Smith studied the structure and dynamics on bi-regular trees and focuses on locally compact instances of this construction.

[Cam90] or [Mac11] for detailed discussions on oligomorphic groups.

However, oligomorphicity alone will not be sufficient for our purposes. As in the finite arity case, our classification theorem will also require 2-transitivity of the local action. Fortunately, it is well understood what makes a Fraïssé limit $\aleph_0$ -categorical/2-transitive or not. Examples where both properties hold include the separable infinite dimensional projective space $\mathbb{P}^\infty(q)$ on a finite field $\mathbb{F}_q$ or the $\aleph_0$ -categorical hypergraphs from [EP02].

The study of the irreducible unitary representations of any subgroup G of $\text{Aut}(\mathbf{T})$ splits into two very distinct parts. In the classical terminology, an irreducible representation of G is called *spherical* if it admits a non-zero vector that is invariant under the action of a vertex stabilizer. Ol'sanskiĭ's classification of spherical irreducible representations of $\text{Aut}(\mathbf{T})$ extends with the same proof to every metrically 2-transitive subgroup of $\text{Aut}(\mathbf{T})$ [Ol'80, Sec. 5.3–5.5]. In particular, it applies to $\mathbf{U}(\Gamma)$ provided that $\Gamma \curvearrowright \Omega$ is 2-transitive. In fact, the spherical part of the unitary dual of $\mathbf{U}(\Gamma)$ is independent of the choice of the 2-transitive closed permutation group Γ . We still give a brief description of these representations for the interested reader that is familiar with the GNS construction. Fix $x \in \mathbf{T}$, let $c \in [-1, 1]$ and denote by d the graph distance in $\mathbf{T}$. The map $\varphi_c: \mathbf{U}(\Gamma) \rightarrow \mathbb{C}$ defined by $\varphi_c(g) = c^{d(g(x), x)}$ is positive definite. Moreover, the representation π_c obtained from φ_c by the GNS construction is irreducible and spherical. In fact, every spherical irreducible representation of $\mathbf{U}(\Gamma)$ is isomorphic to π_c for some unique $c \in [-1, 1]$. It also follows from Ol'sanskiĭ's results that the process of restricting representations of $\text{Aut}(\mathbf{T})$ to $\mathbf{U}(\Gamma)$ induces a one-to-one correspondence between the spherical irreducible representations of $\text{Aut}(\mathbf{T})$ and those of $\mathbf{U}(\Gamma)$.

In this paper, we will be studying the *aspherical* representations of $\mathbf{U}(\Gamma)$, i.e. the representations such that no vertex stabilizer fixes a non-zero vector. The main result of this paper is a full description of the aspherical representations of $\mathbf{U}(\Gamma)$, irreducible or not, provided that $\Gamma \curvearrowright \Omega$ is oligomorphic and 2-transitive. The core of it is contained in the following statement and a more precise description of the aspherical irreducible representations of $\mathbf{U}(\Gamma)$ will be given in Section 5.3, Theorem 5.3.19. Inputting $\Gamma = \text{Sym}(\Omega)$ yields $\mathbf{U}(\Gamma) = \text{Aut}(\mathbf{T})$ hence we recover Ol'sanskiĭ's classification as a special case. See Subsection 5.4.3 for reminders about commensurators.

Theorem 5.1.1. *Let Γ be an oligomorphic and 2-transitive closed permutation group on a countable set Ω .*

1. *Every spherical irreducible representation of $\mathbf{U}(\Gamma)$ is the restriction of a uniquely determined spherical irreducible representation of $\text{Aut}(\mathbf{T})$.*
2. *Every aspherical unitary representation of $\mathbf{U}(\Gamma)$ splits as a sum of aspherical irreducible subrepresentations.*
3. *Every aspherical irreducible representation of $\mathbf{U}(\Gamma)$ is of the form $\text{Ind}_{C(V)}^{\mathbf{U}(\Gamma)}(\sigma)$ where $V \leq \mathbf{U}(\Gamma)$ is an open subgroup of $\mathbf{U}(\Gamma)$, $C(V)$ is its commensurator, $V \trianglelefteq C(V)$ with finite index, and σ is an irreducible representation of the finite group $C(V)/V$.*

This theorem is very similar to Tsankov's [Tsa12, Th. 4.2]. However, the representation theory of $\mathbf{U}(\Gamma)$ significantly differs from that of oligomorphic groups

because of the spherical case. Indeed, oligomorphic groups only have countably many irreducible representations and all their representations split as sums of irreducibles. The same holds for $\mathbf{U}(\Gamma)$ if one forbids sphericity but both parts fail in general. Recall also that oligomorphic groups have Property (T) [ET16, Th. 1.1] whereas $\mathbf{U}(\Gamma)$ does not.

The notion of *Type I* groups is an important dividing line in classical representation theory, see the end of Subsection 5.3.3 for a definition. It aims at gauging the complexity of the representation theory of the group. In the line of Nebbia's conjecture, we deduce from our classification that the representation theory of $\mathbf{U}(\Gamma)$ is tame. The proof closely follows that of analogue result in finite arity [Ol'80, Th. 5.7], see Theorem 5.3.21 below.

Theorem 5.1.2. *Let Γ be an oligomorphic and 2-transitive closed permutation group on a countable set Ω . Then $\mathbf{U}(\Gamma)$ is of Type I.*

The author is presently not aware of a process that allows decomposition of arbitrary representations of $\mathbf{U}(\Gamma)$ into direct integrals of irreducibles.

Remark 5.1.3. Recall that a permutation group Γ on a countable set Ω is *dissociated* in the sense of Chapter 4 if for every representation $\Gamma \curvearrowright \mathcal{H}$ and all finite subsets $A, B \subseteq \Omega$, we have

$$\mathcal{H}_A \perp_{\mathcal{H}_{A \cap B}} \mathcal{H}_B,$$

i.e. $p_A p_B = p_{A \cap B}$ where p_A is the orthogonal onto $\mathcal{H}_A$ (and similarly for B and $A \cap B$). The previous two results also apply if one replaces "oligomorphic" with "dissociated". Indeed, this is precisely the assumptions one needs to carry out the proofs in [Ol'80, Sec. 5] without requiring the additional techniques developed in the present paper.

In a separate discussion, we also obtain the following geometrical result in Section 5.4. It illustrates the fact that the large scale geometry of these group is well captured by its representation theory, after passing to the following index-2 simple subgroup:

$$\mathbf{U}(\Gamma)^+ := \{g \in \mathbf{U}(\Gamma) : \forall x \in \mathbf{T}, d(g(x), x) \text{ is even}\}.$$

Theorem 5.1.4. *Let Γ be an oligomorphic and 2-transitive closed permutation group on a countable set Ω . Then $\mathbf{U}(\Gamma)^+$ has the Howe-Moore property.*

We emphasize on the fact that the proof of this result does not use the classification. However, combining the results of sections 5.3 and 5.4 provides a deep understanding of the open subgroups of $\mathbf{U}(\Gamma)$, see Corollary 5.4.10.

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5.2 Preliminaries

5.2.1 Permutation groups and colored trees

Given Ω a countable set, $\text{Sym}(\Omega)$ denotes the group of all bijections of Ω . We equip it with the *permutation group topology*, i.e. the Polish group topology given by pointwise convergence on Ω discrete. A *closed permutation group* on Ω is a closed subgroup $G \subseteq \text{Sym}(\Omega)$. Given $A \subseteq \Omega$, we will write G_A for the *pointwise stabilizer* of A in G and $G_{\{A\}}$ for the *setwise stabilizer* of A in G , that is:

$$G_A := \{g \in G : \forall a \in A, g(a) = a\} \quad \& \quad G_{\{A\}} = \{g \in G : g(A) = A\}.$$

Trees, i.e. acyclic connected graphs, will be identified with the set of their vertices. In particular, if X is a tree, " $x \in X$ " will always mean that x is a vertex of X . The set of vertices of X that are adjacent to $x \in X$, called the *neighbors* of x , will be denoted $V_X(x)$. We will write $x \sim y$ if x and y are adjacent vertices of a tree. If x has exactly one neighbor in X , it will be called a *leaf* of X . We will denote by E_X the set of *unoriented edges* of X , i.e. the set of pairs $\{x, y\}$ of adjacent vertices in X , and by $E_X(x)$ the set of all edges of X that involve x .

A *path* from x to y is a tuple $(x_0, \dots, x_n)$ of vertices of X such that $x_0 = x$, $x_n = y$ and $x_i \sim x_{i+1}$ for every $i < n$. In this setting, $n \in \mathbb{N}$ is the *length* of this path. For all $x, y \in \mathbf{T}$, there exists a unique *shortest path* $[x, y]$ from x to y and we define $d(x, y)$ to be the length of $[x, y]$. Then (X, d) is a metric space and $\text{Aut}(X) = \text{Iso}(X, d)$. We will write, for every $n \in \mathbb{N}$,

$$B_X(x, n) = \{z \in X : d(x, z) \leq n\} \quad \text{and} \quad S_X(x, n) = \{z \in X : d(x, z) = n\}.$$

For a subset or tuple $E \subseteq X$, $\langle E \rangle$ will denote the subtree of X generated by E , meaning:

$$\langle E \rangle = \bigcup \{[z, z'] : z, z' \in E\}.$$

$\mathbf{T}$ will denote the *regular tree with countably infinite arity*. Given a countably infinite set Ω , an Ω -*coloring* of $\mathbf{T}$ is a labeling of its unoriented edges by elements of Ω . More precisely, it is a map $\chi : E_{\mathbf{T}} \rightarrow \Omega$ such that for every $x \in \mathbf{T}$, the restriction $\chi : E_{\mathbf{T}}(x) \rightarrow \Omega$ is a bijection. A back and forth argument shows that there exists a unique Ω -coloring of $\mathbf{T}$ up to conjugation by an element of $\text{Aut}(\mathbf{T})$.

Given $g \in \text{Aut}(\mathbf{T})$ and $x \in \mathbf{T}$, a coloring χ allows us to define the *local action* g_x of g at x . Indeed, let g_x be the unique bijection of Ω such that the following diagram commutes:

$$\begin{array}{ccc} E_{\mathbf{T}}(x) & \xrightarrow{g} & E_{\mathbf{T}}(g(x)) \\ \chi \downarrow & & \downarrow \chi \\ \Omega & \xrightarrow{g_x} & \Omega \end{array}$$

Note that the map $(g, x) \in \text{Aut}(\mathbf{T}) \times \mathbf{T} \mapsto g_x \in \text{Sym}(\Omega)$ is a *cocycle*:

$$\forall g, h \in \text{Aut}(\mathbf{T}), \forall x \in \mathbf{T}, (gh)_x = g_{h(x)} \circ h_x.$$

Definition 5.2.1. Let Γ be a closed permutation group on a countably infinite set Ω and let χ be an Ω -coloring of $\mathbf{T}$. The *Burger-Mozes universal group with prescribed local action* $\Gamma \curvearrowright \Omega$ is the following closed subgroup of $\text{Aut}(\mathbf{T})$:

$$\mathbf{U}(\Gamma) := \{g \in \text{Aut}(\mathbf{T}) : \forall x \in \mathbf{T}, g_x \in \Gamma\}.$$

This group clearly depends on the choice of the coloring χ . However, if χ' is another Ω -coloring of $\mathbf{T}$, the groups obtained using χ and χ' are conjugate in $\text{Aut}(\mathbf{T})$ by any element of $\text{Aut}(\mathbf{T})$ changing χ into χ' , which always exists.

This construction is the infinite-arity analogue of the one carried out in the seminal paper [BM00a, Sec. 3]. The terminology is motivated by the following (adapted from the original result [BM00a, Prop. 3.2.2.] in the finite arity case, to which we refer for the proof). Two actions $G \curvearrowright \Omega$ and $G' \curvearrowright \Omega'$ are *isomorphic* if there exists a bijection $\Omega \rightarrow \Omega'$ such that the induced isomorphism $\text{Sym}(\Omega) \rightarrow \text{Sym}(\Omega')$ sends the image of G in $\text{Sym}(\Omega)$ onto the image of G' in $\text{Sym}(\Omega')$.

Proposition 5.2.2. *Let Γ be a transitive closed permutation group on a countable set Ω and let $G \leq \text{Aut}(\mathbf{T})$ be a vertex-transitive subgroup such that for every $x \in \mathbf{T}$, the action $G \curvearrowright E_{\mathbf{T}}(x)$ is isomorphic to $\Gamma \curvearrowright \Omega$. Then G is contained in a conjugate of $\mathbf{U}(\Gamma)$ by an element of $\text{Aut}(\mathbf{T})$.*

Interestingly, this construction predates [BM00a] by at least two decades. Indeed, it appears in [Ol'80] where Ol'sanskiĭ classifies the irreducible continuous unitary representations of these groups in the finite-arity case assuming 2-transitivity. The purpose of the present paper is to get a similar classification for infinite-arity Burger-Mozes universal groups.

Convention. Whenever we write " $\text{Let } \Gamma \curvearrowright \Omega \text{ be a closed permutation group}$ ", we mean that Ω is a countable set and that Γ is a closed subgroup of $\text{Sym}(\Omega)$. We also assume that an Ω -coloring χ of $\mathbf{T}$ is given and still denote by $\mathbf{T}$ the resulting colored tree. As χ is essentially unique, we suppress it from all notations.

An Ω -colored tree $\mathbf{T}$ is represented in Figure 5.1a where $x, y, z \in \mathbf{T}$, $a, b \in \Omega$ and $\chi(x, y) = a$, $\chi(x, z) = b$. In this setting, we will call $\Gamma \curvearrowright \Omega$ the *local action* and $\mathbf{U}(\Gamma) \curvearrowright \mathbf{T}$ the *global action*.

We start the study of these groups with a general method for building elements of $\mathbf{U}(\Gamma)$. All one needs to provide is the image of an arbitrary vertex and a coherent collection of local actions:

Lemma 5.2.3. *Let $\Gamma \curvearrowright \Omega$ be a closed permutation group. For all $x, y \in \mathbf{T}$ and every $(\gamma_u)_{u \in \mathbf{T}} \in \Gamma^{\mathbf{T}}$, the following assertions are equivalent:*

- (i) *there exists a unique $g \in \mathbf{U}(\Gamma)$ such that $g(x) = y$ and $g_u = \gamma_u$ for all $u \in \mathbf{T}$,*
- (ii) *for all $u \sim v$ in $\mathbf{T}$, $\gamma_u(\chi(u, v)) = \gamma_v(\chi(u, v))$.*

Proof. The implication (i) $\Rightarrow$ (ii) is trivial. For the converse, let $(\gamma_u)_{u \in \mathbf{T}}$ be as in (ii). We define g inductively.

- Set $g(x) = y$.

– Suppose g is defined on $B_{\mathbf{T}}(x, n)$ for some $n \in \mathbb{N}$ in such a way that:

1. $g: B_{\mathbf{T}}(x, n) \longrightarrow B_{\mathbf{T}}(y, n)$ is a graph isomorphism,
2. $\forall u \in B_{\mathbf{T}}(x, n) \setminus S_{\mathbf{T}}(x, n), g_u = \gamma_u$.

Given $u \in \mathbf{T}$ such that $d(u, x) = n + 1$, there exists a unique $v \in B_{\mathbf{T}}(x, n)$ which is connected to u . Define $g(u)$ as the unique $u' \in V_{\mathbf{T}}(g(v))$ such that $\chi(u', g(v)) = \gamma_v(\chi(u, v))$. Once this is done for every u at distance $n + 1$ from x , g is defined on $B_{\mathbf{T}}(x, n + 1)$. Because $(\gamma_z)_{z \in \mathbf{T}}$ satisfies (ii), Items 1 and 2 above still hold after replacing n with $n + 1$.

This proves existence. Uniqueness is straightforward. □

We will make extensive use of the following refinement of the previous result.

Corollary 5.2.4. *Let $\Gamma \curvearrowright \Omega$ be a closed permutation group. Let $x, y \in \mathbf{T}$ and let X be a subtree of $\mathbf{T}$. Let $(\gamma_z)_{z \in X} \in \Gamma^X$ be such that for all $u \sim v$ in X , $\gamma_u(\chi(u, v)) = \gamma_v(\chi(u, v))$. There exists $g \in \mathbf{U}(\Gamma)$ such that $g(x) = y$ and $g_u = \gamma_u$ for all $u \in X$.*

Proof. If X is empty, apply Lemma 5.2.3 to $(\text{id}_{\Omega})_{u \in \mathbf{T}}$. If not, we extend $(\gamma_u)_{u \in X}$ to $\mathbf{T}$ by propagation: Given $u \in \mathbf{T} \setminus X$, let $p(u)$ be the unique element of X such that $d(u, p(u)) = d(u, X)$ and set $\gamma_u := \gamma_{p(u)}$. It now suffices to check that $(\gamma_u)_{u \in \mathbf{T}}$ satisfies condition (ii) of Lemma 5.2.3. To that aim, let $u \sim v$ be adjacent vertices in $\mathbf{T}$. There are three cases:

- If both u and v are in X , then $\gamma_u(\chi(u, v)) = \gamma_v(\chi(u, v))$ by assumption.
 - If neither are in X , then $p(u) = p(v)$ since $u \sim v$, hence $\gamma_u = \gamma_v$.
 - Otherwise, we can assume that $u \notin X$ and $v \in X$. Then $p(u) = v$ and $\gamma_u = \gamma_v$.
-

Here are the different notions of transitivity we will be working with.

Definition 5.2.5. A permutation group Γ on a countable set Ω is said to be

1. *transitive* if the action $\Gamma \curvearrowright \Omega$ has only one orbit.
2. *2-transitive* if for all $a \neq b, a' \neq b' \in \Omega$, there exists $\gamma \in \Gamma$ such that $\gamma(a) = a'$ and $\gamma(b) = b'$.
3. *oligomorphic* if for every $n \in \mathbb{N}$, the diagonal action $\Gamma \curvearrowright \Omega^n$ only has finitely many orbits.

In the special case of a subgroup G of $\text{Aut}(\mathbf{T})$, the action of G on $\mathbf{T}$ is said to be:

1. *vertex transitive* if it has only one orbit.
2. *edge transitive* if for all $x, y, x', y' \in \mathbf{T}$ with $x \sim y$ and $x' \sim y'$, there exists $g \in G$ such that $h(x) = x'$ and $h(y) = y'$.

3. *metrically 2-transitive* if for all $x, y, x', y' \in \mathbf{T}$ with $d(x, y) = d(x', y')$, there exists $g \in G$ such that $g(x) = x'$ and $g(y) = y'$.

Let $\Gamma \curvearrowright \Omega$ be a closed permutation group. We will be going back and forth between the local action $\Gamma \curvearrowright \Omega$ and the global action $\mathbf{U}(\Gamma) \curvearrowright \mathbf{T}$. Given $x \in \mathbf{T}$, a set $A \subseteq E_{\mathbf{T}}(x)$ of edges around x identifies through the coloring χ with the subset $\chi(x, A)$ of Ω . **We will make extensive use of this identification, most often implicitly.** Next is how we will make identifications in the other direction:

Definition 5.2.6. Let $\Gamma \curvearrowright \Omega$ be a closed permutation group. Given $a \in \Omega$ and $x \in \mathbf{T}$, we will denote by a_x the unique vertex of $\mathbf{T}$ adjacent to x such that $\chi(x, a_x) = a$. Similarly, if A is a subset of Ω , it identifies at x with the following star-shaped subtree of $\mathbf{T}$:

$$A_x := \{x\} \cup \{y \in V_{\mathbf{T}}(x) : \chi(x, y) \in A\}.$$

We will make extensive use of this definition. For example, $V_{\mathbf{T}}(x) = \Omega_x$ for every $x \in \mathbf{T}$.

Lemma 5.2.7. Let $\Gamma \curvearrowright \Omega$ be a closed permutation group.

1. $\mathbf{U}(\Gamma) \curvearrowright \mathbf{T}$ is always vertex transitive.
2. $\mathbf{U}(\Gamma) \curvearrowright \mathbf{T}$ is edge transitive if and only if $\Gamma \curvearrowright \Omega$ is transitive.
3. $\mathbf{U}(\Gamma) \curvearrowright \mathbf{T}$ is metrically 2-transitive if and only if $\Gamma \curvearrowright \Omega$ is 2-transitive.

Proof.

1. Let $x, y \in \mathbf{T}$. Simply apply Corollary 5.2.4 to $X = \emptyset$ while requiring x to be sent to y .
2. Suppose $\Gamma \curvearrowright \Omega$ is transitive. Let $x \sim y, x' \sim y' \in \mathbf{T}$. By hypothesis, there exists $\gamma \in \Gamma$ such that $\gamma(\chi(x, y)) = \chi(x', y')$. Set $X := \{x\}$ and $\gamma_x := \gamma$. Apply Corollary 5.2.4 while requiring x to be sent to x' . The converse is straightforward.
3. Suppose $\mathbf{U}(\Gamma) \curvearrowright \mathbf{T}$ is metrically 2-transitive and let $a \neq b, a' \neq b' \in \Omega$. Fix any $x \in \mathbf{T}$. Then:

$$d(a_x, b_x) = 2 = d(a'_x, b'_x),$$

hence there exists $g \in \mathbf{U}(\Gamma)$ such that $g(a_x) = a'_x$ and $g(b_x) = b'_x$. Necessarily, $g(x) = x$ and g_x is an element of Γ that sends a to a' and b to b' .

Conversely, assume $\Gamma \curvearrowright \Omega$ is 2-transitive. Let $x, y, x', y' \in \mathbf{T}$ be vertices satisfying $d(x, y) = d(x', y')$. Write $[x, y] = (x_0, \dots, x_n)$ and $[x', y'] = (x'_0, \dots, x'_n)$. For every $i \in \{1, \dots, n\}$, let $a_i := \chi(x_{i-1}, x_i)$ and $a'_i := \chi(x'_{i-1}, x'_i)$. By hypothesis on Γ , there exists for every $i \in \{1, \dots, n-1\}$ some element $\gamma_i \in \Gamma$ such that $\gamma_i(a_i) = a'_i$ and $\gamma_i(a_{i+1}) = a'_{i+1}$.

We now apply Corollary 5.2.4 to $X = \{x_1, \dots, x_{n-1}\}$ and $\gamma_{x_i} = \gamma_i$ for every $i \in \{1, \dots, n-1\}$ while requiring x_1 to be sent to x'_1 . Visibly, the output g sends both x to x' and y to y' .

□

Unfortunately, the action $\mathbf{U}(\Gamma) \curvearrowright \mathbf{T}$ is not rich enough to fully capture the representation theory of $\mathbf{U}(\Gamma)$ in general. In fact, the lack already resides in the local action $\Gamma \curvearrowright \Omega$, which might not capture all the open subgroups of Γ as stabilizers of finite subsets. Capturing enough of the open subgroups is essential as every open subgroup U of Γ gives rise to a unitary representation of Γ , namely $\Gamma \curvearrowright \ell^2(\Gamma/U)$. This can be always be ensured by adding points to Ω , one for every open subgroup of Γ . To this end, let $\text{Sub}_0(\Gamma)$ be the set of open subgroups of Γ . Let $\text{Cos}_0(\Gamma) := \bigsqcup_{U \in \text{Sub}_0(\Gamma)} G/U$ be the set of left translates of open subgroups of Γ called its *open cosets*. Since Γ is second countable, Γ/U is countable for every open subgroup U of Γ . Moreover, if $\Gamma \curvearrowright \Omega$ is oligomorphic, $\text{Sub}_0(\Gamma)$ is countable [Tsa12, Cor. 2.5] hence so is $\text{Cos}_0(\Gamma)$. Note that Γ acts on $\text{Cos}_0(\Gamma)$ by left translations and the induced permutation group on $\text{Cos}_0(\Gamma)$ is topologically isomorphic to Γ (see e.g. [ET16, Lem. 1.9] or Proposition 5.2.10 below).

Remark 5.2.8. If Γ is the automorphism group of an $\aleph_0$ -categorical structure $\mathfrak{M}$, then $\text{Cos}_0(\Gamma)$ is essentially the same as $\mathfrak{M}^{\text{eq}}$, the structure which contains $\mathfrak{M}$ and the imaginaries of $\mathfrak{M}$. An *imaginary element* on $\mathfrak{M}$ is an equivalence class of a Γ -invariant equivalence relation on $\mathfrak{M}^n$ for some $n \in \mathbb{N}$. This is a standard construction in model theory.

This alternative presentation of Γ as a permutation group has the property that the stabilizer map $\text{Cos}_0(\Gamma) \rightarrow \text{Sub}_0(\Gamma)$ is surjective. Indeed, $\Gamma_U = U$ for every $U \in \text{Sub}_0(\Gamma)$. If $\Gamma \curvearrowright \Omega$ is oligomorphic, the whole representation theory of Γ is encoded in the action $\Gamma \curvearrowright \text{Cos}_0(\Gamma)$. Indeed, every representation of Γ is a subrepresentation of a direct sum of copies of $\Gamma \curvearrowright \ell^2(\text{Cos}_0(\Gamma))$ [JT22, Fact 3.1].

In order to implement this construction into $\mathbf{T}$, it will be convenient to keep the original set Ω . To this end, we will view Γ as a permutation group either on Γ or on $\Omega_\Gamma := \Gamma \cup \text{Cos}_0(\Gamma)$. Of course, the enriched action $\Gamma \curvearrowright \Omega_\Gamma$ also captures all the representation theory of Γ and this is precisely why we pass from Ω to Ω_Γ . See also Remark 5.3.1 below.

Definition 5.2.9. Let $\Gamma \curvearrowright \Omega$ be a closed permutation group. We define the following set:

$$\Omega_\Gamma := \Gamma \cup \text{Cos}_0(\Gamma)$$

on which Γ naturally acts.

Proposition 5.2.10. Let $\Gamma \curvearrowright \Omega$ be a closed permutation group. Then Γ is naturally a closed permutation group on the (possibly uncountable) set Ω_Γ which has the property that for every open subgroup $U \leq \Gamma$, there exists $e \in \Omega_\Gamma$ such that $U = \Gamma_e$.

Proof. The action $\Gamma \curvearrowright \Omega_\Gamma$ yields a group morphism $\iota: \Gamma \rightarrow \text{Sym}(\Omega_\Gamma)$ which trivially injective and easily seen to be continuous. To see that ι is a topological embedding, it suffices to prove that it is open onto its image. Now, if $U \leq \Gamma$ is an open subgroup, then $\iota(U) = \iota(\Gamma)_U$ which is indeed open in $\iota(\Gamma)$. We can thus indeed view Γ as a permutation on Ω_Γ . If Ω_Γ happens to be countable, then ι is a topological embedding between Polish hence automatically has a closed image. The more general argument is very straightforward. $\square$

Convention. When writing "Let $\Gamma \curvearrowright \Omega$ be a closed permutation group", we will always define $\mathbf{U}(\Gamma)$ by viewing Γ as a permutation group on the original set Ω and never on Ω_Γ . The main issue being that all transitivity properties are lost when passing from Ω to Ω_Γ .

We now implement this construction into $\mathbf{T}$.

Definition 5.2.11. Let $\Gamma \curvearrowright \Omega$ be a closed permutation group. We define the tree $\mathbf{T}_\Gamma$ obtained from $\mathbf{T}$ as follows. For every $x \in \mathbf{T}$ and every $C \in \text{Cos}_0(\Gamma)$, we add a leaf around x and label the associated edge with C . The coloring χ is thus extended to an Ω_Γ -coloring of $\mathbf{T}_\Gamma$ in such a way that:

$$\forall x \in \mathbf{T}, \left[\chi: E_{\mathbf{T}_\Gamma}(x) \longrightarrow \Omega_\Gamma \text{ is a bijection} \right].$$

The resulting colored tree will still be denoted $\mathbf{T}_\Gamma$ and is represented in Figure 5.1b. We extend Definition 5.2.6 to that context. For example, $V_{\mathbf{T}_\Gamma}(x) = \Omega_{\Gamma x}$ for every $x \in \mathbf{T}$. The action $\mathbf{U}(\Gamma) \curvearrowright \mathbf{T}$ naturally extends to an action $\mathbf{U}(\Gamma) \curvearrowright \mathbf{T}_\Gamma$.

Remark 5.2.12. There two types of vertices in this new tree $\mathbf{T}_\Gamma$:

- the original vertices of $\mathbf{T}$, which have infinite arity and form the skeleton of $\mathbf{T}_\Gamma$,
- the added leaves, of arity 1.

Whenever needed, we will view $\mathbf{U}(\Gamma)$ as a closed permutation group on $\mathbf{T}_\Gamma$. Depending on Γ , the tree $\mathbf{T}_\Gamma$ might be uncountable. However, the underlying topological group $\mathbf{U}(\Gamma)$ does not change and remains Polish. Moreover, the main results will only be applied to instances of this construction Γ is oligomorphic, in which case $\mathbf{T}_\Gamma$ is countable. The worried reader can thus make the standing assumption that Γ has only countably many distinct open subgroups.

As briefly described above, enriching the action $\Gamma \curvearrowright \Omega$ into $\Gamma \curvearrowright \Omega_\Gamma$ captures all the open subgroups of Γ and the whole representation theory of Γ . In turn, we will show in this paper that enriching the action $\mathbf{U}(\Gamma) \curvearrowright \mathbf{T}$ into $\mathbf{U}(\Gamma) \curvearrowright \mathbf{T}_\Gamma$ captures most of the open subgroup of $\mathbf{U}(\Gamma)$ (Corollary 5.4.10) and all of the aspherical representations of $\mathbf{U}(\Gamma)$ (Theorems 5.3.19). This is the only reason why we enrich the actions. In the end, and as in the introduction, the main results of this paper can be stated without referring to Ω_Γ or $\mathbf{T}_\Gamma$.

Convention. In graphical representations such as in Figure 5.1, solid vertices and thick edges will always correspond to object in $\mathbf{T}$ with labels in Ω . On the other hand, hollowed vertices and thin edges will always correspond to leaves in $\mathbf{T}_\Gamma \setminus \mathbf{T}$ with labels in $\Omega_\Gamma \setminus \Omega = \text{Cos}_0(\Gamma)$.

Proposition 5.2.13. Let $\Gamma \curvearrowright \Omega$ be a closed permutation group. Let X be a subtree of $\mathbf{T}_\Gamma$ and let $x \in X \cap \mathbf{T}$. Let $Z := X \cup V_{\mathbf{T}_\Gamma}(x)$ and let $A := E_X(x)$, seen as a subset of Ω_Γ . The restriction morphism

$$r_x: \mathbf{U}(\Gamma)_X \longrightarrow \Gamma_A, \quad g \mapsto g_x$$

is continuous, open and surjective. In particular, $\mathbf{U}(\Gamma)_X / \mathbf{U}(\Gamma)_Z \simeq \Gamma_A$ (as topological groups).

Proof. Continuity is clear. Openness will follow from surjectivity and the open mapping theorem for Polish groups. The kernel of r_x being precisely $\mathbf{U}(\Gamma)_Z$, the last claim also follows. We thus prove surjectivity. Let $\gamma \in \Gamma_A$. Set $\gamma_x := \gamma$ and $\gamma_y := \text{id}_\Omega$ for every $y \in X \setminus \{x\}$. Then $(\gamma_y)_{y \in X}$ satisfies the conditions of Corollary 5.2.4. We thus get $g \in \mathbf{U}(\Gamma)$ such that $g(x) = x$ and for every $y \in X$, $g_y = \gamma_y$. In particular, $g_x = \gamma$ and, visibly, $g \in \mathbf{U}(\Gamma)_X$. $\square$

Corollary 5.2.14. *Let $\Gamma \curvearrowright \Omega$ be a closed permutation group. Let X be a subtree of $\mathbf{T}_\Gamma$, and $x \in X \cap \mathbf{T}$. Let $A := E_X(x)$ be seen as a subset of Ω_Γ . For every $y \in V_{\mathbf{T}_\Gamma}(x)$, the map*

$$\mathbf{U}(\Gamma)_X \cdot y \longrightarrow \Gamma_A \cdot \chi(x, y), \quad g(y) \mapsto g_x(\chi(x, y))$$

is a bijection.

Let $\Gamma \curvearrowright \Omega$ be a closed permutation group. As in the finite-arity case [BM00a, Sec. 3.2], one can give a precise structural description of vertex stabilizers in $\mathbf{U}(\Gamma)$. To this end, fix $x \in \mathbf{T}$ and, for every $n \geq 1$, let $\Gamma_n \leq \text{Sym}(B_{\mathbf{T}}(x, n))$ be the permutation group induced by the action of $\mathbf{U}(\Gamma)_x$ on $B_{\mathbf{T}}(x, n)$. For every $y \in S_{\mathbf{T}}(x, n)$, let a^y be the label in Ω of the unique edge going from y towards x . The group Γ_n acts on $\prod_{y \in S_{\mathbf{T}}(x, n)} \Gamma_{a^y}$ according to the *wreath-product* action. It is defined as follows: for all $\gamma \in \Gamma_n$ and $k = (k_y)_{y \in S_{\mathbf{T}}(x, n)} \in \prod_{y \in S_{\mathbf{T}}(x, n)} \Gamma_{a^y}$,

$$\gamma \cdot k = \left(u_y \cdot k_{\gamma^{-1}(y)} \cdot u_y^{-1} \right)_{y \in S_{\mathbf{T}}(x, n)}$$

where for every $y \in S_{\mathbf{T}}(x, n)$, u_y is the local action of γ at the unique neighbor of $\gamma^{-1}(y)$ in $[x, \gamma^{-1}(y)]$.

Proposition 5.2.15. *Let $\Gamma \curvearrowright \Omega$ be a closed permutation group. We have the following topological group isomorphisms.*

1. $(\Gamma_n)_{n \geq 1}$ is naturally an inverse sequence of which $\mathbf{U}(\Gamma)_x$ is the inverse limit,
2. $\Gamma_1 \simeq \Gamma$ and for every $n \geq 1$,

$$\Gamma_{n+1} \simeq \Gamma_n \ltimes \prod_{y \in S_{\mathbf{T}}(x, n)} \Gamma_{a^y}$$

for the wreath-product action and where a^y is the label in Ω of the unique edge going from y towards x for every $y \in S_{\mathbf{T}}(x, n)$.

Proof.

1. The natural morphism $\mathbf{U}(\Gamma)_x \longrightarrow \varprojlim \Gamma_n$ is clearly injective and continuous. By the open mapping theorem for Polish groups, it suffices to show that it is surjective. Let $g \in \varprojlim \Gamma_n$ and let $\bar{g}$ be the map $\mathbf{T} \longrightarrow \mathbf{T}$ it defines. Clearly, $g \in \text{Aut}(\mathbf{T})$ and fixes x . Moreover, if $y \in \mathbf{T}$ and $n := d(y, x)$, there exists $h \in \mathbf{U}(\Gamma)_x$ such that $h|_{B_{\mathbf{T}}(x, n+1)} = g|_{B_{\mathbf{T}}(x, n+1)}$. In particular $g_y = h_y \in \Gamma$ hence $g \in \mathbf{U}(\Gamma)_x$.
2. For $n = 1$, there is a continuous injection $\Gamma_1 \longrightarrow \text{Sym}(\Omega)$ through the coloring χ which takes values in Γ . Applying Corollary 5.2.4 to $X = \{x\}$, it is onto Γ . Thus $\Gamma_1 \longrightarrow \text{Sym}(\Omega)$ is a topological group isomorphism by the open mapping theorem for Polish groups.

Let $n \geq 1$. We will find copies of Γ_n and $K := \prod_{y \in S_{\mathbf{T}}(x, n)} \Gamma_{a^y}$ in Γ_{n+1} and show that Γ_{n+1} is the internal semi-direct product of these copies associated with the wreath-product action.

The most straightforward copy to find is that of K , as it identifies with the pointwise stabilizer $\bar{K} := (\Gamma_{n+1})_{B_{\mathbf{T}}(x, n)}$. Indeed, towards using Corollary 5.2.4, if $k \in K$, let $(\gamma_y)_{y \in B_{\mathbf{T}}(x, n)}$ be as follows:

- If $y \in B_T(x, n-1)$, let $\gamma_y := \text{id}_\Omega$.
- If $y \in S_T(x, n)$, let $\gamma_y := k_y$.

Let g be the output of Corollary 5.2.4 applied to $(\gamma_y)_{y \in B_T(x, n)}$ while requiring x to be fixed. Finally, let $\bar{k} := g|_{B_T(x, n+1)}$. Then $k \mapsto \bar{k}$ is an injective and continuous group morphism $K \rightarrow \Gamma_{n+1}$. Its image is visibly included in $\bar{K}$ and easily seen to be equal to $\bar{K}$ using Corollary 5.2.4. By the open mapping theorem for Polish groups, our map $K \rightarrow \bar{K}$ is a topological group isomorphism and we have found our copy of K . Note that $B_T(x, n)$ is stabilized as a set by the action of Γ_{n+1} hence $\bar{K}$ is a normal subgroup of Γ_{n+1} .

Next, let $\gamma \in \Gamma_n$ and let $g \in \mathbf{U}(\Gamma)_x$ be such that $g|_{B_T(x, n)} = \gamma$. We will extend γ to a $B_T(x, n+1)$ by propagating the outermost local actions of γ . This is achieved by means of Corollary 5.2.4, for which we define $(\gamma_y)_{y \in B_T(x, n)}$ as follows:

- If $y \in B_T(x, n-1)$, $\gamma_y := g_y$.
- If $y \in S_T(x, n)$, let z be the unique neighbor of y in $B_T(x, n-1)$ and let $\gamma_y := g_z$.

Let $\bar{g}$ be the output of Corollary 5.2.4 applied to $(\gamma_y)_{y \in B_T(x, n)}$ while requiring x to be fixed. Let $\bar{\gamma} := \bar{g}|_{B_T(x, n+1)} \in \Gamma_{n+1}$. This propagation process $\gamma \mapsto \bar{\gamma}$ defines a continuous and injective morphism $\Gamma_n \rightarrow \Gamma_{n+1}$. Observe through Corollary 5.2.4 that its image is exactly the subgroup $\bar{\Gamma}_n$ of elements $\gamma \in \Gamma_{n+1}$ such that for every $y \in S_T(x, n)$ and $z \in S_T(x, n-1)$ with $y \sim z$, the local actions of γ at y and z are equal. This is a closed condition hence $\bar{\Gamma}_n$ is Polish. By the open mapping theorem for Polish groups, our map $\Gamma_n \rightarrow \bar{\Gamma}_n$ is a topological group isomorphism and we have found our copy of Γ_n . By the same observation, $\bar{K} \cap \bar{\Gamma}_n = \{1_{\Gamma_{n+1}}\}$.

We now show that $\Gamma_{n+1} = \bar{K} \cdot \bar{\Gamma}_n$ and that the action by conjugation $\bar{\Gamma}_n \curvearrowright \bar{K}$ corresponds to the wreath-product action. Let $\gamma \in \Gamma_{n+1}$ and let $\gamma' := \gamma|_{B_T(x, n)} \in \bar{\Gamma}_n$. Then $\gamma'|_{B_T(x, n)} = \gamma|_{B_T(x, n)}$ hence $\gamma'^{-1}\gamma \in \bar{K}$ and $\gamma \in \bar{K} \cdot \bar{\Gamma}_n$. Next, let $\gamma \in \Gamma_n$ and $k \in K$. There exists $g, h \in \mathbf{U}(\Gamma)_x$ such that $g|_{B_T(x, n+1)} = \bar{\gamma}$ and $h|_{B_T(x, n+1)} = \bar{k}$. Then h fixes pointwise all of $B_T(x, n)$ and, for every $y \in S_T(x, n)$, we have in Γ :

$$(ghg^{-1})_y = g_{hg^{-1}(y)} \cdot h_{g^{-1}(y)} \cdot (g^{-1})_y = g_{g^{-1}(y)} \cdot k_{\gamma^{-1}(y)} \cdot (g_{g^{-1}(y)})^{-1}.$$

Since $g|_{B_T(x, n)} = \bar{\gamma}$ and by definition of the propagation morphism $\Gamma_n \rightarrow \bar{\Gamma}_n$, the element $g_{g^{-1}(y)}$ is equal to the local action of γ at the unique neighbor of $\gamma^{-1}(y)$ in $B_T(x, n-1)$.

We have shown that

$$\Gamma_{n+1} \simeq \Gamma_n \ltimes \prod_{y \in S_T(x, n)} \Gamma_{a^y}$$

for the described action, it only remains to show that the isomorphism is topological. Continuity is clear and we conclude by the open mapping theorem for Polish groups.

□

5.2.2 Roelcke precompactness, and coarse geometry

The topological manifestation of oligomorphicity is Roelcke precompactness. Recall that a subset E of a topological group G is *Roelcke precompact* if for every open neighborhood U of the identity in G , there exists a finite subset $F \subseteq G$ such that $E \subseteq UFU$. A group is *Roelcke precompact* if it is Roelcke precompact as a subset of itself and it is *locally Roelcke-precompact* if it admits a Roelcke-precompact non-empty open subset. Oligomorphic permutation groups are Roelcke precompact [Tsa12, Th.2.4].

Roelcke precompactness is a generalization of compactness that seems to retain most of its geometrical properties. Within representation theory, this is witnessed by Tsankov's classification Theorem [Tsa12] which is reminiscent of the Peter-Weyl Theorem for compact groups. Another strong similarity is that every Roelcke precompact Polish group has Kazhdan's Property (T) [Iba21]. The duality theory developed by Tannaka and Krein for compact groups using abstract harmonic analysis also extends to Roelcke precompact automorphism groups [Bar25]. Recall that if $G \leq \text{Aut}(\mathbf{T})$, $G^+ = \{g \in G : \forall x \in \mathbf{T}, d(g(x), x) \text{ is even}\}$.

Proposition 5.2.16. *Let $\Gamma \curvearrowright \Omega$ be an oligomorphic and transitive closed permutation group. Then $\mathbf{U}(\Gamma)$ and $\mathbf{U}(\Gamma)^+$ are locally Roelcke-precompact. More precisely, the stabilizers of vertices and edges in $\mathbf{U}(\Gamma)$ are Roelcke precompact.*

Proof. For vertex stabilizers, this is a consequence of Proposition 5.2.15. Indeed, direct products, semi-direct products and inverse limits of Roelcke precompact groups are Roelcke precompact [Tsa12, Prop. 2.2].

Let $x, y \in \mathbf{T}$ be connected. Since $\Gamma \curvearrowright \Omega$ is transitive, $\mathbf{U}(\Gamma) \curvearrowright \mathbf{T}$ is edge transitive by Item 2 of Lemma 5.2.7 and there exists $g \in \mathbf{U}(\Gamma)$ that exchanges x and y . Then $\mathbf{U}(\Gamma)_{\{x,y\}} = \mathbf{U}(\Gamma)_{\{x,y\}} \sqcup g\mathbf{U}(\Gamma)_{\{x,y\}} \leq \mathbf{U}(\Gamma)_x \cup g\mathbf{U}(\Gamma)_x$ hence $\mathbf{U}(\Gamma)_{\{x,y\}}$ is Roelcke precompact as a subset of a finite union of Roelcke precompact sets. □

The above proposition highlights another benefit of inputting oligomorphic permutation groups: it produces particularly well-behaved groups in terms of Rosendal's *large scale geometry* [Ros22]. Recall that an *écart* on a topological group G is a map $\delta: G^2 \rightarrow \mathbb{R}_+$ such that for all $x, y, z \in G$, we have $\delta(x, x) = 0$ and $\delta(x, z) \leq \delta(x, y) + \delta(y, z)$. It is *left-invariant* if $\delta(xy, xz) = \delta(y, z)$ for all $x, y, z \in G$. A subset $E \subseteq G$ is *coarsely bounded* (in the left coarse-structure) if its δ -diameter $\delta(E)$ is finite for every continuous left-invariant écart δ on G .

It is easily seen that Roelcke precompact subsets are always coarsely bounded. In particular, Roelcke precompact groups have trivial large scale geometry. It turns out, however, that algebra and topology blend most harmoniously in the large scale geometry of locally Roelcke-precompact Polish groups [Ros22, Chap. 3]. For one thing, these groups are characterized by having a locally compact Roelcke completion [Zie21], see also Subsection 5.4.1 below. What is more, a subset of a locally Roelcke-precompact Polish group G is coarsely bounded if and only if it is Roelcke precompact. Finally, such a group G always admits a *coarsely proper*

écart, i.e. a left-invariant and continuous écart δ that generates the left coarse-structure in the following sense:

$$\forall E \subseteq G, [E \text{ is coarsely bounded} \iff \delta(E) < \infty].$$

In our case, the écarts naturally arising from the action $\mathbf{U}(\Gamma) \curvearrowright \mathbf{T}$ plays such a role. Indeed, if $\Gamma \curvearrowright \Omega$ is a 2-transitive and oligomorphic closed permutation group, the next proposition implies that for every $x \in \mathbf{T}$, the écart $\delta_x: \mathbf{U}(\Gamma) \times \mathbf{U}(\Gamma) \rightarrow \mathbb{N}$ given by $\delta_x(g, h) = d(g(x), h(x))$ is coarsely proper.

Proposition 5.2.17. *Let $\Gamma \curvearrowright \Omega$ be a 2-transitive and oligomorphic closed permutation group. Given K be a subset of $\mathbf{U}(\Gamma)$ (resp. $\mathbf{U}(\Gamma)^+$), the following properties are equivalent:*

- (i) K is coarsely bounded in $\mathbf{U}(\Gamma)$ (resp. $\mathbf{U}(\Gamma)^+$),
- (ii) K is Roelcke precompact in $\mathbf{U}(\Gamma)$ (resp. $\mathbf{U}(\Gamma)^+$),
- (iii) $K \cdot x$ is bounded in $\mathbf{T}$ for every vertex $x \in \mathbf{T}$,
- (iv) $K \cdot x$ is bounded in $\mathbf{T}$ for some vertex $x \in \mathbf{T}$,

If K is a subgroup of $\mathbf{U}(\Gamma)$ (or $\mathbf{U}(\Gamma)^+$), the previous properties are moreover equivalent to:

- (v) $K \leq \mathbf{U}(\Gamma)_x$ for some vertex $x \in \mathbf{T}$ or $K \leq \mathbf{U}(\Gamma)_{\{\{x,y\}\}}$ for some connected vertices $x, y \in \mathbf{T}$,
- (vi) $K \leq \mathbf{U}(\Gamma)_{\{X\}}$ for some non-empty finite subtree $X \subseteq \mathbf{T}$.

Proof. The proofs for $\mathbf{U}(\Gamma)$ and $\mathbf{U}(\Gamma)^+$ are the same since $\mathbf{U}(\Gamma)_A = \mathbf{U}(\Gamma)_A^+$ as soon as $A \subseteq \mathbf{T}$ is non-empty, hence we only prove the equivalences for $\mathbf{U}(\Gamma)$.

(i) $\iff$ (ii): Since $\mathbf{U}(\Gamma)$ is locally Roelcke-precompact by Proposition 5.2.16, we refer to [Zie21, Th. 14] for this equivalence.

(ii) $\implies$ (iii): Assume K is Roelcke precompact in $\mathbf{U}(\Gamma)$ and let $x \in \mathbf{T}$. Then, $\mathbf{U}(\Gamma)_x$ being open, there exists a finite subset F of K such that $K \subseteq \mathbf{U}(\Gamma)_x \cdot F \cdot \mathbf{U}(\Gamma)_x$. Let $M := \max\{d(f(x), x) : f \in F\}$. If $k \in K$, there exists $u, v \in \mathbf{U}(\Gamma)_x$ and $f \in F$ such that $k = ufv$. Then $d(k(x), x) = d(fv(x), u^{-1}(x)) = d(f(x), x) \leq M$.

(iii) $\implies$ (iv): Trivial.

(iv) $\implies$ (ii): Let $x \in \mathbf{T}$ be such that $K \cdot x$ is bounded. We first observe that, for every $g \in \mathbf{U}(\Gamma)$, the double coset $\mathbf{U}(\Gamma)_x \cdot g \cdot \mathbf{U}(\Gamma)_x$ is determined by $d(gx, x)$. Indeed, suppose $g, h \in \mathbf{U}(\Gamma)$ satisfy $d(gx, x) = d(hx, x)$. Then, since $\mathbf{U}(\Gamma) \curvearrowright \mathbf{T}$ is metrically 2-transitive by Item 3 of Lemma 5.2.7, there exists $u \in \mathbf{U}(\Gamma)$ such that $ug(x) = h(x)$ and $u(x) = x$. In particular, $h^{-1}ug \in \mathbf{U}(\Gamma)_x$ and $g \in \mathbf{U}(\Gamma)_x \cdot h \cdot \mathbf{U}(\Gamma)_x$ i.e. $\mathbf{U}(\Gamma)_x \cdot g \cdot \mathbf{U}(\Gamma)_x = \mathbf{U}(\Gamma)_x \cdot h \cdot \mathbf{U}(\Gamma)_x$.

Now, since $K \cdot x$ is bounded, the set $\{d(kx, x) : k \in K\}$ is finite. Fix $F \subseteq K$ finite covering all the possibilities. By what precedes, $K \subseteq \mathbf{U}(\Gamma)_x \cdot F \cdot \mathbf{U}(\Gamma)_x$. Now, $\mathbf{U}(\Gamma)$ is locally Roelcke-precompact by Proposition 5.2.16 hence products of Roelcke precompact subsets of $\mathbf{U}(\Gamma)$ are Roelcke precompact [Zie21, Prop. 12]. Since $\mathbf{U}(\Gamma)_x$ is Roelcke precompact in $\mathbf{U}(\Gamma)$ by Proposition 5.2.16 and since finite sets are always Roelcke precompact, so is K .

We now assume that K is a subgroup of $\mathbf{U}(\Gamma)$. The implications $(v) \Rightarrow (vi) \Rightarrow (iv)$ being obvious if even K is not a subgroup, it only remains to prove $(iv) \Rightarrow (v)$. Suppose $x \in \mathbf{T}$ is such that $K \cdot x$ is bounded. Let $X := \langle K \cdot x \rangle$ be the tree generated by $K \cdot x$. This tree has finite diameter and is stable under the action of the group K . If X has diameter 0 or 1, then (v) holds. If not, let X' be the tree obtained from X by removing all of its leaves. Then X' remains stable under the action of K while the diameter has decreased by 2. We conclude by induction on the diameter of X . $\square$

In Section 5.4, we will see that the large scale geometry of $\mathbf{U}(\Gamma)$ is well captured by its representation theory, provided that $\Gamma \curvearrowright \Omega$ is oligomorphic and 2-transitive. More precisely, $\mathbf{U}(\Gamma)^+$ has the Howe-Moore property in this case.

5.2.3 Algebraicity

The notion of algebraicity will be central in our study. We will be working with its permutation group theoretic definition.

Definition 5.2.18. Let G be a permutation group on a set N and let A be a subset of N .

1. An element $x \in N$ is said to be *algebraic over A* if there exists a finite subset $B \subseteq A$ such that the orbit $G_B \cdot x$ is finite. The *algebraic closure* of A , denoted $\text{acl}_N(A)$, is the set of elements of N that are algebraic over A . The set A is *algebraically closed* if it is equal to its algebraic closure. Algebraic closures are always algebraically closed.
2. A is said to be *definable* if $G_{\{D\}}$ is open in G .
3. A is said to be *locally G -finite* if its intersection with every orbit of the action of G on N is finite.
4. We will write $\mathcal{C}(\Omega)$ for the set of definable, locally G -finite and algebraically closed subsets of N , called the *classifying subsets*.

The following two lemmas contain well-known facts about algebraicity which we will use extensively and often implicitly.

Lemma 5.2.19. Let G be a permutation group on set N and let $A = \{a_1, \dots, a_n\}$ be a finite subset of N .

1. For every subset $E \subseteq G$, there is a canonical bijection

$$EG_A/G_A \longrightarrow E \cdot (a_1, \dots, a_n)$$

where $E \cdot (a_1, \dots, a_n) := \{(g(a_1), \dots, g(a_n)) : g \in E\}$.

2. If B is a subset of A , then $G_A \leq G_B$ and

$$[G_B : G_A] < +\infty \iff A \subseteq \text{acl}_N(B).$$

Proof.

1. The map $\phi: G \rightarrow G \cdot (a_1, \dots, a_n)$ is surjective. Moreover, for all $g, h \in G$, we have $\phi(g) = \phi(h)$ if and only if $g^{-1}h \in G_A$, i.e. $gG_A = hG_A$. Thus ϕ factorizes into a bijection $G/G_A \rightarrow G \cdot (a_1, \dots, a_n)$ which sends EG_A/G_A onto $E \cdot (a_1, \dots, a_n)$.
2. This assertion follows from the previous Item together with the observation that, A being finite, $G_B \cdot (a_1, \dots, a_n)$ if and only if $G_B \cdot a_i$ is finite for every $i \leq n$.

□

Lemma 5.2.20. *Let G be a permutation group on set N and let $A, B \subseteq N$.*

1. $A \subseteq \text{acl}_N(A)$.
2. If $A \subseteq B$, then $\text{acl}_N(A) \subseteq \text{acl}_N(B)$.
3. $\text{acl}_N(A) = \text{acl}_N(\text{acl}_N(A))$, i.e. $\text{acl}_N(A)$ is algebraically closed.
4. If A and B are algebraically closed, then so is $A \cap B$.
5. For every $g \in G$, we have $g(\text{acl}_N(A)) = \text{acl}_N(g(A))$.

Proof.

1. Trivial.
2. Trivial.
3. The $\subseteq$ inclusion follows from Item 1. For the reverse inclusion, let $a \in \text{acl}_N(A)$. There exists $A' \subseteq \text{acl}_N(A)$ finite such that $a \in \text{acl}_N(A')$ is finite. In turn, there exists $A'' \subseteq A$ finite such that $A' \subseteq \text{acl}_N(A'')$. Since $a \in \text{acl}_N(A'')$, $G_{A''} \cdot a$ where, by Item 1 of Lemma 5.2.19, we have $G_{A''} \cdot a \simeq G_{A''}G_a/G_a$. Thus, $G_{A''}$ is contained in a finite union of left cosets of G_a . We can thus fix $g_1, \dots, g_n \in G$ such that $G_{A''} \subseteq \cup_i g_i G_a$. Similarly, there exists $h_1, \dots, h_m \in G$ such that $G_{A'} \subseteq \cup_j h_j G_{A''}$. Then:

$$G_{A''} \subseteq \bigcup_{i,j} h_j g_i G_a$$

and $G_{A''} \cdot a \subseteq \{h_j g_i(a) : i \leq n \text{ \& } j \leq m\}$ is finite. Thus $a \in \text{acl}_N(A'') \subseteq \text{acl}_N(A)$.

4. By Items 2 and 3 above, $\text{acl}(A \cap B) \subseteq \text{acl}(A) \cap \text{acl}(B) = A \cap B$. The reverse inclusion follows from Item 1.
5. Observe first that for all $g \in G$ and $C \subseteq N$, we have $gG_B g^{-1} = G_{g(B)}$. Thus, for every $A' \subseteq A$ finite and every $a \in N$, we have:

$$\begin{aligned} a \in \text{acl}_N(g(A')) &\iff G_{g(A')} \cdot a \text{ is finite} \iff gG_{A'} g^{-1} \cdot a \text{ is finite} \\ &\iff g^{-1}(a) \in \text{acl}_N(A') \iff a \in g(\text{acl}_N(A')). \end{aligned}$$

□

Let $\Gamma \curvearrowright \Omega$ be an oligomorphic closed permutation group. Recall that Γ can also be seen as a permutation group over Ω_Γ which contains Ω and the countable collection of the open cosets of Γ . Then $(\mathcal{C}(\Omega_\Gamma), \subseteq, \wedge, \vee)$ where $A \wedge B = A \cap B$ and $A \vee B = \text{acl}_{\Omega_\Gamma}(A \cup B)$ for all $A, B \in \mathcal{C}(\Omega_\Gamma)$ is an ordered lattice [Tsa25, Th. 2.4]. In this case, understanding $\mathcal{C}(\Omega_\Gamma)$ is key to understanding the representation theory of Γ , see the end of the next subsection. Below are the properties of this lattice that will be useful to us, extracted from [Tsa25].

Proposition 5.2.21 ([Tsa25]). *Let $\Gamma \curvearrowright \Omega$ be a 2-transitive and oligomorphic closed permutation group.*

1. *The elements of $\mathcal{C}(\Omega_\Gamma)$ are exactly the subsets of Ω_Γ of the form $\text{acl}_{\Omega_\Gamma}(e)$ for some $e \in \Omega_\Gamma$ (equivalently, of the form $\text{acl}_{\Omega_\Gamma}(A)$ for some finite subset A of Ω_Γ).*
2. *For every $A \in \mathcal{C}(\Omega_\Gamma)$, the intersection $A \cap \Omega$ is finite.*
3. *The ordered lattice $(\mathcal{C}(\Omega_\Gamma), \subseteq, \wedge, \vee)$ is well-founded, i.e. every non-empty subset of $\mathcal{C}(\Omega_\Gamma)$ admits a minimal element.*

Proof.

1. This is the combination of Lemmas 2.1 and 2.2 in [Tsa25].
2. Since $\Gamma \curvearrowright \Omega$ is oligomorphic, Ω splits into finitely many Γ -orbits. This item then follows from the fact that elements of $\mathcal{C}(\Omega_\Gamma)$ are locally $\mathbf{U}(\Gamma)$ -finite.
3. This is an application of [Tsa25, Th. 2.4].

□

Let $\Gamma \curvearrowright \Omega$ be an oligomorphic closed permutation group. The smallest element of $\mathcal{C}(\Omega_\Gamma)$ is obviously $\text{acl}_{\Omega_\Gamma}(\emptyset)$. If $\Gamma \curvearrowright \Omega$ is moreover 2-transitive, every element of

$$\mathcal{C}_1(\Omega_\Gamma) := \{\text{acl}_{\Omega_\Gamma}(a) : a \in \Omega\}$$

is a direct successor of $\text{acl}_{\Omega_\Gamma}(\emptyset)$. We end this subsection with the proof of this fact.

Lemma 5.2.22. *Let N be an infinite countable set and let $(E_i)_{i \in I}$ be a countably infinite collection of pairwise distinct finite subsets of N . Suppose there exists $n \in \mathbb{N}$ such that $|E_i \cap E_j| = n$ for all $i \neq j \in I$. Then:*

$$\forall i \neq j \in I, E_i \cap E_j = \bigcap_{k \in I} E_k.$$

Proof. Let $i_1 \in I$. The set $\{E_{i_1} \cap E_j : j \in I\}$ is an infinite collection of subsets of the finite set E_{i_1} . By the pigeonhole principle, there exists an infinite subset $J \subseteq I$ such that for all $j, j' \in J$, $E_{i_1} \cap E_j = E_{i_1} \cap E_{j'}$. We will denote by F this common intersection.

Now, for all $j \neq j' \in J$,

$$F = F \cap F = E_{i_1} \cap E_j \cap E_{i_1} \cap E_{j'} \subseteq E_j \cap E_{j'}.$$

Since moreover $|F| = n = |E_j \cap E_{j'}|$, we have:

$$\forall j \neq j' \in J, E_j \cap E_{j'} = F.$$

Let $i_2 \in I \setminus \{i_1\}$. The set $\{E_{i_2} \cap E_j, j \in J\}$ is an infinite collection of subsets of the finite set E_{i_2} . By the pigeonhole principle, there exists an infinite subset $K \subseteq J$ such that for all $k, k' \in K$, $E_{i_2} \cap E_k = E_{i_2} \cap E_{k'}$. It follows from the argument in the previous paragraph that this common intersection is precisely F .

Finally, K being infinite, we can pick $k \in K \setminus \{i_1, i_2\}$. Recalling that $K \subseteq J$, we have:

$$F = F \cap F = E_{i_1} \cap E_k \cap E_{i_2} \cap E_k \subseteq E_{i_1} \cap E_{i_2}.$$

By cardinality, $E_{i_1} \cap E_{i_2} = F$. Since i_1 and i_2 are arbitrary, the conclusion follows. $\square$

Proposition 5.2.23. *Let $\Gamma \curvearrowright \Omega$ be a 2-transitive and oligomorphic closed permutation group. Let $a \in \Omega$ and $A \in \mathcal{C}(\Omega_\Gamma)$ be such that $\text{acl}_{\Omega_\Gamma}(\emptyset) \subseteq A \subseteq \text{acl}_{\Omega_\Gamma}(a)$. Then either $A = \text{acl}_{\Omega_\Gamma}(\emptyset)$ or $A = \text{acl}_{\Omega_\Gamma}(a)$.*

Proof. Let $C := \bigcap_{c \in \Omega} \text{acl}_{\Omega_\Gamma}(c)$. We will use the previous combinatorial Lemma to show that $\text{acl}_{\Omega_\Gamma}(a) \cap \text{acl}_{\Omega_\Gamma}(b) = C$ for all $a \neq b \in \Omega$.

Let U be an open subgroup of Γ and let $N := \Gamma/U$. Fix $a \in \Omega$ and let $V := \Gamma_a \cap U$ which is open in Γ . Since Γ is Roelcke precompact by [Tsa12, Th. 2.4], there exists $F \subseteq \Gamma$ finite such that $\Gamma = V F V$. Then, seen in the action $\Gamma \curvearrowright N$, we have

$$N = \Gamma \cdot U = (V F V) \cdot U = \bigcup_{f \in F} V \cdot f U.$$

Thus, the action $\Gamma_a \curvearrowright N$ has finitely many orbits for every $a \in \Omega_\Gamma$.

In particular, $E_a := \{x \in N : \Gamma_a \cdot x \text{ is finite}\}$ is finite for every $a \in \Omega$. What is more, by 2-transitivity of Ω , $|E_a \cap E_b|$ does not depend on the choice of $a \neq b \in \Omega$. By Lemma 5.2.22, neither does $E_a \cap E_b$. We have thus shown that

$$\forall a \neq b \in \Omega, \text{acl}_{\Gamma/U}(a) \cap \text{acl}_{\Gamma/U}(b) \subseteq C \cap (\Gamma/U).$$

By 2-transitivity of $\Gamma \curvearrowright \Omega$, we also have $\text{acl}_\Omega(a) = \{a\}$ for every $a \in \Omega$. Thus, recalling that $\Omega_\Gamma = \Omega \cup \bigcup_U \Gamma/U$, we have, for every $a \neq b \in \Omega$,

$$\begin{aligned} \text{acl}_{\Omega_\Gamma}(a) \cap \text{acl}_{\Omega_\Gamma}(b) &= (\text{acl}_\Omega(a) \cap \text{acl}_\Omega(b)) \cup \\ &\quad \bigcup_{U \in \text{Sub}_0(\Gamma)} \text{acl}_{\Gamma/U}(a) \cap \text{acl}_{\Gamma/U}(b) \\ &\subseteq (\{a\} \cap \{b\}) \cup C = C. \end{aligned}$$

The other inclusion is trivial.

We now claim that $C = \text{acl}_{\Omega_\Gamma}(\emptyset)$. Indeed, let $a \neq b \in \Omega$. By 2-transitivity of $\Gamma \curvearrowright \Omega$, we have $\Gamma_a \Gamma_b \cdot a = \Gamma_a \cdot (\Omega \setminus \{b\}) = \Omega$ thus $\Gamma = \Gamma_a \Gamma_b \Gamma_a$. Suppose now that $c \in C$. Since C is stable under the action of Γ , the set $F_1 := \Gamma_a \cdot c$ is a finite subset of C hence so is $F_2 := \bigcup \Gamma_b \cdot F_1$ and so is $F_3 := \bigcup \Gamma_a \cdot F_2$. Since $F_3 = \Gamma_a \Gamma_b \Gamma_a \cdot c = \Gamma \cdot c$, we indeed get that $c \in \text{acl}_{\Omega_\Gamma}(\emptyset)$. The reverse inclusion is trivial. We have thus shown that

$$\forall a \neq b \in \Omega, \text{acl}_{\Omega_\Gamma}(a) \cap \text{acl}_{\Omega_\Gamma}(b) = \text{acl}_{\Omega_\Gamma}(\emptyset).$$

We are now ready to conclude. Let $e \in \text{acl}_{\Omega_\Gamma}(a)$. It suffices to show that either $a \in \text{acl}_{\Omega_\Gamma}(e)$ or $e \in \text{acl}_{\Omega_\Gamma}(\emptyset)$. Suppose $a \notin \text{acl}_{\Omega_\Gamma}(e)$. In particular, there exists $\gamma \in \Gamma_e$ such that $b := \gamma(a) \neq a$. The equality $e = \gamma(e)$ yields $e \in \text{acl}_{\Omega_\Gamma}(a) \cap \gamma(\text{acl}_{\Omega_\Gamma}(a)) = \text{acl}_{\Omega_\Gamma}(a) \cap \text{acl}_{\Omega_\Gamma}(b) = \text{acl}_{\Omega_\Gamma}(\emptyset)$. $\square$

5.2.4 Hilbert spaces and unitary representations

We recall here some basic results about Hilbert spaces and unitary representations of permutation groups. First, the Alaoglu-Birkhoff Theorem will help us locate invariant vectors. See [BJJ24, Th. 2.3] for a proof of this specific version.

Theorem 5.2.24. *Let $\mathcal{H}$ be a Hilbert and let G be any subgroup of the unitary group $\mathcal{U}(\mathcal{H})$ of $\mathcal{H}$. Let p be the orthogonal projection on $\mathcal{H}^G$, the closed subspace of G -invariant vectors. For every $\xi \in \mathcal{H}$, $p\xi$ is the unique vector of minimal norm in the closed convex hull $\overline{\text{Conv}}(G \cdot \xi)$ of $G \cdot \xi$ in $\mathcal{H}$.*

Next is a convergence theorem for projections associated with a directed collection of subspaces of a Hilbert space, which will allow us to carry out approximation arguments.

Lemma 5.2.25. *Let $\mathcal{H}$ be a Hilbert space, $(I, \leq)$ a directed set and $(\mathcal{K}_i)_{i \in I}$ a directed union of closed subspaces of $\mathcal{H}$. For every $i \in I$, denote by p_i the orthogonal projection onto $\mathcal{K}_i$. Let $\mathcal{K} := \overline{\bigcup_{i \in I} \mathcal{K}_i}$ and let p be the orthogonal projection onto $\mathcal{K}$. Then $(p_i)_{i \in I}$ converges to p in the strong operator topology, that is:*

$$\forall \xi \in \mathcal{H}, \|p_i \xi - p \xi\| \xrightarrow{i} 0.$$

Proof. Pick first $\xi \in \mathcal{K}$ so that $p\xi = \xi$ and $\varepsilon > 0$. There exists $\xi_\varepsilon \in \mathcal{K}_i$ for some $i \in I$ such that $\|\xi - \xi_\varepsilon\| < \varepsilon$. Then, for every $j \geq i$, we have $p_j \xi_\varepsilon = \xi_\varepsilon$ and

$$\|p_j \xi - p \xi\| \leq \|p_j \xi - \xi_\varepsilon\| + \|\xi_\varepsilon - p \xi\| = \|p_j \xi - p_j \xi_\varepsilon\| + \|\xi_\varepsilon - \xi\| < 2\varepsilon.$$

We have shown that $\|p_i \xi - p \xi\| \xrightarrow{i} 0$ for every $\xi \in \mathcal{K}$. For $\xi \in \mathcal{H}$ arbitrary, observe that $p_i p = p p_i = p_i$ since $\mathcal{H}_i \subseteq \mathcal{H}$ and apply what precedes to $p\xi$. $\square$

We now recall classical definitions and facts about unitary representations. A *unitary representation* of a topological group G is a linear isometric action of G on a Hilbert space $\mathcal{H}$. It is *continuous* if for all $\xi, \eta \in \mathcal{H}$, the map $g \mapsto \langle \pi(g)\xi, \eta \rangle$ is continuous. It is *non-zero* if $\mathcal{H} \neq 0$. Throughout this paper, "representation" will always mean "continuous unitary representation". Let $\pi: G \curvearrowright \mathcal{H}$ and $\pi': G \curvearrowright \mathcal{H}'$ be two representations of G . If H is a subgroup of G , we will denote by $\pi|_H$ the restriction of π to H . An *isomorphism* between π and π' is a G -equivariant surjective isometry $\mathcal{H} \rightarrow \mathcal{H}'$. We will say that π' is (isomorphic to) a *subrepresentation* of π , and write $\pi' \subseteq \pi$, if there exist a G -equivariant isometric embedding $\mathcal{H}' \rightarrow \mathcal{H}$. The subrepresentation π' is *proper* if $\mathcal{H}'$ identifies with a proper subspace of $\mathcal{H}$ via the embedding. The representation π is *irreducible* if $\mathcal{H} \neq 0$ and π admits no non-zero proper subrepresentation. The *unitary dual* of G is the set of isomorphism classes of irreducible representations of G .

Induction is a central technique in abstract harmonic analysis. Suppose U is an open subgroup of a separable topological group G , then G/U is countable discrete and can be endowed with the counting measure. If moreover $\sigma: U \curvearrowright \mathcal{H}$ is a unitary representation of U , let $E := \{\xi: G \rightarrow \mathcal{H} \mid \forall u \in U, \forall g \in G, \xi(gu) = \sigma(u^{-1}) \cdot \xi(g)\}$. Then, for all $\xi, \eta \in E$, the maps defined on G by

$$g \mapsto \langle \xi(g), \eta(g) \rangle \quad \& \quad g \mapsto \|\xi(g)\|$$

are constant on every U -coset. Fix a transversal $(g_i)_{i \in I}$ of G/U and define:

$$\mathcal{E} := \{\xi \in E: \sum_{i \in I} \|\xi(g_i)\|^2 < +\infty\}.$$

Then $\mathcal{E}$ is a Hilbert space with the following inner product:

$$\forall \xi, \eta \in \mathcal{E}, \langle \xi, \eta \rangle := \sum_{i \in I} \langle \xi(g_i), \eta(g_i) \rangle.$$

In this setting, we will write $\xi \in \ell^2(G/U)$ to refer to the fact that if $\xi \in \mathcal{E}$, then $\sum_{i \in I} \|\xi(g_i)\|^2 < +\infty$. The *induced representation* $\text{Ind}_U^G(\sigma)$ is the action $G \curvearrowright \mathcal{E}$ defined by

$$\forall g \in G, \forall \xi \in \mathcal{E}, \quad g \cdot \xi: x \mapsto \xi(g^{-1}x).$$

Note that this construction does not depend on the choice of a transversal $(g_i)_{i \in I}$. This notion is usually defined for locally compact groups but remains particularly relevant for permutation groups, thanks to their rich lattice of open subgroups.

In what follows, the *support* of a function $\xi: E \rightarrow \mathcal{H}$ where E is a set and $\mathcal{H}$ a Hilbert space is the set $\text{supp } \xi := \{x \in E: \xi(x) \neq 0\}$. The next lemma is a well known fact about induced representations.

Lemma 5.2.26. *Let G be a topological group and let $U \leq G$ be an open subgroup. Let $\sigma: U \curvearrowright \mathcal{K}$ be a unitary representation of U on a Hilbert space $\mathcal{K}$. Consider the representation $\pi := \text{Ind}_U^G(\sigma)$ of G , denote by $\mathcal{H}$ its underlying Hilbert space, and let*

$$\mathcal{H}_0 := \{\xi \in \mathcal{H}: \text{supp } \xi \subseteq U\}.$$

Then $\mathcal{H}_0$ is closed and stable under the action of U . Moreover, $\pi|_U: U \curvearrowright \mathcal{H}_0$ is isomorphic to σ .

Proof. $\mathcal{H}_0$ being stable under the action of U is trivial. To see that $\mathcal{H}_0$ is closed, simply observe that if $\xi \in \mathcal{H}$ satisfies $\xi(g) \neq 0$ for some $g \notin U$, then

$$\inf \{\|\xi - \eta\|: \eta \in \mathcal{H}_0\} \geq \|\xi(g)\| > 0.$$

Finally, the isomorphism is the map $\xi \mapsto \xi(1_G)$. □

When G is a permutation group on a set N , the richness of the lattice of open subgroups makes induction particularly relevant. Moreover, it provides much control on the representation theory of G . If $\pi: G \curvearrowright \mathcal{H}$ is a unitary representation of G and A is a finite subset of N , we will write $\mathcal{H}_A$ for the closed subspace of $\mathcal{H}$ consisting of G_A -invariant vectors. That is

$$\mathcal{H}_A := \{\xi \in \mathcal{H}: \forall g \in G_A, \pi(g)\xi = \xi\}.$$

If A is infinite, we set:

$$\mathcal{H}_A = \overline{\bigcup_{B \subseteq A \text{ finite}} \mathcal{H}_B}.$$

If $A = \{a\}$ is a singleton, we will write $\mathcal{H}_a$ instead. Finally, p_A (or p_a) will denote the orthogonal projection onto $\mathcal{H}_A$. When several representations are at play, we will specify them in the notation and write p_A^π instead. These subspaces always determine the whole representation. See e.g. [Tsa12, Lem. 3.1] for a proof of this fundamental result.

Lemma 5.2.27. *Let G be a permutation group on a set N and let $\pi: G \curvearrowright \mathcal{H}$. Then:*

$$\bigcup_{A \subseteq N \text{ finite}} \mathcal{H}_A \text{ is dense in } \mathcal{H}.$$

Let $\Gamma \curvearrowright \Omega$ be an oligomorphic closed permutation group. Up to viewing Γ as a permutation group on Ω_Γ (Proposition 5.2.10), these subspaces $\mathcal{H}_A$ moreover live coherently inside their underlying space.

Theorem 5.2.28 ([Tsa12]). *Let G be a Roelcke-precompact closed permutation group.*

1. *Every representation of G splits as a sum of irreducible subrepresentations.*
2. *Every irreducible representation of G is of the form $\text{Ind}_{C(V)}^G(\pi)$ where V is an open subgroup of G , $C(V)$ is its commensurator, V is normal in $C(V)$ with finite index and σ an irreducible representation of $C(V)$ that factorizes through the finite group $C(V)/V$.*
3. *Every irreducible representation of G is isomorphic to a subrepresentation of $\lambda_{G/U}: G \curvearrowright \ell^2(G/U)$ for some open subgroup $U \leq G$.*

Lemma 5.2.29. *Let G be a permutation group on a countable set N and let $\mathcal{D}$ be the class of representations π of G such that $p_A^\pi p_B = p_{A \cap B}^\pi$ for all algebraically closed subsets $A, B \subseteq N$.*

1. *For every $n \in \mathbb{N}$, the representation $\lambda_n: G \curvearrowright \ell^2(N^n)$ is in $\mathcal{D}$.*
2. *$\mathcal{D}$ is stable under taking subrepresentations and arbitrary sums.*

Proof.

1. Let $n \in \mathbb{N}$ and let $\mathcal{H} := \ell^2(N^n)$. We first show that for every algebraically closed subsets $A \subseteq N$,

$$\mathcal{H}_A = \{\xi \in \mathcal{H}: \text{supp } \xi \subseteq A^n\}. \quad (5.1)$$

Indeed, let $\xi \in \mathcal{H}_A$. The set on the right side above being closed, we can assume that $\xi \in \mathcal{H}_{A_0}$ for some finite subset $A_0 \subseteq A$. Given $x \in \text{supp } \xi$, $\|\xi\|$ is constant and non-zero on $G_{A_0} \cdot x$ which must then be finite since $\xi \in \ell^2(N^n)$. In other words, $x \in \text{acl}_N(A_0)^n \subseteq A^n$.

Conversely, let $\xi \in \mathcal{H}$ be such that $\text{supp } \xi \subseteq A^n$ and let $\varepsilon > 0$. There exists a finite subset $E \subseteq A^n$ such that $\sum_{x \in A^n \setminus E} \|\xi(x)\|^2 < \varepsilon^2$. Let $\xi_\varepsilon \in \ell^2(N^n)$ be the

function defined by $\xi_\varepsilon(x) = \xi(x)$ if $x \in E$ and 0 otherwise. We will show that $\xi_\varepsilon \in \mathcal{H}_{A_0}$ where $A_0 := \bigcup E$ is a finite subset of A . Indeed, let $g \in G_{A_0}$ and $x \in N^n$. If $x \in E$, $g(x) = x$ and $\lambda_n(g)\xi_\varepsilon(x) = \xi_\varepsilon(g^{-1}x) = \xi_\varepsilon(x)$. Otherwise, $x \notin E$, $g(x) \notin E$ and $\lambda_n(g)\xi_\varepsilon(x) = 0 = \xi_\varepsilon(x)$. Thus $\xi_\varepsilon \in \mathcal{H}_{A_0} \subseteq \mathcal{H}_A$. Since $\mathcal{H}_A$ is closed and $\|\xi - \xi_\varepsilon\| < \varepsilon$, letting $\varepsilon \rightarrow 0$ yields $\xi \in \mathcal{H}_A$.

Now that (5.1) is proved, we see that p_A is the multiplication by the indicator map $\mathbb{1}_{A^n}$ of A^n whenever A is algebraically closed. Since intersections of algebraically closed sets are algebraically closed (Item 4 of Lemma 5.2.20), the Lemma follows.

2. Let $\pi \in \mathcal{D}$ and let ρ be a subrepresentation of G . Let $\mathcal{K} \subseteq \mathcal{H}$ be the underlying Hilbert spaces of ρ and π respectively, let q be the orthogonal projection onto $\mathcal{K}$, and let $A, B \subseteq N$ be algebraically closed subsets. Because $\mathcal{H}_A \cap \mathcal{K} = \mathcal{K}_A$, it can be checked pointwise that

$$p_A^\pi q = q p_A^\pi = p_A^\rho$$

and similarly for B and $A \cap B$. Thus, we also get by checking pointwise that

$$p_A^\rho p_B^\rho = p_A^\pi q p_B^\pi q = p_A^\pi p_B^\pi q = p_{A \cap B}^\pi q = p_{A \cap B}^\rho.$$

Let $(\pi_i : G \curvearrowright \mathcal{H}_i)_{i \in I}$ be a collection of representations in $\mathcal{D}$ and let $\pi := \bigoplus_{i \in I} \pi_i : G \curvearrowright \mathcal{H}$ where $\mathcal{H}$ is the Hilbert sum of $\mathcal{H}_i$'s. We first show that

$$\mathcal{H}_A = \overline{\bigoplus_{i \in I} \mathcal{H}_{iA}}. \quad (5.2)$$

Clearly, $\bigoplus_{i \in I} \mathcal{H}_{iA_0} \subseteq \mathcal{H}_{A_0} \subseteq \mathcal{H}_A$ for every finite subset $A_0 \subseteq A$. Since $\mathcal{H}_A$ is closed, the right to left inclusion holds. Let $\xi = (\xi_i)_{i \in I} \in \mathcal{H}_A$. Since $\overline{\bigoplus_{i \in I} \mathcal{H}_{iA}}$ is closed, we can assume that $\xi \in \mathcal{H}_{A_0}$ for some finite subset A_0 . Then, for every $g \in G_{A_0}$ and $i \in I$,

$$\pi_i(g)\xi_i = (\pi(g)\xi)_i = \xi_i$$

whence $\xi_i \in \mathcal{H}_{iA_0} \subseteq \mathcal{H}_{iA}$. Equation (5.2) is proved.

For every $i \in I$, let p_i be the orthogonal projection onto $\mathcal{H}_i$. Using Equation (5.2), one can check pointwise on $\mathcal{H}$ that:

$$p_A^\pi p_i = p_i p_A^\pi = p_A^{\pi_i}$$

and similarly for B and $A \cap B$. Moreover, $\text{id}_{\mathcal{H}} = \bigoplus_{i \in I} p_i$ and $p_i p_j = 0$ for all $i \neq j \in I$. Thus:

$$\begin{aligned} p_A^\pi p_B^\pi &= p_A^\pi \cdot \bigoplus_{i \in I} p_i \cdot p_B^\pi \cdot \bigoplus_{i \in I} p_i = \bigoplus_{i, j \in I} p_A^\pi p_B^\pi p_i p_j = \bigoplus_{i \in I} p_A^\pi p_i p_B^\pi p_i \\ &= \bigoplus_{i \in I} p_A^{\pi_i} p_B^{\pi_i} = \bigoplus_{i \in I} p_{A \cap B}^{\pi_i} = \bigoplus_{i \in I} p_{A \cap B}^\pi p_i = p_{A \cap B}^\pi \cdot \bigoplus_{i \in I} p_i = p_{A \cap B}^\pi \end{aligned}$$

which concludes the proof.

□

The next result is a corollary of the classification Theorem 5.2.28 and a reformulation of [JT22, Prop. 3.2].

Corollary 5.2.30. *Let $\Gamma \curvearrowright \Omega$ be a Roelcke precompact closed permutation group and let $C \subseteq \Omega_\Gamma$ be a finite subset. Let $\sigma: \Gamma \curvearrowright \mathcal{H}$ be a unitary representation of Γ_C . For all algebraically closed subsets A, B of Ω_Γ containing C ,*

$$\mathcal{H}_A \perp_{\mathcal{H}_{A \cap B}} \mathcal{H}_B,$$

i.e. $p_A p_B = p_{A \cap B}$.

Proof. Let $\pi := \text{Ind}_{\Gamma_C}^\Gamma(\sigma)$ and let Λ be the the unitary representation $\Gamma \curvearrowright \ell^2(\Omega_\Gamma)$. Since Γ is a Roelcke precompact closed permutation group, π is isomorphic to a subrepresentation of a direct sum of copies of Λ [JT22, Fact 3.1]. Moreover, $\sigma \subseteq \pi|_{\Gamma_C}$ by Lemma 5.2.26. Thus, using Item 2 of Lemma 5.2.29, we can assume that $\sigma = \Lambda|_{\Gamma_C}$. Item 1 of Lemma 5.2.29 yields

$$p_A^\Lambda p_B^\Lambda = p_{A \cap B}^\Lambda$$

for all algebraically closed subsets A, B of Ω_Γ . We conclude by observing that for every subset $D \subseteq \Omega_\Gamma$ containing C , we have $p_D^\sigma = p_D^\Lambda$. □

5.3 Aspherical representations of $U(\Gamma)$

Recall that a representation $\pi: U(\Gamma) \curvearrowright \mathcal{H}$ is aspherical if $\mathcal{H}_x = 0$ for every (equivalently, some) $x \in \mathbf{T}$. The content of this section is devoted to classifying the aspherical representations of $U(\Gamma)$, irreducible or not, when $\Gamma \curvearrowright \Omega$ is oligomorphic and 2-transitive. Let us briefly describe the strategy we will adopt to classify aspherical representations.

Remark 5.3.1. Let $\Gamma \curvearrowright \Omega$ be a Roelcke precompact closed permutation group. The last result of the previous section says that, if one is willing to see Γ as a permutation group on Ω_Γ instead, the class $\mathcal{D}$ defined in the Lemma 5.2.29 is the class of all the representations of Γ . This property, combined with the fact that $\mathcal{C}(\Omega_\Gamma)$ is a well founded lattice, is enough to recover the full classification Theorem from [Tsa12]. We will carry out a similar argument to classify the aspherical representations of $U(\Gamma)$ assuming that Γ is moreover 2-transitive. Indeed, we will achieve the following:

- In Subsection 5.3.1, we study the classifying subtrees, i.e. the definable, locally $U(\Gamma)$ -finite, and algebraically closed subtrees of $\mathbf{T}_\Gamma$. We show in particular that $\mathcal{C}(\mathbf{T}_\Gamma)$ is a well founded lattice (Proposition 5.3.7).
- In Subsection 5.3.2, we anticipate on an ingredient we will require later by describing the representation of a certain Roelcke precompact closed permutation.
- In Subsection 5.3.3, we will see that, up to viewing $U(\Gamma)$ as permutation group on $\mathbf{T}_\Gamma$ instead of $\mathbf{T}$, the class $\mathcal{D}$ defined in Lemma 5.2.29 contains all the aspherical representations of $U(\Gamma)$ (Lemma 5.3.17 and Corollary 5.3.20). We then state and prove the main result of this paper, Theorem 5.3.19.

5.3.1 The lattice of classifying subtrees

In order to capture the aspherical irreducible representations of $U(\Gamma)$, we start with the study of $\mathcal{C}(\mathbf{T}_\Gamma)$. Note that for every $X \subseteq \mathbf{T}_\Gamma$, any graph automorphism of $\mathbf{T}_\Gamma$ that fixes X also fixes the tree X generates. In particular, algebraically closed subsets of $\mathbf{T}_\Gamma$ are automatically trees. We will thus refer to elements of $\mathcal{C}(\mathbf{T}_\Gamma)$ as the *classifying subtrees* of $\mathbf{T}_\Gamma$.

Remark 5.3.2. Let $\Gamma \curvearrowright \Omega$ be an oligomorphic and 2-transitive closed permutation group. Then $(\mathcal{C}(\mathbf{T}_\Gamma), \subseteq, \wedge, \vee)$ where $X \wedge Y = X \cap Y$ and $X \vee Y = \text{acl}_{\mathbf{T}_\Gamma}(X \cup Y)$ is an ordered lattice. We postpone the proof of this fact to Proposition 5.3.7 below and start with some very elementary observations.

1. Let $X \in \mathcal{C}(\mathbf{T}_\Gamma)$ be non-empty. Then:

- $X \cap \mathbf{T} \neq \emptyset$,
- the set of leaves of X is non-empty,
- and all the leaves of X are in $\mathbf{T}_\Gamma \setminus \mathbf{T}$.

Indeed, any graph automorphism of $\mathbf{T}_\Gamma$ that fixes some vertex in $\mathbf{T}_\Gamma \setminus \mathbf{T}$ must also fix its unique neighbor. Since X is algebraically closed, it shows that $X \cap \mathbf{T} \neq \emptyset$. Moreover, by construction of Ω_Γ , we have $\Gamma_a \in \text{acl}_{\Omega_\Gamma}(a) \setminus \Omega$ for every $a \in \Omega_\Gamma$. Thus, since X is algebraically closed, X indeed has leaves and none of them can be in $\mathbf{T}$.

2. The smallest element of the lattice is the empty set. It is indeed algebraically closed by transitivity of $U(\Gamma) \curvearrowright \mathbf{T}$ (Item 1 of Lemma 5.2.7).
3. The atoms of $\mathcal{C}(\mathbf{T}_\Gamma) \setminus \{\emptyset\}$ are precisely the elements of

$$\mathcal{C}_0(\mathbf{T}_\Gamma) := \{\text{acl}_{\Omega_\Gamma}(\emptyset)_x : x \in \mathbf{T}\}.$$

More precisely, the elements of $\mathcal{C}_0(\mathbf{T}_\Gamma)$ are the direct successors of the empty tree and every non-empty classifying subtree of $\mathbf{T}_\Gamma$ contains some element of $\mathcal{C}_0(\mathbf{T}_\Gamma)$.

4. $\mathcal{C}(\mathbf{T}_\Gamma)$ and $\mathcal{C}_0(\mathbf{T}_\Gamma)$ are stable under the action of $U(\Gamma)$.

Proposition 5.3.3. *Let $\Gamma \curvearrowright \Omega$ be a closed permutation group. For every subtree X of $\mathbf{T}_\Gamma$ such that $X \cap \mathbf{T} \neq \emptyset$, the following equivalences hold:*

1. X is locally $U(\Gamma)$ -finite $\iff X \cap \mathbf{T}$ is finite and $E_X(x)$ is locally Γ -finite in Ω_Γ for every $x \in X \cap \mathbf{T}$.
2. X is algebraically closed $\iff E_X(x)$ is algebraically closed in Ω_Γ for every $x \in X \cap \mathbf{T}$.

Moreover, if $X \cap \mathbf{T}$ is finite:

3. X is definable $\iff E_X(x)$ is definable in Ω_Γ for every $x \in X \cap \mathbf{T}$.

Proof.

1. Assume X is locally $\mathbf{U}(\Gamma)$ -finite. Then $X \cap \mathbf{T}$ is indeed finite since $\mathbf{T}$ is a single orbit under the action of $\mathbf{U}(\Gamma)$ (Item 1 of Lemma 5.2.7). Let $x \in X \cap \mathbf{T}$ and let A be the subset of Ω_Γ corresponding to $E_X(x)$. Let us show that A is locally Γ -finite in Ω_Γ .

Fix $a \in \Omega_\Gamma$. Then $\Gamma \cdot a \simeq \mathbf{U}(\Gamma)_x \cdot a_x$ by Corollary 5.2.14, hence:

$$\Gamma \cdot a \cap A \simeq \mathbf{U}(\Gamma)_x \cdot a_x \cap V_X(x) \subseteq \mathbf{U}(\Gamma) \cdot a_x \cap X.$$

Since X is locally $\mathbf{U}(\Gamma)$ -finite, $\mathbf{U}(\Gamma) \cdot a_x \cap X$ is finite and so is $\Gamma \cdot a \cap A$.

Conversely, assume that $X \cap \mathbf{T}$ is finite and that $E_X(z)$ is locally Γ -finite in Ω_Γ for every $z \in X \cap \mathbf{T}$. Let $x \in \mathbf{T}_\Gamma$. Since $X \cap \mathbf{T}$ is finite and $\mathbf{T}$ is $\mathbf{U}(\Gamma)$ -stable, it suffices to deal with the case $x \in \mathbf{T}_\Gamma \setminus \mathbf{T}$. Then x is a leaf, adjacent to a unique vertex $y \in \mathbf{T}$. Setting $a := \chi(x, y)$, we have:

$$\mathbf{U}(\Gamma) \cdot x \cap X = \bigcup_{z \in X \cap \mathbf{T}} \mathbf{U}(\Gamma) \cdot x \cap V_X(z) = \bigcup_{z \in X \cap \mathbf{T}} (\Gamma \cdot a)_z \cap V_X(z),$$

where the last equality is given by Corollary 5.2.14. Recalling the assumption we made on X , $\mathbf{U}(\Gamma) \cdot x \cap X$ is a finite union of finite sets hence finite.

2. Assume X is algebraically closed with respect to the action $\mathbf{U}(\Gamma) \curvearrowright \mathbf{T}_\Gamma$. Fix $x \in X$ and let A be the subset of Ω_Γ corresponding to $E_X(x)$. Let us show that A is algebraically closed.

Let $A' \subseteq A$ be finite. For every $a \in \text{acl}_{\Omega_\Gamma}(A')$, Corollary 5.2.14 yields:

$$\Gamma_{A'} \cdot a \simeq \mathbf{U}(\Gamma)_{A'_x} \cdot a_x.$$

In particular, $\text{acl}_{\Omega_\Gamma}(A')_x \subseteq \text{acl}_{\mathbf{T}_\Gamma}(A'_x) \subseteq \text{acl}_{\mathbf{T}_\Gamma}(X) = X$. Thus $\text{acl}_{\Omega_\Gamma}(A')_x \subseteq V_X(x) \cup \{x\}$ i.e. $\text{acl}_{\Omega_\Gamma}(A') \subseteq A$.

Conversely, assume that $E_X(x)$ is algebraically closed in Ω_Γ for every $x \in X \cap \mathbf{T}$. Let X' be a finite subset of X . Up to replacing X' with a larger finite subset of X , we assume it is a subtree of $\mathbf{T}_\Gamma$ and that $X' \cap \mathbf{T} \neq \emptyset$. Let $x \in \text{acl}_{\mathbf{T}_\Gamma}(X')$. Let y be the unique vertex of X' witnessing $d(x, y) = d(x, X')$. We reason by induction on $n = d(x, y)$.

– $n = 0$:

In this case, $x \in X'$ and there is nothing to prove.

– $n > 0$:

Let A be the subset of Ω_Γ corresponding to $E_X(y)$ and let A' be the one corresponding to $E_{X'}(y)$. Let z be the unique vertex of $\mathbf{T}_\Gamma$ adjacent to y that satisfies $d(z, x) = n - 1$. Since $\mathbf{U}(\Gamma)_{X'} \cdot x$ is finite and $\mathbf{T}_\Gamma$ is a tree, $\mathbf{U}(\Gamma)_{X'} \cdot z$ has to be finite too. Moreover, by Corollary 5.2.14,

$$\mathbf{U}(\Gamma)_{X'} \cdot z \simeq \Gamma_{A'} \cdot \chi(y, z).$$

Thus $\chi(y, z) \in \text{acl}_{\Omega_\Gamma}(A') \subseteq A$ i.e. $z \in X$ and we can add z to X' . The induction hypothesis now applies, yielding $x \in X$.

3. Assume that X is definable (the finiteness hypothesis is not needed for this direction). Let $x \in X \cap \mathbf{T}$ and let A be the subset of Ω_Γ corresponding to $E_X(x)$. By Proposition 5.2.13, the image of $\mathbf{U}(\Gamma)_{\{X\}} \cap \mathbf{U}(\Gamma)_x$ under the restriction map $g \mapsto g_x$ is open. Since, clearly, it is included in $\Gamma_{\{A\}}$, A is definable in Ω_Γ .

Conversely, let X be a subtree of $\mathbf{T}_\Gamma$ such that $X \cap \mathbf{T}$ is non-empty and finite with $E_X(x)$ definable in Ω_Γ for every $x \in X \cap \mathbf{T}$. Note that:

$$\mathbf{U}(\Gamma)_{\{X\}} \supseteq \bigcap_{x \in X \cap \mathbf{T}} \mathbf{U}(\Gamma)_{\{V_X(x) \cup \{x\}\}}.$$

The intersection being finite, it suffices to show that for every definable subset $A \subseteq \Omega_\Gamma$, $\mathbf{U}(\Gamma)_{\{A_x\}}$ is open in $\mathbf{U}(\Gamma)$. Now, if A is a definable subset of Ω_Γ , then $\mathbf{U}(\Gamma)_{\{A_x\}}$ is the inverse image of $\Gamma_{\{A\}}$ under the continuous restriction map $\mathbf{U}(\Gamma)_x \rightarrow \Gamma$.

□

The previous lemma directly implies the following local characterization of classifying subtrees.

Proposition 5.3.4. *Let $\Gamma \curvearrowright \Omega$ be a closed permutation group and let X be a non-empty subtree of $\mathbf{T}_\Gamma$. The following assertions are equivalent:*

- (i) $X \in \mathcal{C}(\mathbf{T}_\Gamma)$,
- (ii) $X \cap \mathbf{T}$ is finite, non-empty, and for every $x \in X \cap \mathbf{T}$, $E_X(x) \in \mathcal{C}(\Omega_\Gamma)$.

Similarly to what happens with elements of $\mathcal{C}(\Omega_\Gamma)$ in Ω_Γ , classifying subtrees of $\mathbf{T}_\Gamma$ are often infinite but are algebraic closures of finite subtrees if $\Gamma \curvearrowright \Omega$ is oligomorphic. In what follows, given a subtree X of $\mathbf{T}_\Gamma$ and a subgroup G of $\mathbf{U}(\Gamma)$, we will say that X is G -stable if $G \leq \mathbf{U}(\Gamma)_{\{X\}}$.

Lemma 5.3.5. *Let $\Gamma \curvearrowright \Omega$ be an oligomorphic closed permutation group and let X be a classifying subtree of $\mathbf{T}_\Gamma$.*

1. *There exists a finite subtree $X_0 \subseteq X$ such that $\text{acl}_{\mathbf{T}_\Gamma}(X_0) = X$.*
2. *Every such X_0 satisfies $\mathbf{U}(\Gamma)_{X_0} \leq \mathbf{U}(\Gamma)_{\{X\}}$ with finite index.*
3. *For every such X_0 , there exists an $\mathbf{U}(\Gamma)_{\{X\}}$ -stable finite subtree $X'_0 \subseteq X$ containing X_0 . In particular, $\mathbf{U}(\Gamma)_{X'_0}$ is a normal subgroup of $\mathbf{U}(\Gamma)_{\{X\}}$ with finite index.*

Proof. Using Item 1 in Proposition 5.2.21, for every $x \in X \cap \mathbf{T}$ there exists $e^x \in \Omega_\Gamma$ such that $E_X(x) = \text{acl}_{\Omega_\Gamma}(e^x)_x$. Define:

$$X_0 := (X \cap \mathbf{T}) \cup \{e^x_x : x \in X \cap \mathbf{T}\}.$$

Since $X \cap \mathbf{T}$ is finite by Proposition 5.3.4, X_0 is a finite subtree of X . Fix $x \in X \setminus X_0$. We will show that $x \in \text{acl}_{\mathbf{T}_\Gamma}(X_0)$.

Necessarily, x is an element of $\mathbf{T}_\Gamma \setminus \mathbf{T}$ hence a leaf in X . Let y be its unique neighbor, which is an element of $X \cap \mathbf{T}$. Then, using Corollary 5.2.14:

$$\mathbf{U}(\Gamma)_{X_0} \cdot x \simeq \Gamma_{e^y} \cdot \chi(x, y),$$

which is finite by the choice of e^y . Thus $x \in \text{acl}_{\mathbf{T}_\Gamma}(X_0)$ and $X \subseteq \text{acl}_{\mathbf{T}_\Gamma}(X_0)$. The reverse inclusion holds since X is algebraically closed.

The fact that $\text{acl}_{\mathbf{T}_\Gamma}(X_0) = X$ readily implies that $\mathbf{U}(\Gamma)_{X_0} \subseteq \mathbf{U}(\Gamma)_{\{X\}}$. For the finite index assertion, note that for every $x \in X_0$, $\mathbf{U}(\Gamma)_{\{X\}} \cdot x \subseteq F_x := (\mathbf{U}(\Gamma) \cdot x) \cap X$ which is finite. Now, writing $X_0 = \{x_1, \dots, x_n\}$ in an injective way, we have by Item 1 of Lemma 5.2.19 that $\mathbf{U}(\Gamma)_{\{X\}}/\mathbf{U}(\Gamma)_{X_0} \simeq \mathbf{U}(\Gamma)_{\{X\}} \cdot (x_1, \dots, x_n) \subseteq F_{x_1} \times \dots \times F_{x_n}$. In particular, $\mathbf{U}(\Gamma)_{\{X\}}/\mathbf{U}(\Gamma)_{X_0}$ is finite. Thus Items 1 and 2 are proved.

For the last statement,

$$X'_0 := \bigcup (\mathbf{U}(\Gamma)_{\{X\}} \cdot X_0) \subseteq \bigcup_{1 \leq i \leq n} F_{x_i}$$

is finite, included in $\text{acl}_{\mathbf{T}_\Gamma}(X_0)$, and stable under the action of $\mathbf{U}(\Gamma)_{\{X\}}$. Replace X_0 with the larger set X'_0 . □

If $\Gamma \curvearrowright \Omega$ is transitive and has algebraicity over singletons, the algebraic closure in $\mathbf{T}$ of any edge of $\mathbf{T}$ is infinite, preventing the existence of non-empty classifying trees. This is also where infinite arity significantly differs from finite arity: the algebraic closure of any non-empty subset of $\mathbf{T}_d$ is always $\mathbf{T}_d$. However, if $\Gamma \curvearrowright \Omega$ is oligomorphic, the next result shows that this is the only obstruction. Recall that $\langle X \rangle$ denotes the subtree of $\mathbf{T}_\Gamma$ generated by a subset $X \subseteq \mathbf{T}_\Gamma$ and that 2-transitivity is much stronger than having "no algebraicity over singletons".

Proposition 5.3.6. *Let $\Gamma \curvearrowright \Omega$ be an oligomorphic closed permutation group with no algebraicity over singletons, i.e. such that $\text{acl}_\Omega(a) = \{a\}$ for every $a \in \Omega$. For every finite subset $X \subseteq \mathbf{T}_\Gamma$, $\text{acl}_{\mathbf{T}_\Gamma}(X)$ is a classifying subtree.*

More generally, if X is a classifying subtree of $\mathbf{T}_\Gamma$ and $Y \subseteq \mathbf{T}_\Gamma$ is either finite or a subset of a classifying subtree, $\text{acl}_{\mathbf{T}_\Gamma}(X \cup Y)$ is a classifying subtree.

Proof. Let X be a finite subset of $\mathbf{T}_\Gamma$. Since $\text{acl}_{\mathbf{T}_\Gamma}(X) = \text{acl}_{\mathbf{T}_\Gamma}(\langle X \rangle)$ and $\langle X \rangle$ is finite, we may assume that X is a tree. By transitivity of $\mathbf{U}(\Gamma) \curvearrowright \mathbf{T}$, $\text{acl}_{\mathbf{T}_\Gamma}(\emptyset) = \emptyset$ and we may assume that X is non-empty. Moreover, if $X \cap \mathbf{T} = \emptyset$, then X is a singleton $\{x\}$ where $x \in \mathbf{T}_\Gamma \setminus \mathbf{T}$ and there exists a unique $y \in \mathbf{T}$ such that $x \sim y$. Then $\mathbf{U}(\Gamma)_x = \mathbf{U}(\Gamma)_{\{x,y\}}$ hence $\text{acl}_{\mathbf{T}_\Gamma}(X) = \text{acl}_{\mathbf{T}_\Gamma}(X \cup \{y\})$ and we can assume that $X \cap \mathbf{T} \neq \emptyset$.

Claim. *After these simplifications, we have:*

$$\text{acl}_{\mathbf{T}_\Gamma}(X) \cap \mathbf{T} = \bigcup_{x \in X \cap \mathbf{T}} (\text{acl}_{\Omega_\Gamma}(E_X(x)) \cap \Omega)_x,$$

In particular, $\text{acl}_{\mathbf{T}_\Gamma}(X) \cap \mathbf{T}$ is finite.

Proof of the claim. Since $\Gamma \curvearrowright \Omega$ has no algebraicity on singletons, $\mathbf{U}(\Gamma)_X$ assigns an infinite orbit to every vertex $y \in \mathbf{T}$ such that $d(y, X) \geq 2$ (we are using the assumption that X is a tree). Thus, if $y \in (\text{acl}_{\mathbf{T}_\Gamma}(X) \cap \mathbf{T}) \setminus X$, then $d(y, X) = 1$. Since $X \cap \mathbf{T} \neq \emptyset$, we even have $d(y, X \cap \mathbf{T}) = 1$ and there exists a unique $x \in X \cap \mathbf{T}$ such that $x \sim y$. Moreover, seeing $E_X(x)$ as a finite subset of Ω_Γ , we have by Corollary 5.2.14

$$\mathbf{U}(\Gamma)_X \cdot y \simeq \Gamma_{E_X(x)} \cdot \chi(x, y)$$

which is finite by the choice of y . Thus $y \in (\text{acl}_{\Omega_\Gamma}(E_X(x)) \cap \Omega)_x$. The reverse inclusion also follows from Corollary 5.2.14.

Finally, for every $x \in X \cap \mathbf{T}$, $\text{acl}_{\Omega_\Gamma}(E_X(x)) \cap \Omega$ is finite by Items 1 and 2 of Proposition 5.2.21 hence so is $\text{acl}_{\mathbf{T}_\Gamma}(X) \cap \mathbf{T}$. $\square_{\text{claim}}$

By the above claim, we can assume that $\text{acl}_{\mathbf{T}_\Gamma}(X) \cap \mathbf{T} \subseteq X$. Viewing things in Ω_Γ and applying Corollary 5.2.14, we get that for every $x \in \text{acl}_{\mathbf{T}_\Gamma}(X) \cap \mathbf{T}$, $E_{\text{acl}_{\mathbf{T}_\Gamma}(X)}(x) = \text{acl}_{\Omega_\Gamma}(E_X(x))$. We recall Item 1 of Proposition 5.2.21 and conclude with Proposition 5.3.4

Suppose now X is a classifying subtree and Y is a finite subset of $\mathbf{T}_\Gamma$. By Lemma 5.3.5, there exists a finite subset $X_0 \subseteq \mathbf{T}_\Gamma$ such that $X = \text{acl}_{\mathbf{T}_\Gamma}(X_0)$. Then $\text{acl}_{\mathbf{T}_\Gamma}(X \cup Y) = \text{acl}_{\mathbf{T}_\Gamma}(X_0 \cup Y)$ and the first part of the proof applies. Similarly, if Y is a subset of a classifying subtree, apply Lemma 5.3.5 again to get Y_0 finite such that $Y \subseteq \text{acl}_{\mathbf{T}_\Gamma}(Y_0)$. Then $\text{acl}_{\mathbf{T}_\Gamma}(X \cup Y) \subseteq \text{acl}_{\mathbf{T}_\Gamma}(X_0 \cup Y_0)$. It is easily checked by the local characterization of Proposition 5.3.4 that $\text{acl}_{\mathbf{T}_\Gamma}(X \cup Y)$ is classifying. $\square$

We recall again that 2-transitivity is much stronger than having no algebraicity over singletons.

Proposition 5.3.7. *Let $\Gamma \curvearrowright \Omega$ be an oligomorphic closed permutation group with no algebraicity over singletons. Then $(\mathcal{C}(\mathbf{T}_\Gamma), \subseteq, \wedge, \vee)$ is a well-founded ordered lattice.*

Proof. If $X, Y \in \mathcal{C}(\mathbf{T}_\Gamma)$, the local characterization from Proposition 5.3.4 readily implies that $X \cap Y \in \mathcal{C}(\mathbf{T}_\Gamma)$. Moreover, $\text{acl}_{\mathbf{T}_\Gamma}(X \cup Y)$ also belongs to $\mathcal{C}(\mathbf{T}_\Gamma)$ by Proposition 5.3.6. Thus, the ordered lattice $(\mathcal{C}(\mathbf{T}_\Gamma), \subseteq, \wedge, \vee)$ is well defined.

We now turn to well-foundedness. Assume $(X_n)_{n \in \mathbb{N}} \in \mathcal{C}(\mathbf{T}_\Gamma)^{\mathbb{N}}$ is a decreasing sequence. Then $(X_n \cap \mathbf{T})_{n \in \mathbb{N}}$ is a decreasing sequence of finite subsets of $\mathbf{T}$ (Proposition 5.3.4) hence stationary. We can thus assume it is constant and denote by Z its unique value. By Item 3 of Proposition 5.2.21, the sequence $(E_{X_n}(x))_{n \in \mathbb{N}}$ is stationary for every $x \in Z$. Since Z is finite, $(X_n)_{n \in \mathbb{N}}$ itself is stationary. $\square$

5.3.2 An extension of Γ by a compact group

We now discuss the representation theory of a permutation group that will arise in a later proof. **Until the end of this subsection, we fix an arbitrary vertex $x \in \mathbf{T}$ and a 2-transitive oligomorphic permutation Γ on a countable set Ω .**

Definition 5.3.8. We define the following subtree of $\mathbf{T}_\Gamma$:

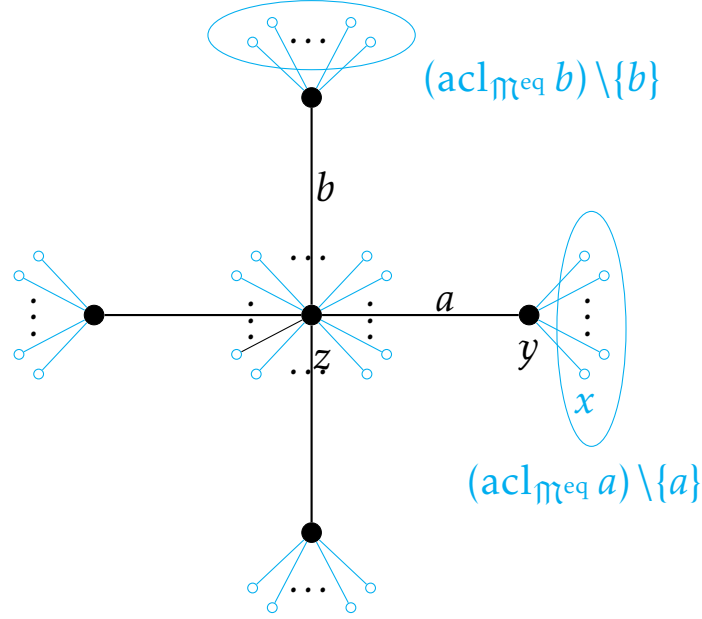
$$\Omega := \text{acl}_{\mathbf{T}_\Gamma}(V_{\mathbf{T}_\Gamma}(x)).$$

The group $U(\Gamma)_x$ acts on Ω and we denote by $\Delta \leq \text{Sym}(\Omega)$ the induced permutation group on Ω . We still denote by χ the restriction of the Ω_Γ -coloring of $\mathbf{T}_\Gamma$ to Ω .

Using Proposition 5.3.3, we also have:

$$\Omega = (\Omega_\Gamma)_x \cup \bigcup_{y \in \mathbf{T}} (\text{acl}_{\Omega_\Gamma}(\chi(x, y)))_y.$$

In particular, since $\text{acl}_{\Omega_\Gamma}(a) \cap \Omega = \{a\}$ for every $a \in \Omega$ by 2-transitivity, the subtree Ω has diameter 4 and $\Omega \cap \mathbf{T}$ has diameter 2. See Figure 5.2.


 Figure 5.2: The structure Ω .

Proposition 5.3.9. *The permutation group Δ on Ω is Roelcke precompact and closed in $\text{Sym}(\Omega)$.*

Proof. The topological group underlying Δ is Roelcke precompact as the continuous image of the Roelcke precompact group $\mathbf{U}(\Gamma)_x$. Let us now show that Δ is closed in $\text{Sym}(\Omega)$. Consider the following continuous restriction morphisms:

- $r_\Omega: \mathbf{U}(\Gamma)_x \longrightarrow \text{Sym}(\Omega)_x$, so that $\Delta = r_\Omega(\mathbf{U}(\Gamma)_x)$,
- $r_x: \text{Sym}(\Omega)_x \longrightarrow \text{Sym}(\Omega_\Gamma)$ which only keeps the local action at x ,
- $r_a: \Gamma_a \longrightarrow \text{Sym}(\text{acl}_{\Omega_\Gamma}(a))$ for every $a \in \Omega$,
- $s_y: \text{Sym}(\Omega)_{x,y} \longrightarrow \text{Sym}(\text{acl}_{\Omega_\Gamma}(\chi(x,y)))$ for every $y \in V_\Gamma(x)$.

Let $(\delta_n)_{n \in \mathbb{N}} \in \Delta^\mathbb{N}$ and $\delta \in \text{Sym}(\Omega)_x$ be such that $\delta_n \longrightarrow \delta$. Then $r_x(\delta_n) \longrightarrow r_x(\delta)$. Now, $r_x(\Delta) \subseteq \Gamma$ which is closed in $\text{Sym}(\Omega_\Gamma)$ by Proposition 5.2.10. Thus $\gamma := r_x(\delta) \in \Gamma$. Let $g \in \mathbf{U}(\Gamma)_x$ be the output of Lemma 5.2.3 applied to the constant sequence $(\gamma)_{y \in \Gamma}$ while requiring x to be fixed, and let $\alpha := r_\Omega(g) \in \Delta$. Then $\delta\alpha^{-1}$ fixes pointwise all of $V_\Omega(x)$.

Next, we claim that $r_a(\Gamma_a)$ is also closed in $\text{Sym}(\text{acl}_{\Omega_\Gamma}(a))$ for every $a \in \Omega$. Indeed, given $a \in \Omega$, every orbit of the action $\Gamma_a \curvearrowright \text{acl}_{\Omega_\Gamma}(a)$ is finite hence $r_a(\Gamma_a)$ is relatively compact in $\text{Sym}(\text{acl}_{\Omega_\Gamma}(a))$. Since Γ_a is Roelcke precompact, we conclude by applying [ET16, Lem. 2.9].

Finally, let $y \in V_\Gamma(x)$ and let $a := \chi(x,y) \in \Omega$. Since $\delta_n\alpha^{-1}$ fixes y for n large enough, the sequence $(s_y(\delta_n\alpha^{-1}))_{n \in \mathbb{N}}$ is eventually well defined in $s_y(\Delta_y)$ and converges in $\text{Sym}(\text{acl}_{\Omega_\Gamma}(a))$. Clearly, $s_y(\Delta_y) \subseteq r_a(\Gamma_a)$ which is closed by the previous paragraph. Thus, there exists $\gamma_y \in \Gamma_a$ such that $r_a(\gamma_y) = s_y(\delta\alpha^{-1})$.

Let $\gamma_x := \text{id}_\Omega$ and let $h \in U(\Gamma)_x$ be the output of Corollary 5.2.4 applied to $(\gamma_y)_{y \in \Omega \cap \Gamma}$ while requiring x to be fixed. Then $r_\Omega(h) = \delta \alpha^{-1}$ hence $\delta = r_\Omega(h) \alpha = r_\Omega(h) r_\Omega(g) \in \Delta$. We have thus shown that Δ is closed. $\square$

Similarly as in Proposition 5.2.15, there is a topological isomorphism

$$\Delta \simeq \Gamma \ltimes \prod_{a \in \Omega} \Gamma_a / \Gamma_{\text{acl}_{\Omega_\Gamma}(a)}$$

for the wreath-product action, i.e. the action such that for all $\gamma \in \Gamma$ and $k = (k_a)_{a \in \Omega} \in \prod_{a \in \Omega} \Gamma_a / \Gamma_{\text{acl}_{\Omega_\Gamma}(a)}$,

$$\gamma \cdot k = \left(\gamma \cdot k_{\gamma^{-1}(a)} \cdot \gamma^{-1} \right)_{a \in \Omega}.$$

The proof of this fact is very similar to that of Proposition 5.2.15. Since we will not use this fact, we omit the proof.

The permutation group Δ being Roelcke precompact and closed, its representation theory is described by the classification recalled in Theorem 5.2.28. In particular, if Ω_Δ is as in Definition 5.2.9, the enriched action $\Gamma \curvearrowright \Omega_\Delta$ captures the whole representation theory of Δ by Corollary 5.2.30. In the proof of our main theorem, we will need the stronger fact that the representation theory of Δ is fully captured by the original action $\Delta \curvearrowright \Omega$.

The goal of this section is thus to prove that, for Δ presented as a permutation group on Ω , the class $\mathcal{D}$ defined in Lemma 5.2.29 is the class of all the representations of Δ , i.e. that we do not need to pass to Ω_Δ . This will be proved in Proposition 5.3.13 below. The classification Theorem 5.2.28 making use of the open subgroup of Δ , our main task is thus to give a description of these open subgroups.

Lemma 5.3.10. *For every open subgroup $U \leq \Delta$, there exists $e \in \Omega_\Gamma$ such that $U \leq \Delta_{e_x}$ with finite index.*

Proof. Let U be an open subgroup of Δ . The restriction morphism $r: \Delta \rightarrow \Gamma$ which only keeps the local action around the center of Ω is open by Proposition 5.2.13. Thus $r(U)$ is an open subgroup of Γ hence of the form Γ_e for some $e \in \Omega_\Gamma$ (Proposition 5.2.10). Then $U \leq \Delta_{e_x}$ and it remains to show that the index is finite. Since $\Delta_{e_x} = r^{-1}(\Gamma_e) = r^{-1}(r(U)) = \ker r \cdot U$, it suffices to show that $\ker r$ is contained in a finite union of left cosets of U . Now, every orbit of the action $\ker r \curvearrowright \Omega$ is finite hence $\ker r$ is compact. To conclude extract a finite subcover from $\ker r \subseteq G \cdot U$. $\square$

In model theoretic terms, the following result essentially shows that Ω admits weak elimination of imaginaries. It gives a precise description of the open subgroups of Δ in terms of the action $\Delta \curvearrowright \Omega$.

Theorem 5.3.11. *Let U be an open subgroup of Δ . There exists $A \subseteq \Delta$ finite such that $\Delta_A \leq U$ with finite index.*

Proof. Since U is open, we start by fixing a finite subtree B of Ω such that $\Delta_B \leq U$. Using Lemma 5.3.10, we also pick $e \in \Omega_\Gamma$ such that $U \leq \Delta_{e_x}$ with finite index.

Using Lemma 5.2.3, we will denote by $\iota_x: \Gamma \rightarrow \Delta$ the continuous group morphism that sends γ to the unique element of Δ for which every local action is γ . Then $\iota_x^{-1}(U)$ is an open subgroup of Γ hence of the form Γ_f for some $f \in \Omega_\Gamma$ (Proposition 5.2.10). Since U has finite index in Δ_{e_x} , so does $\iota_x^{-1}(U)$ in $\iota_x^{-1}(\Delta_{e_x}) = \Gamma_e$. In particular, $f \in \text{acl}_{\Omega_\Gamma}(e)$. Equivalently, by Corollary 5.2.14, $f_x \in \text{acl}_\Omega(e_x)$.

Similarly, for every $y \in V_T(x)$, we will denote by $\iota_y: \Gamma_{\chi(x,y)} \rightarrow \Delta$ the continuous group morphism that sends γ to the unique element of Δ for which every local action is $\text{id}_{\Omega_\Gamma}$, except at a_x where it is γ . For every $y \in B \cap V_T(x)$, $\iota_y^{-1}(U)$ is an open subgroup of $\Gamma_{\chi(x,y)}$ hence of the form Γ_{e^y} for some $e^y \in \Omega_\Gamma$. Again, since U has finite index in Δ_{e_x} , so does $\iota_y^{-1}(U)$ in $\iota_y^{-1}(\Delta_{e_x}) = \Gamma_{\chi(x,y)}$. Thus, $e^y \in \text{acl}_{\Omega_\Gamma}(\chi(x,y))$ and, in particular, $e^y \in \Omega$.

Out of all the e^y defined above, we only keep those for which y belongs to $\text{acl}_\Omega(e_x)$. More precisely, we set

$$A := \{e_x, f_x\} \cup \{e^y: y \in V_T(x) \cap B \cap \text{acl}_\Omega(e_x)\} \cup \{e_x^y: y \in V_T(x) \cap B \cap \text{acl}_\Omega(e_x)\},$$

which is a finite subset of $\text{acl}_\Omega(e_x)$ and show that $\Delta_A \leq U$ with finite index.

Let $g \in \Delta_A$. Step by step, by tweaking the local action of g at every $y \in V_T(x)$, we will produce $u \in U$ such that $gu = \iota_x(g_x)$. Since $g_x \in \Gamma_f = \iota_x^{-1}(U)$, it will imply that $g \in U$.

Fixing $y \in V_A(x) \cap T$, the element $u := \iota_y(g_y^{-1}g_x)$ is well defined. Moreover, $g_y^{-1}g_x \in \Gamma_{e^y}$ hence, by the choice of e^y , we have $u \in U$. Thus, replacing g with gu , we can assume that $g_y = g_x$ without altering any other local action of g . Since A is finite, we will in fact assume that $g_y = g_x$ for every $y \in V_A(x) \cap T$.

Next, using Corollary 5.2.4, let v be the unique element of Δ such that:

- $v(x) = x$,
- $v_x = \text{id}_{\Omega_\Gamma}$,
- $v_y = \text{id}_{\Omega_\Gamma}$ for every $y \in V_B(x) \cap T$,
- $v_y = g_y^{-1}g_x$ for every $y \in V_T(x) \setminus B$.

Visibly, v fixes B hence $v \in U$. We replace g with gv and note that the local actions at elements of $A \cap T$ are left untouched. We thus now have $g_y = g_x$ for every $y \in V_T(x) \setminus (B \setminus A)$. Moreover, g_x was never altered in the process so far.

It only remains to deal with the local action of g at elements of $(B \setminus A) \cap V_T(x)$. Let $A' := A \cap V_\Omega(x)$. Then $e_x \in A' \subseteq \text{acl}_\Omega(e_x)$ hence e_x and A' have the same algebraic closure in Ω and $\Delta_{A'}$ acts on $\Omega \setminus \text{acl}_\Omega(e_x)$. In particular, since $\iota_x(\Gamma_{\chi(x,A')}) \leq \Delta_{A'}$, the action $\iota_x(\Gamma_{\chi(x,A')}) \curvearrowright \Omega \setminus \text{acl}_\Omega(e_x)$ is well defined. Moreover, for every $y \in V_T(x)$,

$$\iota_x(\Gamma_{\chi(x,A')}) \cdot y \simeq \Gamma_{\chi(x,A')} \cdot \chi(x,y)$$

hence every orbits of the action $\iota_x(\Gamma_{\chi(x,A')}) \curvearrowright \Omega \setminus \text{acl}_\Omega(e_x)$ is infinite. Applying Neumann's Lemma [BBMN76, Th. 1] to the finite set $B \setminus A \subseteq \Omega \setminus \text{acl}_\Omega(e_x)$ provides $w \in \iota_x(\Gamma_{\chi(x,A')})$ such that $w(B \setminus A) \cap B \setminus A = \emptyset$. Then w fixes $V_T(x) \cap A$ and gw satisfies:

- $(gw)_x = g_{w(x)}w_x = g_x\gamma$,

- for every $y \in V_A(x) \cap \mathbf{T}$, $(gw)_y = g_{w(y)}w_y = g_y\gamma = g_x\gamma = (gw)_x$,
- for every $y \in V_B(x) \cap \mathbf{T} \setminus A$, we have $w(y) \notin B$ hence $(gw)_y = g_{w(y)}w_y = g_x\gamma = (gw)_x$.

Moreover, since $f \in \chi(x, A')$, we have $w \in \iota_x(\Gamma_f) \subseteq U$. Since both g and w fix f_x , we also have $\iota((gw)_x) \in U$. Finally, as $(gw)_{y \in \Omega \cap \mathbf{T}}$ is constant on B , we have $\iota((gw)_x)^{-1}gw \in \Delta_B \subseteq U$ hence $g \in U$.

We have shown that $\Delta_A \leq U \leq \Delta_{e_x}$. Since A is a finite subset of $\text{acl}_\Omega(e_x)$, Δ_A has finite index in Δ_{e_x} by Item 2 of Lemma 5.2.19. A fortiori, Δ_A has finite index in U . □

Combining the previous result with the classification recalled in Theorem 5.2.28, we show that the representation theory of Δ is fully captured by the action $\Delta \curvearrowright \Omega$. For every $n \in \mathbb{N}$, there is a diagonal action $\Delta \curvearrowright \Omega^n$ and we will denote by λ_n the canonical representation of Δ on $\ell^2(\Omega^n)$.

Corollary 5.3.12. *Let π be a representation of Δ . There exists a collection $(n_i)_{i \in I}$ of integers such that*

$$\pi \subseteq \bigoplus_{i \in I} \lambda_{n_i}.$$

Proof. Recall that Δ is Roelcke precompact by Proposition 5.3.9. By Item 1 of Theorem 5.2.28, we can replace π with an irreducible subrepresentation. By Item 3 of Theorem 5.2.28, we can assume that π is a subrepresentation of $\lambda_{\Delta/U} : \Delta \curvearrowright \ell^2(\Delta/U)$. By Theorem 5.3.11, there exists a finite subset $A \subseteq \Omega$ such that $\Delta_A \leq U$ with finite index. Then $\lambda_{\Delta/U}$ is a subrepresentation of $\lambda_{\Delta/\Delta_A}$ hence we can replace π with $\lambda_{\Delta/\Delta_A}$. Write $A = \{a_1, \dots, a_n\}$ in an injective way and let $\bar{a} := (a_1, \dots, a_n)$. Then $\Delta/\Delta_A \simeq \Delta \cdot \bar{a} \subseteq \Omega^n$ hence $\pi = \lambda_{\Delta/\Delta_A} \subseteq (\Delta \curvearrowright \ell^2(\Delta \cdot \bar{a})) \subseteq (\Delta \curvearrowright \ell^2(\Omega^n)) = \lambda_n$. □

We conclude this subsection with the precise statement we will need in order to prove our main result.

Proposition 5.3.13. *Let C be a finite subtree of Ω and let $\sigma : \Delta_C \curvearrowright \mathcal{H}$ be a unitary representation of Δ_C . For all algebraically closed subsets $A, B \subseteq \Omega$ containing C ,*

$$\mathcal{H}_A \perp_{\mathcal{H}_{A \cap B}} \mathcal{H}_B,$$

i.e. $p_A p_B = p_{A \cap B}$.

Proof. Let $\pi := \text{Ind}_{\Delta_C}^\Delta(\sigma)$. By Lemma 5.2.26, $\sigma \subseteq \pi|_{\Delta_C}$. By Item 2 of Lemma 5.2.29, we can assume that $\sigma = \pi|_{\Delta_C}$. In this case, $p_D^\sigma = p_D^\pi$ for every subset D of Ω containing C . Apply Corollary 5.3.12 and both Items of Lemma 5.2.29 to the representation π of Δ . □

5.3.3 The classification theorem

We now turn to the classification of the aspherical representations of $\mathbf{U}(\Gamma)$ where $\Gamma \curvearrowright \Omega$ is 2-transitive and oligomorphic. We follow the strategy sketched in Remark 5.3.1. Using the classifying subtrees, we first establish the list of the aspherical irreducible representations of $\mathbf{U}(\Gamma)$ and identify repetitions within it. The key to achieving this first step lies in the following lemma, which will allow us to compute invariant spaces associated with induced representations. In what follows, a representation π of a topological group G is trivial on a subgroup H if $H \subseteq \ker \pi$.

Lemma 5.3.14. *Let $\Gamma \curvearrowright \Omega$ be an oligomorphic and 2-transitive closed permutation group. Let $X_0 \subseteq \mathbf{T}_\Gamma$ be a finite subtree, and let*

$$\pi := \text{Ind}_{\mathbf{U}(\Gamma)_{\{X\}}}^{\mathbf{U}(\Gamma)}(\sigma): \mathbf{U}(\Gamma) \curvearrowright \mathcal{H}$$

where $X := \text{acl}_{\mathbf{T}_\Gamma}(X_0)$ and σ is a representation of $\mathbf{U}(\Gamma)_{\{X\}}$ which is trivial on $\mathbf{U}(\Gamma)_{X_0}$. Then, for every $Y \in \mathcal{C}(\mathbf{T}_\Gamma)$,

$$\mathcal{H}_Y = \{\xi \in \mathcal{H}: \forall g \in \mathbf{U}(\Gamma), [\xi(g) \neq 0 \Rightarrow g(X) \subseteq Y]\}.$$

In particular,

$$\mathcal{H}_X = \{\xi \in \mathcal{H}: \text{supp}(\xi) \subseteq \mathbf{U}(\Gamma)_{\{X\}}\}$$

and p_X is the multiplication by the indicator map of $\mathbf{U}(\Gamma)_{\{X\}}$.

Proof. Write $\tilde{\mathcal{H}}_Y := \{\xi \in \mathcal{H}: \forall g \in \mathbf{U}(\Gamma), [\xi(g) \neq 0 \Rightarrow g(X) \subseteq Y]\}$. We first prove that $\mathcal{H}_Y \subseteq \tilde{\mathcal{H}}_Y$. Let $Y' \subseteq Y$ be a finite subtree. Since $\tilde{\mathcal{H}}_Y$ is closed, it suffices to prove that $\mathcal{H}_{Y'} \subseteq \tilde{\mathcal{H}}_Y$.

Let $\xi \in \mathcal{H}_{Y'}$ and assume $g \in \mathbf{U}(\Gamma)$ is such that $\xi(g) \neq 0$. Then $\|\xi(\cdot)\|$ is constant on $\mathbf{U}(\Gamma)_{Y'} \cdot g \cdot \mathbf{U}(\Gamma)_{\{X\}}$. Recalling that ξ is ℓ^2 , we get that $(\mathbf{U}(\Gamma)_{Y'} \cdot g \cdot \mathbf{U}(\Gamma)_{\{X\}}) / \mathbf{U}(\Gamma)_{\{X\}}$ is finite. By Item 2 of Lemma 5.3.5, we have $\mathbf{U}(\Gamma)_{X_0} \leq \mathbf{U}(\Gamma)_{\{X\}}$ with finite index thus $(\mathbf{U}(\Gamma)_{Y'} \cdot g \cdot \mathbf{U}(\Gamma)_{X_0}) / \mathbf{U}(\Gamma)_{X_0}$ is also finite. By Item 1 of Lemma 5.2.19, writing $X_0 = \{x_1, \dots, x_n\}$ in an injective way, we have $(\mathbf{U}(\Gamma)_{Y'} \cdot g \cdot \mathbf{U}(\Gamma)_{X_0}) / \mathbf{U}(\Gamma)_{X_0} \simeq \mathbf{U}(\Gamma)_{Y'} \cdot g(x_1, \dots, x_n)$. Thus every element of $g(X_0)$ has a finite orbit under the action of $\mathbf{U}(\Gamma)_{Y'}$. In other words, $g(X_0) \subseteq \text{acl}_{\mathbf{T}_\Gamma}(Y') \subseteq \text{acl}_{\mathbf{T}_\Gamma}(Y) = Y$ hence $g(X) = \text{acl}_{\mathbf{T}_\Gamma}(g(X_0)) \subseteq Y$.

Turning to the reverse inclusion, let $\xi \in \tilde{\mathcal{H}}_Y$. By definition of induced representations, $\text{supp} \xi$ is a union of $\mathbf{U}(\Gamma)_{\{X\}}$ cosets and cutting it off in larger and larger finite sub-unions yields an approximation of ξ . Since $\mathcal{H}_Y$ is closed, we can thus assume that $\text{supp} \xi$ is a finite union of $\mathbf{U}(\Gamma)_{\{X\}}$ cosets. Because $\mathbf{U}(\Gamma)_{X_0}$ has finite index in $\mathbf{U}(\Gamma)_{\{X\}}$, we even have that $\text{supp} \xi$ is a finite union of $\mathbf{U}(\Gamma)_{X_0}$ -cosets and we can write

$$\text{supp}(\xi) = \bigcup_{1 \leq i \leq n} g_i \mathbf{U}(\Gamma)_{X_0}$$

for some $g_1, \dots, g_n \in \mathbf{U}(\Gamma)$.

Using Lemma 5.3.5, let $Y_0 \subseteq Y$ be a finite subtree such that $\text{acl}_{\mathbf{T}_\Gamma}(Y_0) = Y$ and define

$$Y' = Y_0 \cup \bigcup_{1 \leq i \leq n} g_i(X_0).$$

Since $g_1, \dots, g_n$ belong to $\text{supp}(\xi)$ and hence satisfy $g_i(X) \subseteq Y$ for every $i \leq n$, we have that Y' is a finite subset of Y . Let us show that $\xi \in \mathcal{H}_{Y'}$. To that aim, fix $u \in \mathbf{U}(\Gamma)_{Y'}$ and $g \in \mathbf{U}(\Gamma)$.

– If $g \notin g_1 \mathbf{U}(\Gamma)_{X_0} \cup \dots \cup g_n \mathbf{U}(\Gamma)_{X_0}$:

Then $u^{-1}g \notin u^{-1}g_1 \mathbf{U}(\Gamma)_{X_0} \cup \dots \cup u^{-1}g_n \mathbf{U}(\Gamma)_{X_0}$. Since for every $i \leq n$, $g_i(X_0) \subseteq Y'$, it follows that $u^{-1}g_i \mathbf{U}(\Gamma)_{X_0} = g_i \mathbf{U}(\Gamma)_{X_0}$ hence $u^{-1}g \notin \text{supp } \xi$. In other words:

$$(u \cdot \xi)(g) = \xi(u^{-1}g) = 0 = \xi(g).$$

– If $g \in g_i \mathbf{U}(\Gamma)_{X_0}$ for some $i \leq n$:

In this case, $u^{-1}g \in \mathbf{U}(\Gamma)_{Y'}g = g\mathbf{U}(\Gamma)_{g^{-1}(Y')}$. Since $g(X_0) = g_i(X_0) \subseteq Y'$, we have $X_0 \subseteq g^{-1}(Y')$. Thus $\mathbf{U}(\Gamma)_{g^{-1}(Y')} \subseteq \mathbf{U}(\Gamma)_{X_0}$ and there exists $v \in \mathbf{U}(\Gamma)_{X_0}$ such that $u^{-1}g = gv$. Then:

$$(u \cdot \xi)(g) = \xi(u^{-1}g) = \xi(gv) = \sigma(v^{-1})(\xi(g)) = \xi(g)$$

where the last equality follows from the assumption that σ is trivial on $\mathbf{U}(\Gamma)_{X_0}$.

As for the last statement, assume $X = Y$ and let $g \in \mathbf{U}(\Gamma)$ be such that $g(X) \subseteq X$. It suffices to show that $X \subseteq g(X)$. Let $x \in X$. Since X is locally $\mathbf{U}(\Gamma)$ -finite, $E := (\mathbf{U}(\Gamma) \cdot x) \cap X$ is finite. Now, $g(E) \subseteq E$ hence $g(E) = E$ by cardinality and $x \in g(X)$. $\square$

All we need before giving the complete list of the aspherical irreducible representations of $\mathbf{U}(\Gamma)$ is some more terminology. Let H, H' be conjugate subgroups of a topological group G , say $gHg^{-1} = H'$ for some $g \in G$. If τ is a representation of H' on a Hilbert space $\mathcal{K}$, its *conjugate representation* is $\tau^g: H \curvearrowright \mathcal{K}$ is the representation of H given by $\tau^g(h) = \tau(ghg^{-1})$ for every $h \in H$. The *trivial representation* of a topological group is its trivial action on $\mathbb{C}$. A vector in a representation $\pi: G \curvearrowright \mathcal{H}$ is *G-cyclic* if its orbit under the action of G spans $\mathcal{H}$ as a Hilbert space. A subspace $\mathcal{K} \subseteq \mathcal{H}$ is *G-cyclic* if $\bigcup \{\pi(g)\mathcal{K} : g \in G\}$ spans $\mathcal{H}$ as a Hilbert space.

Proposition 5.3.15. *Let $\Gamma \curvearrowright \Omega$ be an oligomorphic and 2-transitive closed permutation group. Let $X_0 \subseteq \mathbf{T}_\Gamma$ be a finite subtree, and let*

$$\pi := \text{Ind}_{\mathbf{U}(\Gamma)_{\{X\}}}^{\mathbf{U}(\Gamma)}(\sigma)$$

where $X := \text{acl}_{\mathbf{T}_\Gamma}(X_0)$ and σ is a representation of $\mathbf{U}(\Gamma)_{\{X\}}$ which is trivial on $\mathbf{U}(\Gamma)_{X_0}$.

1. The representation π is irreducible if and only if σ is irreducible.
2. Suppose that π is irreducible. Then π is spherical exactly when either
 - X is empty, in which case π is the trivial representation of $\mathbf{U}(\Gamma)$,
 - or $\mathbf{U}(\Gamma)_{\{X\}} = \mathbf{U}(\Gamma)_x$ for some $x \in \mathbf{T}$ and σ is the trivial representation of $\mathbf{U}(\Gamma)_x$, in which case π is isomorphic to $\mathbf{U}(\Gamma) \curvearrowright \ell^2(\mathbf{U}(\Gamma)/\mathbf{U}(\Gamma)_x)$.

3. Let $\rho := \text{Ind}_{\mathbf{U}(\Gamma)_{\{Y\}}}^{\mathbf{U}(\Gamma)}(\tau)$ be of the same form as π . Then ρ is isomorphic to π if and only if there exists $g \in \mathbf{U}(\Gamma)$ such that $g(X) = Y$ and $\sigma \simeq \tau^g$.

Proof.

1. It is a standard fact that induction preserves sums of representations, see e.g. [Bd20, Prop. 1.F.6(2)]. In particular, if σ is not irreducible, then π cannot be irreducible. Conversely, assume that σ is irreducible. We prove the irreducibility of π by showing that every non-zero vector in $\mathcal{H}$ is $\mathbf{U}(\Gamma)$ -cyclic. To this end, let $\xi \in \mathcal{H} \setminus \{0\}$. There exists $g \in G$ such that $\xi(g) \neq 0$. Up to replacing with ξ with $\pi(g)\xi$, we can assume that $\xi(1_{\mathbf{U}(\Gamma)}) \neq 0$. By Lemma 5.3.14, $p_X(\xi) \neq 0$. Now, by Lemma 5.2.25 and the Alaoglu-Birkhoff Theorem 5.2.24, $p_X(\xi)$ lies in the $\mathbf{U}(\Gamma)$ -cyclic hull of ξ . It thus suffices to prove the following claim:

Claim. *Every non-zero vector of $\mathcal{H}_X$ is $\mathbf{U}(\Gamma)$ -cyclic in $\mathcal{H}$.*

Proof of the claim. Using Lemmas 5.3.14 and 5.2.26, the representation

$$\pi|_{\mathbf{U}(\Gamma)_{\{X\}}} : \mathbf{U}(\Gamma)_{\{X\}} \curvearrowright \mathcal{H}_X$$

is isomorphic to σ hence irreducible. Thus, every non-zero vector of $\mathcal{H}_X$ is $\mathbf{U}(\Gamma)_{\{X\}}$ -cyclic in $\mathcal{H}_X$. By Lemma 5.3.14, $\mathcal{H}_X = \{\xi \in \mathcal{H} : \text{supp}(\xi) \subseteq \mathbf{U}(\Gamma)_{\{X\}}\}$ which is $\mathbf{U}(\Gamma)$ -cyclic in $\mathcal{H}$ by the definition of induction. The claim follows. $\square_{\text{claim}}$

2. Suppose π is spherical and let $x \in \mathbf{T}$. Write $A := \text{acl}_{\Omega_\Gamma}(\emptyset)$. Then $\mathcal{H}_x \subseteq \mathcal{H}_{A_x}$. Thus, given $\xi \in \mathcal{H}_x \setminus \{0\}$ and $g \in \mathbf{U}(\Gamma)$ such that $\xi(g) \neq 0$, we have $g(X) \subseteq A_x$ by Lemma 5.3.14. Since X is algebraically closed, either X is empty or $g(X) = A_x$ and $X \in \mathcal{C}_0(\mathbf{T}_\Gamma)$.

In the first case, X_0 is also empty, σ is the trivial representation of $\mathbf{U}(\Gamma)$ and induction does nothing. Thus π is indeed the trivial representation which is indeed spherical.

In the second case, we can assume without loss of generality that $x \in X$, i.e. that $X = A_x$. Then $\mathbf{U}(\Gamma)_{\{X\}} = \mathbf{U}(\Gamma)_x$ and we claim that $\xi(1_{\mathbf{U}(\Gamma)})$ is a non-zero invariant vectors of σ . Indeed, if $u \in \mathbf{U}(\Gamma)_x$, we have since $\xi \in \mathcal{H}_x$: $\sigma(u)(\xi(1_{\mathbf{U}(\Gamma)})) = \xi(u^{-1}) = (\pi(u)\xi)(1_{\mathbf{U}(\Gamma)}) = \xi(1_{\mathbf{U}(\Gamma)})$. Since σ is irreducible and has a fixed non-zero vector, σ must be the trivial representation. Moreover, inducing the trivial representation of $\mathbf{U}(\Gamma)_x$ to $\mathbf{U}(\Gamma)$ yields the quasi-regular representation $\mathbf{U}(\Gamma) \curvearrowright \ell^2(\mathbf{U}(\Gamma)/\mathbf{U}(\Gamma)_x)$ (see e.g. [Bd20, Def. 1.5]) which is indeed spherical.

3. Assume π and ρ are isomorphic and denote by $\mathcal{H}$ and $\mathcal{K}$ their respective underlying Hilbert space. By Lemma 5.3.14, $\mathcal{K}_Y \neq 0$. Since $\mathcal{K}_Y \simeq \mathcal{H}_Y$, there exists $\xi \in \mathcal{H}_Y \setminus \{0\}$. Let $g \in \mathbf{U}(\Gamma)$ be such that $\xi(g) \neq 0$. By Lemma 5.3.14, g satisfies $g(X) \subseteq Y$. By symmetry, we can find $h \in \mathbf{U}(\Gamma)$ such that $h(Y) \subseteq X$. Since X and Y are locally $\mathbf{U}(\Gamma)$ -finite, this implies $g(X) = Y$. In particular,

$g\mathbf{U}(\Gamma)_{\{X\}}g^{-1} = \mathbf{U}(\Gamma)_{\{Y\}}$. Finally, using Lemmas 5.3.14 and 5.2.26 again for the first and last isomorphisms:

$$\begin{aligned}\sigma &\simeq \left(\mathbf{U}(\Gamma)_{\{X\}} \overset{\pi}{\curvearrowright} \mathcal{H}_X \right) \simeq \left(\mathbf{U}(\Gamma)_{\{X\}} \overset{\pi'}{\curvearrowright} \mathcal{K}_X \right) \\ &\simeq \left(\mathbf{U}(\Gamma)_{\{X\}} \overset{\pi'^g}{\curvearrowright} \mathcal{K}_Y \right) = \left(\mathbf{U}(\Gamma)_{\{Y\}} \overset{\pi'}{\curvearrowright} \mathcal{K}_Y \right)^g \simeq \tau^g.\end{aligned}$$

The second isomorphism is the restriction of the one between π and π' . The third is $\pi(g)$.

The converse follows from the classical fact that inducing isomorphic representations yields isomorphic representations, see e.g. [Bd20, Prop. 1.F.6(1)].

□

In order to complete the classification, we need to show in particular that every aspherical irreducible representation of $\mathbf{U}(\Gamma)$ is of the above form, i.e. that the list is complete. We split the argument in two separate lemmas and deal with the most elementary case first, related to elements of $\mathcal{C}_0(\mathbf{T}_\Gamma)$.

Lemma 5.3.16. *Let $\Gamma \curvearrowright \Omega$ be an oligomorphic and 2-transitive closed permutation group and let $A := \text{acl}_{\Omega_\Gamma}(\emptyset)$. Let $\pi: \mathbf{U}(\Gamma) \curvearrowright \mathcal{H}$ be a unitary representation of $\mathbf{U}(\Gamma)$. For all $x \neq y \in \mathbf{T}$,*

$$\mathcal{H}_{A_x} \perp_{\mathcal{H}_x} \mathcal{H}_{A_y},$$

i.e. $p_{A_x}p_{A_y} = p_x$.

Proof. Let $x \neq y \in \mathbf{T}$. Let $A' \subseteq A$ be finite. By Lemma 5.2.25, it suffices to show that $p_{A'_x}p_{A'_y} = p_x$, i.e. that for all $\xi \in \mathcal{H}_{A'_x}$ and $\eta \in \mathcal{H}_{A'_y}$,

$$\langle \xi, p_x \eta \rangle = \langle \xi, \eta \rangle.$$

Let $\varepsilon > 0$. By the Alaoglu-Birkhoff Theorem 5.2.24, there exists $g_1, \dots, g_n \in \mathbf{U}(\Gamma)_x$ and $\lambda_1, \dots, \lambda_n \geq 0$ such that $\lambda_1 + \dots + \lambda_n = 1$ and:

$$\|p_x \eta - \sum_i \lambda_i \pi(g_i) \eta\| < \varepsilon.$$

Claim. *Let $X := A'_x$ and $Y := \langle A'_y \cup \{x\} \rangle$ be the tree generated by A'_y and x . Then:*

$$\mathbf{U}(\Gamma)_x = \mathbf{U}(\Gamma)_X \mathbf{U}(\Gamma)_Y.$$

Proof of the claim. We first show that $\Gamma = \Gamma_{A'} \Gamma_a$ for every $a \in \Omega$. Let $B := \bigcup \Gamma \cdot A'$ which is a finite subset of $\text{acl}_{\Omega_\Gamma}(\emptyset)$ that is stable under the action of Γ . Then Γ_B is normal in Γ hence $\Gamma_B \Gamma_a$ is a subgroup of Γ . Because $\Gamma \curvearrowright \Omega$ is 2-transitive, it is a fortiori primitive hence Γ_a is a maximal subgroup of Γ [Cam99, Th. 1.7]. Thus $\Gamma \subseteq \Gamma_B \Gamma_a \subseteq \Gamma_{A'} \Gamma_a$ since $A' \subseteq B$.

Next, let z be the unique neighbor of x in $[x, y]$ and let $a := \chi(x, z)$. By Proposition 5.2.13, the restriction morphism $r_x: h \mapsto h_x$ satisfies $r_x(\mathbf{U}(\Gamma)_X) = \Gamma_{A'}$ and $r_x(\mathbf{U}(\Gamma)_Y) = \Gamma_a$. Thus, for every $g \in \mathbf{U}(\Gamma)_x$, we have $g_x \in \Gamma = \Gamma_{A'} \Gamma_a = r_x(\mathbf{U}(\Gamma)_X) \cdot r_x(\mathbf{U}(\Gamma)_Y)$ and there exist $h \in \mathbf{U}(\Gamma)_X$ and $k \in \mathbf{U}(\Gamma)_Y$ such that $(hk)_x = g_x$. Visibly, $gk^{-1} \in \mathbf{U}(\Gamma)_X$ thus $g \in \mathbf{U}(\Gamma)_X \cdot k \subseteq \mathbf{U}(\Gamma)_X \cdot \mathbf{U}(\Gamma)_Y$. The reverse inclusion is trivial since $x \in X \cap Y$. □_{claim}

By the claim, there exist $h_1, \dots, h_n \in \mathbf{U}(\Gamma)_X$ and $k_1, \dots, k_n \in \mathbf{U}(\Gamma)_Y$ such that for every $i \leq n$, $g_i = h_i k_i$. It follows that $\langle \xi, \pi(g_i)\eta \rangle = \langle \pi(h_i^{-1})\xi, \pi(k_i)\eta \rangle = \langle \xi, \eta \rangle$ and:

$$|\langle \xi, p_x \eta \rangle - \langle \xi, \eta \rangle| = |\langle \xi, p_x \eta \rangle - \sum_i \lambda_i \langle \xi, \pi(g_i)\eta \rangle| = |\langle \xi, p_x \eta - \sum_i \lambda_i \pi(g_i)\eta \rangle| \leq \|\xi\| \cdot \varepsilon$$

Since ε is arbitrary, the result follows. $\square$

The following result is inspired by [Ol'80, Lem. 5.1] but involves significantly more technicalities. In particular, this is where we will use the results of Section 5.3.2 about the permutation group Δ .

Lemma 5.3.17. *Let $\Gamma \curvearrowright \Omega$ be an oligomorphic and 2-transitive closed permutation group. Let $X, Y \in \mathcal{C}(\mathbf{T}_\Gamma)$ be such that*

$$X \neq \emptyset, \quad X \notin \mathcal{C}_0(\mathbf{T}_\Gamma), \quad \text{and} \quad X \not\subseteq Y.$$

There exists a proper classifying subtree $X' \subseteq X$ such that for every representation $\pi: \mathbf{U}(\Gamma) \curvearrowright \mathcal{H}$ of $\mathbf{U}(\Gamma)$, we have $p_Y p_X = p_Y p_{X'}$.

Proof. First, note that we can always replace Y with a larger classifying subtree. Indeed, suppose $Y' \in \mathcal{C}(\mathbf{T}_\Gamma)$ is such that $Y \subseteq Y'$. Suppose also that there exists $X' \subseteq X$ such that for every representation $\pi: \mathbf{U}(\Gamma) \curvearrowright \mathcal{H}$ of $\mathbf{U}(\Gamma)$, we have $p_Y p_X = p_{Y'} p_{X'}$. Since $Y \subseteq Y'$ we have $p_Y p_{Y'} = p_Y$ and:

$$p_Y p_X = p_Y p_{Y'} p_{X'} = p_Y p_{Y'} p_{X'} = p_Y p_{X'}.$$

We will thus allow ourselves to enlarge Y several times in this proof.

Now, since $X \cap \mathbf{T}$ is finite by Proposition 5.3.4, X has finite diameter and there exists $x \in X \setminus Y$ at maximal distance from Y . Then x is a leaf in X . Since X is algebraically closed, we have $x \in \mathbf{T}_\Gamma \setminus \mathbf{T}$ by Item 1 of Remark 5.3.2. In particular, there exists a unique $y \in X \cap \mathbf{T}$ such that $x \sim y$. We distinguish two cases.

Case 1: $y \in Y$.

Define the subtree $X' \in \mathcal{C}(\mathbf{T}_\Gamma)$ as follows:

$$X' := X \setminus [V_X(y) \setminus Y].$$

In other words, X' is obtained from X by replacing $V_X(y)$ with $V_X(y) \cap V_Y(y)$. All of the vertices so removed are leaves in X by maximality of $d(x, Y)$, hence X' is a tree. Moreover $E_{X'}(y) = E_X(y) \cap E_Y(y)$ and every $x' \in X' \setminus \{y\}$ satisfies $E_{X'}(x') = E_X(x')$. Thus $X' \in \mathcal{C}(\mathbf{T}_\Gamma)$ by Proposition 5.3.4. X' is strictly contained in X since $x \notin X'$.

Claim. *Up to replacing Y with a larger classifying tree, we can assume that $X' = Y$.*

Proof of the claim. Let $Y' := \text{acl}_{\mathbf{T}_\Gamma}(Y \cup X')$. By Proposition 5.3.6, this is a classifying subtree of $\mathbf{T}_\Gamma$. Moreover, since $Y \cup X'$ was already a tree satisfying $E_{Y \cup X'}(y) = E_Y(y)$, applying Corollary 5.2.14 to finite subtrees of $Y \cup X'$ at y yields $E_{Y'}(y) = E_Y(y)$. In particular, we still have $x \notin Y'$. $\square_{\text{claim}}$

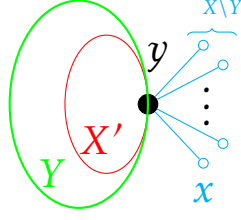


Figure 5.3: Situation after simplifications.

We are thus left with the following simplified situation, illustrated in Figure 5.3: $X \setminus Y$ is only comprised of neighbors of y . Note that the elements of $X \setminus Y$ are all leaves in X hence elements of $T_\Gamma \setminus T$ by Item 1 of Remark 5.3.2.

Using Lemma 5.3.5, let X_0, Y_0 and X'_0 be finite subtrees of X, Y and X' respectively such that:

$$\text{acl}_{T_\Gamma}(X_0) = X, \quad \text{acl}_{T_\Gamma}(Y_0) = Y, \quad \text{and} \quad \text{acl}_{T_\Gamma}(X'_0) = X'.$$

Up to replacing X_0, Y_0, X'_0 with larger finite trees, we can assume that $X_0 \cap Y_0 = X'_0$ and that $y \in X'_0$. Let $A := E_{X_0}(y)$, $B := E_{Y_0}(y)$ and $C := E_{X'_0}(y)$, seen as finite subsets of Ω_Γ . Necessarily, $A \cap B = C$ and:

$$\text{acl}_{\Omega_\Gamma}(A) = E_X(y), \quad \text{acl}_{\Omega_\Gamma}(B) = E_Y(y) \quad \& \quad \text{acl}_{\Omega_\Gamma}(C) = E_{X'}(y).$$

Define:

$$Z := X'_0 \cup (\Omega_\Gamma)_y = X_0 \cup (\Omega_\Gamma)_y.$$

Then $U(\Gamma)_{X'_0}$ leaves Z stable as a set hence acts on $\mathcal{H}_Z$. Since the kernel of the action $U(\Gamma)_{X'_0} \curvearrowright \mathcal{H}_Z$ contains $U(\Gamma)_Z$, the action factorizes into an action of the quotient group $U(\Gamma)_{X'_0}/U(\Gamma)_Z$. Now, $U(\Gamma)_{X'_0}/U(\Gamma)_Z$ identifies with Γ_C through the restriction morphism of Proposition 5.2.13. Factorizing and restricting π thus gives rise to the following representation:

$$\sigma: \Gamma_C \curvearrowright \mathcal{K} := \mathcal{H}_Z.$$

Similarly, $U(\Gamma)_{X_0}/U(\Gamma)_Z \simeq \Gamma_A$ and $U(\Gamma)_{Y_0}/(U(\Gamma)_{Y_0} \cap U(\Gamma)_Z) \simeq \Gamma_B$, both subgroups of Γ_C . Since that $\text{acl}_{\Omega_\Gamma}(A) \cap \text{acl}_{\Omega_\Gamma}(B) = \text{acl}_{\Omega_\Gamma}(C)$, Corollary 5.2.30 applied to the representation $\sigma: \Gamma_C \curvearrowright \mathcal{K}$ yields:

$$p_{\text{acl}_{\Omega_\Gamma}(B)}^\sigma p_{\text{acl}_{\Omega_\Gamma}(A)}^\sigma = p_{\text{acl}_{\Omega_\Gamma}(C)}^\sigma$$

Claim. Projections associated with σ identify with restrictions of projections associated with π as follows:

$$p_{\text{acl}_{\Omega_\Gamma}(A)}^\sigma = p_X^\pi|_{\mathcal{H}_Z}, \quad p_{\text{acl}_{\Omega_\Gamma}(B)}^\sigma = p_Y^\pi|_{\mathcal{H}_Z} \quad \& \quad p_{\text{acl}_{\Omega_\Gamma}(C)}^\sigma = p_{X'}^\pi|_{\mathcal{H}_Z}.$$

Proof of the claim. We only prove it for Y , the other identifications are similar and slightly easier. First, note that:

$$p_Y^\pi \mathcal{H}_Z = \mathcal{H}_Y \cap \mathcal{H}_Z.$$

Indeed, let $\xi \in \mathcal{H}_Z$ and $\varepsilon > 0$. By Lemma 5.2.25, there exists $Y_1 \subseteq Y$ finite such that:

$$\|p_{Y_1}^\pi \xi - p_Y^\pi \xi\| \leq \varepsilon.$$

Up to replacing Y_1 with a larger finite subset of Y , we can assume $X'_0 \subseteq Y_1$. Then $U(\Gamma)_{Y_1}$ leaves Z , and hence $\mathcal{H}_Z$, stable. In particular, using the Alaoglu-Birkhoff Theorem 5.2.24 and the fact that $\mathcal{H}_Z$ is closed, we get:

$$p_{Y_1}^\pi \xi \in \mathcal{H}_Z.$$

Since ε was arbitrarily small and $\mathcal{H}_Z$ is closed, $p_Y^\pi \xi \in \mathcal{H}_Z$. The other inclusion is trivial.

Now, $p_Y^\pi|_{\mathcal{H}_Z}$ and $p_{\text{acl}_{\Omega_\Gamma}(B)}^\sigma$ both are orthogonal projections defined on $\mathcal{H}_Z$. To prove they are equal, it suffices to show they have the same image, i.e. that $\mathcal{H}_Y \cap \mathcal{H}_Z = \mathcal{K}_{\text{acl}_{\Omega_\Gamma}(B)}$.

Let $\xi \in \mathcal{H}_Y \cap \mathcal{H}_Z$ and $\varepsilon > 0$. As above, there exists Y_1 a finite subset of Y containing X'_0 such that $\|p_{Y_1}^\pi \xi - \xi\| \leq \varepsilon$ and $p_{Y_1}^\pi \mathcal{H}_Z \subseteq \mathcal{H}_Z$. Let $B_1 := E_{Y_1}(y)$ seen as a finite subset of Ω_Γ . Let $\gamma \in \Gamma_{B_1}$. Using surjectivity in Proposition 5.2.13, pick $g \in U(\Gamma)_{Y_1}$ such that $g_y = \gamma$. Then:

$$\sigma(\gamma)p_{Y_1}^\pi(\xi) = \pi(g)p_{Y_1}^\pi \xi = p_{Y_1}^\pi \xi,$$

hence $p_{Y_1}^\pi(\xi) \in \mathcal{K}_{B_1} \subseteq \mathcal{K}_{\text{acl}_{\Omega_\Gamma}(B)}$. Since ε was arbitrarily small and $\mathcal{K}_{\text{acl}_{\Omega_\Gamma}(B)}$ is closed, $\xi \in \mathcal{K}_{\text{acl}_{\Omega_\Gamma}(B)}$.

Conversely, let $B_1 \subseteq \text{acl}_{\Omega_\Gamma}(B)$ be finite. Let $Y_1 := X'_0 \cup (B_1)_y$. Clearly, $\mathcal{K}_{B_1} \subseteq \mathcal{H}_Z \cap \mathcal{H}_{Y_1} \subseteq \mathcal{H}_Z \cap \mathcal{H}_Y$. Since B_1 was arbitrary and $\mathcal{H}_Z \cap \mathcal{H}_Y$ is closed, $\mathcal{K}_{\text{acl}_{\Omega_\Gamma}(B)} \subseteq \mathcal{H}_Z \cap \mathcal{H}_Y$. $\square_{\text{claim}}$

Since $X, X' \subseteq Z$, we have $p_X^\pi p_Z^\pi = p_Z^\pi = p_{X'}^\pi p_Z^\pi$. Thus, by the previous claim,

$$p_Y^\pi p_X^\pi = p_Y^\pi p_X^\pi p_Z^\pi = p_{\text{acl}_{\Omega_\Gamma}(B)}^\sigma p_{\text{acl}_{\Omega_\Gamma}(A)}^\sigma p_Z^\pi = p_{\text{acl}_{\Omega_\Gamma}(C)}^\sigma p_Z^\pi = p_X^\pi p_Z^\pi = p_X^\pi.$$

Since $X' \subseteq Y$ we also have $p_X^\pi = p_Y^\pi p_X^\pi$, and the proof of Case 1 is finished.

Case 2: $y \notin Y$.

Let $z \in \mathbf{T}$ be the unique neighbor of y such that $d(z, Y) = d(y, Y) - 1$. Case 1 being settled allows us to make the following two additional assumptions.

Claim. *Up to applying Case 1, we can assume that:*

1. $z \in X$.
2. for every $y' \in (V_X(z) \setminus Y) \cap \mathbf{T}$, $E_X(y') = \text{acl}_{\Omega_\Gamma}(\chi(z, y'))$, i.e. $E_X(y')$ is as small as possible.

Proof of the claim.

1. Suppose that $z \notin X$. Since $d(x, Y)$ is maximal, X must be concentrated around x , i.e. contained in $(\Omega_\Gamma)_y$. There thus exists $A \in \mathcal{C}(\Omega_\Gamma)$ such that $X = A_y$. Since X itself is algebraically closed, all of its leaves are in $\mathbf{T}_\Gamma \setminus \mathbf{T}$ by Item 1 of Remark 5.3.2 hence $A \cap \Omega = \emptyset$. Writing $a := \chi(y, z)$, we distinguish two cases:

- If $A \not\subseteq \text{acl}_{\Omega_\Gamma}(a)$, replace Y with $\text{acl}_{\Gamma_\Gamma}(Y \cup \{y\})$ and x with any element of $(A \setminus \text{acl}_{\Omega_\Gamma}(a))_y$. We are now back into Case 1 and the proof of the lemma terminates.
- If $A \subseteq \text{acl}_{\Omega_\Gamma}(a)$, then in fact $A \subseteq \text{acl}_{\Omega_\Gamma}(a) \setminus \{a\}$ since $A \cap \Omega = \emptyset$. By Proposition 5.2.23, $A = \text{acl}_{\Omega_\Gamma}(\emptyset)$ hence $X \in \mathcal{C}_0(\mathbf{T}_\Gamma)$, a contradiction.

2. Suppose that there exists $y' \in (V_X(z) \setminus Y) \cap \mathbf{T}$ such that

$$E_X(y') \not\subseteq \text{acl}_{\Omega_\Gamma}(\chi(z, y')) .$$

Replacing y with y' and x with any $x' \in V_X(y') \setminus \text{acl}_{\Omega_\Gamma}(\chi(z, y'))_{y'}$, we can assume that $y = y'$. Finally, replacing Y with $\text{acl}_{\Gamma_\Gamma}(Y \cup \{y'\})$, we are again back into Case 1 and the proof of the lemma terminates.

□_{claim}

We now assume that the statements in previous claim hold. Since $\Gamma \curvearrowright \Omega$ is 2-transitive, it has no algebraicity on singletons and $y \notin \text{acl}_{\Gamma_\Gamma}(Y \cup \{z\})$. Thus, up to replacing Y with $\text{acl}_{\Gamma_\Gamma}(Y \cup \{z\})$, we can assume that $z \in Y$.

We can now define X' as follows:

$$X' := X \setminus (B_{\Gamma_\Gamma}(z, 2) \setminus Y),$$

which is indeed a subtree of X by maximality of $d(x, Y)$. In other words, X' is the subtree of $\mathbf{T}_\Gamma$ obtained from X by replacing $V_X(z)$ with $V_X(z) \cap V_Y(z)$ and discarding the disconnected vertices. Since $x, y \notin X'$, this is a proper subtree of X . What is more, it follows from Proposition 5.3.4 that $X' \in \mathcal{C}(\mathbf{T}_\Gamma)$. Up to replacing Y with $\text{acl}_{\Gamma_\Gamma}(Y \cup X')$, we will assume that $X' = X \cap Y$, which preserves the fact that $x, y \notin Y$.

We now conclude similarly as in Case 1. Using Lemma 5.3.5, let X_0, Y_0 and X'_0 be finite subtrees of X, Y and X' respectively such that:

$$\text{acl}_{\Gamma_\Gamma}(X_0) = X, \quad \text{acl}_{\Gamma_\Gamma}(Y_0) = Y, \quad \text{and} \quad \text{acl}_{\Gamma_\Gamma}(X'_0) = X'.$$

Up to replacing X_0, Y_0, X'_0 with larger finite trees, we can assume that $X_0 \cap Y_0 = X'_0$ and that $z \in X'_0$ and $x, y \in X_0$. We define:

$$Z := X_0 \cup \text{acl}_{\Gamma_\Gamma}(V_{\Gamma_\Gamma}(z)).$$

The situation is slightly more complicated than that of Case 1. Instead of seeing Ω_Γ around y , we see the larger structure $\Omega := \text{acl}_{\Gamma_\Gamma}(V_{\Gamma_\Gamma}(z))$ from Subsection 5.3.2 around z . Let $A := X_0 \cap \Omega$, $B := Y_0 \cap \Omega$ and $C := X'_0 \cap \Omega$. Necessarily, $A \cap B = C$ and

$$\text{acl}_\Omega(A) = X \cap \Omega, \quad \text{acl}_\Omega(B) = Y \cap \Omega, \quad \text{and} \quad \text{acl}_\Omega(C) = X' \cap \Omega.$$

Then, $\mathbf{U}(\Gamma)_{X'_0}$ leaves Z stable as a set hence acts on $\mathcal{H}_Z$. Again, this representation is trivial on $\mathbf{U}(\Gamma)_Z$ hence factorizes into a representation of $\mathbf{U}(\Gamma)_{X'_0}/\mathbf{U}(\Gamma)_Z$. The restriction morphism $\mathbf{U}(\Gamma)_Z \rightarrow \Delta$ sends $\mathbf{U}(\Gamma)_{X_0}$, $\mathbf{U}(\Gamma)_{Y_0}$ and $\mathbf{U}(\Gamma)_{X'_0}$ onto Δ_A , Δ_B and Δ_C respectively (surjectivity is easily seen through Corollary 5.2.4). What is more, it factorizes into topological group isomorphisms

$$\mathbf{U}(\Gamma)_{X_0}/\mathbf{U}(\Gamma)_Z \simeq \Delta_A, \quad \mathbf{U}(\Gamma)_{Y_0}/(\mathbf{U}(\Gamma)_{Y_0} \cap \mathbf{U}(\Gamma)_Z) \simeq \Delta_B, \quad \text{and} \quad \mathbf{U}(\Gamma)_{X'_0}/\mathbf{U}(\Gamma)_Z \simeq \Delta_C$$

by the open mapping theorem for Polish groups.

Factorizing and restricting π gives rise to the following representation:

$$\sigma: \Delta_C \curvearrowright \mathcal{K} := \mathcal{H}_Z.$$

Since $\text{acl}_\Omega(A) \cap \text{acl}_\Omega(B) = \text{acl}_\Omega(C)$, Proposition 5.3.13 applied to σ yields:

$$p_{\text{acl}_\Omega(B)}^\sigma p_{\text{acl}_\Omega(A)}^\sigma = p_{\text{acl}_\Omega(C)}^\sigma$$

Just as in Case 1 and with the same proof,

$$p_{\text{acl}_\Omega(A)}^\sigma = p_X^\pi|_{\mathcal{H}_Z}, \quad p_{\text{acl}_\Omega(B)}^\sigma = p_Y^\pi|_{\mathcal{H}_Z} \quad \& \quad p_{\text{acl}_\Omega(C)}^\sigma = p_{X'}^\pi|_{\mathcal{H}_Z}.$$

Hence:

$$p_Y^\pi p_X^\pi = p_Y^\pi p_X^\pi p_Z^\pi = p_{\text{acl}_\Omega(B)}^\sigma p_{\text{acl}_\Omega(A)}^\sigma p_Z^\pi = p_{\text{acl}_\Omega(C)}^\sigma p_Z^\pi = p_{X'}^\pi p_Z^\pi = p_{X'}^\pi,$$

Since $X' \subseteq Y$, we also have $p_{X'}^\pi = p_Y^\pi p_{X'}^\pi$, which completes the proof of Case 2. $\square$

We now have the tools to find irreducible induced representations of the desired form in any aspherical representation of $\mathbf{U}(\Gamma)$. The following result is thus the final step towards the classification theorem.

Proposition 5.3.18. *Let $\Gamma \curvearrowright \Omega$ be an oligomorphic and 2-transitive closed permutation group and let $\pi: \mathbf{U}(\Gamma) \curvearrowright \mathcal{H}$ be a non-zero aspherical representation of $\mathbf{U}(\Gamma)$. There exists*

- a classifying subtree $X \subseteq \mathbf{T}_\Gamma$,
- a finite $\mathbf{U}(\Gamma)_{\{X\}}$ -stable subtree X_0 of X such that $\text{acl}_{\mathbf{T}_\Gamma}(X_0) = X$,
- and an irreducible representation σ of the finite group $\mathbf{U}(\Gamma)_{\{X\}}/\mathbf{U}(\Gamma)_{X_0}$ such that

$$\text{Ind}_{\mathbf{U}(\Gamma)_{\{X\}}}^{\mathbf{U}(\Gamma)}(\sigma) \subseteq \pi.$$

Proof. By Lemma 5.2.27, there exists a finite subtree $Y \subseteq \mathbf{T}$ such that $\mathcal{H}_Y \neq 0$. Then $Z := \text{acl}_{\mathbf{T}_\Gamma}(Y)$ is a classifying subtree of $\mathbf{T}_\Gamma$ by Proposition 5.3.6 such that $\mathcal{H}_Z \supseteq \mathcal{H}_Y \neq 0$. In particular:

$$\{X \in \mathcal{C}(\mathbf{T}_\Gamma): \mathcal{H}_X \neq 0\} \neq \emptyset.$$

By well-foundedness of $\mathcal{C}(\mathbf{T}_\Gamma)$ (Proposition 5.3.7), we can fix a minimal element X in this set.

Claim. *For all $g, h \in \mathbf{U}(\Gamma)$ such that $g\mathbf{U}(\Gamma)_{\{X\}} \neq h\mathbf{U}(\Gamma)_{\{X\}}$, we have:*

$$\pi(g)\mathcal{H}_X \perp \pi(h)\mathcal{H}_X. \tag{5.3}$$

Proof of the claim. Suppose first that $X \in \mathcal{C}_0(\mathbf{T}_\Gamma)$. Let x be the unique element of $X \cap \mathbf{T}$. Then, $X = \text{acl}_{\Omega_\Gamma}(\emptyset)_x$ hence $g(X) = \text{acl}_{\Omega_\Gamma}(\emptyset)_{g(x)}$ and $h(X) = \text{acl}_{\Omega_\Gamma}(\emptyset)_{h(x)}$ with $g(x) \neq h(x)$. In this case, the claim follows directly from Lemma 5.3.16 (recall that π is aspherical). We can thus assume that $X \notin \mathcal{C}_0(\mathbf{T}_\Gamma)$. Let $\xi, \eta \in \mathcal{H}_X$. Then:

$$\begin{aligned} \langle \pi(g)\xi, \pi(h)\eta \rangle &= \langle \pi(h^{-1}g)p_X\xi, p_X\eta \rangle = \langle p_{h^{-1}g(X)}\pi(h^{-1}g)\xi, p_X\eta \rangle \\ &= \langle \pi(h^{-1}g)\xi, p_{h^{-1}g(X)}p_X\eta \rangle. \end{aligned} \quad (5.4)$$

Since X is locally $\mathbf{U}(\Gamma)$ -finite and $h^{-1}g \notin \mathbf{U}(\Gamma)_{\{X\}}$, we get $X \not\subseteq h^{-1}g(X)$. Applying Lemma 5.3.17, there exists a proper classifying subtree $X' \subseteq X$ such that $p_{h^{-1}g(X)}p_X = p_{h^{-1}g(X)}p_{X'}'$. The minimality of X implies $p_{X'}' = 0$ hence $p_{h^{-1}g(X)}p_X = 0$. Equation (5.4) now yields $\langle \pi(g)\xi, \pi(h)\eta \rangle = 0$ and the claim follows. $\square_{\text{claim}}$

Finally, let $X_0 \subseteq X$ be a finite subtree such that $\mathcal{H}_{X_0} \neq 0$. Up to replacing X_0 with a larger set, Lemma 5.3.5 allows us to assume that $\text{acl}_{\mathbf{T}_\Gamma}(X_0) = X$ and that X_0 is $\mathbf{U}(\Gamma)_{\{X\}}$ -stable. Now, $\mathbf{U}(\Gamma)_{\{X\}}$ acts on $\mathcal{H}_{X_0}$, an action which factorizes into a representation σ of the finite group $\mathbf{U}(\Gamma)_{\{X\}}/\mathbf{U}(\Gamma)_{X_0}$. Form $\pi' := \text{Ind}_{\mathbf{U}(\Gamma)_{\{X\}}}^{\mathbf{U}(\Gamma)}(\sigma)$ with underlying space $\mathcal{H}'$. Let $(g_i)_{i \in I}$ be a system of coset representatives of $\mathbf{U}(\Gamma)/\mathbf{U}(\Gamma)_{\{X\}}$. Define a map $\Phi: \mathcal{H}' \rightarrow \mathcal{H}$ in the following way:

$$\forall f \in \mathcal{H}', \Phi(f) = \sum_{i \in I} \pi(g_i)f(g_i).$$

By definition of the norm on $\mathcal{H}'$ and by Equation (5.3), Φ is well defined and isometric. Note that it does not depend on the choice of $(g_i)_{i \in I}$, for if $h^{-1}g \in \mathbf{U}(\Gamma)_{\{X\}}$, then $\pi(g)f(g) = \pi(g)f(hh^{-1}g) = \pi(g)(\sigma(g^{-1}h)f(h)) = \pi(g)\pi(g^{-1}h)f(h) = \pi(h)f(h)$. With similar calculations, it is easily seen that Φ is $\mathbf{U}(\Gamma)$ -equivariant hence an isomorphism of representations onto its image. We have shown that $\pi' \subseteq \pi$.

To conclude, recall that $\mathbf{U}(\Gamma)_{\{X\}}/\mathbf{U}(\Gamma)_{X_0}$ is a finite group hence σ splits as a sum of irreducible subrepresentations. Moreover, by the basic properties of induction, if $\tau \subseteq \sigma$, then $\text{Ind}_{\mathbf{U}(\Gamma)_{\{X\}}}^{\mathbf{U}(\Gamma)}(\tau) \subseteq \text{Ind}_{\mathbf{U}(\Gamma)_{\{X\}}}^{\mathbf{U}(\Gamma)}(\sigma)$ hence σ can indeed be assumed irreducible (see. e.g. [Bd20, Prop. 1.F.6(2)]). $\square$

We are now ready to state and prove our main theorem. Combined with the discussion on commensurators in the next Section (see Proposition 5.4.9), it readily implies Theorem 5.1.1 from the Introduction.

Theorem 5.3.19. *Let $\Gamma \curvearrowright \Omega$ be an oligomorphic and 2-transitive closed permutation group and let $\mathbf{U}(\Gamma) \leq \text{Aut}(\mathbf{T})$ be the associated universal Burger-Mozes group. Let $\mathbf{T}_\Gamma$ be as in Definition 5.2.11.*

1. *Let X_0 be a non-empty finite subtree of $\mathbf{T}_\Gamma$ and let $X := \text{acl}_{\mathbf{T}_\Gamma}(X_0)$. For every irreducible representation σ of $\mathbf{U}(\Gamma)_{\{X\}}$ that is trivial on $\mathbf{U}(\Gamma)_{X_0}$, the representation $\pi := \text{Ind}_{\mathbf{U}(\Gamma)_{\{X\}}}^{\mathbf{U}(\Gamma)}(\sigma)$ of $\mathbf{U}(\Gamma)$ is irreducible.*
2. *A representation of the above form is spherical if and only if either*

- X is empty, in which case π is the trivial representation of $\mathbf{U}(\Gamma)$,
 - or $\mathbf{U}(\Gamma)_{\{X\}} = \mathbf{U}(\Gamma)_x$ for some $x \in \mathbf{T}$ and σ is the trivial representation of $\mathbf{U}(\Gamma)_x$, in which case π is isomorphic to $\mathbf{U}(\Gamma) \curvearrowright \ell^2(\mathbf{U}(\Gamma)/\mathbf{U}(\Gamma)_x)$.
3. Every aspherical irreducible representation of $\mathbf{U}(\Gamma)$ is of the above form.
 4. Two representations of the above form $\text{Ind}_{\mathbf{U}(\Gamma)_{\{X\}}}^{\mathbf{U}(\Gamma)}(\sigma)$ and $\text{Ind}_{\mathbf{U}(\Gamma)_{\{Y\}}}^{\mathbf{U}(\Gamma)}(\tau)$ are isomorphic if and only if there exists $g \in \mathbf{U}(\Gamma)$ such that $g(X) = Y$ and $\tau^g \simeq \sigma$.
 5. Every aspherical representation of $\mathbf{U}(\Gamma)$ splits as a sum of aspherical irreducible subrepresentations.

Proof. Items 1, 2 and 4 correspond to Proposition 5.3.15. Item 5 is a straightforward combination of Proposition 5.3.18 and Zorn's Lemma. It only remains to prove Item 3. To that aim, let π be an aspherical irreducible unitary representation of $\mathbf{U}(\Gamma)$. By Proposition 5.3.18, there exists a π' of the desired form such that $\pi' \subseteq \pi$. Since π is irreducible, we have $\pi' \simeq \pi$. $\square$

It is worth pointing out that our classification theorem has the following consequence. It is the exact analogue of Corollary 5.2.30 and similar to dissociation as defined in [BJJ24].

Corollary 5.3.20. *Let $\Gamma \curvearrowright \Omega$ be an oligomorphic and 2-transitive closed permutation group and let $\pi: \mathbf{U}(\Gamma) \curvearrowright \mathcal{H}$ be an aspherical representation of $\mathbf{U}(\Gamma)$. For all algebraically closed subsets $X, Y \subseteq \mathbf{T}_\Gamma$,*

$$\mathcal{H}_X \perp_{\mathcal{H}_{X \cap Y}} \mathcal{H}_X.$$

Proof. We will use Lemma 5.2.29 several times. By Item 2 of Lemma 5.2.29 and Items 1 and 5 of Theorem 5.3.19, we can assume that $\pi = \text{Ind}_{\mathbf{U}(\Gamma)_{\{X\}}}^{\mathbf{U}(\Gamma)}(\sigma)$ for some σ, X , and X_0 . By Lemma 5.3.5, X_0 can be enlarged to be assumed $G_{\{X\}}$ -stable, ensuring that $\mathbf{U}(\Gamma)_{X_0}$ is a normal subgroup of $\mathbf{U}(\Gamma)_{\{X\}}$ with finite index. Then σ factorizes through the finite group $\mathbf{U}(\Gamma)_{\{X\}}/\mathbf{U}(\Gamma)_{X_0}$. By the Peter-Weyl Theorem, σ is a subrepresentation of the regular representation $\lambda_{\mathbf{U}(\Gamma)_{\{X\}}/\mathbf{U}(\Gamma)_{X_0}}: \mathbf{U}(\Gamma)_{\{X\}}/\mathbf{U}(\Gamma)_{X_0} \curvearrowright \ell^2(\mathbf{U}(\Gamma)_{\{X\}}/\mathbf{U}(\Gamma)_{X_0})$. Then:

$$\begin{aligned} \pi = \text{Ind}_{\mathbf{U}(\Gamma)_{\{X\}}}^{\mathbf{U}(\Gamma)}(\sigma) &\subseteq \text{Ind}_{\mathbf{U}(\Gamma)_{\{X\}}}^{\mathbf{U}(\Gamma)}(\lambda_{\mathbf{U}(\Gamma)_{\{X\}}/\mathbf{U}(\Gamma)_{X_0}}) = \text{Ind}_{\mathbf{U}(\Gamma)_{\{X\}}}^{\mathbf{U}(\Gamma)}\left(\text{Ind}_{\mathbf{U}(\Gamma)_{X_0}}^{\mathbf{U}(\Gamma)_{\{X\}}}(\mathbf{1}_{\mathbf{U}(\Gamma)_{X_0}})\right) \\ &= \text{Ind}_{\mathbf{U}(\Gamma)_{X_0}}^{\mathbf{U}(\Gamma)}(\mathbf{1}_{\mathbf{U}(\Gamma)_{X_0}}) \\ &= \lambda_{\mathbf{U}(\Gamma)/\mathbf{U}(\Gamma)_{X_0}} \end{aligned}$$

where $\lambda_{\mathbf{U}(\Gamma)/\mathbf{U}(\Gamma)_{X_0}}$ is the quasi regular representation $\mathbf{U}(\Gamma) \curvearrowright \ell^2(\mathbf{U}(\Gamma)/\mathbf{U}(\Gamma)_{X_0})$. The inclusion is [Bd20, Prop. 1.F.6(2)], the equalities following it are covered by [Bd20, Def. 1.5] and [Bd20, Prop. 1.F.6(1,3)]. By Item 2 of Lemma 5.2.29, we can thus replace π with $\lambda_{\mathbf{U}(\Gamma)/\mathbf{U}(\Gamma)_{X_0}}$. Now, by Item 1 Lemma 5.2.19, writing $X_0 = \{x_1, \dots, x_n\}$ in an injective way, $\mathbf{U}(\Gamma)/\mathbf{U}(\Gamma)_{X_0} \simeq \mathbf{U}(\Gamma) \cdot (x_1, \dots, x_n) \subseteq \mathbf{T}_\Gamma^n$.

This identification being $\mathbf{U}(\Gamma)$ -equivariant, $\lambda_{\mathbf{U}(\Gamma)/\mathbf{U}(\Gamma)_{x_0}}$ is isomorphic to a subrepresentation of $\lambda_n: \mathbf{U}(\Gamma) \curvearrowright \ell^2(\mathbf{T}_\Gamma^n)$ and we conclude using both Items of Lemma 5.2.29. $\square$

An important consequence of the results in this section is the fact that $\mathbf{U}(\Gamma)$ is of Type I when $\Gamma \curvearrowright \Omega$ is oligomorphic and 2-transitive. Before proving this fact, we formalize this notions. See e.g. [Mac76] or [Bd20, Chap.6] for a complete exposition, the beginning of which does not assume local compactness.

Let G be a topological group and let $\pi: G \curvearrowright \mathcal{H}$ and $\pi': G \curvearrowright \mathcal{H}'$ be two unitary representations of G . We will say that π and π' are *disjoint* if no non-zero subrepresentation of π is isomorphic to a subrepresentation of π' . If π has the property that, whenever $\pi \simeq \pi_1 \oplus \pi_2$, the representations π_1 and π_2 are always disjoint, we will say that π is *multiplicity free*. For instance, every irreducible representation is multiplicity free and, more generally, so is every sum of pairwise non-isomorphic irreducible representations.

The representations π and π' are *quasi-equivalent* if they are isomorphic up to multiplicity, that is if $\oplus_{n \in \mathbb{N}} \pi \simeq \oplus_{n \in \mathbb{N}} \pi'$. Then π is of *Type I* if π is quasi-equivalent to a multiplicity free representation. This is equivalent to the von Neumann algebra $\pi(G)' := \{T \in \mathcal{B}(\mathcal{H}): \forall g \in G, T\pi(g) = \pi(g)T\}$ being of Type I in the usual terminology [Dix96, Prop. 5.4.7]. Moreover, if π is of Type I, the multiplicity free representation to which it is quasi-equivalent is unique up to isomorphism and is a subrepresentation of π . Finally, the group G is of *Type I* if all of its representations are of Type I.

The proof of the following result follows closely Ol'sanskii's proof in the special case $\Gamma = \text{Sym}(\Omega)$, i.e. $\mathbf{U}(\Gamma) = \text{Aut}(\mathbf{T})$ [Ol'80, Th. 5.7].

Theorem 5.3.21. *For every oligomorphic and 2-transitive closed permutation group $\Gamma \curvearrowright \Omega$, the associated universal Burger-Mozes group $\mathbf{U}(\Gamma)$ is of Type I.*

Proof. Let $\pi: \mathbf{U}(\Gamma) \curvearrowright \mathcal{H}$ be a representation of $\mathbf{U}(\Gamma)$. Let $\mathcal{K}$ be the closed subspace of $\mathcal{H}$ generated by $\bigcup_{x \in \mathbf{T}} \mathcal{H}_x$. Let π_1 and π_2 be the restrictions of π to $\mathcal{K}$ and $\mathcal{K}^\perp$ respectively. By [Bd20, Prop. 6.B.13(2)], sums of Type I representations are of Type I. It thus suffices to check that both π_1 and π_2 are of Type I. We will thus assume that either $\pi = \pi_1$ or $\pi = \pi_2$.

We start with the case $\pi = \pi_2$. This representation is aspherical hence splits as a sum of irreducible subrepresentations by Item 1 of 5.3.19. Irreducible representations being multiplicity free, they are of Type I. We conclude with [Bd20, Prop. 6.B.13(2)].

We now assume that $\pi = \pi_1$ and show that $\pi(\mathbf{U}(\Gamma))'$ is a Type I von Neumann algebra. Fixing $x \in \mathbf{T}$, let $\mathcal{A}_x$ be the von Neumann algebra generated by $p_x \cdot \pi(\mathbf{U}(\Gamma)) \cdot p_x$. The operator p_x induces an isomorphism between $\pi(\mathbf{U}(\Gamma))'$ and the commutant of $\mathcal{A}_x$. It thus suffices to show that the commutant of $\mathcal{A}_x$ is of Type I which in turn reduces to showing that $\mathcal{A}_x$ itself is of Type I [Tak02, Cor. 1.30]. In fact, we will show that $\mathcal{A}_x$ is commutative, which is enough to conclude [Tak02, Prop. 1.23].

We now follow the proof of [Ol'80, Sec 5.3], using the mentioned alternative method due to Ismagilov, to show that $\mathcal{A}_x$ is commutative. For every $n \in \mathbb{N}$, let $g_n \in \mathbf{U}(\Gamma)$ be such that $d(g_n(x), x) = n$. Such a sequence can be obtained for instance by transitivity of $\mathbf{U}(\Gamma) \curvearrowright \mathbf{T}$ (Item 1 of Lemma 5.2.7). As seen in the proof of

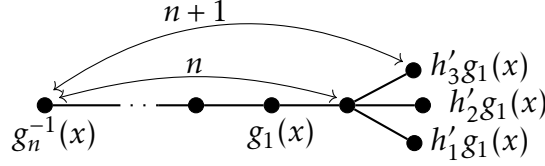


Figure 5.4: Computing the distances $d(h'_i g_1(x), g_n^{-1}(x))$.

Proposition 5.2.17 (first argument in $(iv) \Rightarrow (ii)$), the double coset $\mathbf{U}(\Gamma)_x \cdot g \cdot \mathbf{U}(\Gamma)_x$ for $g \in \mathbf{U}(\Gamma)$ only depends on $d(g(x), x)$. In particular, $a_n := p_x \pi(g_n) p_x$ does not depend on the choice of g_n , and $\{a_n : n \in \mathbb{N}\} = p_x \cdot \pi(\mathbf{U}(\Gamma)) \cdot p_x$. Moreover, since $d(g(x), x) = d(x, g^{-1}(x))$ for every $g \in \mathbf{U}(\Gamma)$, we have $a_n^* = a_n$ for every $n \in \mathbb{N}$. Thus, the linear span of $\{a_n : n \in \mathbb{N}\}$ is dense in $\mathcal{A}_x$ and it suffices to show that the a_n 's pairwise commute. It is a direct consequence of the following claim:

Claim. For every $n \in \mathbb{N}$, $a_{n+1} = a_n a_1$.

Proof of the claim. Let $n \in \mathbb{N}$. Up to replacing g_1 , we can assume that $g_1(x) \in [x, g_n^{-1}(x)]$. Let $\xi \in \mathcal{H}$ and let $\varepsilon > 0$. The Alaoglu-Birkhoff Theorem 5.2.24 applied to $\eta := \pi(g_1) p_x \xi$, yields $\lambda_1, \dots, \lambda_m \geq 0$ with $\lambda_1 + \dots + \lambda_m = 1$ and $h_1, \dots, h_m \in \mathbf{U}(\Gamma)_x$ such that:

$$\|p_x \eta - \sum_{i=1}^m \lambda_i \pi(h_i) \eta\| < \varepsilon. \quad (5.5)$$

By edge transitivity of $\mathbf{U}(\Gamma) \curvearrowright \mathbf{T}$ (Item 2 in Lemma 5.2.7), pick $h \in \mathbf{U}(\Gamma)_x$ such that $h g_1(x) \notin \{h_1 g_1(x), \dots, h_m g_1(x)\}$ and let $h'_i := h^{-1} h_i$ for every $i \leq m$. It satisfies $d(g_n h'_i g_1(x), x) = d(h^{-1} h_i g_1(x), g_n^{-1}(x)) = n + 1$, see Figure 5.4. In particular, $p_x \pi(g_n h'_i g_1) p_x = a_{n+1}$ for every $i \leq m$. Moreover, since $h \in \mathbf{U}(\Gamma)_x$, replacing h_i with h'_i for every $i \leq m$ preserves (5.5). Thus:

$$\begin{aligned} \|a_{n+1} \xi - a_n a_1 \xi\| &= \|a_{n+1} \xi - p_x \pi(g_n) p_x \eta\| \\ &< \varepsilon + \|a_{n+1} \xi - p_x \pi(g_n) \cdot \sum_{i=1}^m \lambda_i \pi(h'_i) \eta\| \\ &< \varepsilon + \sum_{i=1}^m \lambda_i \|a_{n+1} \xi - p_x \pi(g_n h'_i g_1) p_x \xi\| \\ &< \varepsilon + \sum_{i=1}^m \lambda_i \|a_{n+1} \xi - a_{n+1} \xi\| \\ &< \varepsilon. \end{aligned}$$

Since ε is arbitrary, the claim follows. □_{claim}

□

5.4 The Howe-Moore property

This Section is fully independent from Section 5.3 and does not use the expansion of $\mathbf{T}$ into $\mathbf{T}_\Gamma$, with the exception of our very last results, Proposition 5.4.9

and Corollary 5.4.10. Its goal is to show that the representation theory of $\mathbf{U}(\Gamma)$ interacts well with its large scale geometry. More precisely, we will be studying the following subgroup of $\mathbf{U}(\Gamma)$:

$$\mathbf{U}(\Gamma)^+ := \{g \in \mathbf{U}(\Gamma) : \forall x \in \mathbf{T}, d(g(x), x) \text{ is even} \}.$$

We say that a permutation group G on a countable set N is *generated by point stabilizers* if $\bigcup_{a \in \Omega} G_a$ generates G as an abstract group. The universal groups $\mathbf{U}(\Gamma)$ are never generated by point stabilizers, since $\mathbf{U}(\Gamma)_x \subseteq \mathbf{U}(\Gamma)^+ \subsetneq \mathbf{U}(\Gamma)$ for every permutation group $\Gamma \curvearrowright \Omega$ and every $x \in \mathbf{T}$.

Proposition 5.4.1. *Let $\Gamma \curvearrowright \Omega$ be a permutation group.*

1. $\mathbf{U}(\Gamma)^+$ is a normal open subgroup of $\mathbf{U}(\Gamma)$ of index 2.
2. If $\Gamma \curvearrowright \Omega$ is 2-transitive, then $\mathbf{U}(\Gamma)^+$ is the group generated by $\bigcup_{x \in \mathbf{T}} \mathbf{U}(\Gamma)_x$ and is abstractly simple.

Proof.

1. Observe first the following fact: the relation on $\mathbf{T}$ given by $x \sim y \iff [d(x, y) \text{ is even}]$ is an equivalence relation with two classes which is preserved by the action of $\text{Aut}(\mathbf{T})$. Thus, for every $g \in \text{Aut}(\mathbf{T})$ and all $x, y \in \mathbf{T}$, the distances $d(g(x), x)$ and $d(g(y), y)$ have the same parity. In particular, if $g \in \mathbf{U}(\Gamma)$ and $x \in \mathbf{T}$, we have:

$$g \in \mathbf{U}(\Gamma)^+ \iff g(x) \sim x \iff \exists y \in \mathbf{T}, g(y) \sim y. \quad (5.6)$$

In particular, if $g, h \in \mathbf{U}(\Gamma)^+$ and $x \in \mathbf{T}$, applying the to $y := h(x)$ yields $x \sim h(x) \sim gh(x)$ hence $gh \in \mathbf{U}(\Gamma)^+$. Since moreover $d(g^{-1}(x), x) = d(x, g(x))$, $\mathbf{U}(\Gamma)^+$ is indeed a subgroup of $\mathbf{U}(\Gamma)$. And if $u \in \mathbf{U}(\Gamma)$, we have $d(ugu^{-1}(x), x) = d(gu(x), u(x))$, which is even by assumption on G . Thus $\mathbf{U}(\Gamma)^+$ is indeed normal in $\mathbf{U}(\Gamma)$. For openness, simply note that Equation (5.6) implies that $\mathbf{U}(\Gamma)_x \leq \mathbf{U}(\Gamma)^+$.

Finally, since $\mathbf{U}(\Gamma) \curvearrowright \mathbf{T}$ is vertex transitive by Lemma 5.2.7, $\mathbf{U}(\Gamma)^+$ is a proper subgroup of $\mathbf{U}(\Gamma)$. Let $g \in \mathbf{U}(\Gamma) \setminus \mathbf{U}(\Gamma)^+$ and $x \in \mathbf{T}$. For every $h \in \mathbf{U}(\Gamma)^+$, either $h \in \mathbf{U}(\Gamma)^+$ or $g(x) \sim x \sim h(x)$ in which case $g(x) \sim h(x)$ and $g^{-1}h \in \mathbf{U}(\Gamma)^+$. Thus $\mathbf{U}(\Gamma) = \mathbf{U}(\Gamma)^+ \cup g\mathbf{U}(\Gamma)^+$.

2. Let $g \in \mathbf{U}(\Gamma)(\Gamma)^+$ and let $x \in \mathbf{T}$. Since $d(g(x), x)$ is even, $[x, g(x)]$ has a middle point y . Let $y' \in V_{\mathbf{T}}(y) \setminus [x, g(x)]$ and let $z \in \{y, y'\}$ be such that $d(x, z)$ is even. Then $d(x, z) = d(z, g(y))$ and, by metric 2 transitivity of $\mathbf{U}(\Gamma) \curvearrowright \mathbf{T}$ (Item 3 of Lemma 5.2.7), there exists $h \in \mathbf{U}(\Gamma)$ such that $h(z) = z$ and $h(x) = g(x)$. Then $h^{-1}g \in \mathbf{U}(\Gamma)_x$ hence $g \in h\mathbf{U}(\Gamma)_x \subseteq \mathbf{U}(\Gamma)_z \mathbf{U}(\Gamma)_x$.

For simplicity, we refer to [Smi17] and use their terminology. Let $\{\text{id}\}$ be seen as a permutation group on $\{1, 2\}$. By Remark 4 in [Smi17] and in Smith's terminology, $\mathbf{U}(\Gamma)$ is topologically isomorphic to $\mathcal{U}(\Gamma, \text{Sym}(\{1, 2\}))$. One can easily deduce that $\mathbf{U}(\Gamma)^+$ is topologically isomorphic to $\mathcal{U}(\Gamma, \{\text{id}\})$. The trivial group $\{\text{id}\}$ is obviously generated by point stabilizers and so is Γ . Indeed, fix $a \neq b \in \Omega$. By 2 transitivity of $\Gamma \curvearrowright \Omega$, we have $\Gamma_a \cdot b = \Omega \setminus \{b\}$ and $\Gamma_b \Gamma_a \cdot b = \Omega$ hence $\Gamma = \Gamma_b \Gamma_a \Gamma_b$. Thus $\mathbf{U}(\Gamma)^+$ is simple by Item (iv) of [Smi17, Th. 1].

□

In Subsection 5.4.1, we extend the definition of the Howe-Moore property to general topological groups with a special focus on locally Roelcke-precompact Polish groups. In Subsection 5.4.2, we recall classical results about the geometry of $\mathbf{T}$ and provide a proof of Theorem 5.1.4 from the introduction, namely that the normal subgroup $\mathbf{U}(\Gamma)^+$ of index 2 has the Howe-Moore property. We also show that this result combined with those of Section 5.3 provides a deep understanding of the lattice of open subgroups of $\mathbf{U}(\Gamma)$.

5.4.1 The Howe-Moore property for locally Roelcke-precompact Polish groups

Recall that given a representation $\pi: G \curvearrowright \mathcal{H}$ of a topological group G , an *invariant vector* is a fixed point of the action. Moreover, the *matrix coefficients* of G arising from π are the maps $G \rightarrow \mathbb{C}$ of the form $f_{\xi,\eta}^\pi: g \mapsto \langle \pi(g)\xi, \eta \rangle$ for all $\xi, \eta \in \mathcal{H}$. These continuous maps capture the whole representation theory of G [Bd20, Sec. 1.B] and we will show that they also respect the large scale geometry of universal Burger-Mozes groups. Initially defined for locally compact groups, the Howe-Moore property captures precisely that. While it naturally extends to general topological groups using the framework developed in Subsection 5.2.2, it has not yet been studied in a non-locally compact setting before to the best of the author's knowledge, except in [BJJ25].

Definition 5.4.2. Let G be a topological group.

1. A map $f: G \rightarrow \mathbb{C}$ *vanishes at infinity* if for every $\varepsilon > 0$, the set $\{g \in G: |f(g)| \geq \varepsilon\}$ is coarsely bounded. We will denote by $C_0(G)$ the continuous functions that vanish at infinity.
2. The group G has the *Howe-Moore property* if for every representation $\pi: G \curvearrowright \mathcal{H}$ with no non-zero invariant vector, all the matrix coefficients of G arising from π vanish at infinity.

In fact, this notion can often be rephrased using more familiar objects, namely locally compact ones. Given a topological group G , its *Roelcke completion* $\mathcal{R}(G)$ is the completion of G with respect to the meet of the left and right uniformities on G . Another way to describe this object when G is metrizable is as follows. Let δ_L be any left-invariant metric on G that generates its topology and set for all $g, h \in G$:

$$\delta_{L \wedge R}(g, h) = \inf\{\delta_L(g, k) + \delta_L(k^{-1}, h^{-1}): k \in G\}.$$

The metric $\delta_{L \wedge R}$ is no longer left-invariant but it still generates the topology of G . Moreover, $\mathcal{R}(G)$ is the completion of the metric space $(G, \delta_{L \wedge R})$. It is straightforward that a Polish group G is Roelcke precompact if and only if $\mathcal{R}(G)$ is compact. A deeper result due to Zielinski is that G is locally Roelcke-precompact if and only if $\mathcal{R}(G)$ is locally compact [Zie21]. In this case, we define $C_0(\mathcal{R}(G))$ in the classical way, as the set of continuous functions $f: \mathcal{R}(G) \rightarrow \mathbb{C}$ such that for every $\varepsilon > 0$, the set $\{x \in \mathcal{R}(G): |f(x)| \geq \varepsilon\}$ is compact.

Back to unitary representations, it easily checked that every matrix coefficient of a topological group G is *Roelcke-uniformly continuous*, i.e. uniformly continuous with respect to $\delta_{L \wedge R}$. In particular, every matrix coefficient $G \rightarrow \mathbb{C}$ uniquely extends to a continuous function $\mathcal{R}(G) \rightarrow \mathbb{C}$.

Proposition 5.4.3. *Let G be a locally Roelcke-precompact Polish group. The following properties are equivalent:*

- (i) G has the Howe-Moore property,
- (ii) for every unitary representation $\pi: G \curvearrowright \mathcal{H}$ with no non-zero invariant vector, all the matrix coefficients of G arising from π vanish at infinity on the locally compact space $\mathcal{R}(G)$.

Given a sequence $(g_n)_{n \in \mathbb{N}}$ in a topological group G , we will write $g_n \rightarrow \infty$ if $(g_n)_{n \in \mathbb{N}}$ eventually escapes every coarsely bounded subset of G . The large scale geometry of our groups is nice enough so that it is fully captured by sequences.

Lemma 5.4.4. *Let $\Gamma \curvearrowright \Omega$ be an oligomorphic and 2-transitive closed permutation group and let $f: \mathbf{U}(\Gamma)^+ \rightarrow \mathbb{C}$ be a continuous function. The following properties are equivalent:*

- (i) f vanishes at infinity,
- (ii) for every $(g_n)_{n \in \mathbb{N}} \in \mathbf{U}(\Gamma)^{+\mathbb{N}}$, $g_n \rightarrow \infty$ implies $f(g_n) \rightarrow 0$.

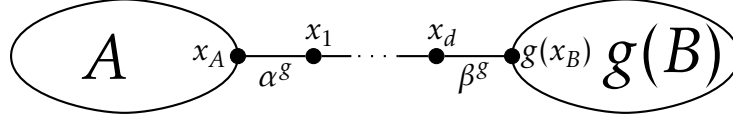
Proof. (i) $\Rightarrow$ (ii) is obviously true in higher generality. Suppose now that f does not satisfy (i). Then, there exists $\varepsilon > 0$ such that $N := \{g \in G^+ : |f(g)| \geq \varepsilon\}$ is not coarsely bounded. By Proposition 5.2.17, $N \cdot x$ is unbounded in $\mathbf{T}$. Thus, for every $n \in \mathbb{N}$, there exists $g_n \in N$ such that $d(g_n(x), x) \geq n$. We have $g_n \rightarrow \infty$ by Proposition 5.2.17 but $|f(g_n)| \geq \varepsilon$, contradicting (ii). $\square$

5.4.2 Ends, translations, and the Howe-Moore property for $\mathbf{U}(\Gamma)^+$

We now briefly recall some terminology and classical facts regarding the geometry of $\mathbf{T}$ itself. See e.g. [FN91, Chap. 1] for more details.

We start by defining the boundary $\partial \mathbf{T}$ of $\mathbf{T}$. Given two infinite injective paths $(x_n)_{n \in \mathbb{N}}, (y_n)_{n \in \mathbb{N}} \in \mathbf{T}^{\mathbb{N}}$, we will write $(x_n)_{n \in \mathbb{N}} \simeq (y_n)_{n \in \mathbb{N}}$ if they eventually coincide up to a shift of the indices. That is, if there exists $k \in \mathbb{Z}$ and $N \in \mathbb{N}$ such that $x_{n+k} = y_n$ for every $n \geq N$. Then $\partial \mathbf{T}$ will denote the set of $\simeq$ -equivalence classes of infinite injective paths. The set $\partial \mathbf{T}$ will be called the *boundary* of $\mathbf{T}$ and its elements the *ends* of $\mathbf{T}$. Given $\omega \neq \omega' \in \partial \mathbf{T}$, there exists a bi-infinite injective path $(x_n)_{n \in \mathbb{Z}}$ such that $(x_n)_{n \in \mathbb{N}} \in \omega'$ and $(x_{-n})_{n \in \mathbb{N}} \in \omega$. Moreover, this sequence is unique up to shift of the indices hence we will denote by $] \omega, \omega' [$ the set of vertices appearing in such a sequence.

A *translation* of $\mathbf{T}$ is a tree automorphism that neither fixes a vertex nor flips an edge. Given a translation $\tau \in \text{Aut}(\mathbf{T})$ and $x \in \mathbf{T}$ such that $d(\tau(x), x)$ is minimal, let $\Delta := \bigcup_{n \in \mathbb{Z}} \tau^n([x, \tau(x)])$. Then, Δ is stable under the action of τ and there exists $\omega \neq \omega' \in \partial \mathbf{T}$ such that $\Delta =] \omega, \omega' [$. The set Δ is called the *axis* of τ and $n := d(\tau(x), x)$ is the *step* of τ . Note that $\tau \in \text{Aut}(\mathbf{T})^+$ if and only if it is of even step. Applying


 Figure 5.5: The local data used for identifying $\mathbf{U}(\Gamma)_A g \mathbf{U}(\Gamma)_B$.

Corollary 5.2.4 to $X =]\omega, \omega'[$ for any $\omega \neq \omega' \in \partial \mathbf{T}$, it is easy to see that $\mathbf{U}(\Gamma)^+$ does contain translations of step 2 when $\Gamma \curvearrowright \Omega$ is 2-transitive.

Our proof that $\mathbf{U}(\Gamma)^+$ has the Howe-Moore property is adapted from the classical one in the finite arity case. The main difference is that since our groups are not locally compact, a finer Cartan decomposition is required. More precisely, we give a description of $\mathbf{U}(\Gamma)_A \backslash \mathbf{U}(\Gamma)^+ / \mathbf{U}(\Gamma)_B$ for all finite subtrees $A, B \subseteq \mathbf{T}$, while the case $A = B = \{x\}$ suffices in finite arity. When $A = B = \{x\}$, the graph distance is enough to describe $\mathbf{U}(\Gamma)_x \backslash \mathbf{U}(\Gamma)^+ / \mathbf{U}(\Gamma)_x$. In the general case, additional information is required, which we now describe.

Let $A, B \subseteq \mathbf{T}$ be two finite and non-empty subtrees. Let $n := |A|$, $m := |B|$ and write $A = \{a_1, \dots, a_n\}$, $\bar{a} := (a_1, \dots, a_n)$, $B = \{b_1, \dots, b_m\}$ and $\bar{b} := (b_1, \dots, b_m)$. The symbol $\wedge$ will denote concatenation of tuples. For $a \in A$, we will write $A(a)$ instead of $E_A(a)$ and similarly for B . Given $g \in \mathbf{U}(\Gamma)^+$ such that $g(B) \cap A = \emptyset$, let x_A^g and x_B^g be the unique vertices of A and B respectively such that $d(A, g(B)) = d(x_A^g, g(x_B^g))$. Let also α^g be the element of Ω coloring the unique edge going from x_A^g toward $g(B)$. Similarly, let β^g be the element of Ω coloring the unique edge around x_B whose image under g goes toward A . Write $\mathcal{O}_A^g := \Gamma_{A(x_A^g)} \cdot \alpha^g$ and $\mathcal{O}_B^g := \Gamma_{B(x_B^g)} \cdot \beta^g$. See Figure 5.5. Finally, define:

$$\Psi(g) := (x_A^g, x_B^g, \mathcal{O}_A^g, \mathcal{O}_B^g, d(A, g(B))).$$

Lemma 5.4.5. *Let $\Gamma \curvearrowright \Omega$ be an oligomorphic and 2-transitive closed permutation group. For all $g, h \in \mathbf{U}(\Gamma)^+$ such that $g(B) \cap A = \emptyset$ and $h(B) \cap A = \emptyset$, the following properties are equivalent:*

- (i) $\mathbf{U}(\Gamma)_A \cdot g \cdot \mathbf{U}(\Gamma)_B = \mathbf{U}(\Gamma)_A \cdot h \cdot \mathbf{U}(\Gamma)_B$,
- (ii) $\Psi(g) = \Psi(h)$.

Proof. Clearly, Ψ is left $\mathbf{U}(\Gamma)_A$ -invariant and right $\mathbf{U}(\Gamma)_B$ -invariant hence (i) $\Rightarrow$ (ii). Conversely, suppose $\Psi(g) = \Psi(h)$. We will use the following simplified notations: $x_A := x_A^g = x_A^h$ and $x_B := x_B^g = x_B^h$. Let $(x_0, \dots, x_{d+1})$ be the shortest path from A to $g(B)$ and let $(y_0, \dots, y_{d+1})$ be the shortest path from A to $h(B)$. Necessarily, we have $x_0 = y_0 = x_A$ as well as $x_{d+1} = g(x_B)$ and $y_{d+1} = h(x_B)$. Let C be the finite subtree of $\mathbf{T}$ generated by A and $g(B)$, i.e. $C = A \cup g(B) \cup \{x_1, \dots, x_d\}$. We define $(\gamma_c)_{c \in C}$ be as follows (see also Figure 5.6):

- Since $\mathcal{O}_A^g = \mathcal{O}_A^h$, there exists $\gamma_{x_A} \in \Gamma_{A(x_A)}$ such that $\gamma_{x_A}(\alpha^g) = \alpha^h$. For every $a \in A \setminus \{x_A\}$, set $\gamma_a := \text{id}_\Omega$.
- Similarly, since $\mathcal{O}_B^g = \mathcal{O}_B^h$, there exists $\gamma \in \Gamma_{B(x_B)}$ such that $\gamma \cdot \beta^g = \beta^h$. Set $\gamma_{g(x_B)} := h_{x_B} \gamma (g_{x_B})^{-1}$ and, for every $b \in B \setminus \{x_B\}$, set $\gamma_{g(b)} := h_b (g_b)^{-1}$.

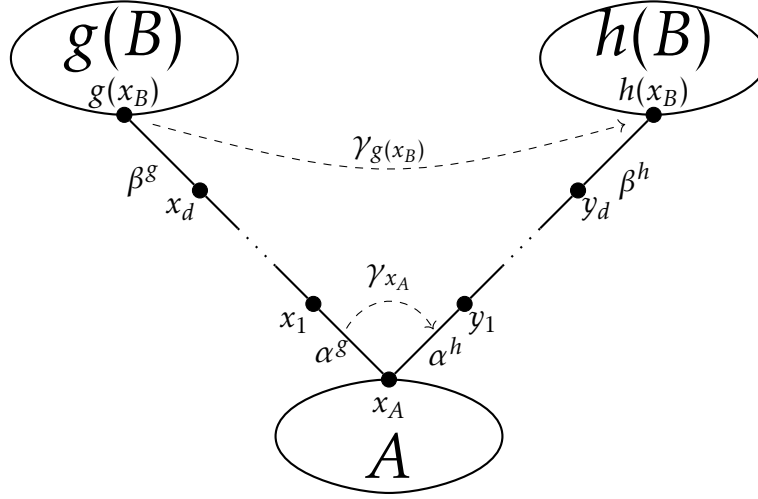


Figure 5.6: Local actions at x_A and $g(x_B)$. At the bottom, γ_{x_A} fixes $A(x_A)$ and sends α^g to α^h . Up above, $\gamma_{g(x_B)}$ reverses the action of g_{x_B} on $B(x_B)$ and replaces it with that of h_{x_B} while sending β^g to β^h .

- By 2-transitivity of $\Gamma \curvearrowright \Omega$, we can pick for every $i \in \{1, \dots, d\}$ some $\gamma_{x_i} \in \Gamma$ such that $\gamma_{x_i} \cdot \chi(x_i, x_{i+1}) = \chi(y_i, y_{i+1})$ and $\gamma_{x_i} \cdot \chi(x_i, x_{i-1}) = \chi(y_i, y_{i-1})$.

Then $(\gamma_c)_{c \in C}$ satisfies the conditions of Corollary 5.2.4 which yields $u \in \mathbf{U}(\Gamma)$ such that $u_c = \gamma_c$ for every $c \in C$ and $u(x_A) = x_A$. By construction, $u \in \mathbf{U}(\Gamma)_A$ and $ug(b) = h(b)$ for every $b \in B$. Thus $(ug)^{-1}h \in \mathbf{U}(\Gamma)_B$ hence $h \in ug \cdot \mathbf{U}(\Gamma)_B \subseteq \mathbf{U}(\Gamma)_A \cdot g \cdot \mathbf{U}(\Gamma)_B$. We have shown that $\mathbf{U}(\Gamma)_A \cdot g \cdot \mathbf{U}(\Gamma)_B = \mathbf{U}(\Gamma)_A \cdot h \cdot \mathbf{U}(\Gamma)_B$. $\square$

Note that if we assume that $\Gamma \curvearrowright \Omega$ is oligomorphic, the first four components of Ψ take values in a finite set. From now on, we fix $\tau \in \mathbf{U}(\Gamma)^+$ a translation of step 2, which will allow us to control the last component of Ψ .

Lemma 5.4.6. *Let $\Gamma \curvearrowright \Omega$ be an oligomorphic and 2-transitive closed permutation group. There exists a finite subset $F \subseteq \mathbf{U}(\Gamma)^+$ such that:*

$$\mathbf{U}(\Gamma)^+ = \mathbf{U}(\Gamma)_A \cdot F \cdot \{\tau^n : n \geq 0\} \cdot F \cdot \mathbf{U}(\Gamma)_B.$$

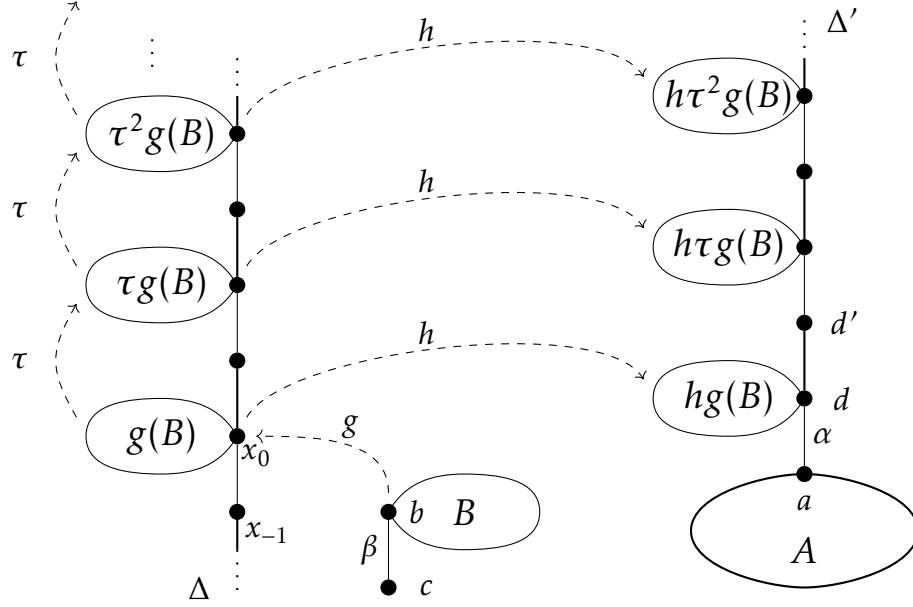
Proof. Let $K := \{g \in \mathbf{U}(\Gamma)^+ : g(B) \cap A \neq \emptyset\}$. Then K is Roelcke precompact in $\mathbf{U}(\Gamma)^+$ by Lemma 5.2.16. Thus, there exists a finite subset $F_0 \subseteq K \subseteq \mathbf{U}(\Gamma)^+$ such that $K \subseteq \mathbf{U}(\Gamma)_A \cdot F_0 \cdot \mathbf{U}(\Gamma)_B$. It remains to deal with $\mathbf{U}(\Gamma)^+ \setminus K$.

With an eye towards Lemma 5.4.5, fix $a \in A$, $b \in B$ as well as $\mathcal{O}_A = \Gamma_{A(a)} \cdot \alpha$ and $\mathcal{O}_B = \Gamma_{B(b)} \cdot \beta$ for some $\alpha \in \Omega \setminus A(a)$ and $\beta \in \Omega \setminus B(b)$ respectively. Let also $\delta = 0$ if $d(a, b)$ is even or $\delta = 1$ if $d(a, b)$ is odd. We will produce $g = g(b, \mathcal{O}_B)$ and $h = h(a, \mathcal{O}_A)$ in $\mathbf{U}(\Gamma)^+$ such that for every $n \in \mathbb{N}$,

$$\Psi(h\tau^n g) = (a, b, \mathcal{O}_A, \mathcal{O}_B, 2(n+1) - \delta).$$

Note that for every $g \in \mathbf{U}(\Gamma)^+$, we have $d(a, g(b)) \bmod 2 = d(a, b) \bmod 2$ hence this will be enough to conclude, after setting:

$$F := F_0 \cup \bigcup_{a, b, \mathcal{O}_A, \mathcal{O}_B} \{g(b, \mathcal{O}_B), h(a, \mathcal{O}_A)\}.$$


 Figure 5.7: Realization of a given value of Ψ .

Since the action $\Gamma \curvearrowright \Omega$ is oligomorphic, this is indeed a finite subset of $\mathbf{U}(\Gamma)^+$.

Let c be the unique neighbor of b such that $\chi(b, c) = \beta$. Let $\Delta := \{x_i : i \in \mathbb{Z}\}$ be the axis of τ , written in such a way that $\tau(x_i) = x_{i+2}$ for every $i \in \mathbb{Z}$ and $d(x_0, b)$ is even. By Item 3 of Lemma 5.2.7, $\mathbf{U}(\Gamma) \curvearrowright \mathbf{T}$ is metrically 2-transitive and we can find $g \in \mathbf{U}(\Gamma)$ such that $g(b) = x_0$ and $g(c) = x_{-1}$. By the choice of x_0 we automatically have $g \in \mathbf{U}(\Gamma)^+$.

Next, let d be the unique neighbor of a such that $\chi(a, d) = \alpha$, let d' be any neighbor of d distinct from a . Pick Δ' a bi-infinite geodesic in $\mathbf{T}$ such that $\Delta' \cap A = \{a\}$ and $d, d' \in \Delta$. Using 2-transitivity of $\Gamma \curvearrowright \Omega$ and Corollary 5.2.4, one can find $h \in \mathbf{U}(\Gamma)$ such that:

- $h(\Delta) = \Delta'$,
- $(h(x_{-1}), h(x_0)) = \begin{cases} (a, d) & \text{if } d(x_0, d) \text{ is even,} \\ (d, d') & \text{otherwise.} \end{cases}$

Again, by parity of $d(h(x_0), x_0)$, we in fact have $h \in \mathbf{U}(\Gamma)^+$. The case where $d(x_0, d)$ is even is represented in Figure 5.7. Since for every $n \in \mathbb{N}$, no edge of $h\tau^n g(B)$ ever belongs to Δ' , we indeed have:

$$\forall n \in \mathbb{N}, \Psi(h\tau^n g) = (x_A, x_B, \mathcal{O}_A, \mathcal{O}_B, 2(n+1) - \delta).$$

□

Just as in the finite arity case, the previous Lemma reduces the problem to studying the vanishing of matrix coefficients of $\mathbf{U}(\Gamma)$ on the abelian semigroup $\{\tau^n : n \in \mathbb{N}\}$.

Corollary 5.4.7. *Let $\pi : \mathbf{U}(\Gamma)^+ \curvearrowright \mathcal{H}$ be a representation of $\mathbf{U}(\Gamma)^+$ and suppose there exists $\xi, \eta \in \mathcal{H}$ such that $f_{\xi, \eta}^\pi$ does not vanish at infinity. Then, there exists $\xi', \eta' \in \mathcal{H}$ such that:*

$$f_{\xi', \eta'}^\pi(\tau^n) \not\rightarrow 0 \text{ as } n \rightarrow +\infty.$$

Proof. Suppose $f_{\xi,\eta}^\pi$ does not vanish at infinity. Approximating ξ and η by means of Lemma 5.2.27 yields a uniform approximation of f . We can thus assume $\xi, \eta \in \mathcal{H}_A$ for some non-empty finite subtree $A \subseteq \mathbf{T}$. By Lemma 5.4.4, there exists $\varepsilon > 0$ and $(g_n) \in \mathbf{U}(\Gamma)^{+\mathbb{N}}$ such that $g_n \rightarrow \infty$ while $|f_{\xi,\eta}^\pi(g_n)| \geq \varepsilon$ for every $n \in \mathbb{N}$. By Lemma 5.4.6, we have $\mathbf{U}(\Gamma)^+ = \mathbf{U}(\Gamma)_A \cdot F \cdot \{\tau^n : n \in \mathbb{N}\} \cdot F \cdot \mathbf{U}(\Gamma)_A$ for some finite subset $F \subseteq \mathbf{U}(\Gamma)^+$ and we can write $g_n = u_n f_n \tau^{k_n} f'_n u'_n$ for some $u_n, u'_n \in \mathbf{U}(\Gamma)_A$, $f_n, f'_n \in F$ and $k_n \in \mathbb{N}$. Up to passing to a subsequence, we can assume that (f_n) and (f'_n) are constant sequences and denote their respective value by f and f' . Observe using Proposition 5.2.17 that $\mathbf{U}(\Gamma)_A$ and F are coarsely bounded and that $k_n \rightarrow +\infty$.

Setting $\xi' := \pi(f')\xi$ and $\eta' := \pi(f^{-1})\eta$ we have for every $n \in \mathbb{N}$:

$$\begin{aligned} f_{\xi,\eta}^\pi(g_n) &= f(u_n f \tau^{k_n} f' u'_n) = \langle \pi(\tau^{k_n})\pi(f')\pi(u'_n)\xi, \pi(f^{-1})\pi(u_n^{-1})\eta \rangle \\ &= \langle \pi(\tau^{k_n})\xi', \eta' \rangle = f_{\xi',\eta'}^\pi(\tau^{k_n}). \end{aligned}$$

We thus have $|f_{\xi',\eta'}^\pi(\tau^{k_n})| \geq \varepsilon$ for every $n \in \mathbb{N}$ while $k_n \rightarrow +\infty$. In particular, $f_{\xi',\eta'}^\pi(\tau^n) \not\rightarrow 0$. $\square$

The classical proof of Lubotzky and Mozes now carries through verbatim. More precisely, Propositions 1 and 2 in [LM92] hold for $\mathbf{U}(\Gamma)^+$. Once combined, they yield the following, which completes the proof of Theorem 5.1.4:

Proposition 5.4.8. *Let $\pi: \mathbf{U}(\Gamma)^+ \curvearrowright \mathcal{H}$ be a representation of $\mathbf{U}(\Gamma)^+$ with no non-zero $\mathbf{U}(\Gamma)^+$ -invariant vector. Then $f_{\xi,\eta}^\pi(\tau^n) \rightarrow 0$ for all $\xi, \eta \in \mathcal{H}$.*

Proof. See Propositions 1 and 2 in [LM92]. $\square$

5.4.3 Commensurators and open subgroups of $\mathbf{U}(\Gamma)$

Combining the results of this section with those of Section 5.3, we can describe the open subgroups of $\mathbf{U}(\Gamma)$, at least up to finite index. In model theoretic terms, the results below imply that every imaginary of $\mathbf{T}$ is inter-algebraic with some finite subtree of $\mathbf{T}_\Gamma$.

In what follows, given two subgroups $H, H' \leq \mathbf{U}(\Gamma)$, we will say that H and H' are *commensurate* if $H \cap H'$ has finite index in both H and H' , which defines an equivalence relation. The *commensurator* of H is the group $\text{Comm}_{\mathbf{U}(\Gamma)}(H) := \{g \in \mathbf{U}(\Gamma) : H \text{ and } gHg^{-1} \text{ are commensurate}\}$. For every $g \in \mathbf{U}(\Gamma)$, the number of left H -cosets in HgH is precisely the index $[H : H \cap gHg^{-1}]$ while the number of right cosets is $[gHg^{-1} : H \cap gHg^{-1}]$. If H and H' are commensurate, it is easily seen that $\text{Comm}_{\mathbf{U}(\Gamma)}(H) = \text{Comm}_{\mathbf{U}(\Gamma)}(H')$. We will say that H is *self-commensurate* in $\mathbf{U}(\Gamma)$ if $H = \text{Comm}_{\mathbf{U}(\Gamma)}(H)$.

Proposition 5.4.9. *Let $\Gamma \curvearrowright \Omega$ be an oligomorphic and 2-transitive closed permutation group. For every $X \in \mathcal{C}(\mathbf{T}_\Gamma)$ the group $\mathbf{U}(\Gamma)_{\{X\}}$ is self-commensurate in $\mathbf{U}(\Gamma)$.*

Proof. Let $g \in \text{Comm}_{\mathbf{U}(\Gamma)}(\mathbf{U}(\Gamma)_{\{X\}})$. Let $X_0 \subseteq X$ be finite and such that $\text{acl}_{\mathbf{T}_\Gamma}(X_0) = X$ (Lemma 5.3.5). Since $\mathbf{U}(\Gamma)_{X_0}$ has finite index in $\mathbf{U}(\Gamma)_{\{X\}}$, we also have $g \in \text{Comm}_{\mathbf{U}(\Gamma)}(\mathbf{U}(\Gamma)_{X_0})$. In particular, the number of left- $\mathbf{U}(\Gamma)_{X_0}$ cosets contained in

$\mathbf{U}(\Gamma)_{X_0} \cdot g \cdot \mathbf{U}(\Gamma)_{X_0}$ is finite. Equivalently, by Item 1 of Lemma 5.2.19, $\mathbf{U}(\Gamma)_{X_0} \cdot g(X_0)$ is finite. Equivalently, $g(X_0) \subseteq \text{acl}_{\mathbf{T}_\Gamma}(X_0) = X$ hence $g(X) \subseteq X$. Since X is locally $\mathbf{U}(\Gamma)$ -finite, we in fact have $g(X) = X$, i.e. $g \in \mathbf{U}(\Gamma)_{\{X\}}$.

Conversely, suppose $g \in \mathbf{U}(\Gamma)_{\{X\}}$. Since $\mathbf{U}(\Gamma)_{X_0}$ has finite index in $\mathbf{U}(\Gamma)_{\{X\}}$, it suffice to show that $g \in \text{Comm}_{\mathbf{U}(\Gamma)}(\mathbf{U}(\Gamma)_{X_0})$. Write $X_0 = \{x_1, \dots, x_n\}$, $\bar{x} := (x_1, \dots, x_n)$ and simply observe that the number of left $\mathbf{U}(\Gamma)_{X_0}$ -cosets in $\mathbf{U}(\Gamma)_{X_0} \cdot g \cdot \mathbf{U}(\Gamma)_{X_0}$ is the cardinal of the orbit $\mathbf{U}(\Gamma)_{X_0} \cdot g(\bar{x})$ by Item 1 of Lemma 5.2.19. Since $g(X_0) \subseteq X = \text{acl}_{\mathbf{T}_\Gamma}(X_0)$, this orbit is finite. Similarly, the number of right $\mathbf{U}(\Gamma)_{X_0}$ -cosets in $\mathbf{U}(\Gamma)_{X_0} \cdot g \cdot \mathbf{U}(\Gamma)_{X_0}$ is the cardinal of the orbit $\mathbf{U}(\Gamma)_{X_0} \cdot g^{-1}(\bar{x})$ which is also finite. $\square$

Corollary 5.4.10. *Let $\Gamma \curvearrowright \Omega$ be an oligomorphic and 2-transitive closed permutation group and let $U \leq \mathbf{U}(\Gamma)$ be a proper open subgroup. Then, either $U = \mathbf{U}(\Gamma)^+$ or U is Roelcke precompact. In the latter case, there exists $X \in \mathcal{C}(\mathbf{T}_\Gamma)$ such that $U \leq \mathbf{U}(\Gamma)_{\{X\}}$ with finite index.*

In particular, the self-commensurate open subgroups of $\mathbf{U}(\Gamma)$ are exactly the subgroups of the form $\mathbf{U}(\Gamma)_{\{X\}}$ for $X \in \mathcal{C}(\mathbf{T}_\Gamma)$.

Proof. If $\mathbf{U}(\Gamma)^+ \subseteq U$ then $\mathbf{U}(\Gamma)^+ = U$ because $\mathbf{U}(\Gamma)^+$ has index 2 in $\mathbf{U}(\Gamma)$. We thus suppose that $\mathbf{U}(\Gamma)^+ \not\subseteq U$. We first show that U is Roelcke precompact. Since $U \cap \mathbf{U}(\Gamma)^+$ has index at most 2 in U , it suffices to prove that $U \cap \mathbf{U}(\Gamma)^+$ is Roelcke precompact. It is a classical consequence of the Howe-Moore property that every proper open subgroup of $\mathbf{U}(\Gamma)^+$ is coarsely bounded, see e.g. [BJJ25, Lem. 2.9] for a proof. In particular, $U \cap \mathbf{U}(\Gamma)^+$ is coarsely bounded hence Roelcke precompact by [Zie21, Th. 14].

As for the second statement, we will first show that $H := \text{Comm}_{\mathbf{U}(\Gamma)}(U)$ is Roelcke precompact. Since U is open, there exists a finite subtree $Z \subseteq \mathbf{T}$ such that $\mathbf{U}(\Gamma)_Z \leq U$. Since $\text{acl}_{\mathbf{T}}(Z)$ is finite (apply e.g. Proposition 5.3.6 and Proposition 5.3.4), we can assume that Z is algebraically closed with respect to $\mathbf{U}(\Gamma) \curvearrowright \mathbf{T}$. Next, Item (v) of Proposition 5.2.17 provides a subtree $Y \subseteq \mathbf{T}$ which is either a single vertex or a single edge and such that $U \leq \mathbf{U}(\Gamma)_{\{Y\}}$. For every $g \in H$, the number of distinct left-cosets of U contained in UgU is $[U, U \cap U^g]$ hence finite. In particular, $UgU \cdot Y$ is finite and, a fortiori, so is $\mathbf{U}(\Gamma)_Z \cdot g(Y)$. Thus, $g(Y) \subseteq \text{acl}_{\mathbf{T}}(Z) = Z$ and $H \cdot Y \subseteq Z$ is bounded. In particular, H is indeed Roelcke precompact by Proposition 5.2.17. Now, $\text{Comm}_{\mathbf{U}(\Gamma)}(U) = \text{Comm}_H(U)$ and [Tsa12, Lem. 2.7] applies, hence U has finite index in H and H is self-commensurate.

It remains to show that $H = \mathbf{U}(\Gamma)_{\{X\}}$ for some $X \in \mathcal{C}(\mathbf{T}_\Gamma)$. It is easily seen that $\lambda_{\mathbf{U}(\Gamma)/H} : \mathbf{U}(\Gamma) \curvearrowright \ell^2(\mathbf{U}(\Gamma)/H)$ is irreducible. We distinguish two cases.

- Suppose that $\lambda_{\mathbf{U}(\Gamma)/H}$ is spherical. Let $x \in \mathbf{T}$ and $\xi \in \mathcal{H}_x \setminus \{0\}$. There exists $g \in \mathbf{U}(\Gamma)$ such that $\xi(gH) \neq 0$. Since $\xi \in \ell^2(\mathbf{U}(\Gamma)/H)$, $\mathbf{U}(\Gamma)_x gH$ contains only finitely many left cosets of H . Then $x' := g^{-1}(x)$ satisfies that $\mathbf{U}(\Gamma)_{x'} H$ contains only finitely many left cosets of H . In other words, $H \cap \mathbf{U}(\Gamma)_{x'}$ has finite index in $\mathbf{U}(\Gamma)_{x'}$. A fortiori, so does $\mathbf{U}(\Gamma)_{\{Y\}} \cap \mathbf{U}(\Gamma)_{x'}$ in $\mathbf{U}(\Gamma)_{x'}$. However, similarly as in Item 1 of Lemma 5.2.19,

$$\mathbf{U}(\Gamma)_{x'} / (\mathbf{U}(\Gamma)_{\{Y\}} \cap \mathbf{U}(\Gamma)_{x'}) \simeq \mathbf{U}(\Gamma)_{x'} \cdot Y.$$

Since $\mathbf{U}(\Gamma) \curvearrowright \mathbf{T}$ is metrically 2-transitive, $\mathbf{U}(\Gamma)_{x'} \cdot Y$ is finite if and only if $Y = \{x'\}$. Thus $H \leq \mathbf{U}(\Gamma)_{x'}$ and the index is finite. Note that $\mathbf{U}(\Gamma)_{x'} = \mathbf{U}(\Gamma)_{\{X'\}}$

where $X' = \text{acl}_{\Omega_\Gamma}(\emptyset)_{x'} \in \mathcal{C}(\mathbf{T}_\Gamma)$ which is self-commensurate by Proposition 5.4.9. Since H is also self-commensurate, we have $H = \mathbf{U}(\Gamma)_{\{X'\}}$.

- Suppose that $\lambda_{\mathbf{U}(\Gamma)/H}$ is aspherical. In this case, there exists $X \in \mathcal{C}(\mathbf{T}_\Gamma)$, a finite subtree $X_0 \subseteq X$ and an irreducible representation of $\mathbf{U}(\Gamma)_{\{X\}}$ that is trivial on $\mathbf{U}(\Gamma)_{X_0}$ such that $\lambda_{\mathbf{U}(\Gamma)/H}$ is isomorphic to $\pi := \text{Ind}_{\mathbf{U}(\Gamma)_{\{X\}}}^{\mathbf{U}(\Gamma)}(\sigma) : \mathbf{U}(\Gamma) \curvearrowright \mathcal{H}$. Since $\lambda_{\mathbf{U}(\Gamma)/H}$ contains non-zero H -invariant vectors, so does π and we can fix such a vector $\xi \in \mathcal{H} \setminus \{0\}$. Let $g \in \mathbf{U}(\Gamma)$ be such that $\xi(g) \neq 0$. Since $\xi \in \ell^2(\mathbf{U}(\Gamma)/\mathbf{U}(\Gamma)_{\{X\}})$, the set $Hg\mathbf{U}(\Gamma)_{\{X\}}$ contains only finitely many left $\mathbf{U}(\Gamma)_{\{X\}}$ cosets. In other words, $X' := g(X)$ satisfies that $H \cap \mathbf{U}(\Gamma)_{\{X'\}}$ has finite index in H . Let $h_1, \dots, h_n \in H$ be such that $H \subseteq \bigcup_i h_i H \cap \mathbf{U}(\Gamma)_{\{X'\}}$.

Denoting $X'_0 := g(X_0)$, we have $\mathcal{H}_{X'_0} = \pi(g)\mathcal{H}_{X_0} \neq 0$ hence there exists a $\mathbf{U}(\Gamma)_{X'_0}$ -invariant function $f \in \ell^2(\mathbf{U}(\Gamma)/H) \setminus \{0\}$. Let $g' \in \mathbf{U}(\Gamma)$ be such that $f(g') \neq 0$. Then $\mathbf{U}(\Gamma)_{X'_0} \cdot g' \cdot H$ contains only finitely many H cosets. Since $\mathbf{U}(\Gamma)_{X'_0}$ has finite index in $\mathbf{U}(\Gamma)_{\{X'\}}$,

$$\mathbf{U}(\Gamma)_{\{X'\}} \cdot g' \cdot H \text{ also contains only finitely many } H\text{-cosets.} \quad (5.7)$$

In other words, there exists $h'_1, \dots, h'_m \in H$ such that $\mathbf{U}(\Gamma)_{\{X'\}} \cdot g' \cdot H \subseteq \bigcup_j h'_j H$. Consequently:

$$Hg'H \subseteq \bigcup_i h_i H \cap \mathbf{U}(\Gamma)_{\{X'\}} g' H \subseteq \bigcup_{i,j} h_i h'_j H.$$

We have thus shown that $Hg'H$ contains only finitely many left cosets of H . We now show that $Hg'H$ contains only finitely many right cosets of H . Indeed, following the proof of [Tsa12, Lem 2.6], we define

$$\mathcal{F}(H) := \{uH : HuH \text{ contains only finitely many left } H\text{-cosets}\}.$$

Suppose $u \in \mathbf{U}(\Gamma)$ is such that $uH \in \mathcal{F}(H)$. A fortiori, $\mathbf{U}(\Gamma)_Z \cdot u \cdot \mathbf{U}(\Gamma)_{\{Y\}}$ contains only finitely many left $\mathbf{U}(\Gamma)_{\{Y\}}$ -cosets. In other words, $\mathbf{U}(\Gamma)_Z \cdot u(Y)$ is finite, i.e. $u(Y) \subseteq \text{acl}_{\mathbf{T}_\Gamma}(Z)$. Since Z is bounded and $\Gamma \curvearrowright \Omega$ is 2-transitive, $\text{acl}_{\mathbf{T}_\Gamma}(Z)$ is also bounded by Proposition 5.3.3. Thus, by Proposition 5.2.17, the set $E := \{u \in \mathbf{U}(\Gamma) : uH \in \mathcal{F}(H)\} \subseteq \{u \in \mathbf{U}(\Gamma) : u(Y) \subseteq \text{acl}_{\mathbf{T}_\Gamma}(Z)\}$ is Roelcke precompact. Since H is open, there exists $F \subseteq E$ finite such that $E \subseteq HFH$. We now see that $\mathcal{F}(H)$ is finite. The proof of [Tsa12, Lem 2.6] now applies verbatim, showing that $Hg'H$ indeed contains only finitely many right cosets of H .

We have thus shown that $g' \in \text{Comm}_{\mathbf{U}(\Gamma)}(H) = H$. Then, (5.7) yields that $\mathbf{U}(\Gamma)_{\{X'\}} \cap H$ has finite index in $\mathbf{U}(\Gamma)_{\{X'\}}$. Therefore H and $\mathbf{U}(\Gamma)_{\{X'\}}$ are commensurate. Since both H and $\mathbf{U}(\Gamma)_{\{X'\}}$ are self-commensurate, $H = \mathbf{U}(\Gamma)_{\{X\}}$.

For the last statement, let $H \leq \mathbf{U}(\Gamma)$ be a self commensurate open subgroup. If $\mathbf{U}(\Gamma)^+ \subseteq H$ then H has finite index in $\mathbf{U}(\Gamma)$ hence $H = \mathbf{U}(\Gamma) = \mathbf{U}(\Gamma)_{\{\emptyset\}}$. Otherwise, by what precedes, there exists $X \in \mathcal{C}(\mathbf{T}_\Gamma)$ such that $H \leq \mathbf{U}(\Gamma)_{\{X\}}$ with finite index. Then $\mathbf{U}(\Gamma)_{\{X\}} \subseteq \text{Comm}_{\mathbf{U}(\Gamma)}(H) = H$ hence $\mathbf{U}(\Gamma)_{\{X\}} = H$. The converse is given by Proposition 5.4.9.

□

Bibliography

- [AB39] L. Alaoglu and G. Birkhoff, *General ergodic theorems*, Proc. Nat. Acad. Sci. U.S.A. **25** (1939), 628–630.
- [ABH⁺26] Andrés Aranda, David Bradley-Williams, Jan Hubička, Miltiadis Karamanlis, Michael Kompatscher, Matěj Konečný, and Micheal Pawliuk, *Ramsey expansions of metrically homogeneous graphs*, European J. Combin. **132** (2026), Paper No. 104255, 57.
- [AKL14] Omer Angel, Alexander S. Kechris, and Russell Lyons, *Random orderings and unique ergodicity of automorphism groups*, J. Eur. Math. Soc. (JEMS) **16** (2014), no. 10, 2059–2095.
- [Ald85] David J. Aldous, *Exchangeability and related topics*, École d’été de Probabilités de Saint-Flour, XIII—1983, Lecture Notes in Math., vol. 1117, Springer, Berlin, 1985, pp. 1–198.
- [AM66] Louis Auslander and Calvin C. Moore, *Unitary representations of solvable Lie groups*, Mem. Amer. Math. Soc. **62** (1966), 199.
- [Bag78] Larry Baggett, *Representations of the Mautner group. I*, Pacific J. Math. **77** (1978), no. 1, 7–22.
- [BaK26] Adam Bartoš and Wiesław Kubiś, *Uncountable homogeneous structures*, Ann. Pure Appl. Logic **177** (2026), no. 1, Paper No. 103649, 19.
- [Bar25] Rémi Barritault, *Tannaka-Kreĭn duality for Roelcke-precompact non-archimedean Polish groups*, Fund. Math. **271** (2025), no. 2, 97–130.
- [BBMN76] B. J. Birch, R. G. Burns, Sheila Oates Macdonald, and Peter M. Neumann, *On the orbit-sizes of permutation groups containing elements separating finite subsets*, Bull. Austral. Math. Soc. **14** (1976), no. 1, 7–10.
- [Bd20] Bachir Bekka and Pierre de la Harpe, *Unitary representations of groups, duals, and characters*, Mathematical Surveys and Monographs, vol. 250, American Mathematical Society, Providence, RI, 2020.
- [BdV08] Bachir Bekka, Pierre de la Harpe, and Alain Valette, *Kazhdan’s property (T)*, New Mathematical Monographs, vol. 11, Cambridge University Press, Cambridge, 2008.

- [Bek03] M. B. Bekka, *Kazhdan's property (T) for the unitary group of a separable Hilbert space*, Geom. Funct. Anal. **13** (2003), no. 3, 509–520.
- [Ber74] Sterling K. Berberian, *Lectures in functional analysis and operator theory*, Graduate Texts in Mathematics, vol. No. 15, Springer-Verlag, New York-Heidelberg, 1974.
- [BJJ24] Rémi Barritault, Colin Jahel, and Matthieu Joseph, *Unitary Representations of the Isometry Groups of Urysohn Spaces*, 2024.
- [BJJ25] ———, *On dissociated infinite permutation groups*, 2025.
- [BJM25] Samuel Braunfeld, Colin Jahel, and Paolo Marimon, *When invariance implies exchangeability (and applications to invariant Keisler measures)*, 2025.
- [BK96] Howard Becker and Alexander S. Kechris, *The descriptive set theory of Polish group actions*, London Mathematical Society Lecture Note Series, vol. 232, Cambridge University Press, Cambridge, 1996.
- [BM00a] Marc Burger and Shahar Mozes, *Groups acting on trees: From local to global structure*, Inst. Hautes Études Sci. Publ. Math. (2000), no. 92, 113–150.
- [BM00b] ———, *Lattices in product of trees*, Inst. Hautes Études Sci. Publ. Math. (2000), no. 92, 151–194.
- [BNT17] David Bryant, André Nies, and Paul Tupper, *A universal separable diversity*, Anal. Geom. Metr. Spaces **5** (2017), no. 1, 138–151.
- [Boc42] S. Bochner, *On a theorem of Tannaka and Krein*, Ann. of Math. (2) **43** (1942), 56–58.
- [BYIT18] Itaï Ben Yaacov, Tomás Ibarlucía, and Todor Tsankov, *Eberlein oligomorphic groups*, Trans. Amer. Math. Soc. **370** (2018), no. 3, 2181–2209.
- [BYT16] Itaï Ben Yaacov and Todor Tsankov, *Weakly almost periodic functions, model-theoretic stability, and minimality of topological groups*, Trans. Amer. Math. Soc. **368** (2016), no. 11, 8267–8294.
- [Cam90] Peter J. Cameron, *Oligomorphic permutation groups*, London Mathematical Society Lecture Note Series, vol. 152, Cambridge University Press, Cambridge, 1990.
- [Cam99] ———, *Permutation groups*, London Mathematical Society Student Texts, vol. 45, Cambridge University Press, Cambridge, 1999. MR 1721031
- [CGM24] Alessandro Carderi, Alice Giraud, and François Le Maître, *Classification of non-free p.m.p. boolean actions of ergodic full groups and applications*, 2024.

- [Che99] Claude Chevalley, *Theory of Lie groups. I*, Princeton Mathematical Series, vol. 8, Princeton University Press, Princeton, NJ, 1999.
- [Cio15] Corina Ciobotaru, *A unified proof of the Howe-Moore property*, J. Lie Theory **25** (2015), no. 1, 65–89.
- [CKM23] Pierre-Emmanuel Caprace, Mehrdad Kalantar, and Nicolas Monod, *A type I conjecture and boundary representations of hyperbolic groups*, Proc. Lond. Math. Soc. (3) **127** (2023), no. 2, 447–486.
- [Con17] Gabriel Conant, *Neostability in countable homogeneous metric spaces*, Ann. Pure Appl. Logic **168** (2017), no. 7, 1442–1471.
- [CT18] Harry Crane and Henry Towsner, *Relatively exchangeable structures*, J. Symb. Log. **83** (2018), no. 2, 416–442.
- [CV06] P. J. Cameron and A. M. Vershik, *Some isometry groups of the Urysohn space*, Ann. Pure Appl. Logic **143** (2006), no. 1-3, 70–78.
- [dF31] B. de Finetti, *Funzione caratteristica di un fenomeno aleatorio.*, Mem. Accad. naz. Lincei, Cl. Sci. fis. mat. nat. (6) **4** (1931), 251–300.
- [Dix96] Jacques Dixmier, *Les C^* -algèbres et leurs représentations*, Les Grands Classiques Gauthier-Villars, Éditions Jacques Gabay, Paris, 1996.
- [DJ08] Persi Diaconis and Svante Janson, *Graph limits and exchangeable random graphs*, Rend. Mat. Appl. (7) **28** (2008), no. 1, 33–61.
- [DS22] Alexandre I. Danilenko and Cesar E. Silva, *Ergodic theory: Nonsingular transformations*, 2022.
- [EGLMM21] Mahmood Etedadialiabadi, Su Gao, François Le Maître, and Julien Melleray, *Dense locally finite subgroups of automorphism groups of ultraextensive spaces*, Adv. Math. **391** (2021), Paper No. 107966, 42.
- [EP02] David M. Evans and M. E. Pantano, *$\aleph_0$ -categorical structures with arbitrarily fast growth of algebraic closure*, J. Symbolic Logic **67** (2002), no. 3, 897–909.
- [ET16] David M. Evans and Todor Tsankov, *Free actions of free groups on countable structures and Property (T)*, Fund. Math. **232** (2016), no. 1, 49–63.
- [FLMMM22] Pierre Fima, François Le Maître, Julien Melleray, and Soyoung Moon, *Homogeneous actions on Urysohn spaces*, Colloq. Math. **167** (2022), no. 1, 21–61.
- [FN91] Alessandro Figà-Talamanca and Claudio Nebbia, *Harmonic analysis and representation theory for groups acting on homogeneous trees*, London Mathematical Society Lecture Note Series, vol. 162, Cambridge University Press, Cambridge, 1991.
- [Fol95] Gerald B. Folland, *A course in abstract harmonic analysis*, Studies in Advanced Mathematics, CRC Press, Boca Raton, FL, 1995.

- [Gla98] Eli Glasner, *On minimal actions of Polish groups*, Topology and its Applications **85** (1998), no. 1, 119–125.
- [Gla12] ———, *The group $\text{Aut}(\mu)$ is Roelcke precompact*, Canad. Math. Bull. **55** (2012), no. 2, 297–302.
- [Gli61] James Glimm, *Type I C^* -algebras*, Ann. of Math. (2) **73** (1961), 572–612.
- [GM14] Eli Glasner and Michael Megrelishvili, *Representations of dynamical systems on Banach spaces*, Recent Progress in General Topology. III, Atlantis Press, Paris, 2014, pp. 399–470.
- [GR43] I. Gel’fand and D. Raikov, *Irreducible unitary representations of locally bicomact groups*, Rec. Math. [Mat. Sbornik] N.S. **13/55** (1943), 301–316.
- [Gro87] M. Gromov, *Hyperbolic groups*, Essays in Group Theory (S. M. Gersten, ed.), Springer New York, New York, NY, 1987, pp. 75–263.
- [Hal20] Andreas Hallbäck, *Automorphism groups of universal diversities*, Topology Appl. **285** (2020), 107381, 19.
- [HK26] Jan Hubička and Matěj Konečný, *Twenty years of Nešetřil’s classification programme of Ramsey classes*, Comput. Sci. Rev. **59** (2026), Paper No. 100814, 39.
- [HM79] Roger E. Howe and Calvin C. Moore, *Asymptotic properties of unitary representations*, J. Functional Analysis **32** (1979), no. 1, 72–96.
- [HM25] Fabien Hoareau and François Le Maître, *Spatial models for boolean actions in the infinite measure-preserving setup*, 2025.
- [HN08] Jan Hubička and Jaroslav Nešetřil, *A finite presentation of the rational Urysohn space*, Topology Appl. **155** (2008), no. 14, 1483–1492.
- [Hod93] Wilfrid Hodges, *Model theory*, Encyclopedia of Mathematics and Its Applications, vol. 42, Cambridge University Press, Cambridge, 1993.
- [HR70] Edwin Hewitt and Kenneth A. Ross, *Abstract harmonic analysis. Vol. II: Structure and analysis for compact groups. Analysis on locally compact Abelian groups*, Die Grundlehren Der Mathematischen Wissenschaften, vol. Band 152, Springer-Verlag, New York-Berlin, 1970.
- [HR19] Cyril Houdayer and Sven Raum, *Locally compact groups acting on trees, the type I conjecture and non-amenable von Neumann algebras*, Comment. Math. Helv. **94** (2019), no. 1, 185–219.
- [Iba16] Tomás Ibarlucía, *The dynamical hierarchy for Roelcke precompact Polish groups*, Israel J. Math. **215** (2016), no. 2, 965–1009.

-
- [Iba21] ———, *Infinite-dimensional Polish groups and Property (T)*, Invent. Math. **223** (2021), no. 2, 725–757.
 - [JJ25] Colin Jahel and Matthieu Joseph, *Stabilizers for ergodic actions and invariant random expansions of non-Archimedean Polish groups*, Int. Math. Res. Not. IMRN (2025), no. 9, Paper No. rnaf111, 22.
 - [JS91] André Joyal and Ross Street, *An introduction to Tannaka duality and quantum groups*, Category Theory (Como, 1990), Lecture Notes in Math., vol. 1488, Springer, Berlin, 1991, pp. 413–492.
 - [JT22] Colin Jahel and Todor Tsankov, *Invariant measures on products and on the space of linear orders*, J. Éc. polytech. Math. **9** (2022), 155–176.
 - [Kal05] Olav Kallenberg, *Probabilistic symmetries and invariance principles*, Probability and Its Applications (New York), Springer, New York, 2005.
 - [Kal21] ———, *Foundations of modern probability*, third ed., Probability Theory and Stochastic Modelling, vol. 99, Springer, Cham, 2021.
 - [Kat88] M. Katětov, *On universal metric spaces*, General Topology and Its Relations to Modern Analysis and Algebra, VI (Prague, 1986), Res. Exp. Math., vol. 16, Heldermann, Berlin, 1988, pp. 323–330.
 - [Kec95] Alexander S. Kechris, *Classical descriptive set theory*, Graduate Texts in Mathematics, vol. 156, Springer-Verlag, New York, 1995.
 - [Kre41] Mark Krein, *On positive functionals on almost periodic functions*, C. R. (Doklady) Acad. Sci. URSS (N.S.) **30** (1941), 9–12.
 - [Kru16] Alex Kruckman, *Infinitary Limits of Finite Structures*, ProQuest LLC, Ann Arbor, MI, 2016.
 - [KSW25] Aleksandra Kwiatkowska, Rob Sullivan, and Jeroen Winkel, *Extensive embeddings into Fraïssé structures and stationary weak independence relations*, 2025.
 - [KT08] Alexander S. Kechris and Todor Tsankov, *Amenable actions and almost invariant sets*, Proc. Amer. Math. Soc. **136** (2008), no. 2, 687–697.
 - [Lie72] Arthur Lieberman, *The structure of certain unitary representations of infinite symmetric groups*, Trans. Amer. Math. Soc. **164** (1972), 189–198.
 - [LM92] Alexander Lubotzky and Shahar Mozes, *Asymptotic Properties of Unitary Representations of Tree Automorphisms*, Harmonic Analysis and Discrete Potential Theory (Massimo A. Picardello, ed.), Springer US, Boston, MA, 1992, pp. 289–298.
 - [Mac49] George W. Mackey, *Imprimitivity for Representations of Locally Compact Groups I*, Proceedings of the National Academy of Sciences of the United States of America **35** (1949), no. 9, 537–545.

- [Mac76] ———, *The theory of unitary group representations*, Chicago Lectures in Mathematics, University of Chicago Press, Chicago, Ill.-London, 1976.
- [Mac11] Dugald Macpherson, *A survey of homogeneous structures*, Discrete Math. **311** (2011), no. 15, 1599–1634.
- [Meg01] Michael G. Megrelishvili, *Every semitopological semigroup compactification of the group $H_+[0, 1]$ is trivial*, Semigroup Forum **63** (2001), no. 3, 357–370.
- [Mel08] Julien Melleray, *Some geometric and dynamical properties of the Urysohn space*, Topology Appl. **155** (2008), no. 14, 1531–1560.
- [MT11] Dugald Macpherson and Katrin Tent, *Simplicity of some automorphism groups*, J. Algebra **342** (2011), 40–52.
- [Mül16] Isabel Müller, *Fraïssé structures with universal automorphism groups*, J. Algebra **463** (2016), 134–151.
- [Neb99] Claudio Nebbia, *Groups of isometries of a tree and the CCR property*, Rocky Mountain J. Math. **29** (1999), no. 1, 311–316.
- [Ner22] Yury A. Neretin, *Groups of isometries of ultrametric Urysohn spaces and their unitary representations*, 2022.
- [Ol'77] G. I. Ol'sanskiĭ, *Classification of the irreducible representations of the automorphism groups of Bruhat-Tits trees*, Funkcional. Anal. i Priložen. **11** (1977), no. 1, 32–42, 96.
- [Ol'80] G. I. Ol'sanskiĭ, *New "large" groups of type I*, Current problems in mathematics, Vol. 16 (Russian), Itogi Nauki i Tekhniki, Akad. Nauk SSSR, Vsesoyuz. Inst. Nauchn. i Tekhn. Inform., Moscow, 1980, pp. 31–52, 228.
- [Ol'91] G. I. Ol'sanskiĭ, *On semigroups related to infinite-dimensional groups*, Topics in Representation Theory, Adv. Soviet Math., vol. 2, Amer. Math. Soc., Providence, RI, 1991, pp. 67–101.
- [Pes07] Vladimir Pestov, *The isometry group of the Urysohn space as a lévy group*, Topology Appl. **154** (2007), no. 10, 2173–2184.
- [Pon34] L. Pontrjagin, *The theory of topological commutative groups*, Ann. of Math. (2) **35** (1934), no. 2, 361–388.
- [PW27] F. Peter and H. Weyl, *Die Vollständigkeit der primitiven Darstellungen einer geschlossenen kontinuierlichen Gruppe*, Math. Ann. **97** (1927), no. 1, 737–755.
- [RD81] Walter Roelcke and Susanne Dierolf, *Uniform structures on topological groups and their quotients*, Advanced Book Program, McGraw-Hill International Book Co., New York, 1981.

- [Ros22] Christian Rosendal, *Coarse geometry of topological groups*, Cambridge Tracts in Mathematics, vol. 223, Cambridge University Press, Cambridge, 2022.
- [Ryl57] C. Ryll-Nardzewski, *On stationary sequences of random variables and the de Finetti's equivalence*, Colloq. Math. **4** (1957), 149–156.
- [Sem23] Lancelot Semal, *Irreducibly represented Lie groups and Nebbia's CCR conjecture on trees*, 2023.
- [Slu12] Konstantin Slutsky, *Non-genericity phenomena in ordered Fraïssé classes*, J. Symbolic Logic **77** (2012), no. 3, 987–1010.
- [Smi17] Simon M. Smith, *A product for permutation groups and topological groups*, Duke Math. J. **166** (2017), no. 15, 2965–2999.
- [SR72] Neantro Saavedra Rivano, *Catégories Tannakiennes*, Lecture Notes in Mathematics, vol. Vol. 265, Springer-Verlag, Berlin-New York, 1972.
- [SZ20] Garrett Stuck and Robert J. Zimmer, *Stabilizers for ergodic actions of higher rank semisimple groups*, Group Actions in Ergodic Theory, Geometry, and Topology—Selected Papers, Univ. Chicago Press, Chicago, IL, 2020, pp. 339–363.
- [Tak02] M. Takesaki, *Theory of operator algebras. I*, Encyclopaedia of Mathematical Sciences, vol. 124, Springer-Verlag, Berlin, 2002.
- [Tan39] Tadao Tannaka, *über den Dualitätssatz der nichtkommutativen topologischen Gruppen*, Tohoku Math. J. **45** (1939), 1–12.
- [Tho68] Elmar Thoma, *Eine Charakterisierung diskreter Gruppen vom Typ I*, Invent. Math. **6** (1968), 190–196.
- [Tsa12] Todor Tsankov, *Unitary representations of oligomorphic groups*, Geom. Funct. Anal. **22** (2012), no. 2, 528–555.
- [Tsa25] ———, *Non-singular and probability measure-preserving actions of infinite permutation groups*, Int. Math. Res. Not. IMRN (2025), no. 23, Paper No. rnaf345, 15.
- [TZ13a] Katrin Tent and Martin Ziegler, *The isometry group of the bounded Urysohn space is simple*, Bull. Lond. Math. Soc. **45** (2013), no. 5, 1026–1030.
- [TZ13b] ———, *On the isometry group of the Urysohn space*, J. Lond. Math. Soc. (2) **87** (2013), no. 1, 289–303.
- [Ury27] Pavel Urysohn, *Sur un espace métrique universel*, Bull. Sci. Math **51** (1927), no. 2, 43–64.
- [Usp90] V. V. Uspenskij, *On the group of isometries of the Urysohn universal metric space*, Comment. Math. Univ. Carolin. **31** (1990), no. 1, 181–182.

- [Usp98] ———, *The Roelcke compactification of unitary groups*, Abelian Groups, Module Theory, and Topology (Padua, 1997), Lecture Notes in Pure and Appl. Math., vol. 201, Dekker, New York, 1998, pp. 411–419.
- [Usp01] ———, *The Roelcke compactification of groups of homeomorphisms*, Topology and Its Applications, vol. 111, 2001, pp. 195–205.
- [Vai15] Leonid Vainerman, *Tannaka-Krein duality for compact quantum group coactions (survey)*, Methods Funct. Anal. Topology **21** (2015), no. 3, 282–298.
- [van35] E. R. van Kampen, *Locally bicomact abelian groups and their character groups*, Ann. of Math. (2) **36** (1935), no. 2, 448–463.
- [Zie21] Joseph Zielinski, *Locally Roelcke precompact Polish groups*, Groups Geom. Dyn. **15** (2021), no. 4, 1175–1196.